

Technical Proof Supplement to Positive Recurrence of Weakly Reversible Binary Reaction Networks with Three Species and Two Linkage Classes

12 August 2026

Abstract

This supplement contains the complete local analytic theorems used by the main article. Each section is a mechanical LaTeX rendering of a byte-frozen Markdown proof that received an independent hostile audit. The SHA-256 manifest shipped with the source package is checked before compilation. The finite executable certificate is used only for finite support, tier, affine, and disjoint-set identities; no stochastic estimate is obtained by enumerating orientations, reaction histories, or population boxes.

Editorial supersession note. The forty authenticated fragments are chronological proof artifacts. Local headers and scope notes saying “pending,” “uncertified,” or “claim false” record the state of the program when that fragment was frozen; they are not the current global verdict. Sections 39–40 and the main article give the completed composition and supersede those historical status statements. Local counterexamples and explicit claim boundaries remain valid at their stated scopes.

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Part I

One active linkage

1 The hard 181 enabled-promotion access-word theorem

Proof-first scoped corollary, 2026-08-12 PDT. Audit status: pending. This note applies the already audited all-reaction bounded-access lemma to the enabled two-active promotion rows on the hard 333 pairs. The finite table checks only that these rows form a subset of the lemma’s 929-row analytic domain.

1.1 1. Generic analytic lemma

Fix a descriptor sequence and let $A_n \rightarrow \infty$ be the common propensity scale of its global top D-tier. Suppose a bounded directed physical word has an enabled first source, all prescribed source propensities comparable with A_n , every preterminal target in the exact top D-tier, and a terminal edge with factorial gap $g_n \rightarrow \infty$. Stop at completion or at the first competing physical reaction.

Retaining all clocks gives a uniform success probability $p_0 > 0$, physical duration $O(A_n^{-1})$, and arbitrary fixed positive endpoint moments. For every fixed correction ℓ ,

$$\mathbb{E}\Delta G_\ell \leq -p_0 g_n + O(1), \quad \sup_n \mathbb{E}\{(\Delta G_\ell)^+\}^r < \infty. \quad (1.1)$$

The all-competitor estimate uses

$$\frac{b}{A} \{1 + \log(A/b)\}^r \leq C_r, \quad (1.2)$$

so no deleted clock or selected-word probability approximation occurs. The exact fourth-power expansion therefore yields

$$\mathbb{E}_x[W_\ell(X_\tau) - W_\ell(x) + \tau] \leq -cG_\ell(x)^3 g_n. \quad (1.3)$$

This lemma, including the arbitrary- ℓ and endpoint-moment quantifiers, was independently audited as Lemma 3.1 of the easy-943 common potential theorem.

1.2 2. Hard-family specialization

Exactly 181 two-active failures on the hard 333 pairs have an enabled source in the proper top-D subset. They are among the 929 seeded rows in the audited lemma. Strong connectivity gives a directed path inside the top-D subset from the enabled source to a top node whose forced outgoing edge leaves that proper subset. Taking a simple path gives bounded length. Every source on it lies in the same exact top D-tier; hence its propensity is comparable with A_n . The final outgoing edge has the divergent factorial gap to the lower tier. These are precisely the four hypotheses of Section 1.

1.2.1 Theorem 2.1

For every one of the 181 hard enabled-promotion incidences, every strong orientation, every fixed positive rate vector, and every fixed common correction ℓ , the raw physical access-word stopping rule satisfies (1.3), arbitrary fixed endpoint moments, and physical-time moments. Its actual endpoint is reclassified under the same W_ℓ .

The corollary is local. It makes no pair or global claim before independent audit and full descriptor composition.

2 The exceptional two-species single-linkage service theorem

Proof-first theorem, 2026-08-12 PDT. This note closes the unique two-dynamic-species support left outside the published pure-multiple and deficiency-zero theorems. The proof is uniform over every strongly connected orientation and every fixed positive rate vector. It uses no orientation or population enumeration.

2.1 1. The theorem

Let

$$\mathcal{C} = \{0, B, 2B, A + B\} \quad (1.1)$$

carry an arbitrary strongly connected directed reaction graph, with arbitrary positive stochastic mass-action rate constants on its edges. Let $X = (A, B)$ be the resulting CTMC.

Theorem 1.1. The chain is nonexplosive and positive recurrent on every closed irreducible population class.

Together with the structural reduction in `proof_first_single_linkage_scope_and_gap.md`, Theorem 1.1 proves the desired one-active-linkage theorem for every classwise projection with at most two dynamic species.

Put

$$V(a, b) = v(a) + v(b), \quad v(m) = m(\log m - 1) + 1, \quad (1.2)$$

with $0 \log 0 = 0$. This is the nonnegative proper entropy used in the Anderson–Kim continuous-time tier theorem.

2.2 2. Generator descent away from the missing-cofactor face

We use the following standard entropy tier estimate. It is included here to make the interface precise.

Lemma 2.1 (top-source entropy estimate). For a fixed finite binary weakly reversible network, suppose an unbounded state sequence has a source complex y which is simultaneously in the top finite S-tier and the top D-tier, and has a reaction $y \rightarrow y'$ with y' strictly below the top D-tier. If every top S-tier complex is in the top D-tier, then $\mathcal{L}V(x_n) \rightarrow -\infty$.

For completeness, the bounded-jump entropy estimate gives

$$V(x + y' - y) - V(x) \leq \log \frac{(x \vee 1)^{y'}}{(x \vee 1)^y} + O(1). \quad (2.1)$$

After division by the maximal propensity, upward lower-tier terms are bounded using $r \log(1/r) \rightarrow 0$, within-tier terms are bounded, and the displayed descending term tends to $-\infty$. This is exactly the proof of the top-S descending-source theorem of Anderson and Kim, *Some network conditions for positive recurrence of stochastically modeled reaction networks*, Theorem 4.2 (arXiv:1710.11263); no structural corollary of that paper is being invoked here.

We now verify its premise analytically for every unbounded sequence with $b \geq 1$. Pass to a subsequence. If $b \rightarrow \infty$, then:

- when a is bounded, $2B$ is the unique top D- and S-complex;
- when $a \rightarrow \infty$, comparison of a/b shows that the common top D- and S-set is $\{2B\}$, $\{A + B\}$, or their pair.

If b is bounded, make it a fixed positive integer. Then necessarily $a \rightarrow \infty$, and $A+B$ is the unique top D- and S-complex. In every case the common top set is a nonempty proper subset of (1.1). Strong connectivity supplies an edge from that set to its complement. Lemma 2.1 therefore gives

$$\lim_{n \rightarrow \infty} \mathcal{L}V(a_n, b_n) = -\infty \quad \text{whenever } a_n + b_n \rightarrow \infty \text{ and } b_n \geq 1. \quad (2.2)$$

Consequently there is a finite set F_1 such that

$$\mathcal{L}V(a, b) \leq -2 \quad (2.3)$$

for every $(a, b) \notin F_1$ with $b \geq 1$.

2.3 3. The origin-launch service block on $B = 0$

Write $O = 0$, $Q = 2B$, and $T = A + B$. Let $\lambda_O > 0$ be the sum of the outgoing rate constants at O , and let $\lambda_T > 0$ be the corresponding sum at T . Both are positive by strong connectivity.

Start at $(A, B) = (n, 0)$. At $B = 0$, only O is enabled. Its first jump has target B, Q , or T .

- If it lands at B or Q , wait for the next T -sourced reaction.
- If it lands at T , wait for the next T -sourced reaction. A landing at O is a neutral return and starts a new attempt. A landing at B or Q is followed by one more T -sourced reaction.

At every positive- B stage, declare the first firing from a lower source O, B, Q to be an included **defect**. Thus all reaction clocks remain in the physical chain. On the complementary clean event, the terminal T -reaction is an unpaired exit and lowers A from n to $n - 1$.

Ignore defects for one moment. The probability that an attempt produces a positive- B lower landing before its neutral return is

$$p_* = \frac{\sum_{O \rightarrow B, Q} \kappa_{Oz}}{\lambda_O} + \frac{\kappa_{OT}}{\lambda_O} \frac{\sum_{T \rightarrow B, Q} \kappa_{Tz}}{\lambda_T}. \quad (3.1)$$

Here a missing edge contributes zero. We have $p_* > 0$. Otherwise every outgoing O -edge would land at T , every outgoing T -edge would land at O , and $\{O, T\}$ would be a closed proper subset of the reaction graph, contradicting strong connectivity. Hence the clean auxiliary attempt count is geometric with parameter p_* and has moments of every order.

At a state (a, b) with $b > 0$, the aggregate T -exit rate is

$$\lambda_T ab, \quad (3.2)$$

whereas the aggregate lower-source propensity is at most $C(1 + b^2)$. Thus the exact relevant-clock race satisfies

$$\mathbb{P}\{\text{defect before the next } T\text{-exit} \mid (a, b)\} \leq \frac{C(1 + b)}{a}. \quad (3.3)$$

Before termination, $a \in \{n, n + 1\}$, and every clean intermediate B -value is at most two. At a clean service endpoint $B \leq 3$. The geometric attempt bound and (3.3), including any fixed polynomial weight in the attempt count or centered post-jump displacement, therefore give

$$\mathbb{P}_n(D) \geq 1 - C/n.$$

$$\mathbb{E}_n[(1 + R)^r; E] \leq C_r/n, \quad r < \infty. \quad (3.4)$$

$$\mathbb{E}_n \tau^r \leq C_r, \quad r < \infty. \quad (3.5)$$

$$A_\tau = n - 1, \quad B_\tau \leq 3 \quad \text{on } D. \quad (3.6)$$

Here R contains the attempt count and the centered population displacement, E is the included first defect, $D = E^c$, and τ is the actual physical stopping time. Equation (3.5) follows because each visit to $B = 0$ waits an exponential time of rate λ_O , while every positive- B race is faster; the number of attempts is geometric.

On D , the mean-value theorem and (3.6) give

$$V(X_\tau) - V(n, 0) \leq -\log n + C. \quad (3.7)$$

On E , the included causing jump and the geometric number of preceding neutral attempts give

$$\mathbb{E}_n[|V(X_\tau) - V(n, 0)|; E] \leq C \frac{\log(n+1)}{n}. \quad (3.8)$$

Combining (3.4)–(3.8) proves the stopped physical-time drift

$$\mathbb{E}_{(n,0)}[V(X_\tau) - V(n, 0) + \tau] \leq -\frac{1}{2} \log n \quad (3.9)$$

for all sufficiently large n . Notice that designated O -launches are not suppressed lower clocks. Only a lower-source firing which competes while T is enabled is labelled a defect, and that firing is included.

2.4 4. Physical-time Foster composition

Increase a finite set $F \supseteq F_1$ so that it contains every $(n, 0)$ for which (3.9) has not yet taken effect. Starting outside F , define one sampling episode as follows.

1. If $B = 0$, use the physical stopping time in Section 3.
2. If $B \geq 1$, wait for the next ordinary reaction time J .

In the second case, if $q(x)$ is the total propensity, (2.3) gives

$$\mathbb{E}_x[V(X_J) - V(x) + J] = \frac{\mathcal{L}V(x) + 1}{q(x)} \leq -\frac{1}{q(x)} < 0. \quad (4.1)$$

In the first case, (3.9) is at most -1 after enlarging F . Iterate these state-dependent episodes until F is hit. At the successive episode endpoints, $V(X) + \text{elapsed physical time}$ is a nonnegative supermartingale. Hence, for every truncation at m episodes,

$$\mathbb{E}_x S_{m \wedge N_F} \leq V(x). \quad (4.2)$$

Every episode contains at least one actual reaction. The process is nonexplosive: a bimolecular-source reaction cannot increase total molecular population in a binary network, while the aggregate population-increasing rate of degree-zero and degree-one sources is at most $C(1 + A + B)$; bounded jumps and a Yule comparison suffice. Therefore infinitely many episodes cannot accumulate in finite physical time. Letting $m \rightarrow \infty$ in (4.2) proves

$$\mathbb{E}_x T_F \leq V(x) < \infty. \quad (4.3)$$

The construction uses only physical reactions and so remains inside the fixed closed irreducible class. Thus each such class meets the finite set F . The standard finite-set return argument (or the countable-state random-time Foster theorem applied to (4.2)) gives a finite mean return time and hence positive recurrence on that class. Absorbing singleton classes are positive recurrent trivially. This proves Theorem 1.1. $\square$

2.5 5. Certification boundary

This theorem closes the exact support (1.1), and therefore every projected one-linkage branch with at most two dynamic species. It does not close the genuinely three-dynamic-species carrier supports. Those require a separate structural carrier-service theorem; no global flag is changed here.

3 Structural reduction for the three-species single-linkage branch

Proof-first note, 2026-08-12 PDT. Status: partial theorem; carrier lemma open. This note gives a non-enumerative classification of every failure of the Anderson–Cappelletti–Kim source/deterministic-tier inclusion in a classwise reduced binary network with one weakly reversible linkage and at most three dynamic species. It also states the exact physical stopped lemma which would finish the branch. The classification proves that the missing theorem cannot be reduced only to one-active faces: a full-rank, two-active counterexample to that reduction is exhibited below.

This is not a counterexample to positive recurrence. The exceptional two-species face is already closed by `proof_first_single_linkage_2d_exception_service_theorem.md`. Section 5 closes the balanced three-species carrier, and Section 6 isolates the remaining separated-scale carrier problem. No atlas, pair, or global certification flag is changed here.

3.1 1. Reduced classwise setting

Fix a closed irreducible population class and perform the exact reduction used in `proof_first_global_t3_2_classwise_composition.md`: delete constant coordinates, delete linkages with no enabled source on the class, and merge projected linkages which share a projected complex. Suppose that the remaining network has one strongly connected linkage, positive fixed rate constants, at most three dynamic species, and complexes of molecularity at most two.

Let x_n be a proper tier sequence in this class. Write

$$D^1 = T_{\{x_n\}}^{D,1}, \quad S^1 = T_{\{x_n\}}^{S,1}, \quad E = \{y : x_n \geq y \text{ eventually}\}. \quad (1.1)$$

After taking a subsequence, every bounded integer coordinate is constant and every source is either eventually enabled or eventually disabled.

3.2 2. The enabled-top criterion

Lemma 2.1. Let $D^1, D^2, \dots$ be the deterministic tiers in decreasing order and let

$$j_* = \min\{j : D^j \cap E \neq \emptyset\}.$$

Then

$$S^1 = D^{j_*} \cap E. \quad (2.1)$$

In particular,

$$S^1 \subseteq D^1 \iff D^1 \cap E \neq \emptyset. \quad (2.2)$$

3.2.1 Proof

For an enabled complex y , stochastic mass action gives

$$\frac{\lambda_y(x_n)}{(x_n \vee 1)^y} = \prod_i \frac{(x_{n,i})_{y_i}}{(x_{n,i} \vee 1)^{y_i}} \longrightarrow c_y, \quad 0 < c_y < \infty. \quad (2.3)$$

For a disabled source the propensity is identically zero. Thus the stochastic order is precisely the deterministic order restricted to the enabled complexes, which proves (2.1) and (2.2). $\square$

This elementary identity locates the entire gap in the published single-linkage argument: its tier inclusion fails exactly when **every** top deterministic complex is disabled. No reaction orientation enters Lemma 2.1.

The top deterministic tier is a proper subset of the complex set. To see this, suppose instead that all complexes were D-equivalent. Put $\ell_n = \log(x_n \vee 1)$, normalize by $\|\ell_n\|$, and pass to a nonzero limit $w \geq 0$. D-equivalence gives $w \cdot (y - z) = 0$ for every two complexes and hence for every reaction vector. Therefore $w \cdot x$ is constant on the fixed stoichiometric class. But a positive component of w belongs to a divergent coordinate, so $w \cdot x_n \rightarrow \infty$, a contradiction. Strong connectivity thus always supplies an edge from the top D-tier to its complement; the issue is whether an enabled top source can take such an edge.

3.3 3. Binary shape of every obstruction

Because the classwise reduction has deleted constant species, every divergent dynamic coordinate occurs in a complex. Consequently

$$\max_y (x_n \vee 1)^y \longrightarrow \infty. \quad (3.1)$$

If an unavailable binary complex has a divergent deterministic monomial, it cannot be $0, S_i$, or $2S_i$: disabling a pure complex forces its monomial, computed with $x \vee 1$, to equal one. It must therefore be

$$S_i + S_j, \quad (3.2)$$

with exactly one of the two populations eventually zero and the other divergent. Lemma 2.1 proves the following theorem.

Theorem 3.1 (structural obstruction theorem). A proper tier sequence of a classwise reduced binary one-linkage network violates $S^1 \subseteq D^1$ if and only if every member of D^1 is a disabled mixed complex of the form (3.2). Each such top complex contains one zero-population cofactor and one divergent carrier species.

This recovers the pure-multiple premise in the published theorem. If a pure complex S_i or $2S_i$ is present for each divergent species, then a disabled $S_i + S_j$ cannot be strictly above all enabled sources: S_i ties its divergent factor or $2S_i$ dominates it.

There is a useful orientation distinction. For a fixed strong graph and top set $K = D^1$, the one-step descending-source criterion holds exactly when the graph contains

$$y \longrightarrow z, \quad y \in K \cap E, \quad z \notin K. \quad (3.3)$$

If arbitrary strong orientations on a fixed vertex support are quantified, (3.3) is forced for every orientation precisely when $K \subseteq E$. Sufficiency is the first exit from K . For necessity, order a directed Hamiltonian cycle so that a chosen disabled top vertex is the sole exit from the consecutive top block. This comparison explains why the orientation-uniform direct criterion is stricter than (2.2).

3.4 4. Exact three-species obstruction families

Relabel a failed tier so that the zero cofactor is $C = 0$.

3.4.1 4.1 One divergent species

If only $A \rightarrow \infty$, then every top complex is among $A+B, A+C$, with the displayed cofactor equal to zero. The pure complexes $A, 2A$ are absent. This is the one-active carrier boundary already targeted by the stopped service theorems in the hard-family library.

3.4.2 4.2 Two divergent species

Let $A, B \rightarrow \infty$ and $C = 0$. Then

$$\emptyset \neq D^1 \subseteq \{A+C, B+C\}. \quad (4.1)$$

There are three symbolic regimes.

1. If $D^1 = \{A+C, B+C\}$, then $A/B \rightarrow c \in (0, \infty)$, and

$$\mathcal{C} \subseteq \{0, C, 2C, A+C, B+C\}, \quad (4.2)$$

with both mixed complexes present. Indeed, A or B would be an enabled top complex, while $2A, A+B, 2B$ would dominate.

2. If $D^1 = \{A + C\}$, then $A/B \rightarrow \infty$ and

$$\mathcal{C} \subseteq \{0, C, 2C, B, 2B, B + C, A + C\}. \quad (4.3)$$

Any present $2B$ satisfies $B^2/A \rightarrow 0$; equality would make it an enabled top complex and divergence would contradict the chosen top.

3. The case $D^1 = \{B + C\}$ is symmetric.

If all three populations diverge, every complex is enabled and Lemma 2.1 closes the tier. Thus (4.2)–(4.3) are the only genuinely new qualitative faces beyond the one-active obstruction.

3.4.3 4.3 A full-rank counterexample to one-active reduction

Take

$$\mathcal{C} = \{0, C, A + C, B + C\}, \quad \text{quad} \quad 0 \rightarrow A + C \rightarrow B + C \rightarrow C \rightarrow 0. \quad (4.4)$$

The support is binary, one-linkage, weakly reversible, and has stoichiometric rank three. The CTMC is irreducible on $\mathbb{Z}_{\geq 0}^3$. If the four reactions are $r_1, \dots, r_4$, the following executable words have the indicated net effects:

$$\begin{array}{c|cl} r_1 r_4 & +A & \\ r_1 r_2 r_4 & +B & \\ r_1 r_2 r_3 & +C & \\ r_1 r_2^2 r_3^2 r_4 & -A & (A > 0) \\ r_1 r_2 r_3^2 r_4 & -B & (B > 0) \\ r_4 & -C & (C > 0). \end{array} \quad (4.5)$$

Thus $x_n = (n, n, 0)$ lies in a single closed irreducible class. Its deterministic monomials are $1, 1, n, n$, whereas only the zero source is enabled, so

$$D^1 = \{A + C, B + C\}, \quad S^1 = \{0\}. \quad (4.6)$$

The only D-descending edge in (4.4) is disabled. Equations (4.4)–(4.6) disprove the proposed claim that every tier obstruction has exactly one divergent species. They do not disprove recurrence.

3.5 5. The balanced mesoscopic carrier theorem

The balanced family (4.2) can be closed without enumerating its supports or orientations.

For a fixed $\ell \in \mathbb{R}^3$, choose K_ℓ so that

$$G_\ell(z) = K_\ell + \sum_{i=A,B,C} \log(z_i!) + \ell \cdot z \geq 1, \quad W_\ell(z) = G_\ell(z)^4 \quad (5.0)$$

on the nonnegative lattice. The stopping rule below is independent of ℓ .

Theorem 5.1 (balanced all-clock carrier service). Fix a support

$$\{A + C, B + C\} \subseteq \mathcal{C} \subseteq \{0, C, 2C, A + C, B + C\} \quad (5.1)$$

with an arbitrary strongly connected directed graph and arbitrary positive fixed rates, and fix one closed irreducible class Γ . Fix $\epsilon > 0$, and suppose Γ contains states

$$x = (a, b, 0), \quad N = a + b, \quad a \wedge b \geq \epsilon N, \quad (5.2)$$

with arbitrarily large N . From every sufficiently large such state there is an included physical stopping time τ such that, for every fixed p, M ,

$$\begin{aligned}\mathbb{P}(D^c) &\leq C/N + C_M N^{-M}, \\ \mathbb{E}[(1 + |X_\tau - x| + \tau)^p; E] &\leq C_p/N, \\ \mathbb{E}[(1 + |X_\tau - x| + \tau)^p; B] &\leq C_{p,M} N^{-M}, \\ \mathbb{E}(1 + |X_\tau - x| + \tau)^p &\leq C_p.\end{aligned}\tag{5.3}$$

Here D is a net service with $A_\tau + B_\tau = N - 1$, E is the included first pure-source competitor during an open carrier window, and B is an included localization boundary. For every fixed ℓ , the actual endpoint obeys

$$\mathbb{E}_x[G_\ell(X_\tau) - G_\ell(x)] \leq -\log N + O_{\epsilon,\ell}(1),\tag{5.4}$$

and

$$\mathbb{E}_x[W_\ell(X_\tau) - W_\ell(x) + \tau] \leq -cG_\ell(x)^3 \log N.\tag{5.5}$$

All constants may depend on the fixed support, reaction graph, rate constants, ϵ, p, M, ℓ , as applicable, but not on a, b, N .

3.5.1 Proof

If $0 \notin \mathcal{C}$, no source is enabled on $C = 0$, so such a state is static and its irreducible class is a singleton, contrary to the hypothesis. Suppose $0 \in \mathcal{C}$. Put

$$\mathcal{K} = \{A + C, B + C\}, \quad \mathcal{P} = \mathcal{C} \cap \{C, 2C\}.\tag{5.6}$$

If $\mathcal{P} = \emptyset$, every reaction preserves $A + B - C$, because that linear functional is zero on every complex in $\{0, A + C, B + C\}$. On $C = 0$, $A + B$ is therefore fixed by the stoichiometric class, again excluding (5.2) along a divergent fixed-class sequence. Hence a relevant balanced face has $\mathcal{P} \neq \emptyset$.

Construct an auxiliary attempt as follows. At $C = 0$, retain the actual first 0-sourced reaction. While $C > 0$, retain only reactions sourced in $\mathcal{K}$. Stop the attempt when either $C = 0$ with no net loss of $A + B$, or $A + B = N - 1$.

During this auxiliary attempt an internal $\mathcal{K}$ -to- $\mathcal{K}$ reaction preserves both C and $A + B$. A $\mathcal{K}$ -to- $\{0, C, 2C\}$ exit lowers $A + B$ by one. The launch raises $A + B$ by zero or one. Consequently a pure launch followed by one mixed exit is a service. A mixed launch followed by a positive-pure exit and one further mixed exit is a service. A mixed launch followed by an exit to 0 is a neutral attempt and restarts at $C = 0$. Before service or neutral return there are at most two mixed exits and

$$1 \leq C \leq 3.\tag{5.7}$$

Localize when the total number of internal mixed reactions and neutral attempts first reaches

$$L_N = \lfloor N^{1/4} \rfloor,\tag{5.8}$$

including the causing reaction. Before this cutoff, $A \wedge B \geq \epsilon N/2$ for all large N . At every open state the aggregate mixed rate is therefore at least $c_\epsilon NC$. Strong connectivity supplies an edge from the proper block $\mathcal{K}$ to its complement. Since both mixed populations are at least $\epsilon N/2$, the conditional probability that the next mixed firing leaves $\mathcal{K}$ is bounded below by a fixed $\delta_\epsilon > 0$. Thus the number of internal mixed firings has a geometric tail.

There is also a fixed positive probability of a successful attempt. If an outgoing 0-edge has target in $\mathcal{P}$, its fixed launch probability already gives this. Otherwise, strong connectivity of the full support forces an edge from $\mathcal{K}$ to $\mathcal{P}$; its conditional probability is bounded below by the same balanced-population comparison. Hence the number of neutral attempts is geometric with a fixed parameter. It follows that the auxiliary history length has an exponential tail and

$$\mathbb{P}\{\text{auxiliary length} \geq L_N\} \leq C e^{-cL_N}.\tag{5.9}$$

Restore every physical clock. By (5.7), the total propensity of sources in $\{0, C, 2C\}$ during an open window is bounded above by a constant, whereas the mixed propensity is at least $c_\epsilon N$. The exact race probability of a pure-source competitor is therefore $O(N^{-1})$ per mixed race. Stop at and include its actual firing. Summing this bound over the exponentially-tailed auxiliary history, with any fixed polynomial bias in the history length and centered endpoint displacement, proves the second line of (5.3), while (5.9) proves the third. At $C = 0$, each neutral attempt waits an exponential time with the fixed aggregate 0-source rate; open waits are faster than an exponential clock of rate $c_\epsilon N$. Geometric-sum moment formulas prove the last line of (5.3). The first line then follows from the exhaustive clean-service/defect/boundary partition. All estimates are for the actual post-jump endpoint.

For definiteness the event labels are applied in the order: included pure-source competitor E , localization boundary B , then clean service D . In fact the causing reactions make these events disjoint, but this priority removes any convention at a simultaneous bookkeeping threshold.

On D , put $\Delta_A = A_\tau - a$ and $\Delta_B = B_\tau - b$. Then

$$\Delta_A + \Delta_B = -1, \quad \mathbb{E}(1 + |\Delta_A| + |\Delta_B|)^p \leq C_p. \quad (5.10)$$

Because $a, b \asymp_\epsilon N$, the factorial finite difference and (5.10) give

$$\log \frac{A_\tau! B_\tau!}{a! b!} = \Delta_A \log a + \Delta_B \log b + O_\epsilon(1 + |\Delta|^2/N) \leq -\log N + O_\epsilon(1 + |\Delta|). \quad (5.11)$$

The terminal C -population is bounded by three on the clean event. The fixed linear correction has increment $O_\ell(|\Delta| + 1)$. The defect contribution to every fixed moment of ΔG_ℓ is $O(N^{-1}(\log N)^p)$, and the boundary contribution is smaller than every power by (5.9). Equations (5.10)–(5.11) prove (5.4) and

$$\mathbb{E}|\Delta G_\ell|^p \leq C_p(\log N)^p. \quad (5.12)$$

Finally $G_\ell(x) \asymp N \log N$. Insert (5.4) and (5.12) in the exact fourth-power identity. Its quadratic, cubic, and quartic remainders are lower order than $G_\ell(x)^3 \log N$, and the bounded duration moments are lower order as well. This proves (5.5). $\square$

The proof uses only the block structure in (5.6), strong connectivity, and the balanced cone in (5.2). It does not inspect an orientation list or select one prescribed physical reaction word.

3.6 6. The exact missing stopped theorem

The structural theorem and Theorem 5.1 leave only the separated-scale families (4.3) and its symmetric counterpart. It is important not to overstate the missing estimate. In particular, requiring order $a + b$ clean services with probability $1 - O((a + b)^{-1})$ is neither needed for Foster drift nor justified: an $O(a^{-1})$ clock accumulated through order a windows need not remain rare.

Here is the exact sufficient random-time contract. It deliberately puts all physical clocks into the stopped kernel rather than declaring a “first defect” to be negligible.

Separated-scale carrier contract (open). Fix a binary support in (4.3), an arbitrary strongly connected reaction graph, arbitrary positive fixed rates, and one closed irreducible class. Let

$$x_n = (a_n, b_n, 0), \quad a_n, b_n \rightarrow \infty, \quad \frac{b_n}{a_n} \rightarrow 0, \quad (6.1)$$

be actual states in that one class. If $2B$ is a complex, also

$$\frac{b_n^2}{a_n} \rightarrow 0. \quad (6.2)$$

For every fixed $\ell \in \mathbb{R}^3$, put

$$G_\ell(x) = K_\ell + \sum_i \log(x_i!) + \ell \cdot x \geq 1, \quad W_\ell = G_\ell^4. \quad (6.3)$$

There must exist one statewise stopping rule $x \mapsto \tau_x$ on this chart, independent of ℓ . It is all-clock and finite almost surely, and its terminal state is the actual post-reaction state (every service, return, promotion, and localization cutoff includes the reaction which first crosses it). Along every sequence (6.1)–(6.2) there is a function $h_n \rightarrow \infty$ such that, with $\tau_n = \tau_{x_n}$,

$$\mathbb{E}_{x_n}[W_\ell(X_{\tau_n}) - W_\ell(x_n) + \tau_n] \leq -c_\ell G_\ell(x_n)^3 h_n. \quad (6.4)$$

The same physical rule for τ_x must work for every fixed ℓ (only the harmless constants may depend on ℓ), and, for every fixed p , it must have uniform polynomial endpoint and duration integrability

$$\mathbb{E}_{x_n}(1 + |X_{\tau_n} - x_n| + \tau_n)^p \leq C_p(1 + a_n + b_n)^{q_p}. \quad (6.5)$$

A terminal cofactor-free promotion may instead be handed to an already proved drift chart, but only at its actual post-jump endpoint and with its elapsed time retained. A cutoff crossed with $C > 0$ is an open boundary and must be estimated inside (6.4), not contracted to a fictitious return. If the cofactor-free trace cannot produce positive net old service, the alternative conclusion must be an exact class invariant/no-history statement or containment in the finite classwise target.

The service count in a prospective proof is canonical. In (4.3) the only complex containing A is $q = A + C$. If $N_q^{\text{in}}(t)$ and $N_q^{\text{out}}(t)$ count actual reactions entering and leaving q , then

$$N_q^{\text{out}}(t) - N_q^{\text{in}}(t) = a - A_t. \quad (6.6)$$

Thus a K -service boundary is the physical hitting time $A_t = a - K$, including the q -sourced firing which hits it; no marked or virtual service is required. A useful candidate kernel follows the exact chain until this boundary, a physical landing at $C = 0$ where an enabled base monomial becomes comparable to A_t (promotion), or an included open localization boundary. The theorem does **not** prescribe that K grow with a ; a fixed or state-dependent K is admissible if the direct drift (6.4) holds.

There is already an exact invariant/service dichotomy; the gap is not service accessibility. Put

$$\text{cal}F = \mathcal{C} \cap \{0, B, 2B\}, \quad \text{cal}P = \mathcal{C} \cap \{C, 2C, B + C\}. \quad (6.7)$$

Every member of $\mathcal{F}$ is enabled for all large b . If $\mathcal{F} = \emptyset$, the cofactor-free state has no enabled source and its closed class is a singleton, so it cannot realize (6.1).

Lemma 6.1 (invariant or physical service). Suppose $\mathcal{F} \neq \emptyset$. If $\mathcal{P} = \emptyset$, then $A - C$ is an exact stoichiometric invariant, so a cofactor-free state in one fixed class has fixed A and cannot realize (6.1). If $\mathcal{P} \neq \emptyset$, then from every sufficiently large cofactor-free state there is an executable physical reaction history of length at most $|\text{cal}C| + 1$, ending with its causing reaction, for which A has net change -1 .

Indeed, in the first case every complex is in $\mathcal{F} \cup \{q\}$, and the functional $A - C$ is zero on all of them. In the second case choose an enabled $f \in \text{cal}F$ and a simple directed path from f to $\mathcal{P}$. A directed complex path is physically executable: after each firing its target complex is present and enables the next source. Stop the simple path at its first $\mathcal{P}$ -vertex. Before that vertex the path visits q at most once. Entries to and exits from q cancel in A , while the terminal $\mathcal{P}$ -state contains a carrier but no extra A . Fire a nonself outgoing q -reaction; it is enabled by the old A and the carrier, and it lowers A by one. This proves the second alternative. The lemma selects no orientation-specific word in the theorem: it uses only a simple path in an arbitrary strong graph to prove a uniform finite accessibility fact.

Two features prevent importing Theorem 5.1 or the existing one-active theorem verbatim.

1. The second cloud coordinate need not be subpower relative to the first. The tier assumptions give only $B/A \rightarrow 0$, and, when $2B$ is present, $B^2/A \rightarrow 0$.
2. The complexes $B, 2B, B + C$ can generate a quadratic countable base trace before an $A + C$ service window. A crude endpoint estimate loses several factors of $\log B$. Those losses cannot be paid merely from $-\log A$ when, for example, $B = A^{0.49}$.

The scale condition does pay every *individual* source-to-target entropy gap. For example an $A + C \rightarrow 2B$ firing contributes $-\log(A/B^2) + O(1)$, which tends to $-\infty$. The unresolved point is to retain this sourcewise payment through the killed base Green kernel. A path such as

$$0 \longrightarrow B + C, \text{quad} A + C \longrightarrow 2B \quad (6.8)$$

has raw endpoint change $(-1, +3, *)$; bounding only its endpoint would charge $3 \log B$, although the initiating zero-source event is suppressed relative to the quadratic base clock when that clock is present. The needed proof must preserve this propensity weight rather than take a worst-case displacement.

Consequently the precise analytic obstruction is a sourcewise killed Green/Feynman–Kac estimate, uniform in the full mesoscopic range (6.1)–(6.2), which controls the positive factorial endpoint and physical duration while preserving the logarithmic source/target ratios. A direct fluid-scale random-time estimate for the entire carrier trace would be an equivalent replacement. A selected reaction word or an orientation-dependent finite population search cannot establish (6.4).

3.7 7. Conditional classwise composition

Assume the separated-scale carrier contract. Every divergent proper tier sequence is then treated as follows.

- If $D^1 \cap E \neq \emptyset$, use the published source/D-tier theorem or the direct generator estimate.
- If every top-D complex is disabled and only one species diverges, use the existing one-active carrier theorem (including its no-history alternative).
- If two species diverge, Theorem 3.1 and Section 4 put the sequence in (4.2) or (4.3). Theorem 5.1 treats (4.2), while (6.4)–(6.5) treat (4.3).
- If all three diverge, every source is enabled and the first case applies.

All branches use the same physical factorial-linear fourth power. Exact returns may be contracted only analytically; their elapsed time, internal boundary hits, and actual endpoint are retained in the stopped estimate. Reclassification therefore has zero comparison toll. The compactness and random-time Foster argument in Sections 5–7 of `hard333_common_w_fixed_class_theorem.md` then gives finite mean hitting time of a finite target and positive recurrence on the closed class.

Without (6.4)–(6.5), that composition is conditional. The structural classification and balanced carrier theorem are complete, but the three-species one-linkage branch is not.

4 Bounded carrier theorem for two disabled mixed tops

Proof-first theorem, 2026-08-12 PDT. This note closes the one-active single-linkage face with two disabled top complexes. It is independent of the separated-scale operator and uses no support or orientation enumeration.

4.1 1. Setting and theorem

Put

$$q_B = A + B, \quad q_C = A + C, \quad \mathcal{P} = \{B, C, 2B, 2C, B + C\}. \quad (1.1)$$

Fix one strongly connected linkage support

$$\{q_B, q_C\} \subseteq \mathcal{C} \subseteq \{0, q_B, q_C\} \cup \mathcal{P}, \quad (1.2)$$

with arbitrary positive labelled rates. The no-fast entrance is

$$x_n = (n, 0, 0). \quad (1.3)$$

For fixed ℓ , put

$$G_\ell(x) = K_\ell + \sum_{i=A,B,C} \log(x_i!) + \ell \cdot x \geq 1, \quad W_\ell = G_\ell^4. \quad (1.4)$$

Theorem 1.1 (bounded two-top service). In one fixed closed irreducible class, exactly one of the following applies.

1. If $0 \notin \mathcal{C}$, (1.3) is absorbing and its class is a singleton.
2. If $0 \in \mathcal{C}$ and $\mathcal{C} \cap \mathcal{P} = \emptyset$, then $A - B - C$ is invariant. The cofactor-free part of a fixed class contains at most one state (1.3).
3. If $0 \in \mathcal{C}$ and $\mathcal{C} \cap \mathcal{P} \neq \emptyset$, there is an included all-reaction stopping time τ_n such that, for every fixed p ,

$$\begin{aligned} \mathbb{P}(E_n) &\leq C/n, \\ \mathbb{E}[(1 + |X_{\tau_n} - x_n| + \tau_n)^p; E_n] &\leq C_p/n, \\ \mathbb{E}(1 + \tau_n)^p &\leq C_p, \\ A_{\tau_n} &= n - 1, \quad B_{\tau_n} + C_{\tau_n} \leq 3 \quad \text{on } E_n^c, \end{aligned} \quad (1.5)$$

and, for all large n ,

$$\mathbb{E}_{x_n}[W_\ell(X_{\tau_n}) - W_\ell(x_n) + \tau_n] \leq -cG_\ell(x_n)^3 \log n. \quad (1.6)$$

Here E_n is the included first lower-source competitor during a mixed-top carrier phase. All physical clocks are retained.

Constants may depend on the graph, rates, class, and ℓ , but not on n .

4.2 2. Frozen and invariant alternatives

The complex weights of

$$H = A - B - C \quad (2.1)$$

are zero on $0, q_B, q_C$, -1 on B, C , and -2 on $2B, 2C, B + C$.

If 0 is absent, no source is enabled at (1.3), so the state is absorbing. If $\mathcal{P}$ is absent, every reaction has zero H -reward. At $B = C = 0$, the invariant value is $H = A$, so a fixed class contains at most one cofactor-free value of A . This proves alternatives 1–2.

4.3 3. The clean geometric attempt kernel

Assume alternative 3. At the no-fast face, 0 is the only enabled source, with aggregate rate

$$\lambda_0 = \sum_{0 \rightarrow z} \kappa_{0z} > 0. \quad (3.1)$$

Choose a simple directed path from 0 to its first vertex in $\mathcal{P}$. Before its terminal vertex, it lies in $\{0, q_B, q_C\}$. Simplicity prevents an intermediate return to 0 . After the launch, every prescribed source is q_B or q_C , and is physically enabled by the preceding target with the same residual population.

Temporarily suppress lower-source clocks during a top phase. A carrier moves on the finite transient graph $\{q_B, q_C\}$, absorbed at $\{0\} \cup \mathcal{P}$. Conditional outgoing probabilities at a top source are independent of n , because the outgoing clocks have their common falling-factorial source. The chosen path therefore gives a fixed

$$p_* > 0 \quad (3.2)$$

chance that an attempt reaches $\mathcal{P}$. Its internal top-event count is phase type and its physical duration has moments $O(n^{-p})$.

Absorption at 0 returns to the exact population $(n, 0, 0)$. Repeat attempts. Their count is dominated by a geometric variable of parameter p_* . Each no-fast visit waits an exponential time of rate λ_0 . Thus the clean attempt count and total duration have every fixed moment.

4.4 4. Restoring every physical clock

During any pre-success top phase, the inactive population is at most one. Every lower source, including 0 and any enabled member of $\mathcal{P}$, therefore has aggregate rate bounded by a graph-dependent constant. The aggregate enabled top rate is at least cn . Hence

$$\mathbb{P}\{\text{a lower firing before clean top absorption}\} \leq C/n \quad (4.1)$$

per attempt. Stop at and include the first such firing.

At a clean $\mathcal{P}$ -target, the inactive population has molecularity one or two. At least one of q_B, q_C is enabled, and its rate is at least cn . Follow top reactions until the first top-to-lower exit. Top-to-top conversions preserve A ; this exit lowers A by one. Before it, inactive population is at most two and the lower-source aggregate rate is bounded. Its included competitor probability is again $O(n^{-1})$.

Sum these races over the geometric attempt count. The causing lower jump has a binary target, so its actual endpoint is bounded relative to the clean carrier state. Geometric-sum moment formulas give the first three lines of (1.5).

On a clean successful attempt, a launch $0 \rightarrow q_B$ or $0 \rightarrow q_C$ raises A by one and its first top exit cancels that entry. Reaching $\mathcal{P}$ leaves an inactive carrier, whose next top exit is unpaired. A direct $0 \rightarrow \mathcal{P}$ launch supplies that unpaired exit immediately. Thus

$$A_{\tau_n} = n - 1, \quad B_{\tau_n} + C_{\tau_n} \leq 3. \quad (4.2)$$

This proves all of (1.5). No lower clock has been erased: it is either a designated no-fast launch or the included event E_n .

4.5 5. Entropy and common fourth power

On the clean event, (4.2) gives

$$G_\ell(X_{\tau_n}) - G_\ell(x_n) \leq -\log n + C_\ell. \quad (5.1)$$

On E_n , the centered displacement before the causing jump has geometric moments, and the causing jump is bounded. Consequently

$$\mathbb{E}[|G_\ell(X_{\tau_n}) - G_\ell(x_n)|^p; E_n] \leq \frac{C_{p,\ell}(\log(n+e))^p}{n}. \quad (5.2)$$

Together with the duration moments, this yields

$$\mathbb{E}[G_\ell(X_{\tau_n}) - G_\ell(x_n) + \tau_n] \leq -\frac{1}{2} \log n \quad (5.3)$$

for all large n .

Since $G_\ell(x_n) \asymp n \log n$, the exact identity

$$(G + z)^4 - G^4 = 4G^3z + 6G^2z^2 + 4Gz^3 + z^4 \quad (5.4)$$

and (5.2) show that the last three expected terms are $O(G_\ell(x_n)^2(\log n)^2)$, lower order than $G_\ell(x_n)^3 \log n$. The physical-duration term is lower order as well. This proves (1.6). $\square$

4.6 6. Scope

This theorem treats exactly the bounded-base one-active shape with both $A + B$ and $A + C$ disabled at $(n, 0, 0)$. It is independent of the mesoscopic separated theorem. It proves arbitrary orientations and rates from a simple strong-graph path, a finite phase-type carrier, and an included all-clock race; it uses no finite enumeration.

5 Full all-clock joint-return theorem for the separated carrier

Proof-first theorem, 2026-08-12 PDT. This note assembles the orientation-free separated-scale theorem from frozen proof components and their hostile audits. It uses no enumeration of supports, orientations, reaction words, or population boxes. Every reaction clock is physical, every terminal crossing reaction is included, and the stopping rule is independent of the factorial-linear correction.

5.1 1. Statement

Put

$$q = A + C, \quad \{q\} \subseteq \mathcal{C} \subseteq \{0, B, 2B, C, 2C, B + C, q\}. \quad (1.1)$$

Let the directed reaction graph on $\mathcal{C}$ be strongly connected, let all rate constants be fixed and positive, and fix one closed irreducible population class Γ . Delete zero-displacement reactions, which do not change the state process. Define

$$\mathcal{F} = \mathcal{C} \cap \{0, B, 2B\}, \quad \mathcal{P} = \mathcal{C} \cap \{C, 2C, B + C\}, \quad (1.2)$$

and

$$m(b) = \begin{cases} 1 + b^2, & 2B \in \mathcal{C}, \\ 1 + b, & 2B \notin \mathcal{C}, \mathcal{C} \cap \{B, B + C\} \neq \emptyset, \\ 1, & B, 2B, B + C \notin \mathcal{C}. \end{cases} \quad (1.3)$$

For fixed $\ell \in \mathbb{R}^3$, choose K_ℓ so that

$$G_\ell(x) = K_\ell + \sum_i \log(x_i!) + \ell \cdot x \geq 1, \quad W_\ell(x) = G_\ell(x)^4. \quad (1.4)$$

Theorem 1.1 (separated all-clock joint return). Let $x_n = (a_n, b_n, 0) \in \Gamma$ satisfy

$$h_n = \log \frac{a_n}{m(b_n)} \longrightarrow \infty. \quad (1.5)$$

Then the following ordered alternatives are exhaustive.

1. If no member of $\mathcal{F}$ is enabled at b_n , then x_n is an absorbing singleton.
2. If an $\mathcal{F}$ -source is enabled but $\mathcal{P} = \emptyset$, then $A - C$ is an exact stoichiometric invariant. A divergent sequence (1.5) therefore cannot occur in the one fixed class Γ .
3. If an $\mathcal{F}$ -source is enabled and $\mathcal{P} \neq \emptyset$, there is one statewise, all-clock physical stopping rule $x \mapsto \tau_x$, independent of ℓ , such that $\tau_x < \infty$ almost surely, every terminal reaction is included, and, for all sufficiently large separated entrances,

$$\mathbb{E}_x[W_\ell(X_{\tau_x}) - W_\ell(x) + \tau_x] \leq -c_\ell G_\ell(x)^3 \log \frac{a}{m(b)}. \quad (1.6)$$

For every fixed integer $r \geq 1$, the rule in Alternative 3 satisfies

$$\mathbb{E}_x \tau_x^r \leq C_r(1+b)^r, \quad (1.7)$$

and there is a finite q_r , depending only on the fixed network and r , such that

$$\mathbb{E}_x(1 + |X_{\tau_x} - x|_1 + \tau_x)^r \leq C_r(1+a+b)^{q_r}. \quad (1.8)$$

Constants may depend on the fixed graph, rates, class, r , and ℓ , but not on the entrance scale.

5.2 2. Exact physical stopping rule

At $x = (a, b, 0)$, put

$$\epsilon = \frac{m(b)}{a} = e^{-h}, \quad \bar{\delta} = \sqrt{\epsilon} = e^{-h/2}. \quad (2.1)$$

Let $p \in \{0, 1, 2\}$ be the spectator degree selected by (1.3), and fix a sufficiently small constant $c_0 > 0$. Give first priority to the included reaction fired from an open state whose endpoint crosses

$$A \notin [a/2, 2a], \quad C \geq c_0 \bar{\delta} a, \quad 1 + B^p \geq c_0 \bar{\delta} a \quad (p \geq 1). \quad (2.2)$$

Call this label B_O . For $p = 0$, the spectator condition is absent.

At $C = 0$, a **clean macro** begins with one reaction sourced at an enabled member of $\mathcal{F}$, thereafter retains only reactions sourced at q , and ends at the first actual return to $C = 0$. Write

$$k = A_{\text{start}} - A_{\text{return}}. \quad (2.3)$$

Repeat unmarked clean completed macros with $k = 0$. Subject to the priority of B_O , an unmarked clean return with $k \geq 1$ is terminal service S .

If a reaction sourced below q fires while open, mark its first occurrence, retain every later physical clock, and stop at the next actual $C = 0$ return as E , unless B_O occurs first. The first marked reaction may itself be the terminal $C = 0$ return.

Only after an unmarked clean $k = 0$ return, apply the completed-base guard

$$1 + B^p \geq \frac{c_0}{2} \bar{\delta} a \quad (p \geq 1), \quad (2.4)$$

and label such an endpoint B_0 . Thus the exact priority is

$$B_O > S \text{ or } E > B_0, \quad (2.5)$$

and

$$\Omega = S \dot{\cup} E \dot{\cup} B, \quad B = B_O \dot{\cup} B_0. \quad (2.6)$$

In particular, a long service or marked genealogy is never relabelled by the narrower base guard, and a marked path never begins another base macro.

5.3 3. Frozen, invariant, and physical-loss branches

If no member of $\mathcal{F}$ is enabled, every possible source requires positive C , so the base is frozen. If $\mathcal{P} = \emptyset$, all complexes lie in $\mathcal{F} \cup \{q\}$, and the functional $A - C$ has the same value on every complex. It is therefore preserved by every reaction. At a cofactor-free state in one fixed class, A is fixed, which excludes (1.5).

A cofactor-free endpoint with no enabled member of $\mathcal{F}$ is absorbing, because every remaining source contains C ; since Γ is closed, every reachable endpoint lies in Γ , and irreducibility then forces Γ to be that absorbing singleton. Hence on the nonfrozen active-loss branch no distinct no- $\mathcal{F}$ base endpoint is reachable.

Suppose now that an $\mathcal{F}$ -source is enabled and $\mathcal{P} \neq \emptyset$. Choose an enabled $f \in \mathcal{F}$ and a simple directed path from f to its first $\mathcal{P}$ -vertex. The path is physically executable because every target creates the next source. At its last vertex q is enabled, and one nonself q -edge lowers A by one.

On any fixed spectator compact, every chosen base edge has a fixed positive conditional probability, while every chosen q -edge shares the common factor AC with its competitors. The bounded path therefore gives a uniform compact killing/active-loss cut. At large B , the maximal cofactor-free source gives the exterior cut in Section 4. This is the exact physical active-loss branch; failure of either cut is contained in the frozen or invariant alternatives above.

This trichotomy, including its fixed-class exclusion, has an independent strict PASS in `proof_first_separated_invariant_no_history_routing_audit.md`, SHA-256 cef9030583f0856a9243b93abece2a3f3eb3bb28912e438774d75946c24af1b1.

5.4 4. Clean completed-return Green theorem

Let

$$d = \max\{c : cB \in \mathcal{F}\}. \quad (4.1)$$

For a clean completed macro sourced at cB , let T be its number of q -firings and let $e = 1$ when its base target is q . Exact active balance and carrier balance give

$$k = T - e \geq 0, \quad B_\tau - b \leq pk + (d - c). \quad (4.2)$$

The exceptional case $p = 1, d = 0$ is included: only $B + C$ can create a spectator molecule, and a completed carrier genealogy contains at most k such targets.

If $k = 0$, every clean completion is exactly

$$cB \longrightarrow jB, \quad \text{or} \quad cB \longrightarrow q \longrightarrow jB, \quad c, j \leq d. \quad (4.3)$$

After literal population returns are contracted, a maximal-source nonself return strictly lowers B . A positive return must be sourced at degree $c < d$. Its source probability and raw factorial tilt obey, for fixed $0 < \theta < 1/2$,

$$(1 + B)^{c-d} \frac{e^{\theta G_\ell(A, B', 0)}}{e^{\theta G_\ell(A, B, 0)}} \leq C(1 + B)^{-(1-\theta)(d-c)}. \quad (4.4)$$

Literal returns have a uniformly bounded diagonal inverse by the directed cut out of $\{dB, q\}$. The physical active-loss history of Section 3 gives a fixed-block spectral gap on the remaining spectator compact. Consequently the clean kernel, killed at service or localization, has a same-exponent Green bound. Its polynomial additive-functional hierarchy also controls every fixed moment of the number of continuing $k = 0$ macros.

On a clean service path, (4.2) pairs every positive spectator molecule with either the lower source-degree factor or an active loss. Before the moving boundary,

$$(a)_{\underline{k}}^{-\theta} (1 + B_\tau)^{\theta pk} \leq C \bar{\delta}^{\theta k}, \quad (4.5)$$

so $k \geq 1$ carries the separated service factor.

The ledger, same-exponent Green theorem, literal-return cut, polynomial hierarchy, and the warning that a critical genealogy need not have an unweighted spectator moment all have an independent PASS in `proof_first_separated_clean_base_green_audit.md`, SHA-256 96c72e11a6105013b8d7b6e2309da7c2dbebccfa0b72640bfb3cfe6cf1608b36.

5.5 5. Full open phase and raw terminal transforms

For a complex y , put

$$M_x(y) = \prod_i (x_i + 1)^{y_i}, \quad \mathcal{M}(x) = M_x(dB) = (1 + B)^d. \quad (5.1)$$

At an open state attach the last-target phase s , with $s = q$ after a lower-to- q firing. Define

$$V_\theta(x, s) = e^{\theta G_\ell(x)} \left(\frac{\mathcal{M}(x)}{M_x(s)} \right)^\theta \quad (C > 0), \quad V_\theta(x) = e^{\theta G_\ell(x)} \quad (C = 0). \quad (5.2)$$

Inside (2.2), every lower monomial ratio $r_y = M_x(y)/M_x(q)$ is $O(\bar{\delta})$. Bounded-degree factorial comparison gives the sourcewise table

$$\frac{\lambda_{yz}(x)}{\lambda_{\text{tot}}(x)} \frac{V_\theta(x + z - y, s')}{V_\theta(x, s)} \leq \begin{cases} Cr_y^{1-\theta} r_s^\theta, & y \neq q, \\ Cr_s^\theta, & y = q. \end{cases} \quad (5.3)$$

A phase- q exit can be free only once, because it sets a lower phase. With $\eta = \min\{\theta, 1 - \theta\}$, the full post-mark open kernel K_{OO} , killed at its next base return or B_O , therefore satisfies

$$\|K_{OO}^2\|_{V_\theta} \leq C\bar{\delta}^\eta, \quad \|(I - K_{OO})^{-1}\|_{V_\theta} \leq C. \quad (5.4)$$

This Green operator retains arbitrary later lower-source reactions and arbitrary carrier branching.

Let Q be the physical clean prefix kernel, killed at S, E, B_O, B_0 . Restrict first marks to causing reactions which do not cross B_O , and split them as

$$R = R_B + R_O, \quad (5.5)$$

where R_B itself lands at $C = 0$, while R_O remains open. Let T_{OE} be the exit from the open post-mark kernel to its next base without an earlier boundary hit. The exact E -terminal operator is

$$(I - Q)^{-1} \{R_B + R_O(I - K_{OO})^{-1} T_{OE}\}. \quad (5.6)$$

Crossing marks and the complementary open exit belong to B_O . Formula (5.6) therefore includes a direct marked reaction such as $C \rightarrow 0$, retains all later physical clocks, and never continues past the completed marked return.

The first mark costs $C\bar{\delta}^{1-\theta}$. Since (5.2) is exactly the raw exponential at both base endpoints, (5.4)–(5.6) yield

$$\mathbb{E}_x[e^{\theta \Delta G_\ell}; E] \leq C\bar{\delta}^{1-\theta} \leq Ce^{-\eta h/2}. \quad (5.7)$$

For clean service, the completed ledger (4.2)–(4.5), rather than a terminal divisor moment, gives

$$\mathbb{E}_x[e^{\theta \Delta G_\ell}; S] \leq C\bar{\delta}^\theta \leq Ce^{-\eta h/2}. \quad (5.8)$$

The phase table has an independent PASS in `proof_first_separated_phase_corrector_open_green_audit.md`, SHA-256 5286d3fbd5d57e92047f8db1339130c228498215bdd23d4d3ebcb1946db26114. The exact physical first-mark resolvent is frozen in `proof_first_separated_first_mark_resolvent_lemma.md`, SHA-256 d4c4baff29ffda942798f28fc69d4b30ab25ee2c8e13d1960a4ee20b6d772506.

The normalized raw-terminal theorem is frozen in `proof_first_separated_normalized_phase_marked_terminal_repair.md`, SHA-256 149c2edd9a8427a442a66e4f99c026be96313bfc4e5072f96d0aa380502ffa77. Its independent hostile audit is frozen in `proof_first_separated_normalized_phase_marked_terminal_repair_independent_audit.md`, SHA-256 5dd855cbb52bd84b83886209b9a0dd96cb91229275037f7f59be7cdac85d6a0b1, with verdict **STRICT PASS**.

5.6 6. Included boundary estimate

Put

$$L = \min\{a, \bar{\delta}a, (\bar{\delta}a)^{1/p} : p \geq 1\}. \quad (6.1)$$

Since $\epsilon \geq a^{-1}$, one has $L \geq ca^{1/4}$. The entrance is an $o(1)$ fraction of the completed-base guard. Every continuing clean $k = 0$ return changes B by at most two, while the same-exponent clean kernel has a strict contraction $\rho < 1$. Hence a B_0 -path contains $\Omega(L)$ contracted continuing returns and has raw endpoint-weighted mass at most

$$C\rho^{cL}. \quad (6.2)$$

An excursion opens below half the spectator threshold. Reaching any component of (2.2) requires $\Omega(L)$ bounded open reactions. Pairing steps in (5.4) gives corrected mass

$$C(C\bar{\delta}^\eta)^{cL/2}. \quad (6.3)$$

At the included open endpoint every dynamic coordinate is $O(a)$, and conversion from V_θ to the raw exponential costs at most $Ca^{2\theta}$. When $p = 0$, species B occurs in no complex and is constant on the fixed class. Thus every fixed polynomial endpoint mark costs only a fixed power of a . Equations (6.2)–(6.3) imply, for every fixed N and every fixed nonnegative polynomial P ,

$$\mathbb{E}_x[e^{\theta\Delta G_\ell} P(X_\tau); B] \leq C_{N,P} a^{-N}. \quad (6.4)$$

This includes an open crossing whose causing reaction lands directly at $C = 0$, because the required pre-jump path is already counted in (6.3) and the base reset is raw.

5.7 7. Almost-sure termination and physical time

At every preterminal open state, the aggregate q -source probability is $1 - O(\bar{\delta})$. Every nontrivial q -firing lowers A by one, whereas a lower-sourced firing raises A by at most one. For a small fixed $t > 0$, the embedded active increment therefore satisfies

$$\mathbb{E}[e^{t(A_{j+1}-A_j)} \mid \mathcal{F}_j] \leq e^{-\gamma} \quad (7.1)$$

at every preterminal open state, with fixed $\gamma > 0$. If ν is the number of state-changing reactions in one open excursion, the included lower active boundary gives

$$\mathbb{P}\{\nu > n\} \leq \exp(Ca - \gamma n), \quad (7.2)$$

and consequently

$$\mathbb{E}\nu^r \leq C_r a^r. \quad (7.3)$$

Thus every open excursion terminates almost surely, even for critical or supercritical carrier offspring laws. At every preterminal open state the total physical rate is at least $cAC \geq ca$. Conditional exponential holding-time bounds and (7.3) imply that the unique terminal long open excursion has uniformly bounded physical-time moments.

Before that excursion, every continuing clean $k = 0$ macro has the bounded form (4.3). After literal returns are contracted, a maximal-source trial is killed or lowers B with fixed probability, while a positive move has probability $O((1+B)^{-1})$. The compact active-loss cut of Section 3 and the exterior contraction imply

$$\mathbb{P}\{N_B > n\} \leq Ce^{tb-\gamma'n}, \quad \mathbb{E}N_B^r \leq C_r(1+b)^r \quad (7.4)$$

for the number N_B of contracted continuing trials. Literal returns expand into uniformly geometric blocks. At every nonfrozen base, the total rate is bounded below by a fixed positive constant. Restoring all base and open holding times proves

$$\boxed{\mathbb{E}_x \tau^r \leq C_r(1+b)^r}. \quad (7.5)$$

The endpoint estimate (1.8) comes directly from the included caps, not from an unweighted moment of a critical genealogy. An S - or E -endpoint occurs below B_O and, by its terminal priority, is not continued beyond its completed return; hence A, C , and the dynamic spectator B are $O(a)$. A B_O - or B_0 -endpoint is the first crossing of the same caps by a bounded reaction, and has the same polynomial size bound. When $p = 0$, B is fixed on Γ . Combining this deterministic endpoint bound with (7.5) proves (1.8).

More importantly, (7.2) proves almost-sure termination of every open macro under the exact boundary priority (2.5), and (7.4) proves that only finitely many continuing unmarked $k = 0$ base macros occur almost surely. The first later mark, service, or included guard is therefore reached in finite physical time. Thus $\tau_x < \infty$ almost surely for the exact rule of Section 2.

The physical-duration theorem is frozen in `proof_first_separated_physical_duration_joint_return.md`, SHA-256 504b87e600c382e9c82b88cf0ea88f87a6a4b6c7783202cc9cc2faa79fefc640. Its hostile audit is frozen in `proof_first_separated_physical_duration_joint_return_audit.md`, SHA-256 c9be124327288bbb45f42cbb005c9b96d854cfb629f94dd4c7cc61cc456ccb56, with verdict **STRICT PASS ON THE PHYSICAL ACTIVE-LOSS BRANCH**. Section 3 is exactly the routing required by that scope qualification.

5.8 8. Common fourth-power drift

The terminal labels S, E, B are disjoint and exhaustive. Equations (5.7)–(5.8) and (6.4) imply, for fixed $c, \theta > 0$,

$$\mathbb{E}_x[e^{\theta \Delta G_\ell}; S] + \mathbb{E}_x[e^{\theta \Delta G_\ell}; E] \leq C e^{-ch}, \quad \mathbb{E}_x[e^{\theta \Delta G_\ell}; B] \leq C_N a^{-N}. \quad (8.1)$$

Here $h \leq \log a$, $h \rightarrow \infty$, and $G_\ell(a, b, 0) \geq c_\ell a \log a$ for all sufficiently large separated entrances. In particular $h = o(G_\ell(x))$. Exponential Markov applied to (8.1) shows that

$$\Delta G_\ell \leq -c'h \quad (8.2)$$

with probability $1 - o(1)$. Since $G_\ell \geq 1$, monotonicity of $t \mapsto t^4$ on the attainable range gives on (8.2)

$$W_\ell(X_\tau) - W_\ell(x) \leq -c'' G_\ell(x)^3 h. \quad (8.3)$$

For the exceptional positive part,

$$((G_\ell + u)^4 - G_\ell^4)^+ \leq C_\theta (1 + G_\ell^3) e^{\theta u/2}, \quad u \geq 0. \quad (8.4)$$

Cauchy–Schwarz and (8.1) make its expectation $o(G_\ell(x)^3 h)$. No moment of a large negative entropy increment is needed; such an endpoint only improves (8.3).

Finally, (1.3), (1.5), and (7.5) give

$$\mathbb{E}_x \tau = o(G_\ell(x)^3 h). \quad (8.5)$$

Combining (8.3)–(8.5) proves (1.6) with the same $W_\ell = G_\ell^4$ for every chart and every terminal label.

The raw-exponential fourth-power lift is frozen in `proof_first_separated_raw_exponential_w4_lift.md`, SHA-256 3badb799468f6912659916ab2bc4ee556f5c113f00fba01d106c88449cc0134. Its independent hostile audit is frozen in `proof_first_separated_raw_exponential_w4_lift_audit.md`, SHA-256 3cf72b58a328bb493aa48f9b1fd11ec164f49c4504f7a40c0dcc3a1d6062326e, with verdict **STRICT PASS**.

5.9 9. Scope, rejected predecessors, and publication interface

The theorem is:

- fixed-class and exact on the frozen/invariant alternatives;
- valid for every strong orientation and every fixed positive rate vector;
- all-clock after the first mark, with arbitrary later marks and arbitrary carrier branching;
- exact at the direct-base first mark and every included boundary;
- almost surely finite and physical-time aware; and
- stated with the common fourth-power workload used by the global classwise composition.

It does **not** assert an unweighted scale-relative terminal spectator moment. Such a moment is false for critical carrier genealogies and is unnecessary: the normalized phase weight is raw at completed returns, the included boundary is paid edgewise, and physical time is controlled by the active-coordinate reaction clock.

Two predecessor constructions are explicitly retired and must not be cited as proofs:

1. `proof_first_single_linkage_level_trace_completion.md`, frozen at SHA-256 08c216dcf5926484e39edcab22df9ab119cd45f63f3f605154d6193a01c9f558, stopped at the wrong logarithmic scale and failed hostile audit.
2. `proof_first_single_linkage_full_all_clock_nested_carrier.md`, frozen at SHA-256 490f42487ec5045e17a7fa0dc1e69f61f836d17d0ee5ae3ce6af304fc7c230ac, stopped after an unmatched positive active entry and is false.

The present theorem replaces both with completed physical joint returns, the exact $B_O > S/E > B_0$ terminal priority, and the normalized raw phase weight. It is the publication-safe separated input to the single-linkage theorem in at most three dynamic species.

6 Classwise recurrence with one active linkage and at most three species

Proof-first composition, 2026-08-12 PDT. This note proves the one-linkage branch needed by the (T3)–(2) theorem. Every local estimate used below is analytic and quantifies over arbitrary strongly connected orientations and arbitrary fixed positive rate constants. Finite support, orientation, reaction-word, and population enumeration are not used.

6.1 1. Theorem

Theorem 1.1. Let every complex of a finite weakly reversible stochastic mass-action network have molecularity at most two. Fix a closed irreducible population class Γ , delete coordinates constant on Γ , delete linkages having no enabled source there, and merge projected linkages which share a projected complex. If the reduced network has at most three dynamic species and exactly one active linkage, then its physical CTMC is nonexplosive and positive recurrent on Γ .

The conclusion holds for every strongly connected orientation of the linkage support and every positive fixed rate vector. An absorbing singleton is included.

6.2 2. Exact reduction and nonexplosion

If $X_i \equiv m_i$ on Γ , closure forces every enabled reaction to have zero i -increment. Starting from an enabled complex, a directed return path is physically executable because each target creates the next source. Hence the i -coordinate is constant across its whole active linkage, and its falling-factorial factor can be absorbed into the labelled rate constants. Projected linkages sharing a complex have strongly connected union; retaining parallel labelled channels preserves the exact generator. Projection is therefore a CTMC conjugacy on Γ .

A degree-two source cannot increase total population because every target also has degree at most two. The aggregate positive-jump rate is at most

$$C(1 + |x|_1). \tag{2.1}$$

Localization at population levels and Gronwall prevent population escape in finite time. Inside a fixed population sublevel there are finitely many states and bounded total rates, so population-neutral quadratic clocks cannot accumulate. The chain is nonexplosive.

6.3 3. Proper tier sequences and the enabled-top episode

Let $x_n \in \Gamma$ be an escaping proper tier sequence. Write D^1 for its top deterministic tier and E for the eventually enabled complexes. For an enabled source y ,

$$\frac{\lambda_y(x_n)}{(x_n \vee 1)^y} \longrightarrow c_y \in (0, \infty), \quad (3.1)$$

whereas a disabled source has zero propensity. Thus the top stochastic source tier is contained in D^1 precisely when $D^1 \cap E \neq \emptyset$.

The top block D^1 is a proper subset of the support. Otherwise, with

$$L_n = \log(x_n \vee 1), \quad w_n = L_n / \|L_n\|_2 \longrightarrow w,$$

tier equivalence gives $w \cdot (y - z) = 0$ for every reaction vector. Here $w \geq 0$, $\|w\|_2 = 1$, and every positive component of w corresponds to a divergent coordinate. Hence $w \cdot X$ is an exact class invariant while $w \cdot x_n \rightarrow \infty$, a contradiction.

If $D^1 \cap E \neq \emptyset$, choose an enabled top vertex and a simple directed path inside D^1 to its first exit. Every prescribed source is physically enabled by the preceding target, every preterminal propensity is comparable with the common top scale $A_n \rightarrow \infty$, and every competitor has propensity at most CA_n . Retain every clock and stop at completion or at the first competitor. The successful last edge has logarithmic gap $g_n \rightarrow \infty$. With

$$G_\ell(x) = K_\ell + \sum_i \log(x_i!) + \ell \cdot x \geq 1, \quad W_\ell = G_\ell^4, \quad (3.2)$$

the exact race bound

$$r\{1 + \log(1/r)\}^q \leq C_q, \quad 0 \leq r \leq 1, \quad (3.3)$$

pays every competing endpoint. For every fixed q , the episode has the required endpoint moments and physical duration, and

$$\mathbb{E}_{x_n}[W_\ell(X_{\tau_n}) - W_\ell(x_n) + \tau_n] \leq -cG_\ell(x_n)^3 g_n. \quad (3.4)$$

This is the generic analytic lemma proved in `proof_first_hard_enabled181_access_word.md`, SHA-256 9be70e2b6c9ce5c4762bf3130246f1ea660bea73f41aa7abdd997853cc0a6b04, and independently audited at SHA-256 4028c026a7d01c1e0930bdbdaa75216a79402078999d6450c283a77eb2a04883. Its finite hard-row specialization is irrelevant here; the preceding paragraph verifies its generic hypotheses directly. In particular, the argument is a fixed-path race estimate, not an orientation enumeration.

6.4 4. Symbolic exhaustion when every top complex is disabled

Suppose $D^1 \cap E = \emptyset$. The common top monomial diverges. A disabled pure complex has monomial one, so every top complex is mixed, with one zero coordinate and one divergent coordinate. This yields the following exhaustive split, up to relabelling.

6.4.1 4.1 At most two dynamic species

With one dynamic species the support contains its unary or binary pure complex, and Anderson–Cappelletti–Kim (2020, Theorem 4.1) applies. With two dynamic species A, B , if $A, 2A$ are absent then

$$\mathcal{C} = \{A + B\} \cup T, \quad \emptyset \neq T \subseteq \{0, B, 2B\}. \quad (4.1)$$

Every case with $|T| \leq 2$ is weakly reversible and deficiency zero. The only residual support is

$$\{0, B, 2B, A + B\}. \quad (4.2)$$

Its all-clock origin-launch theorem proves physical-time entropy descent and classwise positive recurrence for arbitrary orientations and rates. The publication derivative is frozen at SHA-256 9d878860cb6427688995784ed230776d982eca758a6c306be4180c3e8ffaaf03, and its independent audit at 26e97c6edd5566e0fa d9326523b017d86c1391c2b8762ca8eacddc06c41fd52a.

6.4.2 4.2 Three dynamic species

If all three coordinates diverge, every complex is enabled, contrary to the present failure. If $A, B \rightarrow \infty$ and $C = 0$, then

$$\emptyset \neq D^1 \subseteq \{A + C, B + C\}. \quad (4.3)$$

For a singleton top, say $D^1 = \{A + C\}$, one has the separated support

$$\{A + C\} \subseteq \mathcal{C} \subseteq \{0, C, 2C, B, 2B, B + C, A + C\}, \quad (4.4)$$

and

$$\log \frac{A}{m(B)} \rightarrow \infty, \quad m(B) = \begin{cases} 1 + B^2, & 2B \in \mathcal{C}, \\ 1 + B, & 2B \notin \mathcal{C}, \mathcal{C} \cap \{B, B + C\} \neq \emptyset, \\ 1, & B, 2B, B + C \notin \mathcal{C}. \end{cases} \quad (4.5)$$

The full all-clock joint-return theorem gives the exact frozen/invariant/physical-loss trichotomy, an almost surely finite stopping time, physical duration and actual-endpoint integrability, and

$$\mathbb{E}_x[W_\ell(X_\tau) - W_\ell(x) + \tau] \leq -cG_\ell(x)^3 \log \frac{A}{m(B)}. \quad (4.6)$$

Its exact target and hostile audit hashes are respectively

389e3b446006e9313238a0b4b0029f39e0f1cee0c2d90faf6e63cccf38a581e1
096ba806daa3f7f1bc336986d3248976ac8ade084cfbf5a60e524ceec96f75a6.

If both mixed tops tie, then

$$\{A + C, B + C\} \subseteq \mathcal{C} \subseteq \{0, C, 2C, A + C, B + C\}. \quad (4.7)$$

Every proper support in (4.7) is deficiency zero: for $m = 2, 3, 4$, its stoichiometric rank is $m - 1$. The full five-complex support has rank three and deficiency one. Its balanced all-clock theorem proves common W_ℓ drift, duration, and endpoint moments. The publication target is frozen at SHA-256 8a34f9934f9ffdd078850070de561aa3cf3f734a9fbeb2e4f08bc68c5e106262, and its transferred independent audit at 933c14ca d99b8cf5bc1e2237f3be417aebfb344d1751f7c58be2b284c562d5ab.

If only $A \rightarrow \infty$, a singleton top again gives (4.4). The tied top $\{A + B, A + C\}$, with $B = C = 0$, lies in

$$\{A + B, A + C\} \subseteq \mathcal{C} \subseteq \{0, A + B, A + C, B, C, 2B, 2C, B + C\}. \quad (4.8)$$

The bounded two-top theorem gives an exact frozen/invariant/service trichotomy and common fourth-power drift. Its publication derivative and audit hashes are

afd696d3b28709619759936cfb6e4536859300985bb6e594cff840fb4c7db7f9
c96995531a79570d26330f2babeeba2998323c1787bda3627b2c538a38ee7922.

The symbolic proof of this exhaustion is independently recorded at SHA-256 5b64e251035eedb3e5afe1 d37881b3e1f4db45055ac5ab9a9ab165764720f0d1. No orientation enters the classification.

6.5 5. Uniform statewise cover

Set $\ell = 0$, so every stopped branch above and every ordinary passing branch uses the single proper potential

$$W(x) = \left(K + \sum_i \log(x_i!) \right)^4. \quad (5.1)$$

There is a **finite menu of literal statewise rules**. For the enabled-top case, the menu consists of the finitely many simple directed paths from a possible enabled top vertex to the first vertex outside a proposed top block, with a deterministic graph ordering. The exceptional theorems already give statewise rules on their stated start domains. Each rule uses all clocks and its expected drift is an ordinary real-valued function of the starting state.

Fix a small $\eta > 0$ below the duration coefficient in every member of this finite menu. If infinitely many escaping states admitted no rule with drift at most -1 , choose such a sequence and extract a proper tier subsequence with fixed availability, weak monomial order, relabelling, and candidate rule. Sections 3–4 put it in an enabled-top, exceptional two-species, separated, balanced, bounded two-top, invariant, or frozen alternative. In a nontrivial alternative the corresponding fixed rule has coercive negative drift, a contradiction. An invariant or frozen alternative cannot contain an escaping sequence in the fixed irreducible class, by the cited local theorem. Thus, outside a finite set, at least one menu rule works. Select the first working rule in the deterministic order; the state space is countable, so this is a measurable state-selected rule.

Consequently there are $\delta > 0$ and a finite set $K \subset \Gamma$ such that at every state outside K , its selected physical stopping time $\tau_x > 0$ satisfies

$$\mathbb{E}_x[W(X_{\tau_x}) - W(x) + \eta\tau_x] \leq -\delta, \quad \mathbb{E}_x[W(X_{\tau_x}) + \tau_x] < \infty. \quad (5.2)$$

This sequential-coercivity argument, rather than pointwise positivity or compactness alone, supplies the uniform constants. We used the uniform choice -1 only for convenience; decreasing it gives any fixed $\delta \in (0, 1]$.

Published pure-multiple and deficiency-zero branches already prove positive recurrence of the whole class and need no gluing. Hence (5.2) is used only in the residual structural cases where the common workload has been proved explicitly.

6.6 6. Physical-time Foster gluing and finite return

Apply the state-selected physical-time Foster lemma directly to (5.2). Equivalently, if a local theorem supplies a generator-good segment, stop that segment at first entrance into an episode domain or K ; localized Dynkin pays the W -increment plus η times physical duration. At an episode point, append (5.2) and repeat from its actual endpoint. If K is visited inside an episode, record the physical hit immediately but complete that one episode solely for drift accounting; the completed duration upper-bounds T_K .

If S_m is the accounting time after m exceptional episodes and N the terminal index, conditional iteration gives

$$\delta \mathbb{E}(N \wedge m) + \eta \mathbb{E}S_{N \wedge m} \leq W(x) + \delta. \quad (6.1)$$

Monotone convergence (and, when present, the generator-segment inequality) implies $\mathbb{E}_x T_K < \infty$. If the class is not an absorbing singleton, take one ordinary physical jump from each state of the finite target and apply the same hitting bound to its finitely many successors. The resulting finite target return kernel has a recurrent state with finite mean positive physical return. Irreducibility promotes this to positive recurrence of every state of Γ .

This is the state-selected physical-time Foster lemma proved in `proof_first_global_t3_2_classwise_skeleton_scoped_audit.md`, SHA-256 3d45867b4dd07a92ce43054767b7e7a680fa77035b7f2e1021dcb5004097f962. Together with nonexplosion from Section 2, it proves Theorem 1.1.

6.7 7. Exact dependency boundary

This theorem uses only the analytic inputs and exact hashes displayed above. The earlier conditional one-linkage composition and the false separated candidates at SHAs 08c216... and 490f424... are retired and are not proof dependencies. No two-linkage selector, finite orientation catalog, or global certification flag is used here.

Part II

Two active linkages: pre-residual theorems

7 Exact shielded/available seam in physical time

7.1 1. Status, hypotheses, and conclusion

This note closes one exact two-active seam and its complete finite shielded-support generalization. It is **not** a proof of T3-2. Sections 8 and 9 list what this argument does not cover.

Use stochastic mass-action propensities

$$\lambda_{y \rightarrow z}(x) = \kappa_{yz}(x)_y, \quad (x)_y = \prod_i \frac{x_i!}{(x_i - y_i)!}. \quad (1.1)$$

Fix the available linkage support

$$L_1 = \{C, A + C, B + C\}. \quad (1.2)$$

Its labelled reaction graph may have arbitrary orientations and parallel channels, subject only to strong connectivity and positive rate constants. After subtracting the common catalyst C , it is an irreducible immigration/conversion/death network on the vertices $0, A, B$. Every L_1 propensity has a common factor C , and every L_1 reaction leaves C unchanged.

Let L_0 be a disjoint, strongly connected linkage. Assume it is shielded for one of the four two-active workload representatives

$$(1, 1, 0), \quad (2, 3, 0), \quad (1, 2, 0), \quad (1, 3, 0), \quad (1.3)$$

and that its complex differences admit an invariant $q = (q_A, q_B, q_C)$ with $q_A, q_B > 0$. These are exactly the finite hypotheses replayed by `src/exact_shielded_seam.py`.

The scoped conclusion is:

For the fixed available support (1.2), every compatible L_0 satisfying these hypotheses gives a positive recurrent CTMC on every closed irreducible population class, for every strongly connected orientation and every positive rate vector.

The proof uses the physical-time generator directly. It does not condition on an L_1 activation, truncate C , count fast L_0 jumps, or prove recurrence of the raw embedded jump chain.

7.2 2. Exact seven-support reduction

Take the union over (1.3) of all shielded masks having a positive A, B invariant, then impose disjointness from $\{C, A + C, B + C\}$. Up to exchanging A and B , exactly seven supports remain:

$$\begin{aligned} &\{0, 2C\}, \quad \{A, B\}, \quad \{B, 2A\}, \\ &\{2A, 2B\}, \quad \{2A, A + B\}, \quad \{2B, A + B\}, \\ &\{2A, A + B, 2B\}. \end{aligned} \quad (2.1)$$

The original obstruction

$$L_0 = \{2B, A + B\}, \quad L_1 = \{C, A + C, B + C\} \quad (2.2)$$

is the sixth entry. The self-contained executable certificate checks that the full positive-invariant union has 25 supports and that the disjoint filter gives exactly (2.1).

After removing $\{0, 2C\}$, the other six supports contain only A, B . Consequently C is a reaction-wise invariant of the full network. Five are reversible two-complex linkages. The sixth is the arbitrary directed three-complex linkage $\{2A, A + B, 2B\}$. Sections 3–6 handle these six; Section 7 handles $\{0, 2C\}$ by an autonomous-clock product law.

7.3 3. Reversible two-complex fast linkage

7.3.1 Lemma 3.1: exact factorial-ratio identity

Let L_0 be $y \rightleftharpoons z$. Aggregate parallel channels in each direction, writing $\alpha > 0$ for $y \rightarrow z$ and $\beta > 0$ for $z \rightarrow y$. Put $\zeta = z - y$, and choose $\theta \in (0, \infty)^3$ satisfying

$$\alpha\theta^y = \beta\theta^z. \quad (3.1)$$

One scalar equation in $\log \theta$ always has a solution because $y \neq z$. Define the proper factorial potential with a network-dependent linear corrector

$$F_\theta(x) = \sum_{i=A,B,C} \log(x_i!) - \sum_i x_i \log \theta_i. \quad (3.2)$$

Whenever $y \rightarrow z$ is enabled at x ,

$$F_\theta(x + \zeta) - F_\theta(x) = \log \frac{\beta(x + \zeta)_z}{\alpha(x)_y}. \quad (3.3)$$

Indeed, the factorial part of the increment is

$$\log \prod_i \frac{(x_i - y_i + z_i)!}{x_i!},$$

while (3.1) gives $\beta/\alpha = \theta^{y-z} = \theta^{-\zeta}$. The reverse identity follows identically.

7.3.2 Lemma 3.2: positive fast drift is at most linear

There is a finite constant K_0 , depending only on y, z, α, β , such that

$$\mathcal{L}_0 F_\theta(x) \leq K_0(1 + |x|_1) \quad (3.4)$$

at every population state.

To prove this, put $\lambda(x) = \alpha(x)_y$ and $\mu(x) = \beta(x)_z$. For an enabled forward channel, (3.3) and $\log u \leq u - 1$ give

$$\lambda(x)[F_\theta(x + \zeta) - F_\theta(x)] \leq \mu(x + \zeta) - \lambda(x). \quad (3.5)$$

For an enabled reverse channel,

$$\mu(x)[F_\theta(x - \zeta) - F_\theta(x)] \leq \lambda(x - \zeta) - \mu(x). \quad (3.6)$$

Only enabled terms are retained. Away from the finitely many boundary faces, their sum is

$$\beta[(x + \zeta)_z - (x)_z] + \alpha[(x - \zeta)_y - (x)_y]. \quad (3.7)$$

Each bracket is a first finite difference of a falling-factorial polynomial of degree at most two, hence is affine. On a boundary face, a shift can newly supply one or two missing factors of a bimolecular source. If

it supplies both, the resulting value is bounded; if it supplies one, the other factor is at most linear in $|x|_1$. Thus the positive part of each enabled right-hand side in (3.5)–(3.6) is bounded by $K_0(1 + |x|_1)$. This proves (3.4).

This applies for arbitrary positive rates to the five two-vertex supports

$$\{A, B\}, \quad \{B, 2A\}, \quad \{2A, 2B\}, \quad \{2A, A + B\}, \quad \{2B, A + B\}. \quad (3.8)$$

For the exact seam (2.2), let α be the rate of $2B \rightarrow A + B$ and β the rate of $A + B \rightarrow 2B$. One may take

$$F(a, b, c) = \log(a!) + \log(b!) + \log(c!) + a \log(\beta/\alpha). \quad (3.9)$$

Its exact L_0 drift is

$$\begin{aligned} & \alpha b(b-1) \log \frac{\beta(a+1)}{\alpha b} \\ & + \beta ab \log \frac{\alpha(b+1)}{\beta a}. \end{aligned} \quad (3.10)$$

Thus the fast neutral linkage is handled without averaging its jump chain.

7.4 4. Arbitrary directed three-complex fast linkage

Now take $L_0 = \{2A, A + B, 2B\}$. Index its complexes by their number of A particles:

$$Y_i = iA + (2-i)B, \quad i = 0, 1, 2. \quad (4.1)$$

Aggregate parallel $Y_i \rightarrow Y_j$ channels into a rate $\kappa_{ij} > 0$ whenever that directed edge is present. The graph is arbitrary subject to strong connectivity.

Define

$$\begin{aligned} p(r) &= \sum_{i \rightarrow j} \kappa_{ij} r^i (1-r)^{2-i} (j-i) \\ &= \gamma_0 (1-r)^2 + \gamma_1 r (1-r) + \gamma_2 r^2. \end{aligned} \quad (4.2)$$

Every edge leaving Y_0 has positive A increment, and strong connectivity supplies at least one such edge. Every edge leaving Y_2 has negative A increment, again with at least one present edge. Hence

$$\gamma_0 > 0, \quad \gamma_2 < 0. \quad (4.3)$$

After setting $z = r/(1-r)$, the sign of $p(r)$ is the sign of

$$\gamma_0 + \gamma_1 z + \gamma_2 z^2. \quad (4.4)$$

Its coefficient sequence has exactly one sign change, regardless of the sign of γ_1 . Descartes' rule and (4.3) give a unique positive root. Thus there is a unique $r_* \in (0, 1)$ with

$$p(r) > 0 \quad (r < r_*), \quad p(r) < 0 \quad (r > r_*). \quad (4.5)$$

Put

$$d = -\log \frac{r_*}{1-r_*}, \quad F_*(a, b) = \log(a!) + \log(b!) + da. \quad (4.6)$$

For $n = a + b$, $r = a/n$, and

$$\Phi(r) = \log \frac{r(1-r_*)}{(1-r)r_*}, \quad (4.7)$$

the leading fluid contribution is $n^2 p(r) \Phi(r)$. Equations (4.5)–(4.7) imply

$$p(r) \Phi(r) \leq 0. \quad (4.8)$$

7.4.1 Lemma 4.1: global discrete upper bound

For every strongly connected orientation and every positive rate vector, there is a finite K_* such that

$$\mathcal{L}_0 F_*(a, b) \leq K_*(1 + a + b) \quad (4.9)$$

for all $a, b \in \mathbb{N}_0$.

For an edge $i \rightarrow j$, put $k = j - i$, so $|k| \leq 2$. On an interior strip $\varepsilon n \leq a, b \leq (1 - \varepsilon)n$, exact factorial increments and $|\log(1 + u)| \leq 2|u|$ for $|u| \leq 1/2$ give

$$F_*(a + k, b - k) - F_*(a, b) = k[\log(a/b) + d] + R_{ij}(a, b), \quad (4.10)$$

with

$$|R_{ij}(a, b)| \leq K_\varepsilon(1/a + 1/b). \quad (4.11)$$

Also,

$$(a)_i(b)_{2-i} = n^2 r^i (1 - r)^{2-i} + O_\varepsilon(n). \quad (4.12)$$

Summing over the finite edge set yields

$$\mathcal{L}_0 F_*(a, b) = n^2 p(r) \Phi(r) + O_\varepsilon(n). \quad (4.13)$$

The leading term is nonpositive by (4.8), proving (4.9) on each fixed interior strip.

It remains to make the estimate uniform at the two faces. By (4.3), choose $\varepsilon > 0$ small enough that

$$p(r) \geq \gamma_0/2 \quad (0 \leq r \leq 2\varepsilon), \quad p(r) \leq \gamma_2/2 \quad (1 - 2\varepsilon \leq r \leq 1). \quad (4.14)$$

For $a \leq \varepsilon n$, every enabled edge from $Y_0 = 2B$ moves inward and, for a rate-dependent constant K ,

$$\Delta_{0j} F_* \leq -j \log \frac{b}{a+2} + K. \quad (4.15)$$

The aggregate Y_0 -source rate is at least $k_0 b(b-1)$ for some $k_0 > 0$. The positive parts of all Y_1 - and Y_2 -source terms satisfy

$$\sum_{\substack{i=1,2 \\ i \rightarrow j}} \kappa_{ij} (a)_i (b)_{2-i} (\Delta_{ij} F_*)^+ \leq K(ab + a^2) \left(1 + \log \frac{b+2}{a+1}\right). \quad (4.16)$$

Choose ε smaller, if necessary, so the right side of (4.16) is absorbed by half the negative $b(b-1)$ term from (4.15) whenever $b/(a+2)$ is above a fixed rate-dependent constant R_0 . For this sufficiently small fixed ε , there is a finite $n_0 = n_0(R_0, \varepsilon)$ such that $a \leq \varepsilon n$, $b = n - a$, and $n \geq n_0$ force that ratio above R_0 . This same absorption includes $a = 1$; when $a = 0$, every Y_1 - and Y_2 -source propensity vanishes and (4.15) applies directly. Enlarge K_* over the remaining finite set. The symmetric argument, using $\gamma_2 < 0$, handles $b \leq \varepsilon n$. Together with (4.13), this proves (4.9).

This argument uses only endpoint signs forced by strong connectivity. It does not assume reversibility, detailed balance, deficiency zero, or a special orientation of the three-complex linkage.

7.5 5. Coercivity of the available linkage

Remove the common catalyst C from (1.2), and let $\mathcal{M}$ be the resulting generator on $(a, b) \in \mathbb{N}_0^2$. Its reaction graph on $0, A, B$ is strongly connected. For every fixed $\ell = (\ell_A, \ell_B) \in \mathbb{R}^2$, define

$$G_\ell(a, b) = \log(a!) + \log(b!) + \ell_A a + \ell_B b. \quad (5.1)$$

7.5.1 Lemma 5.1: superlinear negative physical-time drift

There are constants $k > 0$ and $K < \infty$, depending only on the graph, rates, and ℓ , such that

$$\mathcal{M}G_\ell(a, b) \leq -kn \log \log(n + e^e) + Kn, \quad n = a + b. \quad (5.2)$$

The exact increments are

$$\begin{array}{l|l} 0 \rightarrow A & \log(a+1) + \ell_A \\ 0 \rightarrow B & \log(b+1) + \ell_B \\ A \rightarrow 0 & -\log a - \ell_A \\ B \rightarrow 0 & -\log b - \ell_B \\ A \rightarrow B & \log(b+1) - \log a + \ell_B - \ell_A \\ B \rightarrow A & \log(a+1) - \log b + \ell_A - \ell_B. \end{array} \quad (5.3)$$

Let $m = \max(a, b)$, $s = \min(a, b)$, and call their species D, E , respectively. Thus $m \geq n/2$. Fixed linear-corrector contributions are at most $K_1 n$, immigration contributions are at most $K_1 \log(n+2)$, and a conversion from the minor to the dominant species has positive part bounded by

$$s \log \frac{m+1}{s \vee 1} \leq m+1. \quad (5.4)$$

If $D \rightarrow 0$ is present, its death term gives

$$\mathcal{M}G_\ell(a, b) \leq -k_1 m \log m + K_2 n, \quad (5.5)$$

which is stronger than (5.2).

Suppose $D \rightarrow 0$ is absent. Strong connectivity on the three vertices forces both $D \rightarrow E$ and $E \rightarrow 0$: without the first edge D cannot leave its vertex, and without the second no path from D can reach zero. Their combined negative contribution, after absorbing fixed linear terms, is a positive constant times

$$R(m, s) = m \log \frac{m}{s+1} + s \log(s \vee 1). \quad (5.6)$$

The first summand is bounded below by -1 , because $s \leq m$. Put $L = \log(m + e^e)$. If

$$s+1 \leq m/\sqrt{L},$$

then the first summand is at least $\frac{1}{2}m \log L$. Otherwise, for all sufficiently large m ,

$$\begin{aligned} s \log(s \vee 1) &\geq (m/\sqrt{L} - 1) \log(m/\sqrt{L} - 1) \\ &\geq c_0 m \sqrt{\log m} \geq c_0 m \log \log(m + e^e). \end{aligned} \quad (5.7)$$

Equations (5.4)–(5.7), with $m \geq n/2$, prove (5.2) after enlarging K over a finite set.

This includes every immigration event at its natural rate and holds for the linear corrector selected by either Section 3 or Section 4.

7.6 6. Foster conclusion for six supports

For each of the six A, B -only supports in (2.1), $C = c$ is fixed on a closed class.

If $c = 0$, every L_1 propensity vanishes. The positive invariant of L_0 has positive A, B coefficients, so its population shells are finite. Every closed irreducible class is finite.

Suppose $c \geq 1$. The full generator is

$$\mathcal{L} = \mathcal{L}_0 + c\mathcal{M}. \quad (6.1)$$

Choose F from Lemma 3.1 for a two-complex L_0 , or from (4.6) for the three-complex support. Add the constant-in-class term $\log(c!)$. Factorial growth dominates every fixed linear corrector, so F , after an additive shift, is nonnegative and proper. Lemmas 3.2 or 4.1 and Lemma 5.1 give

$$\mathcal{L}F(x) \leq K_0(1+n) + c[-kn \log \log(n + e^e) + Kn] \longrightarrow -\infty \quad (6.2)$$

as $n = A + B \rightarrow \infty$. Thus $\mathcal{L}F \leq -1$ outside a finite subset of each closed class.

The process is nonexplosive. In these supports, every population-increasing channel has propensity at most a constant times $1 + A + B$; quadratic channels preserve molecular count or point inward. Comparison with a linear pure-birth process suffices.

The standard countable-state continuous-time Foster theorem applied to the proper F gives finite mean physical return to a finite set, hence positive recurrence on each closed irreducible class. No expected bound on the number of L_0 jumps is asserted or needed.

This is a global factorial potential, not the marked residual potential. If

$$V_{\text{res}}(x, t) = \sum_i \log((x_i - t_i)!),$$

then

$$F_\ell(x) = V_{\text{res}}(x, t) + \log(x)_t + \ell \cdot x. \quad (6.3)$$

Thus the seam closes by a direct, unconditioned physical-time generator argument. No residual-only target episode is claimed to have negative reward.

7.7 7. The remaining compatible support $\{0, 2C\}$

Let the aggregate rates of $0 \rightarrow 2C$ and $2C \rightarrow 0$ be $\alpha, \beta > 0$. The C process is autonomous. On parity class $p \in \{0, 1\}$, its stationary law is

$$\pi_C(p + 2k) = \frac{1}{Z_p} \left(\frac{\alpha}{\beta} \right)^k \frac{p!}{(p + 2k)!}, \quad k \geq 0. \quad (7.1)$$

It is summable, and

$$\pi_C(c)\alpha = \pi_C(c + 2)\beta(c + 2)(c + 1). \quad (7.2)$$

Let Y be the unmodulated monomolecular A, B chain obtained from L_1 by removing the common factor C , and let $\mathcal{L}_Y$ be its generator. Strong connectivity gives a positive traffic solution (ρ_A, ρ_B) , and

$$\pi_Y(a, b) = e^{-\rho_A - \rho_B} \frac{\rho_A^a}{a!} \frac{\rho_B^b}{b!} \quad (7.3)$$

is stationary. The joint generator is

$$\mathcal{L} = \mathcal{L}_C + C\mathcal{L}_Y. \quad (7.4)$$

For every finitely supported f ,

$$\begin{aligned} \sum_{c, a, b} \pi_C(c) \pi_Y(a, b) \mathcal{L}f(c, a, b) &= \sum_{a, b} \pi_Y(a, b) \sum_c \pi_C(c) \mathcal{L}_C f(c, a, b) \\ &\quad + \sum_c c \pi_C(c) \sum_{a, b} \pi_Y(a, b) \mathcal{L}_Y f(c, a, b) = 0. \end{aligned} \quad (7.5)$$

Therefore $\pi_C \otimes \pi_Y$ is an invariant probability. Each parity product class is irreducible: move C within its parity class, keep it positive while executing any prescribed finite Y path, then move it to the requested endpoint. That finite sequence has positive probability. Population-increasing rates are at most linear in C , so the process is nonexplosive. Every closed irreducible class is therefore positive recurrent.

This remains valid on the even class containing $C = 0$: the Y clock pauses there, but $0 \rightarrow 2C$ has positive rate, and (7.5) includes the pause exactly.

7.8 8. Exact boundary of the argument

The finite reduction contains 25 unique positive-active-invariant shielded supports. Sixteen have two vertices, so Lemma 3.2 gives their fast generator at-most-linear positive drift after a factorial-linear correction. Lemma 4.1 adds $\{2A, A + B, 2B\}$. Thus the fast-generator bound covers 17 of 25 supports for every rate and orientation.

That fact alone does not prove recurrence for arbitrary second linkages: the other linkage must supply coercive negative drift in every divergent coordinate. The eight multi-vertex supports not covered by the present fast-generator lemma are

$$\begin{aligned} &\{0, A + C, B + C\}, & \{0, C, 2C\}, \\ &\{A, A + C, B + C\}, & \{A, B, A + C\}, & \{A, B, B + C\}, \\ &\{B, 2A, B + C\}, & \{B, A + C, B + C\}, \\ &\{A, B, A + C, B + C\}. \end{aligned} \tag{8.1}$$

Six supports in (8.1) have single-linkage deficiency zero. The two deficiency-one supports are

$$\{0, C, 2C\}, \quad \{A, B, A + C, B + C\}. \tag{8.2}$$

This note therefore does not close all 25 masks in arbitrary pairings and does not establish T3-2.

7.9 9. Signed one-active supports

The non-positive-invariant output of the shielded reduction consists, in the displayed orientation, of

$$\{0, A, B + C\}, \quad \{0, 2A, B + C\}, \quad \{A, 2A, B + C\}, \quad \{0, A, 2A, B + C\}. \tag{9.1}$$

Every support in (9.1) preserves

$$W = B - C. \tag{9.2}$$

This quantity is signed and not proper: the ray $B = C \rightarrow \infty$ lies in one level set. An episode can decrease $B - C$ while every positive workload increases. Therefore neither W descent nor the service endpoint $W_\tau \leq W_0 - 1$ implies positive recurrence by physical-time Foster.

The first three supports in (9.1), in isolation, have deficiency zero; the four-vertex support has deficiency one. Product form for an isolated deficiency-zero linkage is not preserved after arbitrary coupling. The direct argument in this note therefore covers none of (9.1) globally. They do not occur in the exact seam (1.2), because each contains $B + C$, already assigned to L_1 .

7.10 10. Executable certificate

Run:

```
PYTHONPATH=src python3 src/exact_shielded_seam.py
PYTHONPATH=src python3 -m unittest tests/test_exact_shielded_seam.py -v
```

The certificate checks:

1. 25 unique positive-invariant shielded supports;
2. the exact seven-support filter (2.1);
3. the 16 two-vertex plus one directed-triple fast-bound partition;
4. the exact eight supports in (8.1);
5. the three deficiency-one supports;
6. the four signed supports and their $B - C$ invariant;

7. the reversible-pair finite-difference algebra;
8. endpoint signs (4.3) for every strongly connected digraph on three vertices; and
9. the detailed-balance recursion (7.2).

These are finite algebraic checks. The analytic estimates in Sections 3–7 are mathematical proofs, not computational certificates.

8 The signed-service seam: a full physical-time proof

8.1 1. Scope and result

Write $D = B + C$ for the mixed complex and

$$W = B - C. \quad (1.1)$$

This note treats a pure- C linkage L_C with support

$$\{C, 2C\} \quad \text{or} \quad \{0, C, 2C\}, \quad (1.2)$$

and a disjoint mixed linkage L_M whose support is one of

$$\{0, A, D\}, \quad \{0, 2A, D\}, \quad \{A, 2A, D\}, \quad \{0, A, 2A, D\}. \quad (1.3)$$

Every labelled reaction graph is arbitrary subject to strong connectivity, and all rate constants are positive. Parallel channels are aggregated only to simplify notation. Stochastic propensities are falling-factorial mass-action propensities. The $A \leftrightarrow B$ images are obtained by relabeling.

The first support in (1.2) includes the exceptional architecture

$$\{C, 2C\} \ \& \ \{0, A, 2A, B + C\}, \quad (1.4)$$

and the second includes

$$\{0, C, 2C\} \ \& \ \{A, 2A, B + C\}. \quad (1.5)$$

The conclusion is classwise.

Signed-service seam theorem. Every closed irreducible population class of every network in (1.2)–(1.3) is positive recurrent. The process is nonexplosive. No recurrence assertion for the raw embedded jump chain is used.

The proof has two parts. A proper quadratic potential gives ordinary physical-time generator drift whenever A or C is large, or whenever BC is large. The only remaining tube has bounded A , $C = 0$, and $B \rightarrow \infty$. On that tube a fast-phase return cycle has a B -decreasing service probability of order $1/B$, an adverse-service probability of order $1/B^2$, and uniformly bounded physical duration. The quadratic potential converts this into a uniform negative return-cycle drift.

8.2 2. Why $V + \gamma(B - C)$ is not the repair

The linear signed correction is proper when added to a factorial potential, but it is too small at the service bottleneck.

Consider the strongly connected mixed cycle

$$0 \longrightarrow D \longrightarrow 2A \longrightarrow A \longrightarrow 0 \quad (2.1)$$

together with $C \rightleftharpoons 2C$. Start at

$$x_N = (0, N, 0). \quad (2.2)$$

The first mixed reaction $0 \rightarrow D$ produces $(0, N+1, 1)$. At that target, the probability that $C \rightarrow 2C$ fires before the D -source reaction is

$$\frac{\alpha}{d(N+1) + O(1)} = \Theta(N^{-1}), \quad (2.3)$$

where α is the $C \rightarrow 2C$ rate constant and d is the aggregate rate constant out of D . On failure, the mixed cycle returns to the bounded phase in $O(1)$ mean time. On success, two rapid D -source clearings, followed by return of the A -phase, take x_N to x_{N-1} with probability $1 - O(N^{-1})$.

Thus a bounded-mean regeneration cycle has service probability $\Theta(N^{-1})$. For

$$F(x) = \sum_i \log(x_i!) + \gamma(B - C), \quad (2.4)$$

the successful return reward is only

$$F(x_{N-1}) - F(x_N) = -\log N - \gamma. \quad (2.5)$$

Consequently the negative expected reward per regeneration cycle is only $O((\log N)/N)$, while the mean physical duration stays bounded away from zero. No fixed $\eta > 0$ can give

$$\mathbb{E}[\Delta F + \eta\tau] \leq -\varepsilon \quad (2.6)$$

through these cycles. Pointwise drift fails even more plainly: at x_N , the reaction $0 \rightarrow D$ has factorial reward $\log(N+1)$, and the signed term is unchanged.

This does not disprove recurrence. It shows that the proposed factorial-plus-linear signed repair has the wrong growth. A potential $f(B)$ needs

$$f(N) - f(N-1) = \Omega(N) \quad (2.7)$$

to convert a service probability of order $1/N$ into constant return-cycle drift. Quadratic growth is the minimal natural choice.

8.3 3. A common product-form shortcut for three mixed complexes

When $L_C = \{C, 2C\}$ and L_M is one of the first three supports in (1.3), both linkages have deficiency zero. This already closes those three cases.

Indeed, $C \rightleftharpoons 2C$ has complex-balanced value

$$\theta_C = \alpha/\beta. \quad (3.1)$$

The mixed linkage has invariant $B - C$. Its complex-balance equations determine θ_A and the product $\theta_B\theta_C$, but the invariant scaling

$$(\theta_B, \theta_C) \mapsto (e^s \theta_B, e^{-s} \theta_C) \quad (3.2)$$

leaves every mixed monomial unchanged. Hence its complex-balanced vector can always be chosen with the value (3.1). The two linkage balances then hold at one common vector. The product-Poisson measure at that vector, restricted to any irreducible population class, is a summable invariant measure. This proves positive recurrence on every closed class.

The four-complex support in (1.3) has deficiency one, so this shortcut is not available for arbitrary rates. The physical-time proof below treats it and also gives a uniform proof for all supports in (1.3), including the second pure- C support in (1.2).

8.4 4. A rate-dependent positive phase workload

Let

$$u(0) = 0, \quad u(A) = p_A, \quad u(2A) = 2p_A, \quad u(D) = p_B + p_C, \quad (4.1)$$

where $p_A, p_B, p_C > 0$. Put

$$U(a, b, c) = p_A a + p_B b + p_C c. \quad (4.2)$$

The weights can be selected so that every highest-degree mixed source has strictly negative U -drift.

8.4.1 Lemma 4.1 (weight interval)

For every strongly connected orientation and every positive rate vector on one of the supports (1.3), positive weights in (4.2) can be chosen with the following properties.

- If $2A$ is present, the aggregate U -increments out of both $2A$ and D are strictly negative.
- If $2A$ is absent, the aggregate U -increments out of both A and D are strictly negative.

To see this, first suppose $2A$ is present. Normalize

$$u(2A) = 1, \quad u(A) = \frac{1}{2}, \quad u(D) = r. \quad (4.3)$$

Write $\kappa_{ij} = 0$ for an absent edge. The two required inequalities are

$$r < L_2^{\text{up}} := 1 + \frac{\kappa_{2,0} + \kappa_{2,A}/2}{\kappa_{2,D}} \quad (4.4)$$

with $L_2^{\text{up}} = +\infty$ when $\kappa_{2,D} = 0$, and

$$r > L_D^{\text{low}} := \frac{\kappa_{D,A}/2 + \kappa_{D,2}}{\kappa_{D,0} + \kappa_{D,A} + \kappa_{D,2}}. \quad (4.5)$$

Terms for vertices absent from the support are omitted. One has $L_D^{\text{low}} \leq 1 \leq L_2^{\text{up}}$. Equality on both sides would say that $2A$ and D only point to each other, contradicting strong connectivity with the remaining vertex or vertices. Thus the open interval is nonempty.

For $\{0, A, D\}$, normalize $u(A) = 1, u(D) = r$. The interval is

$$\frac{\kappa_{D,A}}{\kappa_{D,0} + \kappa_{D,A}} < r < 1 + \frac{\kappa_{A,0}}{\kappa_{A,D}}, \quad (4.6)$$

with the same convention for a missing denominator. Strong connectivity again makes the interval nonempty. Finally split the selected value $u(D)$ as $p_B + p_C$ with both summands positive.

8.4.2 Lemma 4.2 (phase generator drift)

For the weights from Lemma 4.1 there are constants $K, k > 0$ such that

$$\mathcal{L}U \leq K - k(r_A(A) + BC + C^2), \quad (4.7)$$

where $r_A(A) = A$ for the support $\{0, A, D\}$, and $r_A(A) = A^2$ for the other three supports.

For the mixed linkage this is immediate after collecting propensities by source complex. The 0-source contribution is constant, the A -source contribution is at most linear, and Lemma 4.1 makes the $2A$ - and D -source coefficients strictly negative. If $2A$ is absent, the A -source coefficient itself is strictly negative.

For $L_C = \{C, 2C\}$,

$$\mathcal{L}_C U = p_C[\alpha C - \beta C(C - 1)]. \quad (4.8)$$

For $L_C = \{0, C, 2C\}$, every edge out of $2C$ lowers C , so the quadratic coefficient is again strictly negative; the 0- and C -source terms are at most constant and linear. Absorbing those lower orders proves (4.7).

All jumps of U are bounded. The total reaction rate is at most a constant times

$$1 + r_A(A) + BC + C^2. \quad (4.9)$$

Consequently

$$\mathcal{L}(U^2) = 2U\mathcal{L}U + \sum_r \lambda_r(x)(\Delta_r U)^2. \quad (4.10)$$

Equations (4.7)–(4.10) show that there are $R < \infty$ and $\varepsilon > 0$ such that

$$\mathcal{L}(U^2) \leq -1 - \varepsilon \quad (4.11)$$

outside a finite set and the tube

$$\mathcal{T}_R = \{(a, b, 0) : a \leq R\}. \quad (4.12)$$

Indeed, bounded $r_A(A) + BC + C^2$ forces A, C to be bounded; if $C \geq 1$, it also forces B to be bounded. Thus (4.12) is the only infinite region not handled by pointwise generator drift. Notice that

$$\Phi(x) := 1 + U(x)^2 \quad (4.13)$$

is proper.

8.5 5. The fast trace on a large positive signed shell

This section supplies the uniform statement that was absent from the old finite-shell service argument.

Fix $W = N > 0$. Until a pure- C reaction occurs,

$$B = N + C. \quad (5.1)$$

Let $d_M > 0$ be the aggregate rate constant out of D . At $C \geq 1$, the aggregate mixed D -source rate is exactly

$$d_M(N + C)C. \quad (5.2)$$

Delete D from the mixed complex graph. An edge $i \rightarrow D$, followed by the first $D \rightarrow j$, induces the effective rate

$$\bar{\kappa}_{ij} \supset \kappa_{iD} \frac{\kappa_{Dj}}{d_M}. \quad (5.3)$$

Direct lower-complex edges retain their rates. The resulting graph on

$$\{0, A\}, \quad \{0, 2A\}, \quad \{A, 2A\}, \quad \text{or} \quad \{0, A, 2A\} \quad (5.4)$$

is strongly connected: replace every occurrence of the single deleted vertex in a simple directed path by (5.3).

The population chain of this effective one-species network is positive recurrent on each class. In the four cases it is respectively an immigration-death chain, a parity-preserving pair-immigration/quadratic-death chain, a positive-population linear-birth/quadratic-death chain, or an open one-species binary chain. A power of $1 + A$ has negative generator drift of one degree higher outside a finite set. In particular, return times to the minimal atom

$$a_* = 0, \quad a_* = 0 \text{ or } 1, \quad a_* = 1, \quad a_* = 0 \quad (5.5)$$

for the four cases in (5.4) have finite moments. The choice 0 or 1 in the second case is the conserved A -parity.

8.5.1 Lemma 5.1 (uniform trace and occupation estimates)

Start at

$$e_N = (a_*, N, 0) \quad (5.6)$$

and let σ_N be the next return to the same minimal A, C -phase, after at least one reaction. First suppress pure- C reactions. Then there are constants $K < \infty$, $j > 0$, independent of N , such that

$$\sup_{N \geq N_0} \mathbb{E}_{e_N} \sigma_N^2 \leq K, \quad (5.7)$$

the expected number J_N of entrances into D during the cycle obeys

$$j \leq \mathbb{E} J_N \leq K, \quad (5.8)$$

and

$$\mathbb{E} \int_0^{\sigma_N} C_s ds = \Theta(N^{-1}), \quad \mathbb{E} \int_0^{\sigma_N} (C_s)_2 ds = O(N^{-2}). \quad (5.9)$$

The same estimates hold from any A in a fixed finite set before the first hit of a_* .

Here is a proof, including the tail point needed for uniformity. On a fixed $W = N$ shell, subtract the constant $p_B N$ from U and put

$$R_N = p_A A + (p_B + p_C) C. \quad (5.10)$$

The mixed generator calculation from Lemma 4.2 gives

$$\mathcal{L}_M R_N \leq K - k\{r_A(A) + NC + C^2\}. \quad (5.11)$$

Here is the promised uniform high-power estimate. Put

$$Q_N(A, C) = r_A(A) + NC + C^2.$$

Every jump of R_N is bounded, and the total mixed rate is at most $K(1 + Q_N)$. Taylor's formula for an integer $m \geq 4$, followed by (5.11), therefore gives, outside a phase set independent of N ,

$$\begin{aligned} \mathcal{L}_M(1 + R_N)^m &\leq m(1 + R_N)^{m-1} \mathcal{L}_M R_N \\ &\quad + K_m(1 + Q_N)(1 + R_N)^{m-2} \\ &\leq K_m - c_m Q_N(1 + R_N)^{m-1} \\ &\leq K_m - c'_m(1 + R_N)^m. \end{aligned} \quad (5.11a)$$

The last inequality uses $1 + Q_N \geq c(1 + R_N)$, uniformly for $N \geq N_0$. Stopped Dynkin estimates, first for (5.11a) and then for its lower powers, give uniform moments of A, C and of the time to a fixed bounded phase set. The same stopped estimates, applied with successively higher m , control the integrated lower-source intensities and their required moments.

Inside the bounded phase set, (5.2) makes an induced D -excursion last $O(N^{-1})$, and its exit target has the state-independent probabilities κ_{Dj}/d_M . Each edge of a fixed effective path is therefore realized, in a uniformly bounded time, with probability bounded away from zero. The strong effective graph (5.3) gives a uniform minorization to the atom (5.5). Geometric repetition proves (5.7) and the upper bound in (5.8). A fixed feasible reaction sequence from the atom that enters D has probability bounded away from zero before return, proving the lower bound.

Now apply the stopped estimates over this atom cycle. Because its two endpoints have the same value of R_N , (5.7), (5.11a), and the corresponding lower-power inequalities give

$$\mathbb{E}_{e_N} \int_0^{\sigma_N} Q_N(A_s, C_s)(1 + R_N(A_s, C_s))^{m-1} ds \leq K_m. \quad (5.11b)$$

There is also a useful exact compensator calculation. During the mixed-only cycle, entrances to D raise C by one and D -source exits lower it by one. Since the cycle starts and ends at $C = 0$,

$$d_M \mathbb{E} \int_0^{\sigma_N} (N + C_s) C_s ds = \mathbb{E} J_N. \quad (5.12)$$

Applying the generator to $(C)_2$ gives

$$d_M \mathbb{E} \int_0^{\sigma_N} (N + C_s) C_s (C_s - 1) ds = \mathbb{E} \int_0^{\sigma_N} C_s I(A_s) ds, \quad (5.13)$$

where $I(A)$ is the total lower-source intensity of reactions entering D . It satisfies $I(A) \leq K(1 + A^2) \leq K(1 + R_N)^2$. Taking $m \geq 4$ in (5.11b) and using the term NC in Q_N gives

$$\mathbb{E} \int_0^{\sigma_N} C_s I(A_s) ds = O(N^{-1}). \quad (5.14)$$

This argument uses occupation bounds rather than assigning genealogical partners to entrances and exits. Equation (5.13) and (5.14) first give

$$\mathbb{E} \int_0^{\sigma_N} (C_s)_2 ds = O(N^{-2}). \quad (5.15)$$

Since $C^2 = C + (C)_2$, equation (5.12) can now be written as

$$\mathbb{E} J_N = d_M N \mathbb{E} \int_0^{\sigma_N} C_s ds + O(N^{-1}). \quad (5.16)$$

The two bounds in (5.8) prove the first estimate in (5.9). This explicitly rules out a hidden countable-phase tail: the second factorial occupation is one order smaller than the first, uniformly from the regeneration phases used below.

8.6 6. The two-complex pure- C linkage

Now restore

$$C \xrightarrow{\alpha} 2C, \quad 2C \xrightarrow{\beta} C. \quad (6.1)$$

The first reaction lowers W by one and the second raises it by one. Let $\hat{\sigma}_N$ be the first of (i) the next return to the minimal A, C -phase under the full generator and (ii) exit of W from $[N/2, 3N/2]$. Let M_N^- and M_N^+ be the numbers of the two reactions in (6.1), respectively, before this stopping time.

8.6.1 Lemma 6.1 (interrupted-cycle estimate)

There are constants $0 < c < C < \infty$ such that, for all sufficiently large N ,

$$\frac{c}{N} \leq \mathbb{E} M_N^- \leq \frac{C}{N}, \quad \mathbb{E} M_N^+ \leq \frac{C}{N^2}, \quad (6.2)$$

$$\mathbb{E} (M_N^- + M_N^+)^2 \leq \frac{C}{N}, \quad \mathbb{E} \hat{\sigma}_N^2 \leq C. \quad (6.3)$$

To prove the lemma, cut the path at the first mixed-only phase return or the first pure- C mark, whichever occurs first, and repeat after a mark. Uniformly on shells $n \in [N/2, 3N/2]$, (5.11a)–(5.11b) and (5.15) imply that an attempt started with uniformly bounded phase moments has a downward-mark hazard $O(N^{-1})$, an adverse-mark hazard $O(N^{-2})$ when it starts from the atom, and a uniformly bounded second duration moment. For the lower bound, use the fixed feasible sequence from the end of Lemma 5.1 to enter D with probability bounded away from zero. At the resulting bounded phase with $C = 1$, the probability that the downward clock rings before the D -exit clock is at least c/N . Hence the first attempt has a downward mark with probability at least c/N .

A downward mark changes the phase by one bounded jump. Starting from that marked phase, the chance of any further mark before mixed return is $O(N^{-1})$; in particular, the chance of an adverse mark there is $O(N^{-1})$. Thus an adverse mark following a downward mark has probability $O(N^{-2})$. A first adverse mark already has probability $O(N^{-2})$ by (5.15). Iterating this estimate gives a geometric interruption expansion: the probability of at least one mark is $O(N^{-1})$, while, conditional on a mark, the remaining number of marks and the remaining return duration have uniformly bounded second moments. Leaving the shell window would require at least $N/2$ marks, so its probability is smaller than every power of N^{-1} . The high-power phase bounds make its contribution to every first and second endpoint estimate displayed below smaller than every power as well. This proves (6.2)–(6.3); it does not assume that the first mixed-only cycle is uninterrupted or continue a rare escaped path without control.

Since

$$\Delta W = M_N^+ - M_N^-,$$

Lemma 6.1 gives constants $0 < d_- < D_- < \infty$ such that

$$-\frac{D_-}{N} \leq \mathbb{E} \Delta W \leq -\frac{d_-}{N}, \quad \mathbb{E}(\Delta W)^2 = O(N^{-1}). \quad (6.4)$$

At a minimal phase, $C = 0$, hence $B = W = N$, and

$$U(e_N) = p_A a_* + p_B N. \quad (6.5)$$

On the atom-return event the endpoint has the same A, C -phase, so its U -increment is exactly $p_B \Delta W$. The shell-exit event contributes $o(1)$ to the expected Φ -increment by the endpoint estimate in Lemma 6.1. Thus (6.3)–(6.4) give

$$\mathbb{E}_{e_N}[\Phi(X_{\hat{e}_N}) - \Phi(e_N)] \leq -2p_B^2 d_- + o(1). \quad (6.6)$$

Thus the regeneration drift is bounded above by a negative constant for all sufficiently large N , while its physical duration and endpoint second moments are uniformly bounded.

This is the rigorous form of signed service. It does not wait for an upward record of the autonomous $C \rightleftharpoons 2C$ chain. It only uses the $O(N^{-1})$ -long visits to $C = 1$ created by the physical mixed linkage. The factorial record-time obstruction for starting from large C is therefore irrelevant; large C is handled by (4.11).

If the mixed support has no 0, the states $A = C = 0$, with B fixed, are singleton closed classes. There is one further boundary family: when $A = B = 0$ and $C > 0$, only the pure linkage is enabled. For $\{C, 2C\}$ this is the linear-birth/quadratic-death chain on the positive integers, hence is positive recurrent. Every remaining closed class uses the positive atom $a_* = 1$, and the preceding argument is unchanged.

8.7 7. The three-complex pure- C linkage

Suppose instead that $L_C = \{0, C, 2C\}$. Let ρ_k be the aggregate rate of the edges $0 \rightarrow kC$, for $k = 1, 2$. At least one ρ_k is positive. Every such reaction lowers W by k .

The limiting fast trace can be written explicitly. For every lower target j , put $q_j = \kappa_{Dj}/d_M$. Start with the effective one-species generator from (5.3), and add the bounded batch transitions

$$a \mapsto a + j_1 + \cdots + j_k \quad \text{at rate} \quad \rho_k q_{j_1} \cdots q_{j_k}. \quad (7.1)$$

These transitions represent a $0 \rightarrow kC$ event followed by k successive D -exits. They have constant rates and bounded increments. The original effective generator has linear death in the $\{0, A\}$ case and quadratic death in the other three cases. Hence, for every sufficiently large integer m , the augmented generator $\bar{\mathcal{L}}$ satisfies

$$\bar{\mathcal{L}}(1 + a)^m \leq K_m - c_m(1 + a)^m \quad (7.2)$$

outside a finite set. Thus its classwise minimal atom has return times and endpoint values of every required polynomial moment.

This limiting statement is uniform for the physical chain. Let $\tilde{\sigma}_N$ be the first of the next minimal-phase return and exit of W from $[N/2, 3N/2]$. The calculation leading to (5.11a), with the constant-rate 0-sources and the at-most-linear C -sources included, gives the same high-power phase estimate; every positive linear term is absorbed by NC , C^2 , or the stable lower-phase drift. On a fixed bounded phase set, a newly created C particle is cleared at rate at least $d_M N$. The probability that a nonzero-source pure reaction interrupts all k clearings is therefore $O(N^{-1})$, uniformly on that set. The high-power estimate makes the complement uniformly negligible. Combining this with (7.2) and the same finite-set minorization used in Lemma 5.1 proves

$$\mathbb{E}\tilde{\sigma}_N^2 \leq K. \quad (7.3)$$

Leaving the shell window requires at least $N/4$ pure reactions. The stopped high-power estimates give moments of every order for their count, so the shell-exit probability and its first and second endpoint contributions are smaller than every prescribed power of N^{-1} .

Let $Z_{k,N}$ count the $0 \rightarrow kC$ reactions during this cycle, and let H_N count all pure reactions whose source is C or $2C$. The preceding estimates and their stopped high-power versions give

$$\mathbb{E}H_N = O(N^{-1}), \quad \mathbb{E}H_N^2 = O(N^{-1}), \quad \mathbb{E}\left(\sum_{k=1}^2 kZ_{k,N}\right)^2 \leq K. \quad (7.4)$$

The return kernels converge on bounded phases to (7.1), and (7.2) supplies uniform integrability. Consequently

$$\mathbb{E}\sum_{k=1}^2 kZ_{k,N} = d_0 + o(1) \quad (7.5)$$

for a constant $d_0 > 0$. Strict positivity follows already from the positive probability that the first reaction at the atom is an enabled 0-source reaction. Mixed reactions leave W unchanged. Equations (7.4)–(7.5), with the bounded signed increments of the H_N reactions, therefore give

$$\mathbb{E}\Delta W = -d_0 + o(1), \quad \mathbb{E}(\Delta W)^2 \leq K. \quad (7.6)$$

On the minimal-phase return event the endpoint again has $\Delta U = p_B \Delta W$; the shell-exit contribution is $o(N)$. Equations (7.3) and (7.6) yield

$$\mathbb{E}[\Phi(X_{\tilde{\sigma}_N}) - \Phi(e_N)] \leq -cN \quad (7.7)$$

for some $c > 0$ and all large N .

For the architecture (1.5), the lower effective phase is the positive linear-birth/quadratic-death chain on $\{A, 2A\}$, augmented by the bounded zero-order batches just described. Its minimal recurrent atom is $A = 1$. This proves the previously exceptional second architecture without a finite- C truncation.

If the mixed support has no 0, the boundary $A = B = 0$ carries only the autonomous pure- C chain. Its constant/linear births and strictly negative quadratic $2C$ -source drift make every closed class positive recurrent. All other nontrivial classes use the atom $a_* = 1$.

8.8 8. Hybrid Foster closure

We now combine the pointwise phase estimate with the regeneration estimate. Fix a closed irreducible population class. If it is a finite or singleton boundary class, there is nothing to prove. The autonomous one-dimensional boundary classes identified at the ends of Sections 6 and 7 are already positive recurrent. In every remaining class choose the appropriate minimal phase atom from (5.5).

Outside the tube (4.12), stop on first entrance to the tube or to a fixed finite set. Dynkin's formula and (4.11) charge physical time with a strict negative margin. On the tube, first hit the minimal phase atom and then run one regeneration cycle. The high-power bounds above make the first-hit duration and endpoint moments uniform.

Here is the endpoint check. If the starting phase is $a > a_*$, then the preliminary return changes the linear workload by

$$p_A(a_* - a) + p_B \Delta W.$$

The first term is a fixed negative number. In the two-complex case the positive part of the second term has expectation $O(N^{-2})$, by the adverse-mark bound in Lemma 6.1; in the three-complex case its positive part is $O(N^{-1})$, by (7.4). Thus the quadratic cross term is nonpositive for all large N . If $a = a_*$, the preliminary return is empty. The appended atom cycle then supplies (6.6) or (7.7). Consequently there are $\eta, \delta > 0$ and a finite set K such that the composite tube stopping time τ_x satisfies

$$\mathbb{E}_x[\Phi(X_{\tau_x}) - \Phi(x) + \eta\tau_x] \leq -\delta \quad (8.1)$$

for every tube state outside K .

Choose $\eta > 0$ smaller than the regeneration drift divided by the uniform mean cycle duration. Alternating the two stopping rules gives, up to the first hit of a finite set K ,

$$\mathbb{E}\Phi(X_{T_m}) + \eta\mathbb{E}T_m + \delta\mathbb{E}N_m \leq \Phi(X_0), \quad (8.2)$$

where N_m is the number of completed large- W regeneration cycles and $\delta > 0$. The endpoint estimates in Lemma 6.1 and Section 7, together with the high-power bound in Lemma 5.1, justify bounded optional sampling and passage to the limit. If K were avoided, either infinitely much physical time or infinitely many regeneration cycles would accumulate, contradicting (8.2). Hence K is reached in finite mean physical time.

The finite-set trace in a closed irreducible class has finite mean return, so the class is positive recurrent. This is a physical-time argument. It does not charge the number of rapid D -source reactions.

Finally, the process is nonexplosive. Every population-increasing channel has propensity at most linear: 0-source channels have constant rates, A - and C -source channels have linear rates, and every quadratic source points to a complex of no larger molecularity. Comparison of total population with a linear pure-birth process suffices.

8.9 9. Exact boundary and audit conclusions

The following points are proved above.

1. A linear signed correction to the factorial potential does not close the physical-time seam; the bottleneck requires quadratic shell growth.
2. A rate-dependent positive linear workload U always exists and controls every phase-large state through the proper potential U^2 .
3. The large positive- W fast trace has uniform return-time and endpoint moments. Its C -occupation is $\Theta(W^{-1})$, while its $(C)_2$ -occupation is $O(W^{-2})$.
4. For $C \rightleftharpoons 2C$, downward signed service is order W^{-1} and adverse service is order W^{-2} ; quadratic drift is therefore uniformly negative on regeneration cycles.
5. When 0 is in the pure- C linkage, downward service has order one and the return drift is even stronger.
6. Large C is controlled by quadratic physical death, not by an upward record-time policy.

No null or transient closed class occurs in this signed family. The proof uses the exact supports (1.2)–(1.3); it does not claim a generic countable-phase theorem for other atlas pairings.

8.10 10. Exact limit of the superset interpretation

The preceding theorem is not monotone under adding complexes to the pure- C linkage. The global interface certificate makes the scope gap exact. Among the 358 residual ordered pairs with one of the four signed supports in (1.3), 107 available supports contain both C and $2C$. Of those, 69 are already certified by the global tier criterion and 38 remain. The other 161 tier failures do not even contain the two-complex service support, so no superset version of this note could address them.

The 38 remaining supersets have ten inclusion-minimal ordered pairs:

signed support	minimal available additions to $\{C, 2C\}$
$\{0, 2A, B + C\}$	$A, A + B, A + C$
$\{0, A, 2A, B + C\}$	$A + B, A + C$
$\{0, A, B + C\}$	$2A, A + B, A + C$
$\{A, 2A, B + C\}$	$A + B, A + C$

Thus there are four analytic minimal types: a bounded A - or $2A$ -cofactor, an $A + B$ source, and an $A + C$ source. The B - and $2B$ -single-vertex additions are already tier-certified. Adding 0 alone is the exact three-complex case of Section 7.

Each of the four types changes a load-bearing estimate above. An A - or $2A$ -source can fire with $C = 0$, so W is no longer fixed during the lower-phase return. An $A + B$ -source has rate of order N as soon as one A particle is present, and becomes a second fast clock not contained in the stochastic complement (5.3). An $A + C$ intermediate introduces a second reaction that can have to beat the order- N ($B + C$)-clearing clock.

For example, consider an orientation containing



At the phase $A = 0, C = 1, B = N + 1$, the first arrow beats clearing with probability $\Theta(N^{-1})$. Conditional on that arrow, the second again beats clearing with probability $\Theta(N^{-1})$. Thus this particular direct service path has probability $\Theta(N^{-2})$, not the one-clock probability used in Lemma 6.1.

This observation alone is not a counterexample to an N^{-1} *averaged* service rate. The irreducible lower phase has a positive, though possibly very small, occupation of $A > 0$; on those visits $A + C \rightarrow 2C$ needs to win only one order- N race. Establishing the resulting coefficient requires a new countable-phase occupation and interruption theorem that includes the added reactions. It is not supplied by Lemmas 5.1 or 6.1.

Accordingly, (10.1) is a countersequence to a direct one-path restoration argument, while the A , $2A$, and $A + B$ cases invalidate the fixed- W , delete-only- D trace itself. None is asserted to be a null-recurrent or transient reaction network. No deletion or restoration monotonicity is used in the theorem above.

9 The residual pair: a physical-time core trace

9.1 1. Scope and result

This note concerns only the two linkage classes

$$\mathcal{L}_0 = \{B, 2A, B + C\}, \quad \mathcal{L}_1 = \{0, A, C\}. \quad (1.1)$$

Each labelled reaction graph is assumed strongly connected, parallel channels in the same direction are aggregated, and every displayed channel rate is positive. A missing directed edge has aggregate rate zero. All propensities are stochastic mass-action propensities.

The purpose of the argument is to avoid both stationary-only averaging and recurrence of the raw jump chain. The chain is sampled after a physical window of length $T/\sqrt{q + 1}$, where

$$q = A + 2B. \quad (1.2)$$

The fast linkage preserves q . A rate-dependent proper linear workload returns the process to a core on which $A = O(\sqrt{q})$ and $C = O(1)$. On the short physical window the exact fast dynamics have a deterministic

Riccati limit, while all positive q -hazards vanish. This gives a strict, shell-uniform negative drift for the core trace. Endpoint moments and the physical duration of the return appendage are included below.

Theorem 1.1. For every choice of strongly connected orientations and positive rates in (1.1), the minimal stochastic mass-action CTMC is nonexplosive. Every state in every closed irreducible population class is positive recurrent.

No finite inactive-coordinate box is assumed. The finite C -bound below is a state-dependent core produced by a proper Foster workload; excursions outside it are retained and charged.

9.2 2. Aggregated rates and two exact identities

Write the six possible rates in $\mathcal{L}_0$ as

edge	$B \rightarrow 2A$	$B \rightarrow B + C$	$2A \rightarrow B$	$2A \rightarrow B + C$	$B + C \rightarrow B$	$B + C \rightarrow 2A$	(2.1)
rate	x	y	s	r	t	v .	

Put $d = t + v$. Strong connectivity gives $d > 0$. Define

$$\zeta = \frac{2v}{d}, \quad \alpha = x + \frac{vy}{d}, \quad \beta = 2 \left(s + \frac{rt}{d} \right). \quad (2.2)$$

Both α and β are strictly positive. For example, $\alpha = 0$ would imply that neither B nor $B + C$ can reach $2A$; the proof for β is the reverse argument.

Set

$$Z = A + \zeta C.$$

Direct substitution gives the exact identity

$$\mathcal{L}_0 Z = 2\alpha B - \beta(A)_2 = \alpha(q - A) - \beta A(A - 1) =: P_q(A). \quad (2.3)$$

The coefficient of BC cancels because $-\zeta t + (2 - \zeta)v = 0$. Let a_q be the positive root of $P_q(a) = 0$. Then

$$\frac{a_q}{\sqrt{q+1}} \rightarrow a_* := \sqrt{\alpha/\beta}. \quad (2.4)$$

For $\mathcal{L}_1$, write its rates as k_{ij} , with $i, j \in \{0, A, C\}$. Then

$$\mathcal{L}_1 q = k_{0A} + k_{CA}C - wA, \quad w := k_{A0} + k_{AC} > 0. \quad (2.5)$$

The strict positivity of w follows from strong connectivity.

9.3 3. A proper workload and an exact core

The return workload is constructed algebraically. Use the conventions $d/v = +\infty$ if $v = 0$, and $(k_{C0} + k_{CA})/k_{CA} = +\infty$ if $k_{CA} = 0$. Strong connectivity implies that the open intervals

$$I_0 = \left(\frac{r}{s+r}, \frac{d}{v} \right), \quad I_1 = \left(\frac{k_{AC}}{k_{A0} + k_{AC}}, \frac{k_{C0} + k_{CA}}{k_{CA}} \right) \quad (3.1)$$

are nonempty and contain values arbitrarily close to one. If an endpoint equals one, the opposite endpoint is strict; equality at both ends would disconnect one of the three-vertex graphs.

Choose $\lambda \in I_0$ and $\rho \in I_1$, sufficiently close to one that $2\rho - \lambda > 0$, and put

$$U = \rho A + (2\rho - \lambda)B + C. \quad (3.2)$$

Define the positive constants

$$\begin{aligned}
c_B &= x\lambda + y, \\
c_2 &= (s+r)\lambda - r, \\
c_{BC} &= d - v\lambda, \\
d_A &= (k_{A0} + k_{AC})\rho - k_{AC}, \\
d_C &= (k_{C0} + k_{CA}) - k_{CA}\rho, \\
K_0 &= k_{0A}\rho + k_{0C}.
\end{aligned} \tag{3.3}$$

The generator identity is

$$\mathcal{L}U = K_0 + c_B B - c_2(A)_2 - c_{BC}BC - d_A A - d_C C. \tag{3.4}$$

This identity is valid at the boundary because falling factorials, rather than ordinary monomials, occur in (3.4).

Put

$$D(A, B, C) = c_2(A)_2 + c_{BC}BC + d_A A + d_C C \tag{3.5}$$

and define

$$\mathcal{R} = \{D \leq 2(K_0 + c_B B + 1)\}. \tag{3.6}$$

Outside $\mathcal{R}$,

$$\mathcal{L}U \leq -1 - \frac{1}{2}D. \tag{3.7}$$

There are explicit constants

$$K_A = 2 + \sqrt{\frac{2(K_0 + c_B + 1)}{c_2}}, \text{quad} K_C = \frac{2(K_0 + c_B + 1)}{c_{BC}} \tag{3.8}$$

and a finite integer q_0 such that every state of $\mathcal{R}$ with $q > q_0$ satisfies

$$A \leq K_A \sqrt{q+1}, \quad C \leq K_C. \tag{3.9}$$

Indeed, (3.9) follows separately from the $(A)_2$ and BC terms in (3.5). When $B = 0$, (3.6) bounds both A and C , so all sufficiently large states in $\mathcal{R}$ have $B \geq 1$.

Choose K and C_* , after q_0 has been fixed, so that

$$\begin{aligned}
K &> K_A + a_* + 2, & C_* &> K_C + 2, \\
K &> 1 + \max_{x \in \mathcal{R}, q(x) \leq q_0} \frac{A(x)}{\sqrt{q(x)+1}}, \\
C_* &> 1 + \max_{x \in \mathcal{R}, q(x) \leq q_0} C(x).
\end{aligned} \tag{3.10}$$

and define the moving core

$$\mathcal{K} = \{(a, b, c) : a \leq K\sqrt{a+2b+1}, c \leq C_*\}. \tag{3.11}$$

The finite maxima in (3.10) exist because (3.6) bounds C when q is bounded. Together with (3.9), they give the global inclusion

$$\mathcal{R} \subset \mathcal{K}. \tag{3.12}$$

Thus (3.7) is a genuine proper-workload return mechanism to $\mathcal{K}$, not a truncation assumption. The positive part of (3.4) is at most $K(1+U)$. Localization and Gronwall therefore give

$$\mathbb{E}_x U(X_{t \wedge \tau_n}) \leq (U(x) + Kt)e^{Kt}, \quad (3.13)$$

where τ_n exits the U -sublevel $\{U \leq n\}$. Since those sublevels are finite, (3.13) proves nonexplosion and endpoint integrability. The same calculation applied to U^p , using the strict negative coefficients at the two quadratic sources, gives finite polynomial moments on every bounded physical interval.

9.4 4. The short physical window

For a starting state in $\mathcal{K}$, write $N = q(x)$, $S_N = \sqrt{N+1}$, and fix $T > 0$. During the physical interval

$$h_N = \frac{T}{S_N}, \quad (4.1)$$

use accelerated time $\tau = S_N t$ and set

$$z_N(\tau) = \frac{A(\tau/S_N) + \zeta C(\tau/S_N)}{S_N}. \quad (4.2)$$

Let $\Phi_\tau(u)$ denote the solution at time τ of

$$\dot{z} = \alpha - \beta z^2, \quad z(0) = u. \quad (4.3)$$

The following is the quantitative finite-time statement needed below.

Lemma 4.1 (uniform fast window). Uniformly over starting states in $\mathcal{K}$ with $q = N$, as $N \rightarrow \infty$,

$$\mathbb{E}_x \sup_{0 \leq \tau \leq T} |z_N(\tau) - \Phi_\tau(z_N(0))|^2 \rightarrow 0. \quad (4.4)$$

For every integer $p \geq 1$,

$$\sup_N \sup_{0 \leq \tau \leq T} \mathbb{E}_x [C(\tau/S_N)^p] < \infty, \quad (4.5)$$

and the number J_N of q -changing reactions in the window has uniformly bounded p -th moment. Finally,

$$\sup_{x \in \mathcal{K}: q(x)=N} \left| \mathbb{E}_x [q(X_{h_N}) - N] + w \int_0^T \Phi_\tau(z_N(0)) d\tau \right| \rightarrow 0. \quad (4.6)$$

9.4.1 Proof of Lemma 4.1

Let ϑ_N be the first accelerated time at which $q \notin [N/2, 2N]$ or $A > 2KS_N$. On $[0, T \wedge \vartheta_N]$, $B \geq N/8$ for all large N . The physical C -birth rate is at most

$$yB + r(A)_2 + k_{0C} + k_{AC}A \leq b_CN, \quad (4.7)$$

where one may take

$$b_C = y + 8rK^2 + k_{0C} + 2Kk_{AC} + 1.$$

The physical C -death rate contains

$$dBC \geq \delta_C NC, \quad \delta_C = d/8. \quad (4.8)$$

Consequently, up to ϑ_N , C is stochastically dominated by the immigration–death chain $\bar{C}$ with immigration rate b_CN and per-particle death rate δ_CN , started at $C(0) \leq C_*$. At every physical time t ,

$$\bar{C}(t) \stackrel{d}{=} \text{Bin}(C(0), e^{-\delta_CN t}) + \text{Pois}\left(\frac{b_C}{\delta_C}(1 - e^{-\delta_CN t})\right). \quad (4.9)$$

In particular, for every $\eta > 0$,

$$\sup_N \sup_{\tau \leq T} \mathbb{E}_x \exp\{\eta C((\tau \wedge \vartheta_N)/S_N)\} \leq \exp\left\{\eta C_* + \frac{b_C}{\delta_C}(e^\eta - 1)\right\} =: M_\eta. \quad (4.10)$$

This is a transient coupling from every core point, not a stationary-start estimate. By (2.3), the predictable drift of (4.2) is

$$\frac{P_q(A)}{N} + O(N^{-1/2})(1 + C) = \alpha - \beta z_N^2 + o_{L^p}(1) \quad (4.11)$$

uniformly up to ϑ_N , for every fixed p . Here we used $A/S_N = z_N - \zeta C/S_N$, (4.10), and the fact that the q -changing intensity on accelerated time is bounded by $K_1(1 + C)$. Its counting process therefore has moments of every fixed order bounded uniformly in N .

If M_N denotes the martingale part of z_N , its jumps have size at most $2/S_N$, and its predictable quadratic variation is bounded by

$$\langle M_N \rangle_T \leq \frac{K_2}{S_N^3} \int_0^{T \wedge \vartheta_N} \{N + (A)_2 + NC\} d\tau. \quad (4.12)$$

The Burkholder–Davis–Gundy inequality, including its jump term, and (4.10) imply, for every integer $m \geq 1$,

$$\mathbb{E}_x \sup_{\tau \leq T \wedge \vartheta_N} |M_N(\tau)|^{2m} \leq C_m S_N^{-m}. \quad (4.13)$$

The same estimate applied to the integral equation (4.11), followed by Gronwall on the compact stopped range, gives, for every $\epsilon > 0$,

$$\mathbb{P}_x \left\{ \sup_{\tau \leq T \wedge \vartheta_N} |z_N(\tau) - \Phi_\tau(z_N(0))| > \epsilon \right\} \leq C_{\epsilon, m} S_N^{-m}. \quad (4.14)$$

The Riccati solutions starting in $[0, K + 1]$ stay a fixed distance below $2K$. Exiting the q -band requires at least $N/2$ bounded q -jumps. The all-order count bounds and (4.14) therefore yield

$$\mathbb{P}_x \{\vartheta_N \leq T\} \leq C_m S_N^{-m} \quad \text{for every } m. \quad (4.15)$$

On this exceptional event, the polynomial endpoint estimates from (3.13) and Hölder's inequality let us remove the stopping in (4.10)–(4.14). This proves (4.4), (4.5), and the moment assertion for J_N .

On accelerated time, (2.5) now gives

$$\mathbb{E}[q(X_{h_N}) - N] = \int_0^T \mathbb{E} \left[\frac{k_{0A} + k_{CA}C}{S_N} - w \frac{A}{S_N} \right] d\tau. \quad (4.16)$$

The positive terms vanish by (4.5), and $A/S_N = z_N - \zeta C/S_N$. Equation (4.4) gives (4.6). $\square$

The Riccati flow is order preserving. Its smallest solution over all nonnegative starting values is the solution from zero,

$$z_0(\tau) = a_* \tanh(\sqrt{\alpha\beta} \tau).$$

Consequently

$$I_T := \inf_{z(0) \geq 0} \int_0^T z(\tau) d\tau = \frac{1}{\beta} \log \cosh(\sqrt{\alpha\beta} T) > 0. \quad (4.17)$$

Lemma 4.1 therefore gives, uniformly on the core,

$$\mathbb{E}_x[q(X_{h_N}) - N] \leq -\frac{3}{4}wI_T \quad (4.18)$$

for all sufficiently large N .

9.5 5. Returning to the core costs ($o(1)$) workload

Let $Y = X_{h_N}$, and from Y continue the original, unmodified CTMC until its first entrance into $\mathcal{K}$; call the additional time σ_N . If $Y \in \mathcal{K}$, set $\sigma_N = 0$.

Lemma 5.1 (return appendage). Uniformly for core starting states,

$$\mathbb{E}\sigma_N = o(1), \quad \mathbb{E}[(q(X_{h_N+\sigma_N}) - q(Y))^+] = o(1). \quad (5.1)$$

All endpoint variables in (5.1) are integrable, and the full increment $q(X_{h_N+\sigma_N}) - q(Y)$ has a uniformly bounded second moment.

9.5.1 Proof of Lemma 5.1

Because $K > a_*$, order preservation of the Riccati flow gives

$$\Delta := K - \sup_{0 \leq u \leq K} \Phi_T(u) = K - \Phi_T(K) > 0. \quad (5.2)$$

Put $\delta = \Delta/4$. Equations (4.10), (4.14), and (4.15), together with $q(Y) - N = O_{L^p}(1)$, imply that for every integer $m \geq 1$,

$$\begin{aligned} \mathbb{P}_x(G_N^c) &\leq C_m S_N^{-m}, \\ G_N &:= \{3N/4 \leq q(Y) \leq 5N/4, \quad A(Y) \leq (K - 2\delta)\sqrt{q(Y) + 1}, \quad C(Y) \leq S_N^{1/4}\}. \end{aligned} \quad (5.3)$$

On the no-stop part of G_N , $C(Y)$ has the uniform exponential moment (4.10). The exceptional estimate in (5.3) is super-polynomial; the stronger $\exp\{-cS_N\}$ estimate is not needed.

Suppose now that $Y = y \in G_N$ and $C(y) > C_*$. Stop the cleanup at the first of

$$\gamma = \inf\{t : C(t) \leq C_*\}, \quad \chi = \inf\{t : A(t) > K\sqrt{q(t) + 1} \text{ or } q(t) \notin [N/2, 2N]\}. \quad (5.4)$$

Before $\gamma \wedge \chi$, the bounds (4.7)–(4.8) remain valid. After the physical-time change $u = Nt$, the dominating chain in (4.9) has constant immigration b_C and per-particle death δ_C . The explicit binomial–Poisson representation, followed by fixed-length geometric trials after the initial population has contracted, gives, for every integer $m \geq 1$,

$$\begin{aligned} \mathbb{E}_y(\gamma \wedge \chi)^m &\leq \frac{C_m[1 + \log(1 + C(y))]^m}{N^m}, \\ \mathbb{E}_y\left(\int_0^{\gamma \wedge \chi} C(t) dt\right)^m &\leq \frac{C_m[1 + C(y)]^m}{N^m}. \end{aligned} \quad (5.5)$$

For completeness, these same bounds control the probability of the stop χ . Until $\gamma \wedge \chi$, reactions that can change A , other than reactions consuming one C , have total rate at most K_3N . The number of reactions consuming C is pathwise at most $C(y)$ plus the number of reactions producing C . The latter has intensity at most b_CN . If $V_{A,q}$ denotes the total variation of (A, q) before $\gamma \wedge \chi$, thinning the corresponding Poisson clocks and using (5.5) gives

$$\mathbb{E}_y V_{A,q}^m \leq C_m \mathbb{E}_y\{1 + C(y) + N(\gamma \wedge \chi)\}^m \leq C_m\{1 + C(y) + \log(1 + C(y))\}^m. \quad (5.6)$$

On G_N , $C(y) \leq S_N^{1/4}$, whereas the A -boundary is at least δS_N away and leaving the q -band requires order N bounded jumps. Markov's inequality applied to an arbitrarily high moment in (5.6) therefore gives

$$\sup_{y \in G_N} \mathbb{P}_y\{\chi < \gamma\} \leq C_m S_N^{-m} \quad \text{for every } m. \quad (5.7)$$

On $\{\gamma < \chi\}$ the endpoint lies in $\mathcal{K}$. The positive q -jump intensity is $k_{0A} + k_{CA}C$, so (5.5) and the exponential moment of $C(Y)$ give

$$\mathbb{E}[\gamma; \gamma < \chi] = O(N^{-1}), \quad \mathbb{E}N_q^+(\gamma; \gamma < \chi) = O(N^{-1}), \quad (5.8)$$

where N_q^+ counts positive q -jumps. The negative intensity is $wA \leq 2wKS_N$ before χ ; (5.5) therefore also gives uniformly bounded second moments for the full q -increment on the typical cleanup.

It remains to charge G_N^c and $\{\chi < \gamma\}$. By the global inclusion (3.12), (3.7) holds everywhere outside $\mathcal{K}$. If $\tau_{\mathcal{K}}$ is the core hitting time, localized Dynkin estimates give

$$\mathbb{E}_y \tau_{\mathcal{K}} + \frac{1}{2} \mathbb{E}_y \int_0^{\tau_{\mathcal{K}}} D(X_t) dt \leq U(y). \quad (5.9)$$

Both positive and negative q -intensities are bounded by $K_4(1+D)$. Moreover, outside $\mathcal{K}$ there is a $\kappa > 0$ such that $D \geq \kappa(1+U)$: if $C > C_*$, use the BC, A, C terms in D ; if $C \leq C_*$, then $A > K\sqrt{q+1}$, so the $(A)_2$ term bounds q and hence U . Also the total reaction intensity is at most $K_5(1+B+D)$, and outside $\mathcal{R}$ the B -term is bounded by K_6D . All jumps of U are bounded by a rate-dependent constant. Hence, for each integer $p \geq 2$, the bounded-jump Taylor formula gives, outside $\mathcal{K}$,

$$\begin{aligned} \mathcal{L}(1+U)^p &= p(1+U)^{p-1} \mathcal{L}U + R_p, \\ |R_p| &\leq C_p D(1+U)^{p-2}, \\ \mathcal{L}(1+U)^p &\leq C_p - c_p D(1+U)^{p-1}. \end{aligned} \quad (5.10)$$

For the last line, absorb the remainder when U exceeds a p -dependent constant and enlarge C_p on the remaining finite sublevel. Localized Dynkin's formula applied successively to $(1+U)^j$, $1 \leq j \leq 2p$, bounds the moments of $\tau_{\mathcal{K}}$ and of $\int_0^{\tau_{\mathcal{K}}} (1+D) dt$. The counting-martingale inequality then gives

$$\mathbb{E}_y [\tau_{\mathcal{K}}^p + |q(X_{\tau_{\mathcal{K}}}) - q(y)|^{2p}] \leq C_p (1+U(y))^{r_p} \quad (5.11)$$

for a finite integer r_p depending only on p and the rates.

Over the preceding window, (3.13) gives polynomial moments $\mathbb{E}_x(1+U(Y))^p \leq C_p(1+N)^{C_p}$. At the cleanup stop the same bound follows from (5.5)–(5.6). Hölder's inequality, (5.3), (5.7), and arbitrarily high choices of m make the duration, positive q -cost, and second-moment contribution of both exceptional sets $o(1)$. Together with (5.8), this proves (5.1). $\square$

Combining (4.18) and Lemma 5.1 gives a constant

$$\varepsilon = \frac{1}{2} w I_T > 0 \quad (5.12)$$

and an integer N_0 such that the complete core-to-core episode obeys

$$\mathbb{E}_x [q(X_{\tau}) - q(x)] \leq -\varepsilon, \text{ } q \text{ } \text{quadr} \text{ } x \in \mathcal{K}, \text{ } q(x) > N_0. \quad (5.13)$$

Its expected physical duration is uniformly finite (in fact $T/S_N + o(1)$), and its endpoint q -increment has uniformly bounded second moment.

9.6 6. Trace Foster argument and positive recurrence

Start outside $\mathcal{K}$ by running until the core is hit. Since $\mathcal{R} \subset \mathcal{K}$, (3.7), localization, and Dynkin's formula show that this time has finite expectation. At a core state with $q > N_0$, run the complete episode of Sections 4–5. Stop the trace once $q \leq N_0$.

Let Q_n be the shell values of the resulting core trace and $\nu = \inf\{n : Q_n \leq N_0\}$. Summing (5.13) only through $m \wedge \nu$, for finite m , is legitimate by the endpoint moment bounds. Since $q \geq 0$,

$$\varepsilon \mathbb{E}_x(m \wedge \nu) \leq q(x), \quad \mathbb{E}_x \nu \leq \frac{q(x)}{\varepsilon}. \quad (6.1)$$

The episode durations are uniformly integrable, so the physical time to hit

$$F = \mathcal{K} \cap \{q \leq N_0\} \quad (6.2)$$

has finite expectation. The set F is finite because q bounds A, B and the core bounds C .

Fix a closed irreducible population class Γ . The preceding hitting argument is pathwise inside Γ , so $F_\Gamma = F \cap \Gamma$ is nonempty. Record successive visits to F_Γ . The finite trace chain is irreducible on its visited subset. From a nonabsorbing point of F_Γ , take one ordinary jump and apply the preceding finite-mean hitting bound to its finitely many successors; an absorbing singleton is already positive recurrent. Thus the finite trace has finite mean positive return time, and at least one population state in Γ has finite mean positive physical return time. Irreducibility gives positive recurrence of the entire closed class. Nonexplosion was proved in (3.13), completing the proof of Theorem 1.1.

9.7 7. Adversarial checks

1. **No stationary-start assumption.** The Poisson shell law motivates the scale but is not used in Lemma 4.1. The C -moment estimate (4.9) and the Riccati limit follow from stopped generator identities from every core point.
2. **Huge initial C .** Such a state is outside the core. It is returned by the proper workload (3.7). A large C produced at the end of a fast window is charged by (5.5)–(5.11); it is not discarded as a tail-box error.
3. **Boundary $B = 0$.** The bad set (3.6) is finite on $B = 0$, because the negative $(A)_2$ and C terms remain. Hence no large shell with $B = 0$ is silently included in the Riccati core.
4. **Positive $C \rightarrow A$ events.** Their rate appears explicitly in (2.5). It vanishes on the accelerated core window by (4.5), and its full occupation during the return appendage is charged in (5.5), (5.8), and (5.9).
5. **Fast neutral jump counts.** No estimate counts $\mathcal{L}_0$ jumps. The proof uses physical quadratic variation over a prescribed physical interval and a physical hitting-time appendage.
6. **Endpoint integrability.** It follows from the proper linear workload and localized moment estimates, before any trace telescoping is used.
7. **Rate and orientation degeneracies.** Missing edges are allowed. The only strict constants used are $\alpha, \beta, d, c_2, c_{BC}, d_A, d_C, w$, all of which are forced positive by strong connectivity and the explicit interval choices (3.1).

10 Corrected S-tier-superlevel cut and the affine151 branch

Status: publication dependency replacing the cut/count portions of `global_atlas_interface_closure.md` and `stoichiometric_gate_feasibility.md`. The audited legacy files are retained unchanged for provenance.

Date: 2026-08-12 PDT.

Retirement declaration. Section 8.2 of the frozen `global_atlas_interface_closure.md` and the old stoichiometric incidence totals 12886/9913/2973 are retired as publication dependencies. The corrected totals are 12678/9709/2969. The tier-certified and residual pair fingerprints, the exact 2,511 pair residual, and the affine151 recurrence branch remain unchanged and are repinned below.

10.1 1. Exact scope

Consider the finite binary complex universe

$$\mathcal{C}_2 = \{0, A, B, C, 2A, 2B, 2C, A + B, A + C, B + C\}$$

and an ordered pair of disjoint linkage supports, each equipped with an arbitrary strongly connected directed reaction graph and arbitrary positive rates. This note supplies:

1. the exact symbolic condition under which **every** such pair of strongly connected graphs has an Anderson–Kim descending source for a fixed tier descriptor;
2. the exact finite pair split after the pre-tier support branches;
3. the corrected affine-feasibility incidence totals; and
4. the class-local recurrence conclusion for the unchanged 151-pair set.

There is no enumeration of orientations, stochastic paths, population states, or rate values. Finite computation only deduplicates rational tier types and checks support/tier/affine-subspace identities.

10.2 2. Exact universal strong-orientation theorem

Fix a tier descriptor. Write $E = T^{S,1}$ for the global top S-tier among the network complexes. If $E \neq \emptyset$, all its complexes occupy a single D-tier; call its level r . For each linkage support L , define

$$U_L(r) = \{y \in L : y \text{ lies at or above D-level } r\}. \quad (2.1)$$

Theorem 2.1 (S-tier-superlevel cut). Every strongly connected directed graph on each linkage support has a D-descending reaction whose source lies in E if and only if, for some linkage L ,

$$\emptyset \neq U_L(r) \subsetneq L, \quad U_L(r) \subseteq E. \quad (2.2)$$

10.2.1 Proof

If (2.2) holds, strong connectivity gives a directed path from $U_L(r)$ to its complement. The first exiting edge has source in $U_L(r) \subseteq E$ and target below D-level r . It is the required descending reaction.

Conversely, suppose (2.2) fails. Work independently in each linkage. If $U_L(r)$ is empty or all of L , choose any directed Hamiltonian cycle. Otherwise choose $b \in U_L(r) \setminus E$, which exists by failure of (2.2), and order a directed Hamiltonian cycle as

$$U_L(r) \setminus \{b\}, \quad b, \quad L \setminus U_L(r). \quad (2.3)$$

Its only edge from $U_L(r)$ to its complement is sourced at $b \notin E$. Every vertex of $E \cap L$ has its successor within $U_L(r)$, where no target is below level r . Hence this strongly connected cycle has no E-sourced descending edge. Combining the cycles proves the converse. $\square$

The superlevel in (2.1), not merely the global top D-tier, is essential. For

$$L_1 = \{C, 2C\}, \quad L_2 = \{A, 2A, A + B, A + C\}, \quad w = (0, 3, 1), \quad c = (0, 2, 2),$$

the global top D-tier is the disabled singleton $\{A + B\}$, whereas the top S-tier is $E = \{2C\}$. Here $U_{L_1} = \{2C\}$, so every strongly connected graph on L_1 supplies a $2C$ -sourced descent. A test using only the global top D-tier incorrectly rejects this descriptor.

10.3 3. Completeness of the finite tier list

For a tier sequence put

$$u_n = (\log(x_{n,A} \vee 1), \log(x_{n,B} \vee 1), \log(x_{n,C} \vee 1)).$$

Project away the span of every D-equivalent complex difference and every coordinate vector belonging to an eventually bounded coordinate. The projection removed from u_n is bounded. Strict D-comparisons therefore remain strict, D-equalities become exact equalities, active coordinates stay positive, and bounded coordinates become zero. The resulting nonempty relatively open rational cone contains a rational weight. Thus an ordinary nonnegative rational weight represents every D-preorder, including multiscale preorders; no polynomial relation among the original population scales is assumed.

The 21 projectively distinct comparison hyperplanes on the normalized nonnegative simplex have 37 vertices. Vertices, edge midpoints, and triangle centroids give 5,128 rational candidates and 193 distinct (D-preorder, active-coordinate set) types. An eventually bounded integer coordinate is eventually constant, and binary-source availability depends only on whether that constant is 0, 1, or at least 2. Adding these exact caps yields 259 descriptors.

Theorem 2.1 reduces the universal quantifier over strongly connected graphs to the finite set containment (2.2). The program evaluates that containment only; it does not construct or enumerate orientations.

10.4 4. Pre-tier support branches and the exact 2,511 residual

The ordered support branches are applied in this order: finite strict invariant, active-chart invariant, full deficiency zero, exact signed-service seam, exact residual pair, exact seven-support seam, then tier certification.

family	unique pairs	finite	active chart	deficiency zero	seven- support only	signed- service only	residual- pair only	pre-tier residual
positive	4,761	187	110	924	6	2	1	3,531
signed	408	0	0	50	0	0	0	358

The exact S-tier-superlevel cut gives:

family	pre-tier	tier-certified	residual	certified SHA-256	residual SHA-256
positive	3,531	1,219	2,312	744d87292030 9c361d6d7f80 6f140a696e3f c3ae0f75d760 d8a07f304d56 2b6b	0297ba35311c 757cd5c6ec54 8d2af18410df d37e791c7679 de932fe4bf38 695b
signed	358	159	199	7f59ea94fe87 6205ccb72dc9 7b026b2954fe ac6237512263 4aafa3180844 28ee	1a9c06123645 855d3b4f23d4 886b0ada3c3f f3614fc94a7d 22c01f411c13 55c8

Therefore the exact post-tier residual is

$$2312 + 199 = 2511. \quad (4.1)$$

Although the legacy sufficient cut falsely rejected 208 individual pair-descriptor incidences, every affected pair has another genuine failed descriptor. Consequently the pair sets and fingerprints in this section are unchanged.

10.5 5. Corrected affine-feasibility calculation

For a primitive descriptor weight w , at each positive level r put

$$E_r = \{i : w_i = r\}, \quad L_r = \{i : w_i < r\}.$$

For a rational stoichiometric subspace S , the descriptor is realizable in some affine class exactly when, for every r , there is $v_r \in S$ with

$$(v_r)_i = 0 \ (i \in L_r), \quad (v_r)_i > 0 \ (i \in E_r). \quad (5.1)$$

For a specified affine class one also needs a base point having the exact eventual values 0, 1, or at least 2 in the zero-weight coordinates.

Necessity of (5.1) follows from Gordan’s alternative: failure produces an affine invariant zero above level r , nonnegative and nonzero at level r , which contradicts constancy after normalization by that scale. Sufficiency follows from the polynomial flag

$$x(n) = b + \sum_r n^r v_r.$$

Applying this exact test only to the genuine failures from Theorem 2.1 gives:

family	corrected failures	feasible	infeasible	legacy false failures removed
positive	12,250	9,349	2,901	200
signed	428	360	68	8
total	12,678	9,709	2,969	208

The 151-pair classwise branch is unchanged:

family	certified	remaining	certified SHA-256	remaining SHA-256
positive	143	2,169	f48882aa1ff52c1 594a71fd217fa55 9492c7010e95028 5a9fa2e60e02b48 7b76	6763a44c9c312c4 40997a054f79663 47d101e3236cdef 9ecb90599226de1 0458
signed	8	191	aead73fd44d0878 9019326cffcd706 a776addf0cbc841 979a3d54e8c80c5 f88d	f5c7a694bec0241 a67b5cf588e1d07 4c11e00fb9ae3fb c1cee5570f84e9b 4483

The combined certified-set fingerprint is 55e243945f86d106b920a27e2249a20b7077b5dc718ec06918cca4368e4a6c96.

10.6 6. Class-local entropy-Foster consequence

Fix one of the 151 support pairs, arbitrary strongly connected reaction graphs on its two linkage supports, arbitrary positive rates, and a closed irreducible population class Γ . If Γ is infinite, it lies in one affine stoichiometric class.

Suppose a tier sequence in Γ failed the top-S descending-source condition. By Theorem 2.1, its descriptor fails the exact superlevel cut. Because the sequence lies in one affine class, necessity of (5.1) makes that failed descriptor affine-feasible. This contradicts the definition of the 151-pair set. Thus every tier sequence contained in Γ satisfies

$$T_{\{x_n\}}^{S,1} \cap D_{\{x_n\}} \neq \emptyset. \quad (6.1)$$

The cited publication result is David F. Anderson and Jinsu Kim, *Some Network Conditions for Positive Recurrence of Stochastically Modeled Reaction Networks*, **SIAM Journal on Applied Mathematics** 78(5) (2018), 2692–2713, doi:10.1137/17M1161427, **Theorem 9**. Theorem 4.2 is the numbering only in arXiv v3.

Journal Theorem 9 is stated with an ambient-space premise. Its contradiction proof restricts as follows. If the entropy drift were not at most -1 outside a finite subset of Γ , properness would give an unbounded sequence $x_n \in \Gamma$ with drift greater than -1 . Extracting the tier subsequence and using (6.1) repeats the proof of Theorem 9 and forces the drift to $-\infty$, a contradiction. Closedness makes the restricted generator equal to the original generator at every state of Γ .

The chain is nonexplosive independently: a reaction with a quadratic source cannot increase total population because its target is also binary. Every population-increasing reaction therefore has source degree at most one, total increasing intensity $O(1 + |x|)$, and bounded jump size. A linear pure-birth comparison bounds total population on finite time intervals; each population sublevel contains finitely many states and has bounded total rates. Continuous-time Foster applied to the closed irreducible class now proves positive recurrence.

10.7 7. Reproducibility and frozen derivative bytes

Run only the symbolic/set-identity derivative:

```
PYTHONPATH=src python3 src/s_tier_superlevel_interface.py
PYTHONPATH=src python3 -m unittest \
    tests.test_s_tier_superlevel_interface -v
```

The derivative source does not call the legacy orientation-witness constructor. Its finite certificate hash is

77c7ce0d2325379acfed7b13a44f9577454279275918ee14f968e313b488a7e0.

At creation, the exact derivative code bytes were:

file	SHA-256
src/s_tier_superlevel_interface.py	1a4e27fcf40af76cac6281f8830b7644bf086b3c05d97a963ce9f5bac736ad57
tests/test_s_tier_superlevel_interface.py	4d9f960d89a361a27d9dadbd765297783755b9ec09c60f32301229447f51af40

The accompanying hash manifest pins this note and the hostile audit after their final byte pass.

Part III

Two active linkages: residual pair theorems

11 Certified global return theorem for the fourteen rank-two partners

11.1 1. Scope and audit status

This note treats

$$L_0 = \{B, 2A, B + C\} \tag{1.1}$$

and any of the fourteen compatible lower supports certified by `two_active_phase_gate.py`. Each linkage has an arbitrary fixed strongly connected orientation and arbitrary positive rates. Propensities are stochastic mass-action propensities.

The construction uses no fixed box for C , no stationary start, and no raw-fast-jump recurrence:

1. a proper linear workload returns every large- q state to a moving core;
2. a physical-time regenerative block treats the four bounded- q , $C \rightarrow \infty$ faces not controlled pointwise by that workload; and
3. the audited transient Riccati window supplies strict core descent.

Theorem 1.1. For every one of the fourteen supports and every strongly connected orientation with positive rates, the minimal CTMC is nonexplosive and every closed irreducible population class is positive recurrent.

The stopping-time composition in Sections 5–7 and all six obligations listed in the prior candidate version have passed independent proof audit. The theorem is certified at exactly this fourteen-support scope. It is not a global T3-2 theorem.

11.2 2. A common outer workload

Use the fast-linkage rate notation

$$\begin{array}{c|cccccc} \text{edge} & B \rightarrow 2A & B \rightarrow B + C & 2A \rightarrow B & 2A \rightarrow B + C & B + C \rightarrow B & B + C \rightarrow 2A \\ \hline \text{rate} & x & y & s & r & t & v. \end{array} \quad (2.1)$$

Put $d = t + v$. Strong connectivity makes

$$I_0 = \left(\frac{r}{s+r}, \frac{d}{v} \right) \quad (2.2)$$

nonempty and gives values in I_0 arbitrarily close to one; a missing denominator is interpreted as $+\infty$.

For a lower complex $z = (z_A, z_C)$, set

$$\ell_\rho(z) = \rho z_A + z_C, \quad b_z(\rho) = \sum_{z \rightarrow z'} k_{zz'} \{ \ell_\rho(z') - \ell_\rho(z) \}. \quad (2.3)$$

The only quadratic lower complexes are $2C$ and $A + C$. At $\rho = 1$, ℓ_1 is molecularity, so

$$b_{2C}(1) \leq 0, \quad b_{AC}(1) \leq 0. \quad (2.4)$$

There is a $\rho > 0$, arbitrarily close to one, for which every quadratic complex present has a strictly negative coefficient. Equality in (2.4) at one quadratic source means that all its outgoing edges target the other quadratic complex. Both equalities would make $\{2C, AC\}$ a closed directed set. If only $b_{2C}(1) = 0$, take $\rho < 1$, because $2C \rightarrow AC$ has increment $\rho - 1$. If only $b_{AC}(1) = 0$, take $\rho > 1$, because $AC \rightarrow 2C$ has increment $1 - \rho$. Existing strict inequalities survive a sufficiently small perturbation. If only AC is present, we may and do choose

$$0 < \rho < 1. \quad (2.5)$$

Choose $\lambda \in I_0$, close enough to one that

$$p_B = 2\rho - \lambda > 0, \quad (2.6)$$

and put

$$U = \rho A + p_B B + C. \quad (2.7)$$

Direct substitution gives

$$\mathcal{L}_0 U = c_B B - c_2(A)_2 - c_{BC} BC, \quad (2.8)$$

where

$$c_B = x\lambda + y, \quad c_2 = (s+r)\lambda - r, \quad c_{BC} = d - v\lambda \quad (2.9)$$

are strictly positive. From (2.3)–(2.4),

$$\mathcal{L}_- U \leq K_0 + k_A A + k_C C - e_2(C)_2 - e_{AC} AC, \quad (2.10)$$

where $K_0, k_A, k_C \geq 0$, and $e_h > 0$ exactly when the quadratic complex h is present. Every one of the fourteen supports contains $2C$ or AC , so $e_2 + e_{AC} > 0$.

Put

$$\begin{aligned}
P &= K_0 + c_B B + k_A A + k_C C, \\
D &= c_2(A)_2 + c_{BC} BC + e_2(C)_2 + e_{AC} AC, \\
\mathcal{R} &= \{D \leq 2(P+1)\}.
\end{aligned} \tag{2.11}$$

Everywhere outside $\mathcal{R}$,

$$\mathcal{L}U \leq -1 - \frac{1}{2}D. \tag{2.12}$$

All jumps of U are bounded. Outside $\mathcal{R}$, every reaction intensity is bounded by KD . The point requiring care is that P need not contain A or C . The term $(A)_2$ controls A except on the finite set $A \leq 1$, and B occurs in P . If $2C$ is present, $(C)_2$ controls C . If only AC is present, then $A + B = 0$ implies $D = 0$ and hence cannot occur in $\mathcal{R}^c$; otherwise either AC or BC controls C . Consequently, on $\mathcal{R}^c$,

$$1 + A + B + C \leq K(1 + P + D) \leq K'D, \tag{2.13}$$

and each actually present quadratic propensity is already one of the terms in D . Thus the full intensity is at most $K''D$. After enlarging a finite sublevel if necessary, the bounded-jump Taylor formula gives, for every integer $p \geq 2$,

$$\mathcal{L}(1 + U)^p \leq C_p - c_p D(1 + U)^{p-1}. \tag{2.14}$$

For the finitely many powers used below, enlarge $\mathcal{R}$ by the corresponding finite U -sublevels. This leaves Lemma 3.1 unchanged and ensures, in particular,

$$\mathcal{L}(1 + U)^2 \leq 0 \quad \text{on } \mathcal{R}^c. \tag{2.15}$$

The total-population generator is at most affine: a bimolecular source cannot increase molecularity in a binary network. Localized Yule comparison proves nonexplosion and finite-time population moments before optional sampling is used.

11.3 3. Geometry and the first outer return

Lemma 3.1 (large- q core). There are finite $Q, K, C_* > 0$ such that

$$x \in \mathcal{R}, \quad q(x) := A + 2B > Q \implies A \leq K\sqrt{q+1}, \quad C \leq C_*. \tag{3.1}$$

Indeed, (2.11) implies, after changing constants,

$$aA^2 + bBC + eC^2 + fAC \leq K_1(1 + q + C), \tag{3.2}$$

where $a, b > 0$, $e, f \geq 0$, and $e + f > 0$. If $e > 0$, first $C = O(\sqrt{q})$, then $A = O(\sqrt{q})$; hence $B \geq q/4$ for large q , and the BC term gives $C = O(1)$. If $e = 0$, then $f > 0$. When $A \geq 1 + 2K_1/f$, the AC term absorbs the linear C term, gives $A = O(\sqrt{q})$, and then BC bounds C . For bounded A , $B \asymp q$ and BC bounds C directly. Enlarging the constants covers the remaining finite range.

Define

$$\mathcal{K} = \{A \leq K\sqrt{q+1}, \quad C \leq C_*\}, \quad \mathcal{V} = \{q \leq Q\}, \quad \mathcal{B} = \mathcal{R} \cap \mathcal{V}. \tag{3.3}$$

Then $\mathcal{R} \subset \mathcal{K} \cup \mathcal{B}$. For $\tau_{\mathcal{R}} = \inf\{t : X_t \in \mathcal{R}\}$, localized Dynkin estimates give

$$\begin{aligned}
\mathbb{E}_x \tau_{\mathcal{R}} + \frac{1}{2} \mathbb{E}_x \int_0^{\tau_{\mathcal{R}}} D(X_t) dt &\leq U(x), \\
\mathbb{E}_x U(X_{\tau_{\mathcal{R}}}) &\leq U(x), \\
\mathbb{E}_x [\tau_{\mathcal{R}}^p + U(X_{\tau_{\mathcal{R}}})^p] &\leq C_p(1 + U(x))^{r_p}.
\end{aligned} \tag{3.4}$$

No inactive coordinate is truncated: $\mathcal{B}$ is the exact infinite bad-set portion to be treated on the vertical clock.

11.4 4. The four vertical strips

If $2C$ is present, $e_2(C)_2$ makes $\mathcal{B}$ finite. The only supports needing another argument are

$$\{0, A, AC\}, \quad \{0, C, AC\}, \quad \{A, C, AC\}, \quad \{0, A, C, AC\}. \quad (4.1)$$

Use $0 < \rho < 1$. On $q \leq Q$, $E = (A, B)$ is a genuinely finite phase and C is the sole unbounded coordinate.

11.4.1 4.1 The three supports containing C

Aggregate lower rates as

edge	$C \rightarrow AC$	$C \rightarrow A$	$C \rightarrow 0$		$AC \rightarrow C$	$AC \rightarrow A$	$AC \rightarrow 0$
rate	u	a	b		v_1	w	z

(4.2)

On physical time $t = \tau/n$, from $C = n + O(1)$, the reactions with a C -containing source converge to a linear phase process Y . Its first moments and its C -service intensity satisfy

$$\begin{aligned} \frac{d}{d\tau} \mathbb{E}Y_B &= -v\mathbb{E}Y_B, \\ \frac{d}{d\tau} \mathbb{E}Y_A &= u + a - (v_1 + z)\mathbb{E}Y_A + 2v\mathbb{E}Y_B, \\ s(\tau) &= a + b + (w + z)\mathbb{E}Y_A + (t + v)\mathbb{E}Y_B. \end{aligned} \quad (4.3)$$

Here t, v are the rates $BC \rightarrow B, 2A$ from (2.1). The expected leading-workload increment is exactly

$$F_T(a_0, b_0) = \rho\{\mathbb{E}Y_A(T) - a_0\} + p_B\{\mathbb{E}Y_B(T) - b_0\} - \int_0^T s(\tau) d\tau. \quad (4.4)$$

For every fixed Q ,

$$\max_{a_0 + 2b_0 \leq Q} F_T(a_0, b_0) \longrightarrow -\infty \quad (T \rightarrow \infty). \quad (4.5)$$

If $v_1 + z > 0$, (4.3) has a finite limiting mean and the limiting service rate is strictly positive. Otherwise $a = b = w = z = 0$ would leave only the neutral pair $C \rightleftharpoons AC$, a closed directed subset. If $v_1 + z = 0$, then $w > 0$. When $u + a > 0$, the mean of Y_A grows linearly and $\int w\mathbb{E}Y_A$ grows quadratically; when $u + a = 0$, the source C must have a direct edge to 0, so $b > 0$. If Y_B does not decay because $v = 0$, strong connectivity of L_0 gives $t > 0$, and tY_B supplies service. These alternatives prove (4.5), uniformly on the finite initial phase set.

Choose $T, \epsilon > 0$ so the maximum in (4.5) is at most -4ϵ . Stop the T/n full-chain window if

$$C \notin [n/2, 2n] \quad \text{or} \quad A + B > n^{1/4}. \quad (4.6)$$

The phase process is linear and has exponential moments on fixed τ -intervals. Non- C -source reactions have total physical rate at most $K(1 + (A + B)^2)$, whose integral on T/n is $o_{L^p}(1)$. Random time changes and the Burkholder inequality therefore give, uniformly over $q \leq Q$,

$$\mathbb{E}[U(X_{T/n}) - U(X_0)] \leq -3\epsilon. \quad (4.7)$$

$$\sup_n \mathbb{E}|U(X_{T/n}) - U(X_0)|^p < \infty. \quad (4.8)$$

The stop is removable: the number of leading reactions has all fixed moments, (4.6) has super-polynomially small probability, and the total-population Yule bound charges its endpoint.

11.4.2 4.2 The sole dormant atom

For $L_- = \{0, A, AC\}$, suppose first that $A + B > 0$. The enabled C -order sources are AC and BC . Every AC -edge targets 0 or A , and therefore has U -increment $-(\rho + 1)$ or -1 . At source BC , the rate-weighted mean increment is strictly negative by $c_{BC} > 0$. Since $q \leq Q$, conditioning on the first AC - or BC -reaction gives constants $\epsilon_1, c_1 > 0$, uniform over all nonzero phases, such that

$$\mathbb{E}\Delta U \leq -4\epsilon_1, \quad \mathbb{E}\sigma^p + \mathbb{E}|\Delta U|^p \leq C_p \quad (4.9)$$

for the leading block, whose mean physical duration is at most c_1/C . The probability that a non- C -source reaction occurs first is $O(1/C)$, so the same bounds, with $-3\epsilon_1$, hold for the full chain at all large C .

The only phase with no C -order source is

$$(A, B) = (0, 0). \quad (4.10)$$

At (4.10), wait for the first reaction sourced at 0. Its mean time is the reciprocal of the fixed positive total 0-exit rate. A target A , followed by the first AC -reaction, has net U -increment at most $\rho - 1 < 0$. A target AC followed by $AC \rightarrow 0$ is a neutral trial. If instead $AC \rightarrow A$ occurs, one more AC -reaction makes the net increment at most $\rho - 1$. Strong connectivity says that either $0 \rightarrow A$ or $AC \rightarrow A$ has positive rate; otherwise $\{0, AC\}$ cannot reach A .

Thus each trial has probability $p_* > 0$ of decreasing U by at least

$$\delta_* = 1 - \rho > 0, \quad (4.11)$$

and otherwise returns with zero leading reward. The trial count is geometrically dominated. During an AC -part, service has rate $\Theta(C)$, while all interfering reactions have bounded rate at the bounded phase. Their probability is $O(1/C)$, with the same conclusion for fixed endpoint moments. Hence, for large C , the complete dormant block satisfies

$$\mathbb{E}\Delta U \leq -3\epsilon_0, \quad \mathbb{E}\sigma^p + \mathbb{E}|\Delta U|^p \leq C_p. \quad (4.12)$$

This regeneration includes all excursions of the finite phase; it does not stop at a fixed C -box boundary.

11.5 5. Global return to the core or a finite set

Choose M so that

$$F_V = \{x \in \mathcal{B} : U(x) \leq M\} \quad (5.1)$$

is finite and contains all finite- C exceptions in Section 4. For the ten supports containing $2C$, enlarge M so that $\mathcal{B} \subset F_V$; no vertical episode is then required. Starting from $\mathcal{B} \setminus F_V$, run (4.7), (4.9), or (4.12), obtaining Y . If $Y \in \mathcal{R}$, stop; otherwise append $\tau_{\mathcal{R}}$ from (3.4). Call the result σ_V .

The appendage has nonpositive expected U -increment by (3.4). Thus

$$\mathbb{E}_x[U(X_{\sigma_V}) - U(x)] \leq -\epsilon_V \quad (5.2)$$

for a common $\epsilon_V > 0$, and $X_{\sigma_V} \in \mathcal{R}$.

The power estimate is equally important. By (2.15), optional sampling makes the outer appendage nonincreasing for $(1 + U)^2$. Equations (4.8), (4.9), and (4.12) then yield

$$\mathbb{E}_x[(1 + U(X_{\sigma_V}))^2 - (1 + U(x))^2] \leq -c_V(1 + U(x)) + C_V. \quad (5.3)$$

Iterate (5.2) at endpoints in $\mathcal{B} \setminus F_V$, and stop when $\mathcal{K} \cup F_V$ is reached. Summing (5.2) and (5.3), and using (3.4) for appended durations, gives

$$\begin{aligned}
\mathbb{E}_x N_V &\leq \frac{1 + U(x)}{\epsilon_V}, \\
\mathbb{E}_x \sum_{j < N_V} (1 + U_j) &\leq C(1 + U(x))^2, \\
\mathbb{E}_x \tau_{\mathcal{K} \cup F_V} &\leq C(1 + U(x))^2.
\end{aligned} \tag{5.4}$$

Repeating the bounded-jump Taylor and counting-martingale estimates at higher powers gives, for each fixed p ,

$$\mathbb{E}_x \left[\tau_{\mathcal{K} \cup F_V}^p + U(X_{\tau_{\mathcal{K} \cup F_V}})^p + |q(X_{\tau_{\mathcal{K} \cup F_V}}) - q(x)|^p \right] \leq C_p (1 + U(x))^{p^2}. \tag{5.5}$$

This is a global physical-time return theorem with endpoint integrability, not a tightness-to-finite-phase assertion. From an arbitrary initial state, first run $\tau_{\mathcal{R}}$. Its endpoint lies in $\mathcal{K} \cup \mathcal{B}$; if it lies in $\mathcal{B} \setminus F_V$, apply the preceding vertical trace. Thus (5.4)–(5.5), with changed polynomial constants, hold for the global hitting time of $\mathcal{K} \cup F_V$.

11.6 6. Core episode

Start at $x \in \mathcal{K}$, put $N = q(x)$, and run the full chain for

$$h_N = \frac{T}{\sqrt{N+1}}. \tag{6.1}$$

The audited Proposition 5.1 in `two_active_promotion_phase.md`, including its exponential C -domination and scaled corrector $2C/(d\sqrt{N})$, proves, uniformly on the core,

$$\mathbb{E}_x [q(X_{h_N}) - N] \leq -\epsilon_R \tag{6.2}$$

for all large N . It also gives all fixed moments of the q -change count, a fixed Riccati margin below the $A = K\sqrt{q}$ boundary, and a uniform exponential moment for C .

From $Y = X_{h_N}$, append the global return of Section 5 until first entrance into $\mathcal{K} \cup F_V$, stopping permanently at F_V . The needed all-fourteen C -tail estimate is as follows; it is not inferred merely from the core-window proposition. On the typical Riccati event,

$$q(Y) \in [N/2, 2N], \quad A(Y) \leq K_0\sqrt{N}, \quad C(Y) = c_0 \leq N^{1/8}, \tag{6.3}$$

with a fixed margin below the $A = K\sqrt{q}$ boundary. Stop the cleanup when $C \leq C_*$ or one of the first two bounds in (6.3), with half its margin, is violated. Throughout this stop, $B = (q - A)/2 \geq cN$, so the fast BC -service rate is at least $c_1 NC$. Fast C -immigration from B and $2A$ is at most $C_1 N$. The added lower sources can increase C at total rate at most

$$C_2 N + C_2 \sqrt{N} C. \tag{6.4}$$

Here A -sourced births cost $O(\sqrt{N})$, and the only potentially linear per- C addition, $AC \rightarrow 2C$, costs $O(\sqrt{N} C)$; it is absorbed by the $c_1 NC$ death for large N . A stopped comparison with an immigration–death chain of immigration $C_3 N$ and per-particle death $c_3 N$ therefore gives

$$\begin{aligned}
\mathbb{E} \sigma_C &\leq \frac{C(1 + \log(1 + c_0))}{N}, \\
\mathbb{E} \int_0^{\sigma_C} (1 + C_t + C_t^2) dt &\leq \frac{C(\log N + c_0 + c_0^2)}{N} = o(1).
\end{aligned} \tag{6.5}$$

The same comparison gives every fixed polynomial moment of these quantities. The only positive- q lower sources are $0, C, 2C$, whose total propensity is at most $C(1 + C + C^2)$. Hence their expected count, and their second-moment contribution, are $o(1)$ by (6.5). The $AC \rightarrow 2C$ channel is q -nonpositive and has already been

absorbed in (6.4). The fixed Riccati margin, the counting-martingale bound, and (6.5) make the probability of hitting either other cleanup stop smaller than every fixed power of $N^{-1/2}$. Thus on the typical event the cleanup duration, positive q -cost, and full q -increment second moment are $o(1)$.

The complementary event has probability smaller than every power of $N^{-1/2}$. The polynomial return estimate (5.5), Hölder's inequality, and arbitrarily high moments make its duration, positive q -cost, and endpoint second-moment contribution $o(1)$. Thus the complete core episode σ_K satisfies

$$\begin{aligned} X_{\sigma_K} &\in \mathcal{K} \cup F_V, \\ \mathbb{E}_x[q(X_{\sigma_K}) - q(x)] &\leq -\epsilon_K, \\ \sup_{x \in \mathcal{K}, q(x) > N_0} \mathbb{E}_x \sigma_K &< \infty, \\ \sup_{x \in \mathcal{K}, q(x) > N_0} \mathbb{E}_x |q(X_{\sigma_K}) - q(x)|^2 &< \infty \end{aligned} \tag{6.6}$$

for some $\epsilon_K > 0, N_0 < \infty$.

11.7 7. Random-time Foster composition

Let

$$F = F_V \cup (\mathcal{K} \cap \{q \leq N_0\}). \tag{7.1}$$

It is finite. From an arbitrary state, Section 5 reaches $\mathcal{K} \cup F_V$ in finite mean physical time. If F_V is reached, stop. Otherwise run the core episodes until F_V is reached or the core shell has $q \leq N_0$. For successive shell values Q_j and trace hitting index ν , stopped telescoping of (6.6) gives

$$\epsilon_K \mathbb{E}_x(m \wedge \nu) \leq q(x), \quad \mathbb{E}_x \nu \leq \frac{q(x)}{\epsilon_K}. \tag{7.2}$$

The duration and endpoint second-moment bounds in (6.6) justify the telescoping and give finite mean physical hitting time of F .

Fix a closed irreducible population class Γ . The construction stays in Γ , so $F \cap \Gamma \neq \emptyset$. From a state in the finite set $F \cap \Gamma$, take one ordinary jump and apply the finite-mean hitting bound to its finitely many successors. The finite trace has finite mean positive return time (an absorbing singleton is already positive recurrent). Irreducibility gives positive recurrence of the whole closed class. Together with nonexplosion after (2.14), this proves Theorem 1.1.

11.8 8. Independent audit and branch arithmetic

The independent replay checked:

1. the one-sided ρ choice in every missing-edge strong orientation;
2. the exhaustive long-time alternatives in (4.5);
3. removal of (4.6), including slow-interference endpoint moments;
4. both first-target cases in the dormant block;
5. the squared-workload iteration (5.3)–(5.5); and
6. the exceptional-event use of (5.5) in the core appendage.

No orientation/rate counterexample or load-bearing gap was found. The finite certificate identifies 42 descriptor incidences on exactly fourteen ordered support pairs. The phase classifier is applied after the earlier exact residual branch has been removed. Thus, although the family has the same top support, it does not contain the already certified partner

$$\{B, 2A, B + C\} / \{0, A, C\}.$$

The prior-branch overlap is therefore zero, and the overlap-free table contribution is

$$14 - 0 = 14$$

new positive-invariant ordered support pairs and zero signed pairs. Relative to the previously certified remainder, the exact update is

$$(2169, 191) \mapsto (2155, 191). \quad (8.1)$$

This arithmetic does not claim closure of any rank-one, one-active, or all-active interface.

The scoped finite support certificate is reproduced by

```
PYTHONPATH=src python3 -B src/rank_two_return_certificate.py
PYTHONPATH=src python3 -B -m unittest \
  tests/test_rank_two_return_certificate.py -v
```

It freezes 42 incidences, 14 support pairs/lower supports, the split $10 + 4$, the vertical split $3 + 1$, and support hash

ec552cc5f008cbb881c52dfc054d4ea1034357bebe525c6be06b389dd019540c

The executable records analytic `_theorem_certified` as true to mirror the independent proof audit; its finite checks do not substitute for that audit.

12 All-active-only reversible-top Foster theorem

12.1 1. Exact scope and status

This note isolates a pair-level consequence of the independently audited all-active generator theorem. It concerns residual support pairs for which

1. every affine-stoichiometrically feasible failed descriptor is three-active;
2. the fixed whole-top linkage has two complexes, rank one, and deficiency zero; and
3. every failed descriptor satisfies the curvature-cofactor hypothesis of Proposition 5.2 in *three_active_shell_gluing_gate.md*.

The exact selector contains 51 ordered support pairs. All 51 lie in the positive-invariant table and none lies in the signed table. Its SHA-256 is

cc1d4b0941588f7b664a3266076789e548ae1f675924854eff18c9552d86e3ea

The finite selector is executable. The analytic theorem below has been independently replayed against the exact selector, including the boundary descriptors and the classwise Foster step. It is not a global T3-2 claim.

12.2 2. One potential on every divergent sequence

Fix one selected support pair, one closed irreducible population class Γ , arbitrary strongly connected orientations, and positive rates. Let the fixed whole-top linkage be $T = \{y, z\}$. Strong connectivity on two vertices supplies both directions (after aggregating parallel labelled channels). Choose $\theta > 0$ so that

$$\kappa_{yz}\theta^y = \kappa_{zy}\theta^z. \quad (2.1)$$

Use the rate-adjusted entropy

$$V_\theta(x) = \sum_i \left[x_i \left(\log \frac{x_i}{\theta_i} - 1 \right) + \theta_i \right]. \quad (2.2)$$

with $0 \log 0 = 0$. It is proper and bounded below on $\mathbb{N}_0^3$. This is exactly the potential in Proposition 5.2 of the all-active note, which proves, for every affine-feasible failed sequence of this pair,

$$\mathcal{L}V_\theta(x_n) \longrightarrow -\infty. \quad (2.3)$$

The curvature-cofactor premise is exactly what makes the positive discrete remainder of the reversible top reaction no faster than the other linkage’s maximal source tier. The strict lower-linkage exit then dominates it. No stationary start or shell mixing assertion is used.

Now take any divergent sequence in Γ which is not in one of those failed descriptors. Pass to an exact D-tier subsequence. If its descriptor were affine infeasible, it could not be realized in the fixed affine class containing Γ . Hence it satisfies the ordinary top-S descending-source condition. The Anderson–Kim entropy estimate applies. The difference between (2.2) and ordinary entropy is the fixed affine function

$$-\sum_i x_i \log \theta_i + \sum_i (\theta_i - 1). \quad (2.3a)$$

Every reaction therefore changes the correction by a bounded constant. The forced negative logarithmic exit diverges, while positive lower-tier terms are bounded by the standard ge^{-g} estimate. This argument is unchanged when some coordinates are capped: an exact boundary descriptor that failed the descending-source condition would be an affine-feasible failed descriptor, whereas the selector certifies that every such descriptor is three-active. Therefore (2.3) also holds along this subsequence.

We have proved the sequence statement

$$\text{every divergent sequence in } \Gamma \text{ has a subsequence on which } \mathcal{L}V_\theta \rightarrow -\infty. \quad (2.4)$$

If $\{\mathcal{L}V_\theta > -1\} \cap \Gamma$ were infinite, local finiteness of $\mathbb{N}_0^3$ would select a divergent sequence inside it, contradicting (2.4). Thus

$$\mathcal{L}V_\theta \leq -1 \quad \text{outside a finite subset of } \Gamma. \quad (2.5)$$

12.3 3. Classwise recurrence

Shift V_θ by its finite lower bound so that it is nonnegative. The CTMC is nonexplosive. In a binary network, a reaction sourced at a bimolecular complex cannot increase total population; every population-increasing reaction therefore has source molecularity at most one, and its total rate is affine. Localized Dynkin applied to (2.5), then monotone removal of the finite-sublevel and time localizations using nonexplosion and nonnegativity, gives

$$\mathbb{E}_x \tau_{K_\Gamma} \leq V_\theta(x) < \infty \quad (3.1)$$

for the finite exceptional set (enlarged by one state if it is empty). From a state in this finite set, the first holding time and the finitely many possible successor states have finite expected cost to hit the set again. The finite trace therefore has a positive recurrent state; irreducibility of Γ promotes this to every state of Γ . Hence every closed class is positive recurrent.

Theorem 3.1 (independently audited). Every one of the 51 selected support pairs is positive recurrent on every closed irreducible class, for every strongly connected orientation and every positive vector of present rates.

This theorem uses one potential throughout; it has no chart-switch toll. It neither covers the twelve all-active curvature-seam pairs nor promotes a pair with a feasible one- or two-active failure.

12.4 4. Reproduction

```
PYTHONPATH=src python3 -B src/all_active_only_recurrence.py
PYTHONPATH=src python3 -B -m unittest \
  tests/test_all_active_only_recurrence.py -v
```

The executable certifies the selector and its disjoint support count. In particular it checks all 209 selected failed incidences against the all-active incidence table, the fixed whole-top support, and the exact curvature-cofactor predicate. It does not computationally prove the analytic Foster argument.

12.5 5. Independent audit record

The proof replay checked the following three load-bearing implications.

1. Every one of the 209 failed incidences selected by the script occurs in the certified all-active table, has the pair's fixed two-node rank-one deficiency-zero whole-top linkage, and satisfies Proposition 5.2's cofactor hypothesis. There is no selected boundary failure.
2. On every passing descriptor, including every realizable boundary descriptor, (2.3a) has bounded reaction increments and hence cannot cancel the diverging logarithmic exit in the Anderson–Kim argument.
3. The subsequence assertion (2.4) makes the bad-generator set finite on each class, and the stopped Dynkin argument above yields finite mean return. No uniform finite box for an inactive coordinate and no embedded-jump occupation estimate enters this proof.

The original audit-pending draft wrote (2.2) with $\log(x_i!)$. That formula is not literally the potential in Proposition 5.2: its difference from ordinary entropy is not affine. The certified statement uses the continuous entropy (2.2), for which both the failed-cone proposition and the passing-cone bounded-correction argument apply exactly.

13 The rank-one no-promotion pair selector

13.1 1. Exact split and claim boundary

The two-active atlas has 310 ordered support pairs with at least one closed rank-one flat phase. Exactly 77 of them also have an affine-feasible two-active promotion failure. Removing those gives

$$310 - 77 = 233. \tag{1.1}$$

For all 233 pairs, every affine-feasible failed descriptor with at least two active coordinates is now handled by one common rate-corrected factorial potential:

- every two-active failure is a rank-one flat episode covered by Theorem 1.1 of *rank_one_corrected_factorial_endpoint.md*;
- there is no two-active promotion failure and no rank-two failure;
- the all-active branch is either one of 154 safe reversible rate-adjusted pairs, one of 67 directed-triple factorial-linear pairs, or absent for twelve pairs; and
- on all-active passing descriptors the fixed linear correction has bounded jump increments and does not change the Anderson–Kim tier descent.

The all-active/two-active compatibility certificate proves that the whole-top support is the same in both dimensions. The finite selector also checks the stronger fact needed here: each of the 310 rank-one pairs has one common whole-top mask across all its rank-one descriptors.

Thus the 233-pair statement is a certified **dimension-at-least-two common-potential theorem**. It is not automatically a recurrence theorem because 92 of the 233 pairs still have an affine-feasible one-active failed descriptor. The recently refuted universal one-active old-debt theorem is not used here.

13.1.1 1.1 Discrete transfer on the 154 reversible all-active pairs

The all-active Proposition 5.2 of *three_active_shell_gluing_gate.md* is written for the continuous rate-adjusted entropy

$$U_\theta(x) = \sum_i \left[x_i \left(\log \frac{x_i}{\theta_i} - 1 \right) + \theta_i \right], \quad (1.2)$$

whereas the rank-one endpoint theorem uses the exact discrete potential

$$\mathcal{F}_\theta(x) = \sum_i \{ \log(x_i!) - x_i \log \theta_i \}. \quad (1.3)$$

The following transfer is needed before these can be called one common potential.

There is an exact discrete proof which is stronger than a formal Stirling comparison. Write $z = y + \zeta$, let $\alpha = \kappa_{yz}$, $\beta = \kappa_{zy}$, and put

$$\lambda(x) = \alpha(x)_{\underline{y}}, \quad \mu(x) = \beta(x)_{\underline{z}}. \quad (1.4a)$$

Detailed balance of the correction in (1.3), the exact jump identity (2.2) of *rank_one_corrected_factorial_endpoint.md*, and $\log u \leq u - 1$ give

$$\begin{aligned} \lambda(x) \Delta_\zeta \mathcal{F}_\theta(x) &\leq \mu(x + \zeta) - \lambda(x), \\ \mu(x) \Delta_{-\zeta} \mathcal{F}_\theta(x) &\leq \lambda(x - \zeta) - \mu(x). \end{aligned} \quad (1.4b)$$

Therefore, retaining only enabled terms,

$$\mathcal{L}_T \mathcal{F}_\theta(x) \leq \beta \{ (x + \zeta)_{\underline{z}} - (x)_{\underline{z}} \} + \alpha \{ (x - \zeta)_{\underline{y}} - (x)_{\underline{y}} \}. \quad (1.4c)$$

Each bracket is a bounded first finite difference of a falling-factorial polynomial of degree at most two. Its positive part is a finite sum of the changed-coordinate curvature-cofactor monomials (with a bounded boundary remainder). Thus Proposition 5.2's cofactor hypothesis makes (1.4c) $O(\beta_n)$ directly. The forced lower-linkage exit, evaluated with the exact factorial jump identity, has drift divided by β_n tending to minus infinity. This already proves the discrete all-active theorem.

Lemma 1.1 (Stirling finite-difference transfer). Let $D_\theta = \mathcal{F}_\theta - U_\theta$, up to an irrelevant additive constant. For every reaction jump ν in the binary complex set,

$$|D_\theta(x + \nu) - D_\theta(x)| \leq C_\nu \sum_{i: \nu_i \neq 0} \frac{1}{x_i \vee 1}. \quad (1.4)$$

Consequently every reversible all-active pair satisfying the curvature-cofactor hypothesis of Proposition 5.2 also satisfies

$$\mathcal{L} \mathcal{F}_\theta(x_n) \longrightarrow -\infty \quad (1.5)$$

on each of its failed all-active exact-tier sequences. The same $\mathcal{F}_\theta$ retains the Anderson–Kim descent on every passing sequence.

To prove (1.4), put

$$d(m) = \log(m!) - m \log m + m, \quad d(0) = 0. \quad (1.6)$$

For a unit increment,

$$d(m+1) - d(m) = 1 - m \log(1 + 1/m) = O((m \vee 1)^{-1}), \quad (1.7)$$

and the unit decrement is the same estimate shifted by one. Every binary reaction changes a coordinate by at most two, so telescoping proves (1.4), including the finite boundary values after enlarging C_ν .

Now let $T = \{y, z\}$ be the flat reversible all-active top, let $\alpha_n = x_n^y \asymp x_n^z$, and let $\beta_n = \max_{u \in R} x_n^u$ for the lower linkage. By (1.4),

$$|\mathcal{L}_T D_\theta(x_n)| \leq C \alpha_n \sum_{i: (z-y)_i \neq 0} \frac{1}{x_{n,i} \vee 1}. \quad (1.8)$$

For every changed coordinate, the corresponding $\alpha_n/(x_{n,i} \vee 1)$ is, up to a bounded flat-tier factor, the curvature-cofactor monomial in Proposition 5.2. Its certified hypothesis makes (1.8) $O(\beta_n)$. Every lower propensity is at most order β_n , and (1.4) is bounded globally, so

$$(\mathcal{L}_R D_\theta(x_n))^+ = O(\beta_n). \quad (1.9)$$

Equivalently, the continuous-entropy proof supplies the forced lower exit

$$\frac{\mathcal{L}_R U_\theta(x_n)}{\beta_n} \longrightarrow -\infty, \quad (1.10)$$

while $\mathcal{L}_T U_\theta = O(\beta_n)$. Equations (1.8)–(1.10) prove (1.5).

On a passing exact-tier sequence, the forced descending-source edge has negative reward of order $-g_n$ after normalization by its maximal source propensity. Equation (1.4) adds only a bounded normalized term; every positive lower-tier edge is still controlled by ge^{-g} . Thus replacing U_θ by $\mathcal{F}_\theta$ does not change the ordinary Anderson–Kim conclusion. This proves that the potential used by the two-active endpoint is literally the potential used in the all-active and passing regions.

13.2 2. The 141-pair recurrence theorem

The feasible-failure active-count profiles of the 233 pairs are

active counts appearing	pairs
$\{1, 2, 3\}$	92
$\{2, 3\}$	129
$\{2\}$	12.

(2.1)

Therefore exactly

$$233 - 92 = 141 \quad (2.2)$$

pairs have no one-active failed descriptor. Their all-active split is

$$72 + 57 + 12 = 141, \quad (2.3)$$

for safe reversible, directed triple, and no all-active failure, respectively.

13.2.1 Theorem 2.1 (pair-level common-potential composition)

For each of these 141 pairs, fix the potential

$$\mathcal{F}_*(x) = \sum_i \log(x_i!) + \ell_* \cdot x \quad (2.4)$$

selected by its unique rank-one top mask and actual rates. Fix a closed irreducible population class Γ . Then every divergent exact-tier subsequence in Γ has one of the following alternatives:

1. it is a two-active failed flat sequence, and the audited physical episode has expected $\mathcal{F}_*$ -drift tending to $-\infty$;
2. it is an all-active failed sequence, and the audited all-active generator theorem, together with Lemma 1.1 in the reversible case, gives $\mathcal{L}\mathcal{F}_* \rightarrow -\infty$; or

3. it is a passing descriptor, and the ordinary descending-source argument is unchanged by the fixed linear correction.

There is no feasible one-active failed sequence by definition of the 141-pair selector. A zero-active population sequence lies in a finite set. The ten possible *zero_boundary_phase_only* incidences do not create a fourth alternative in Γ : at their displayed inactive cap the lower linkage is disabled, the top linkage preserves that zero coordinate and H_w , and Proposition 4.1 of *two_active_promotion_phase.md* confines Γ to one finite top shell. Thus no divergent sequence in a fixed Γ realizes such an incidence.

Lemma 3.1 of *physical_entropy_gluing_lemma.md* turns the sequence-local flat episodes into a finite exceptional set, and Theorem 2.1 there glues them to the generator-good region using the same proper potential (2.4).

Population-increasing reactions in a binary network have at most affine propensity: a bimolecular source cannot increase total molecular count. Comparison with a linear pure-birth chain gives nonexplosion. The classwise corollary of the common-entropy gluing theorem therefore gives positive recurrence on every closed irreducible class.

An independent replay checked the pair selector, unique common top mask, the discrete reversible transfer in Lemma 1.1, the directed-triple branch, all passing descriptors, the sequence-to-finite-exception step, end-point and duration integrability, common-potential gluing, nonexplosion, and the classwise positive-return conclusion. It found no remaining gap. It also checked exact disjointness from the 151 affine, fourteen rank-two, 51 all-active-only, and twelve H_b -seam selectors.

Therefore every closed irreducible class of each of these 141 support pairs is positive recurrent. The other 92 pairs in the 233-pair local branch are not included: each retains at least one affine-feasible one-active failed descriptor.

13.3 3. Exact hashes and reproduction

The 233-pair dimension-local fingerprint is

```
afc4f8e121cdd6893f31edfdc5461f4d5f4d5b8340e37b64a222adcd7994114c
```

The 141-pair recurrence fingerprint is

```
bc3540674c5ec8eef96fe4272e15c1f3d220a06fe7ad890189d2f745e6c22e67
```

The full selector-row fingerprint is

```
6b3bc0cfb7dfae535f40d90a6d8faa6e7056902f75e3a578567652397a008809
```

Run

```
PYTHONPATH=src python3 -B src/rank_one_no_promotion_branch.py
PYTHONPATH=src python3 -B -m unittest \
  tests/test_rank_one_no_promotion_branch.py -v
```

The certificate turns on the audited dimension-at-least-two common-potential and exact 141-pair recurrence flags. Global T3-2 remains uncertified.

14 The post-rank-one one-active repair

14.1 1. Scope and status

This note treats only the 92 pairs left out of the 141-pair theorem in *rank_one_no_promotion_pair_branch.md*. They have 272 affine-feasible one-active failed flags. The finite classification, arbitrary-orientation zero-contest lemma, corrected-factorial endpoint, and pair-level composition below have passed independent audit. None uses the false uniform old-debt lemma.

The mechanism is simpler than the universal reflected-debt proposal. On these 92 pairs, every nonfrozen one-active face has a physical priority episode with net active reward -1 . A lower-source clock is used only when no active-source clock is enabled. Thus an unresolved positive reward requires at least one slow-before-fast contest, whereas service requires no such contest. Nested entries do not invalidate this ordering.

This is a scoped branch, not a theorem for T3-2. In particular, none of the 1,227-pair one-active flags outside this post-rank-one intersection is included.

14.2 2. Exact support partition

Relabel the active species as $X = C$, and quotient the exchange of the two inactive species A, B . The 272 rows have twelve top/cap profiles and exactly two support forms.

14.2.1 Family I: an inactive quadratic linkage (200 rows)

The first linkage is one of

$$\{2A, 2B\}, \quad \{2A, AB\}, \quad \{2B, AB\}, \quad \{2A, 2B, AB\}. \quad (2.1)$$

It preserves $M = A + B$. The other linkage has active-degree-one set

$$K = \{AC, BC\} \quad \text{or} \quad K = \{C, AC, BC\}, \quad (2.2)$$

and its remaining vertices form a nonempty subset of $\{0, A, B\}$. If $C \notin K$ and 0 is present, that same linkage also contains A or B .

14.2.2 Family II: a mixed reversible kill (72 rows)

The first linkage is exactly

$$\{2A, AC\}, \quad (2.3)$$

and is therefore reversible in every strong orientation. The second linkage has active set $\{BC\}$ or $\{C, BC\}$, while its lower vertices are a nonempty subset of

$$\{0, A, B, 2B, AB\}. \quad (2.4)$$

If C is absent and 0 is present, (2.4) contains a nonzero vertex.

At the displayed capped face the split is

case	rows
an active source is already enabled	230
only the zero-source seed starts the priority word	32
no source is enabled	10.

(2.5)

The executable certificate is `src/one_active_kinetic_depth.py`.

14.3 3. The arbitrary-orientation zero-contest lemma

Fix any strongly connected directed reaction graph on each displayed linkage and positive rates. Call a finite physical word *zero-contest* if it never fires a lower-source reaction while an active-source reaction is enabled.

Lemma 3.1. From every state on one of the 272 faces, exactly one of the following holds.

1. There is a zero-contest word with net active reward -1 .
2. No reaction is enabled; the state belongs to a singleton closed class and X is classwise fixed.

14.3.1 Proof

Suppose first that an active complex k is enabled. In its linkage, strong connectivity gives a simple directed path from k to a lower complex. Truncate at the first lower target. Every source before the last edge is active. Starting from the actual physical copy of k , the usual residual-plus-current-complex lift enables the whole path. Its active reward is -1 , so it is a zero-contest service word.

It remains to consider a state with no enabled active complex. In both families the active set contains complexes with cofactors A and B , so this forces $A = B = 0$; pure C is absent. The quadratic linkage in (2.1) and the reversible linkage (2.3) are then disabled.

If 0 is absent, every remaining source contains an inactive molecule. No reaction is enabled, giving alternative 2. If 0 is present, the support classification supplies a nonzero lower vertex j in the same mixed linkage. Choose a simple directed path from 0 to j . Follow its lower part only while no active source is enabled. If a lower target first creates a cofactor, the first paragraph supplies service. If the path first enters the active set, follow its active segment to the first lower target. The entry and exit cancel. Simplicity prevents that lower target from being 0 , so it retains a cofactor; the first paragraph then supplies one surplus exit. All lower steps occurred with no active clock enabled. This proves alternative 1. $\square$

The argument permits missing reaction edges and uses only strong connectivity. In particular, it is not an inference from Hamilton-cycle orientations.

14.4 4. Kinetic-depth consequence

Let $N = X$, and assign a cost of one to a lower-source reaction fired while an active-source clock is enabled, and cost zero to every other reaction. For a prescribed finite word, this is its power of N^{-1} , up to subpower inactive cofactors.

Corollary 4.1. On every nonfrozen row, old-debt service has minimum contest depth zero. Every finite word which starts with zero reflected debt, ends with positive reflected debt at a state with no active clock, and does not first decrease $H = X - D$, has contest depth at least one.

Indeed, at zero contest depth every lower-to-active entry immediately enables its active target. Before another lower edge can fire, an active path must end at a lower complex; that exit cancels the entry or creates a surplus negative reward. Lower-to-lower and active-to-active reactions have zero active reward. Hence a zero-contest path cannot leave unresolved positive debt at its next inactive base.

The finite regression additionally enumerates every failing directed Hamilton-cycle orientation pair. There are 2,660 such orientation pairs. Among the finite creation words in the certified box, all service depths are zero and all creation depths lie in $\{1, \dots, 8\}$; no row has creation depth at most service depth. This numerical statement is only a regression for Lemma 3.1, not its proof.

14.5 5. Corrected-factorial endpoint

Let

$$\mathcal{F}_*(x) = \sum_i \log(x_i!) + \ell_* \cdot x \tag{5.1}$$

be the same rate-corrected potential selected by the unique rank-one top linkage in *rank_one_corrected_factorial_endpoint.md*. Consider a one-active source-rate sequence with $X = N \rightarrow \infty$.

Section 8.1 of *global_atlas_interface_closure.md* proves that a coordinate has positive weight in the refined rational flag exactly when its population tends to infinity. Consequently a genuinely one-active subsequence has both inactive coordinates bounded. Passing to a subsequence fixes their two exact integer values. If an inactive coordinate diverges, even subpower relative to N , the sequence has at least two active coordinates and belongs to the already audited rank-one multi-active theorem. This deterministic subsequence split is not tightness implying finite support.

Lemma 5.1 (priority endpoint). On each of the 262 nonfrozen rows there is a physical stopping time τ_N , ending at the service word of Lemma 3.1 or at the first lower interruption of an enabled active phase, such that

$$\mathbb{P}\{\text{lower interruption}\} = O(N^{-1}), \quad \sup_N \mathbb{E}\tau_N < \infty, \quad (5.2)$$

and

$$\mathbb{E}[\mathcal{F}_*(X_{\tau_N}) - \mathcal{F}_*(X_0)] = -\log N + O(1). \quad (5.3)$$

14.5.1 The countable active phase

Strip the common active molecule from C, AC, BC . Before the first active exit, the phase is a unimolecular immigration/conversion chain on $\{0, A, B\}$, run on the N -clock. In Family II the additional vertex AC has the direct active exit $AC \rightarrow 2A$. Every sink component of the retained active graph has an edge to the killed lower set; otherwise the original linkage would not be strongly connected.

Starting from the fixed inactive state selected above, this killed linear chain has an N -independent law after the time change $s = Nt$, finite exponential endpoint moments, and an integrable killed duration. Here is a direct proof which includes open phases. Retained reactions among $\{0, A, B\}$ are immigrations, deaths, and conversions; there is no branching. Use the standard independent-particle construction: every initial or immigrant particle follows a finite-state conversion/death chain, and zero-source immigrations are independent Poisson processes.

If 0 is absent, total particle number is fixed and positive. Strong connectivity gives every particle type a path with positive probability to a killing edge in a fixed scaled-time block. Hence the first killed particle has an exponential time tail. If 0 is present, either its active edge is itself a killing edge or strong connectivity gives a positive-rate immigration/conversion route to one. Every resulting particle has a fixed positive probability to reach a killing edge before it dies. Thinning the immigration process again gives an exponential tail, including from the empty state. The number of immigrants before that exponential time has an exponential moment. Thus the total population at the killed endpoint has an exponential moment. This argument is uniform from the fixed finite set of initial inactive states selected on the subsequence.

The same construction, including the geometric number of seed trials, gives for every fixed q

$$\mathbb{E} \int_0^{\hat{\tau}} (1 + |E_s|)^q \{1 + \log(1 + |E_s|)\} ds < \infty, \quad (5.4a)$$

where E_s is the stripped phase on scaled time and

$$\hat{\tau} := \sum_j N(\beta_j - \alpha_j)$$

is the sum of the scaled lengths of the concatenated active windows $[\alpha_j, \beta_j]$; lower-only waiting intervals are not included. This occupation estimate, rather than endpoint moments alone, controls the size-biased lower-interruption law.

It follows directly from (5.1) that

$$\sup_N \mathbb{E}[|\Delta \mathcal{F}_{*,\text{inactive}}| \text{ ; priority phase}] < \infty. \quad (5.4)$$

No finite phase box is asserted: the open unimolecular endpoint is countable, and (5.4) follows from its exponential moment.

14.5.2 Lower interruptions and the active reward

Every lower propensity is a polynomial of degree at most two in the inactive phase. The occupation bound (5.4a) gives

$$\mathbb{E} \int_0^{\tau_N} \lambda_{\text{lower}}(X_s) \mathbf{1}_{\{\text{an active clock is enabled}\}} ds \leq \frac{C}{N}. \quad (5.5)$$

Hence the first assertion of (5.2) follows from the compensator formula. A lower interruption has one bounded population jump; (5.4) and the factorial jump identity make its expected positive cost $O((\log N)/N) = o(1)$. On the complementary priority episode, the net active change is exactly -1 , and hence

$$\log((N-1)!) - \log(N!) = -\log N. \quad (5.6)$$

Equations (5.4)–(5.6) prove (5.3). The lower-only seed part has finite geometric moments: a fixed simple path from 0 to its nonzero lower vertex has positive rate-dependent probability, and every return to 0 restarts the same trial. Waiting at a no-fast state is bounded by an exponential clock with a fixed positive rate; retained active windows take only $O(N^{-1})$ mean physical time. A lower competitor which fires while no active clock exists is part of the finite seed chain, rather than an omitted interruption; a competitor which fires during an active phase is charged by (5.5). Hence the full duration and endpoint are uniformly integrable, proving (5.2) and Lemma 5.1.

14.6 6. Pair-level composition

The 92 pairs already have the following audited common-potential facts.

1. Every feasible failure with at least two active coordinates is handled by the common $\mathcal{F}_*$ theorem for the 233 no-promotion rank-one pairs.
2. Every passing source-rate sequence remains Anderson–Kim descending after the fixed linear correction.
3. The rank-one mask, hence ℓ_* , is the same in every multi-active phase of a fixed pair.

Lemma 5.1 supplies the missing one-active alternative with the same potential. The ten frozen rows are singleton closed classes, not divergent return obstructions. Every sequence with an unbounded inactive coordinate, including subpower divergence, is a refined multi-active flag by (8.6) of *global_atlas_interface_closure.md*. Thus every divergent sequence in a fixed non-singleton closed class has either generator drift tending to $-\infty$ or a physical $\mathcal{F}_*$ -episode with expected drift tending to $-\infty$.

The common-potential gluing theorem then gives a finite exceptional set; nonexplosion is the same affine pure-birth comparison used for the 141-pair branch. An independent replay checked the arbitrary strong-digraph path contraction, the countable killed-linear endpoint and occupation moments, the exact refined-weight split, the common correction, frozen classes, and the classwise gluing. Consequently the argument extends that branch from 141 to all 233 no-promotion rank-one pairs.

The executable certificates set the local analytic and exact 92-pair recurrence flags true. Global T3-2 remains false.

14.7 7. Reproduction

Run

```
PYTHONPATH=src python3 -B src/one_active_kinetic_depth.py
PYTHONPATH=src python3 -B -m unittest \
    tests/test_one_active_kinetic_depth.py -v
```

15 The 36 promotion-only support pairs

15.1 1. Exact scope and certified status

After the four already certified disjoint branches are removed, the exact promotion selector contains 36 support pairs with no affine-feasible one-active failure. Every selected pair has exactly one feasible failed descriptor, and that descriptor is two-active. The disjoint split is

failed phase	incidences	pairs
enabled top seed	20	20
dormant finite shell	16	16

(1.1)

There are 32 positive-invariant pairs and four signed pairs. Their pair fingerprint is

`f2ad8cbe4b9ca7f36c39bed4bfe5aaafc6a9152eaf300390b5c25ba546519137`

This note proves the physical-time theorem for exactly these 36 pairs. Two independent audits checked the stopping rules, endpoint estimates, arbitrary strong orientations, and common-potential composition. The narrowly scoped analytic and pair-level flags are true. Global T3-2 remains uncertified, and its executable flag remains false.

15.2 2. One common proper potential

For a fixed selected network, put

$$\mathcal{F}_\ell(x) = \sum_{i=1}^3 \log(x_i!) + \ell \cdot x. \quad (2.1)$$

Choose $\ell = 0$ in the eight seeded rows with no wholly top linkage. In the other rows there is one reversible two-complex whole-top linkage. Choose ℓ by detailed balance on that linkage:

$$\ell \cdot (z - y) = \log \frac{\kappa_{zy}}{\kappa_{yz}}. \quad (2.2)$$

The supports which occur are

$$\{A, B + C\}, \{B, A + C\} \quad \text{in twelve seeded rows,} \quad (2.3)$$

and

$$\{A, B\} \quad (5 \text{ dormant rows}), \quad \{B, 2A\} \quad (11 \text{ dormant rows}). \quad (2.4)$$

After adding a network-dependent constant, (2.1) is nonnegative and proper on $\mathbb{N}_0^3$. A fixed linear correction does not change any strict D-tier logarithmic gap. Consequently the usual Anderson–Kim source-tier argument applies to every passing descriptor for the same $\mathcal{F}_\ell$. Only the single failed descriptor listed in (1.1) needs a stopped episode.

15.3 3. The twenty enabled-seed rows

All network complexes in these rows have descriptor weight zero or one. More strongly, the exact descriptor partition places every occurring top source of both linkages in one D-tier block. Along an exact realizing sequence write N for any one of their source monomials. Exact D-tier equivalence, not merely equality of normalized logarithmic weights, gives finite positive ratios between *all* these source monomials. The inactive coordinate has cap zero and stays bounded along a finite actual-target path, while every enabled lower source has rate $O(1) = o(N)$. Thus a whole-top competitor cannot hide a subpower refinement inside the displayed top tier.

Choose an enabled vertex y_0 in the proper top subset of one linkage. Strong connectivity supplies a simple directed path

$$y_0 \longrightarrow y_1 \longrightarrow \cdots \longrightarrow y_j, \quad (3.1)$$

stopped at the first target outside the global top tier. If an intermediate top target needs the inactive species, it is an actual target and therefore is enabled for the next edge. Along the prescribed path the inactive count never exceeds a support-dependent constant and active populations change by only $O(1)$. Thus every prescribed edge has rate at least cN , and the total rate of all top-sourced competitors is at most

CN . The path length is at most the number of complexes in its linkage. It follows from the successive exponential races that

$$\mathbb{P}\{(3.1) \text{ fires before every competitor}\} \geq p > 0. \quad (3.2)$$

Stop on completion of (3.1) or on the first competing reaction. The duration is stochastically bounded by a fixed sum of exponentials of mean C/N . Every top-to-top competitor changes $\mathcal{F}_\ell$ by $O(1)$: exact tier equivalence controls its factorial ratio, including a newly created inactive molecule. A top-to-lower competitor is already favorable. A lower-source competitor has probability $O(N^{-1})$; its positive factorial cost is at most $O(\log N)$, and the sharper propensity-times-log estimate

$$u \log^+(v/u) \leq v/e \quad (3.3)$$

also covers a refined subpower subsequence.

The final edge in (3.1) has a top source and a weight-zero target. Hence

$$\Delta_{y_{j-1} \rightarrow y_j} \mathcal{F}_\ell = -\log N + O(1). \quad (3.4)$$

Combining (3.2)–(3.4) gives a physical stopping time τ_N with

$$\mathbb{E} \Delta \mathcal{F}_\ell \leq -p \log N + O(1), \quad \mathbb{E} \tau_N^m = O(N^{-m}) \quad (3.5)$$

for every fixed m . The number of population jumps before the endpoint is pathwise bounded, so every endpoint moment required for localization is automatic after scaling the two active coordinates.

15.4 4. The sixteen dormant finite-shell rows

The inactive coordinate is C in all sixteen rows. The exact support certificate proves:

1. the wholly top linkage is one of the two supports in (2.4);
2. the proper linkage contains $2C$;
3. at $C = 0$, its enabled source set is one of $\{0\}, \{A\}, \{0, A\}$, or is empty; and
4. the empty case occurs in exactly two incidences.

15.4.1 4.1 Whole-shell endpoint and interior estimates

For $A \leftrightarrow B$, put $H = A + B$ and use the correction (2.2). The exact generator identity and $\log u \leq u - 1$ give

$$\mathcal{L}_* \mathcal{F}_\ell \leq \kappa_{AB} + \kappa_{BA}. \quad (4.1)$$

The chain is an Ehrenfest birth–death chain on its finite H -shell. An exponential supermartingale in the two boundary strips gives, from every exact-tier compact interior start and for every fixed L, m ,

$$\mathbb{P}\{(A_t/H, B_t/H) \text{ leaves a fixed larger interior set for some } t \leq L \log H\} \leq C_{L,m} H^{-m}. \quad (4.2)$$

For $B \leftrightarrow 2A$, put $H = A + 2B$ and $N \asymp \sqrt{H}$. The independently audited estimates (3.4), (6.4), and (6.5) of *rank_one_multichannel_carrier.md* give the corresponding statements: uniform scaled endpoint moments at independent bounded-rate clocks, super-polynomial interior retention through $L \log N$, and uniformly bounded expected net $\mathcal{F}_\ell$ -cost. Thus, in both templates,

$$\mathbb{E}[\mathcal{F}_\ell(X_S) - \mathcal{F}_\ell(X_0)] \leq C \quad (4.3)$$

at a bounded finite sequence of independent 0-source waits or killed A -source occupation windows. Equation (4.1) proves this directly for the unimolecular shell; the stopped semigroup estimate proved in the corrected-factorial endpoint theorem gives it for $\{B, 2A\}$.

15.4.2 4.2 The finite priority macrochain

The sixteen rows have the exact disjoint split

$$7 \text{ with unique enabled source } 0, \quad 7 \text{ with enabled source } A, \quad 2 \text{ disabled.} \quad (4.4)$$

The seven unique-0 rows have proper support contained in $\{0, C, 2C, AC, BC\}$. Four have the equal-scale whole shell $\{A, B\}$; the other three have $\{B, 2A\}$ and therefore are not invocations of the earlier weight-one cap-zero theorem. All seven are covered directly by the priority proof below. The seven A -enabled rows also have whole shell $\{B, 2A\}$. The final two rows are handled in Section 4.3.

The two whole-shell templates have different physical scales. We contract only the conservative whole-shell motion and retain every proper-linkage reaction. If the successive proper reactions are $y_k \rightarrow z_k$, define the reflected workload debt

$$D_{k+1} = (D_k + h(z_k) - h(y_k))^+, \quad h(y) = w \cdot y, \quad D_0 = 0. \quad (4.5)$$

A negative increment larger than the current D_k is a *surplus exit*. Before the first surplus exit the cumulative proper-linkage workload increment equals D_k , whereas a surplus endpoint has cumulative increment at most -1 . Whole-shell reactions preserve h , remain present in physical time, and do not change D .

For $A \leftrightarrow B$, every present AC - or BC -source clock has order N whenever $C > 0$. Let $\mathcal{M}$ be the nonempty set of these workload-one vertices. Since the proper linkage is strongly connected and $\mathcal{M}$ is a proper set, no nonempty subset of $\mathcal{M}$ reachable from an actual target can be closed. Its first edge to workload zero has an exponential tail on the N -clock, with all internal $\mathcal{M}$ -edges and all whole-shell jumps retained. More explicitly, on the interior tube the aggregate exit hazard is at least cN , whereas the total internal proper-linkage hazard is at most CN . The number of internal $\mathcal{M} \rightarrow \mathcal{M}$ reactions before each exit consequently has a uniform geometric tail and moments of every fixed order. These internal reactions preserve C and H_w , and exact D-tier equivalence gives an $O(1)$ corrected-factorial increment per reaction. Present C -only source clocks have order one, so their probability of interrupting one such block is $O(N^{-1})$.

The only nonabsorbing macrostates needed in this case are

$$R = (C = 0, D = 0), \quad P = (C > 0, D = 1), \quad Q = (C > 0, D = 0). \quad (4.6)$$

At R , the only enabled proper source is 0. Its target either is C -only, giving Q , or lies in $\mathcal{M}$, giving P . The first $\mathcal{M}$ -exit from P either consumes the last C and returns to R , or retains C and reaches Q . From Q , the next $\mathcal{M}$ -exit is surplus. If every trial from R returned to R , then 0 together with all reachable vertices of $\mathcal{M}$ would be a closed proper subset of the directed proper linkage, omitting $2C$. Strong connectivity rules this out. Neutral restarts therefore have a geometric tail with a network-dependent parameter bounded away from zero.

For $B \leftrightarrow 2A$,

$$A \asymp N, \quad B \asymp N^2. \quad (4.7)$$

Here BC is the unique workload-two proper vertex. Whenever $C > 0$, every BC -source channel has order N^2 , and each of its targets has workload at most one. Consequently

$$h(z) - h(BC) \leq -1 \quad (4.8)$$

for every such edge. At $C = 0$, take A , rather than 0, as the base source whenever A is present: its clocks have order N , while a 0-source clock has order one. If A is absent, 0 is the only enabled base source. Thus an order-one wait never hides an enabled order- N proper reaction.

An entry from 0 creates at most two debt units, and an entry from A creates at most one. Equation (4.8) confines the nonabsorbing priority macrochain to

$$R_0 = (C = 0, D = 0), \quad R_1 = (C = 0, D = 1), \quad P_d = (C > 0, D = d), \quad d = 0, 1, 2, \quad (4.9)$$

where R_1 occurs only when A is present. If a BC -reaction retains C , at most two such reactions clear all old debt and the next one is surplus. If it consumes the last C , its target is 0, with workload drop two, or A , with workload drop one, so the endpoint is R_0 or R_1 .

Any closed service-free macroclass would be contained in the base/reset set generated by $\{0, A, BC\}$; all other targets either give surplus or enter a BC -priority run which does. Such a class would project to a proper closed subset of the proper-linkage digraph. This is impossible because the digraph is strongly connected and contains $2C$. Equivalently, the possible neutral resets $b \rightarrow BC \rightarrow b$, $b \in \{0, A\}$, and mixtures of them always have an outgoing edge. Every edge leaving A has the same order- N scale as the other A -source edges, every edge leaving 0 has the same order-one scale as the other 0 -source edges, and every edge leaving BC has the same order- N^2 scale as the other BC -source edges. Thus each visit has a fixed positive escape probability. An A - or AC -source reaction while BC is enabled has probability $O(N^{-1})$ before the next BC -reaction. It is retained as a physical interruption and is not counted as service. Likewise, when A is the leading cap-zero source, a simultaneous 0 -source reaction interrupts its $O(N^{-1})$ window with probability $O(N^{-1})$. All still lower $0, C, 2C$ clocks during a BC -block have smaller probability.

In either shell, the finite priority macrochain has an absorbing surplus state accessible from every non-disabled starting state and no other closed class. Hence there are constants $C < \infty$ and $q \in (0, 1)$, depending on the network and rates but not on N , such that

$$\mathbb{P}\{\text{no surplus in the first } K \text{ macrotransitions}\} \leq Cq^K + O(N^{-1}). \quad (4.10)$$

Choose K so the first term is below $1/8$, and stop at surplus, at the first lower-priority interruption, or after K macrotransitions. Before surplus the positive endpoint workload is at most two. Enlarging K , if necessary, gives some $\delta > 0$ such that, for all large N ,

$$\mathbb{E}\Delta H_w \leq -\delta, \quad |\Delta H_w| \leq C_K. \quad (4.11)$$

All clocks remain physical. A 0 -source wait occurs only when no proper A -source is present and has fixed exponential moments. An A -source wait is an $O(N^{-1})$ killed occupation window; an equal-shell top exit has duration $O(N^{-1})$; and a BC -priority exit has duration $O(N^{-2})$. The whole shell evolves throughout. With K fixed, the duration has moments of every fixed order, the inactive count is bounded by the number of retained proper jumps, and the whole-shell endpoint has every fixed scaled active moment. No inactive-coordinate truncation is used.

For completeness, a killed BC -window has uniformly bounded whole-shell factorial cost without any scale matching. The exact $\{B, 2A\}$ corrected-factorial inequality gives

$$\mathcal{L}_* \mathcal{F}_\ell \leq C(1 + A) = O(N) \quad (4.12)$$

on the interior tube. Its duration is $O(N^{-2})$, so Dynkin's formula charges $O(N^{-1})$. The A -windows and independent 0 -clock endpoints are covered by (4.3). Thus the scale-separated contraction has not hidden a positive top-shell endpoint cost.

On the interior event,

$$\log(A \vee 1) = \log N + O(1), \quad \log(B \vee 1) = \begin{cases} \log N + O(1), & L_* = \{A, B\}, \\ 2\log N + O(1), & L_* = \{B, 2A\}. \end{cases} \quad (4.13)$$

Hence every bounded lower jump has

$$\Delta \mathcal{F}_\ell = \Delta H_w \log N + O(1). \quad (4.14)$$

Equations (4.2)–(4.3), the deterministic transition cap, and (4.11)–(4.14) give

$$\mathbb{E}\Delta \mathcal{F}_\ell \leq -c \log N + O(1). \quad (4.15)$$

The exceptional interior-exit probability is super-polynomial, while the factorial oscillation on the finite whole shell is polynomial times $\log N$; its endpoint contribution is therefore $o(1)$. Every lower-priority interruption has a bounded workload jump and factorial cost $O(\log N)$; its $O(N^{-1})$ probability therefore contributes $O(\log N/N) = o(1)$.

15.4.3 4.3 The two disabled rows

If the enabled set in item 3 is empty, the proper linkage is identically disabled at $C = 0$, and the whole linkage preserves $C = 0$ and its finite workload shell. On a fixed closed irreducible class, that workload has one fixed value. The class is therefore finite, so no divergent sequence in the class realizes the displayed descriptor.

15.5 5. Classwise composition

Theorem 5.1. Give either linkage of any of the 36 selected support pairs an arbitrary strongly connected orientation and arbitrary positive rates on its present edges. Then every closed irreducible population class is positive recurrent.

Fix a closed irreducible class Γ . If a divergent sequence in Γ lies in a passing source-tier cone, the standard generator estimate for the fixed potential (2.1) tends to $-\infty$. If it lies in the unique failed cone, Section 3 or Section 4 gives a physical episode with

$$\mathbb{E}_x\{\Delta\mathcal{F}_\ell + \eta\tau\} \leq -1 \quad (5.1)$$

outside a finite set; the disabled alternative is not divergent in Γ . A bad-sequence contradiction turns these sequence estimates into one finite exceptional set. The common-entropy physical-time gluing theorem then gives finite mean hitting of that set.

Nonexplosion is elementary here and in the full binary class: a reaction which increases total population has a source of molecularity at most one, so the positive part of the total-population generator is bounded by $C(1 + |x|)$. Local finiteness and finite mean return from the finite target give positive recurrence of Γ .

The two independent audits certify the exact selector and show that it is disjoint from all earlier ordered branches. The ordered residual arithmetic is therefore

$$(1871, 191) \mapsto (1839, 187). \quad (5.2)$$

15.6 6. Audit obligations

An independent replay should check, in this order:

1. that every prescribed seeded path retains all top competitors and that its lower-source positive logarithmic cost is uniformly integrable;
2. the exact unimolecular endpoint inequality (4.1) and the boundary supermartingale (4.2);
3. that the dormant actual-target path can be restarted after every faster top exit without silently deleting a reaction;
4. the reflected-debt coefficient in (4.5) and (4.10)–(4.11), including base states with source A rather than 0;
5. the lift from (4.11) to the common factorial drift (4.15); and
6. the fixed-class treatment of the two disabled rows and the disjoint 36-pair selector.

Both independent replays passed all six checks. In particular they verified the exact $7 + 7 + 2$ dormant split, the priority SCC argument for arbitrary strong digraphs, and the $O(\log N/N)$ interruption charge. No counterorientation or adverse rate choice remains at this scope.

16 The suppressed four-pair promotion orbit

16.1 1. Exact scope and claim boundary

Up to the four species relabellings present in the ordered residual table, consider

$$L_* = \{U, I + V\}, \quad L_- = \{0, I, 2I, I + U\}. \quad (1.1)$$

The first linkage is reversible, and the second is given an arbitrary strongly connected directed graph and arbitrary positive present rates. The exact finite certificate finds four positive-invariant support pairs in this orbit, with no overlap with any previously certified branch. Across those four pairs there are exactly 28 affine-feasible failed incidences:

(w_I, w_U, w_V)	incidences	whole top
(0, 0, 1)	12	no
(0, 1, 1)	4	yes
(0, 1, 2)	4	no
(0, 1, 3)	4	no
(0, 4, 5)	4	no.

(1.2)

The twelve one-active rows have fixed caps $(I, U) = (0, 0), (0, 1), (0, 2)$, once per support pair.

Throughout, a profile means an **exact D-tier** profile, not merely a normalized logarithmic weight. Thus, in the $(0, 1, 2)$ profile, M^2/D stays in a compact subset of $(0, \infty)$. If a subpower factor makes this ratio tend to zero or infinity, the sequence refines the D-tier and falls respectively into the $(0, 1, 3)$ or $(0, 4, 5)$ profile. This three-way split is load-bearing below.

This note proves recurrence for the exact four-pair orbit. Two independent audits replayed the analytic stopping argument, exact selector, and pair-level composition. No broader promotion family is claimed.

16.2 2. One potential and one exact shell invariant

Write the rates of $U \rightarrow I + V$ and $I + V \rightarrow U$ as $\alpha, \mu > 0$. Choose a fixed vector ℓ satisfying

$$\ell_I + \ell_V - \ell_U = \log(\mu/\alpha), \quad (2.1)$$

and put

$$\mathcal{F}_\ell(x) = \log(I!) + \log(U!) + \log(V!) + \ell \cdot x. \quad (2.2)$$

After adding a constant, this is nonnegative and proper. The affine correction (2.1) is the detailed-balance correction for L_* , and it does not alter any strict D-tier logarithmic gap.

The top linkage preserves

$$Q = U + 2V - I. \quad (2.3)$$

The four lower complexes have Q -levels

$$q(0) = q(I + U) = 0, \quad q(I) = -1, \quad q(2I) = -2. \quad (2.4)$$

Thus the apparently bad $0 \leftrightarrow I + U$ resets are exactly Q -neutral. Every edge from $\mathbf{N} := \{0, I + U\}$ to $\mathbf{D} := \{I, 2I\}$ strictly lowers Q . Strong connectivity of L_- guarantees at least one edge across this cut.

Start a top excursion from the core state

$$(I, U, V) = (0, M, D). \quad (2.5)$$

If one lower reaction fires during the excursion and all remaining inactive molecules are then cleared by $I + V \rightarrow U$, its exact core endpoint and factorial reward are as follows:

lower edge	$(U_{\text{end}}, V_{\text{end}})$	$\Delta \mathcal{F}_\ell$
$0 \rightarrow I$	$(M+1, D-1)$	$\log \frac{M+1}{D} + \ell_U - \ell_V$
$0 \rightarrow 2I$	$(M+2, D-2)$	$\log \frac{(M+1)(M+2)}{D(D-1)} + 2(\ell_U - \ell_V)$
$0 \rightarrow I+U$	$(M+2, D-1)$	$\log \frac{(M+1)(M+2)}{D} + 2\ell_U - \ell_V$
$I+U \rightarrow 0$	$(M-2, D+1)$	$\log \frac{D+1}{M(M-1)} + \ell_V - 2\ell_U$
$I+U \rightarrow I$	$(M-1, D)$	$-\log M - \ell_U$
$I+U \rightarrow 2I$	$(M, D-1)$	$-\log D - \ell_V$

(2.6)

These identities do not assume that the lower clock fires at $I = 1$. Before that clock, the top shell has

$$U = M - I, \quad V = D + I. \quad (2.7)$$

Substitution shows that clearing the remaining I 's gives exactly (2.6), independently of the transient value of I . The first two and last two rows lower Q ; the middle reset pair preserves it.

16.3 3. The transient killed-shell occupation lemma

The only genuinely averaged flag is $(w_I, w_U, w_V) = (0, 1, 2)$; the same estimate also repairs the originally suppressed $(0, 4, 5)$ flag.

Lemma 3.1 (top-shell occupation). Suppose $M, D \rightarrow \infty$, $D/M \rightarrow \infty$, and $R := M^2/D \geq r_0 > 0$. Run only L_* from (2.5), and set

$$h = \frac{T}{R} = \frac{TD}{M^2}, \quad J_h = \int_0^h I_t(M - I_t) dt. \quad (3.1)$$

Then, for each fixed $T > 0$,

$$J_h \longrightarrow \frac{\alpha T}{\mu} \quad \text{in } L^1, \quad \mathbb{E} \int_0^h \{I_t + (I_t)_2\} dt = O(M^{-1}). \quad (3.2)$$

The transient I -population at a timeout or at a clock biased by $I_t(M - I_t)$ has uniform exponential moments (the claim concerns shell excess, not the full U, V populations).

16.3.1 Proof

On the top shell, I is the birth-death chain

$$i \longrightarrow i+1 \quad \text{at rate } \alpha(M-i), \quad i \longrightarrow i-1 \quad \text{at rate } \mu i(D+i). \quad (3.3)$$

It is stochastically dominated by immigration-death with immigration αM and per-particle death μD . Starting from zero, the dominating law is Poisson with mean at most $\alpha M/(\mu D)$. Hence, for every fixed $p \geq 1$,

$$\sup_{t \geq 0} \mathbb{E} I_t^p \leq C_p \frac{M}{D}, \quad \mathbb{E} \int_0^h I_t^p dt \leq \frac{C_{p,T}}{M}. \quad (3.4)$$

Let $\mathcal{M}_t$ be the martingale in the coordinate equation for I . At time h ,

$$\mu D \int_0^h I_t dt = \alpha M h - \alpha \int_0^h I_t dt - \mu \int_0^h I_t^2 dt + \mathcal{M}_h - I_h. \quad (3.5)$$

Multiplying by $M/(\mu D)$ and subtracting $\int_0^h I_t^2 dt$ gives

$$J_h = \frac{\alpha M^2 h}{\mu D} - \frac{\alpha M}{\mu D} \int_0^h I_t dt - \left(1 + \frac{M}{D}\right) \int_0^h I_t^2 dt + \frac{M}{\mu D} (\mathcal{M}_h - I_h). \quad (3.6)$$

The first term is $\alpha T/\mu$. Equations (3.4) make all deterministic errors $o_{L^1}(1)$. Moreover

$$\mathbb{E}\langle \mathcal{M} \rangle_h \leq 2\alpha M h = O(D/M), \quad (3.7)$$

so the martingale term in (3.6) has variance $O(M/D) = o(1)$. This proves (3.2). The same Poisson domination, followed by compensation with a slightly smaller exponential parameter, gives the timeout and size-biased endpoint assertion. $\square$

Let Λ_0 be the aggregate rate of 0-source lower edges, Λ_U the aggregate coefficient of $I + U$ -source edges, and Λ_D the remaining I - and $2I$ -source propensity. Conditional on the top path,

$$\int_0^h \Lambda_U(t) dt = \kappa_U J_h, \quad \mathbb{E} \int_0^h \Lambda_D(t) dt = O(M^{-1}). \quad (3.8)$$

Thus the killed shell retains every lower clock: no reaction is deleted.

16.4 4. Physical cleanup and endpoint cost

Lemma 4.1 (return to the core). Under the hypotheses of Lemma 3.1, after one bounded lower jump the chain reaches $I = 0$ by physical $I + V \rightarrow U$ reactions before any further lower reaction, outside an event of probability $o(1)$. The cleanup duration has every fixed scaled moment, and the exceptional positive factorial cost is uniformly integrable.

Indeed, while $I \geq 1$, the clearing rate is at least cDI , whereas the aggregate $I + U$ -source rate is at most CMI , and the $0, I, 2I$ -source rates are at most $C(1 + I^2)$. Poisson domination from Lemma 3.1 gives a geometric number of clearing steps and therefore

$$\mathbb{P}\{\text{lower interference during cleanup}\} \leq C \frac{M}{D} + \frac{C}{D} = o(1). \quad (4.1)$$

At an $I + U$ -biased endpoint, the compensation estimate

$$\mathbb{E} \int_0^h e^{\theta I_t} I_t (M - I_t) dt \leq C_T \quad (4.2)$$

holds for a sufficiently small $\theta > 0$. For I - and $2I$ -source interference, (3.4) gives probability $O(M^{-1})$; multiplying its factorial cost by $\log(M + D)$ still gives $o(1)$ in every profile in (1.2). These estimates prove the claimed endpoint uniform integrability.

16.5 5. Every failed flag

16.5.1 5.1 The twelve one-active rows

Here $D = V \rightarrow \infty$, while $M = U \in \{0, 1, 2\}$ and $I = 0$. Strong connectivity gives a positive aggregate 0-source rate. Retain all clocks and stop at the first lower reaction, then append physical top cleanup. The desired event is that this first reaction has source 0. Top excursions make the expected aggregate hazard of all nonzero-source lower interruptions only $O(D^{-1})$. Every such bounded interruption has positive reward at most $O(\log D)$; in particular, the only row of (2.6) with possibly positive reward is $I + U \rightarrow 0$. Their total expected positive contribution is therefore $O(D^{-1} \log D) = o(1)$. Whatever the 0-source target, the exact table (2.6) gives

$$\Delta \mathcal{F}_\ell \leq -\log D + O(1). \quad (5.1)$$

Here the cleanup estimate is elementary and does not use the $M, D \rightarrow \infty$, $R \geq r_0$ hypotheses of Lemma 4.1. Immediately after the bounded target, $I \leq 4$ and $U = O(1)$. Each present I has top death rate at least cD , whereas all top births and all lower competitors together have rate $O(1 + I^2)$. Thus cleanup precedes interference with probability $1 - O(D^{-1})$, its duration has moments $O(D^{-m})$, and the exceptional positive factorial contribution is $O(D^{-1} \log D) = o(1)$. The initial wait has fixed exponential moments, and the cleaned active endpoint changes by only $O(1)$.

16.5.2 5.2 The whole-top profile $(0, 1, 1)$

Now $M \asymp D \asymp N$, and $R = M^2/D \asymp N$. From the core, the first top birth occurs before a 0-source edge with probability $1 - O(N^{-1})$. Starting at $I = 1$, run until either I returns to zero or a lower reaction fires; an $I + U$ -source edge is the successful lower event, while every other lower event is retained as an exceptional endpoint. All top rates and the $I + U$ -source rate are of order N , so

$$\mathbb{P}\{I + U\text{-source edge before } I = 0\} \geq p > 0, \quad \mathbb{E}\tau^m = O(N^{-m}). \quad (5.2)$$

This follows already by requiring the very next reaction to be an $I + U$ -source edge; additional top excursions only increase the success probability. The rescaled birth–death chain has uniform exponential moments for the transient I population (equivalently, for the excesses of U, V relative to the shell parameters M, D); no uniform moment is claimed for the full active populations. Stop **at** the successful $I + U$ -source reaction; no top-only cleanup is imposed in this equal-depth regime, because further $I + U$ clocks would be of the same order. At a pre-jump state on the top shell,

$$\mathcal{F}_\ell(i, M - i, D + i) - \mathcal{F}_\ell(0, M, D) = O\{1 + i \log(i + 1)\}, \quad (5.3)$$

uniformly for M/D in a compact positive interval. The exponential endpoint bound therefore makes the expected top-shell cost $O(1)$. Each possible $I + U$ -target has direct jump reward $-\log N + O\{1 + \log(i + 1)\}$. Consequently

$$\mathbb{E}\Delta\mathcal{F}_\ell \leq -p \log N + O(1). \quad (5.4)$$

A failed excursion returns to the exact starting core and costs zero. Other lower sources have order one during an $O(N^{-1})$ attempt and contribute $o(1)$.

16.5.3 5.3 The balanced killed shell $(0, 1, 2)$

Here $M \asymp N$, $D \asymp N^2$, and $R = M^2/D$ stays in a compact subset of $(0, \infty)$. Put

$$a_0 = \sum_{z \in \mathbf{D}} \kappa_{0z}, \quad a_U = \sum_{z \in \mathbf{D}} \kappa_{I+U,z}. \quad (5.5)$$

Strong connectivity implies $a_0 + a_U > 0$. In the Lemma 3.1 window, the cumulative service hazard is

$$A_s(h) = a_0 h + a_U J_h. \quad (5.6)$$

It is bounded below with positive probability, while the total $\mathbf{N}$ -source cumulative hazard is bounded above with probability arbitrarily close to one. Conditional Poisson clock calculus therefore gives

$$\mathbb{P}\{\text{the first lower event is an } \mathbf{N} \rightarrow \mathbf{D} \text{ service}\} \geq p > 0. \quad (5.7)$$

Explicitly, choose constants $a, B > 0$ for which $\mathbb{P}\{A_s(h) \geq a, A_{\text{all}}(h) \leq B\} \geq c > 0$. Conditional on the top path, the first-service probability is $\int_0^h e^{-A_{\text{all}}(t)} dA_s(t)$, hence it is at least $e^{-B}a$ on this event. This proves (5.7) without an independence assumption.

The I - and $2I$ -source event probability is $o(1)$ by (3.8). Neutral $0 \leftrightarrow I + U$ endpoints preserve Q and have $O(1)$ factorial reward because $R \asymp 1$. Every service endpoint satisfies, by (2.6),

$$\Delta\mathcal{F}_\ell \leq -\log N + O(1). \quad (5.8)$$

Thus one all-reactions-retained block has expected drift tending to $-\infty$, fixed physical duration, and uniform endpoint moments.

16.5.4 5.4 The source-dominant profile (0, 1, 3)

Here $R = M^2/D \rightarrow 0$ and $D/M \rightarrow \infty$. Wait for the first lower reaction and append physical top cleanup; the desired first clock is the independent aggregate 0-source clock S . Poisson domination gives

$$\mathbb{E} \int_0^S I_t(M - I_t) dt \leq CR = o(1) \quad (5.9)$$

for the independent exponential 0-clock S . Hence a lower $I + U$ -source edge intervenes with probability $O(R)$. Every 0-target is favorable:

$$0 \rightarrow I + U : \quad \Delta \mathcal{F}_\ell = \log R + O(1), \quad 0 \rightarrow D : \quad \Delta \mathcal{F}_\ell \leq -\log(D/M) + O(1). \quad (5.10)$$

The only potentially positive interruption is $I + U \rightarrow 0$, with cost $\log(1/R) + O(1)$; its expected contribution is $O(R \log(1/R)) = o(1)$. Therefore the expected factorial drift tends to $-\infty$.

16.5.5 5.5 The originally suppressed profile (0, 4, 5)

Now $R = M^2/D \rightarrow \infty$. Use the short window $h = T/R$ of Lemma 3.1. The $I + U$ -source cumulative hazard converges to a positive constant, whereas the 0-source hazard is $O(R^{-1})$ and the $I, 2I$ -source hazard is $o(1)$. Thus, for some $p > 0$,

$$\mathbb{P}\{\text{an } I + U\text{-source edge fires by } h\} \geq p. \quad (5.11)$$

All of its possible targets are favorable:

target	$\Delta \mathcal{F}_\ell$
0	$-\log R + O(1)$
I	$-\log M + O(1)$
$2I$	$-\log D + O(1)$.

(5.12)

Timeout followed by cleanup returns to the exact core and costs zero. Equations (3.8), (4.1), and (5.11)–(5.12) give expected factorial drift tending to $-\infty$, duration $O(R^{-1})$, and uniform scaled endpoint moments.

16.6 6. Exact pair theorem

Theorem 6.1. For either ordering of any support pair in the exact four-pair orbit (1.1), give both linkages arbitrary strongly connected directed graphs and arbitrary positive present rates. Then every closed irreducible population class is positive recurrent.

Fix one closed irreducible class Γ . The finite certificate proves that every affine-feasible failed descriptor is one of the profiles in (1.2). Sections 5.1–5.5 give a physical stopping episode for all of them, using the single proper potential (2.2). Every other realizable descriptor passes the ordinary source-tier criterion, and the fixed affine correction ℓ preserves its strict logarithmic exit.

If the required drift and endpoint bounds failed outside every finite subset of Γ , a divergent bad sequence would have a subsequence realizing one of the finitely certified descriptors, contradicting the corresponding estimate above. The common-potential physical-time gluing theorem therefore gives finite mean hitting of one finite subset of Γ .

Nonexplosion is immediate: every reaction which increases total population has source molecularity at most one, so the positive total-population drift is bounded by $C(1 + |x|)$. Local finiteness and finite mean return from the finite target give positive recurrence.

The exact four pairs are positive-invariant and disjoint from all prior ordered branches. The two independent audits certify the disjoint arithmetic

$$(1839, 187) \mapsto (1835, 187). \quad (6.1)$$

Global T3-2 remains uncertified.

16.7 7. Independent-audit record

Both independent replays checked:

1. the exact macro endpoints and rewards in (2.6), including transient $I > 1$;
2. the martingale identity (3.6), its uniform error bounds, and size-biased exponential moments;
3. physical cleanup with every lower clock retained;
4. the cut-hazard lower bound (5.7) for every strong orientation;
5. propensity-times-log uniform integrability in the $R \rightarrow 0$ and $R \rightarrow \infty$ regimes;
6. the common- $\mathcal{F}_\ell$ passing-cone composition and exact four-pair disjointness.

All six obligations passed. One review is recorded in `research_notes/suppressed_promotion_orbit_independent_audit.md`; the second independently replayed the current theorem and executable selector. The analytic and exact pair-level flags are true only for these four pairs; the global flag remains false.

17 The positive- Q trace for the fifteen equal-depth one-active pairs

17.1 1. Scope and claim boundary

This note treats the coefficient-sensitive part of the fifteen support pairs selected by `src/one_active_remaining_structure.py`. It proves the physical sign on the seventy-five active- C equality rows and composes the eight companion rows in Section 7. Two independent audits passed the stopped trace and pair-level theorem. The exact selector is frozen in `src/critical_one_active_q_trace_certificate.py`.

After relabelling the active species to C , every equality row has

$$L_0 = \{0, A + C, B + C\}, \quad L_1 = T \subseteq \{A, B, 2A, A + B, 2B\}. \quad (1.1)$$

The directed graph on each displayed support is arbitrary and strongly connected, and all labelled channel rates are positive. Each of the fifteen possible supports T contains at least one unary and at least one quadratic complex. Propensities use falling factorials.

The finite regression reports a positive-debt word and a first-surplus word of the same slow-before-fast depth two. That equality does **not** give a critical reflected random walk. One unary-to-quadratic reaction already lowers positive debt by one at depth one. A second such reaction is needed only because the regression insists on crossing $D = 0$ and recording a surplus exit.

17.2 2. The exact level

Put

$$M = A + B, \quad Q = C - A - B = C - M. \quad (2.1)$$

Every reaction in L_0 preserves Q . A reaction $y \rightarrow z$ in T changes it by

$$\Delta Q = |y| - |z| \in \{-1, 0, 1\}. \quad (2.2)$$

Thus unary-to-quadratic reactions move Q toward the origin on a large positive shell, while quadratic-to-unary reactions move it away.

For $N \geq 0$, the L_0 -shell $Q = N$ is

$$\Gamma_N = \{(a, b, N + a + b) : a, b \in \mathbb{N}_0\}. \quad (2.3)$$

Its unique base is $z_N = (0, 0, N)$. The base is regenerative for the L_0 -chain: wait for a reaction sourced at 0, then follow the full busy excursion until $M = 0$ again.

17.3 3. Exact shell law and the sign before perturbation

The linkage L_0 is weakly reversible and deficiency zero. Let $K_0, K_{AC}, K_{BC} > 0$ be its directed matrix-tree weights and set

$$u = K_{AC}/K_0, \quad v = K_{BC}/K_0, \quad s = u + v.$$

The conditional product-form law on Γ_N is

$$\pi_N(a, b, N + a + b) = \frac{1}{Z_N} \frac{u^a v^b}{a!b!(N + a + b)!}, \quad Z_N = \sum_{m \geq 0} \frac{s^m}{m!(N + m)!}. \quad (3.1)$$

For fixed $r, t \geq 0$,

$$\mathbb{E}_{\pi_N}[(A)_r(B)_t] = u^r v^t \frac{Z_{N+r+t}}{Z_N}. \quad (3.2)$$

In particular,

$$\mathbb{E}_{\pi_N} A = \frac{u}{N} + O(N^{-2}), \quad \mathbb{E}_{\pi_N} B = \frac{v}{N} + O(N^{-2}), \quad (3.3)$$

whereas every quadratic falling-factorial moment is $O(N^{-2})$.

Let $\mathcal{E}_{12}$ be the set of directed channels in T from a unary to a quadratic complex. Strong connectivity and the presence of both degrees imply $\mathcal{E}_{12} \neq \emptyset$. If a unary source is A , respectively B , write $r_y = u$, respectively v , and put

$$a_- := \sum_{y \rightarrow z \in \mathcal{E}_{12}} \kappa_{yz} r_y > 0. \quad (3.4)$$

The stationary T -hazard therefore satisfies

$$\mathbb{E}_{\pi_N} \mathcal{L}_T Q = -\frac{a_-}{N} + O(N^{-2}). \quad (3.5)$$

Every rate and orientation changes a_- , but cannot change its strict positivity. Quadratic-to-unary channels contribute only to the $O(N^{-2})$ term. Hence no choice of positive rational rates can reverse the leading sign.

17.4 4. Regenerative perturbation with all reactions retained

Let $\Lambda > 0$ be the aggregate rate of the channels sourced at 0 in L_0 . For the L_0 -chain, a cycle from z_N consists of an $\text{Exp}(\Lambda)$ base wait and a busy part. Stripping the common C molecule turns the busy part into a two-state transient unimolecular particle system. Its per-particle clocks are multiplied by $C = N + M \geq N$. Strong connectivity of L_0 makes the single-particle killing chain phase type. Consequently, for every fixed p ,

$$\mathbb{E}(\text{busy time})^p = O(N^{-p}), \quad \mathbb{E} \int (M)_1 dt = O(N^{-1}), \quad \mathbb{E} \int (M)_2 dt = O(N^{-2}). \quad (4.1)$$

The last estimate includes overlapping 0-source immigrants: a second immigrant must arrive during an $O(N^{-1})$ busy interval. More generally, the probability of reaching $M \geq k$ before returning to zero has a geometric-factorial bound uniform in N . This supplies uniform integrability of every fixed endpoint and occupation moment.

Now retain every reaction in T . Fix a sufficiently small, rate-dependent $\rho \in (0, 1/4)$, and define τ_N to be the first of

$$\{\text{the positive return to } M = 0\}, \quad \{M \geq \rho N\}, \quad \{Q \leq N/2\}, \quad \{Q \geq 2N\}. \quad (4.2)$$

The last three endpoints are retained physical states. They are genuine multi-scale exits, not finite-box truncations: an inactive coordinate is of active order, the positive level has already made a macroscopic descent, or the level has made a macroscopic upward excursion which is charged explicitly. Equivalently,

before any boundary one may stop first at a degree-changing T -reaction and append the physical return or boundary hit; no clock is suppressed during the append.

Only unary-to-quadratic reactions can increase M . Their total rate is at most linear in M ; quadratic reactions cannot increase M . Here is a uniform comparison which also retains the degree-preserving quadratic clocks. Strip the common C from AC, BC , and let R be the transient two-state single-particle generator, with absorption at 0. Since $-R$ is a nonsingular M-matrix, there are $r_A, r_B > 1$ and $c > 0$ such that the multiplicative particle corrector

$$G(a, b) = r_A^a r_B^b$$

has top drift at most $-cCMG$. Every quadratic degree-preserving reaction changes G by a bounded factor and contributes at most KM^2G ; every extra 0-immigration or unary-to-quadratic reaction contributes at most $K(1+M)G$. Choose the rate-dependent $\rho > 0$ in (4.2) so small that $KM^2 \leq cNM/4$ for $M \leq \rho N$. The stopped multiplicative drift is then strictly negative for $1 \leq M < \rho N$, after increasing the finite threshold in N .

One explicit choice is $w = (-R)^{-1}\mathbf{1} > 0$ and $r_i = 1 + \varepsilon w_i$. For small $\varepsilon > 0$, the logarithmic drift bracket of one phase- i particle is $\varepsilon(Rw)_i + O(\varepsilon^2) \leq -\varepsilon/2$, which gives the displayed product drift after summing over particles.

The corresponding linear corrector $W = w_A A + w_B B$ gives the block-time bound directly. Suppress only the two slow creation clocks until the next particle absorption. Then M is constant, the quadratic label-change error is at most KM^2 , and $\mathcal{L}W \leq -cNM$ below the chosen ρN . Since $W \asymp M$, stopped Dynkin yields $\mathbb{E}(\text{next absorption time}) \leq C/N$, uniformly in the current label composition.

Equivalently, expose successive phase-type particle-absorption blocks. Even after the quadratic label changes, a block has mean duration at most C/N . Its aggregate extra-immigration/unary-birth rate is at most $K(1+M)$. With ρ as above, the probability of one more creation before the next absorption is bounded by a fixed $q < 1/4$, uniformly below the boundary; shrink the rate-dependent ρ once more to ensure this. At block endpoints, M is therefore dominated by the nearest-neighbor walk with up probability q and down probability $1 - q$, killed at zero. Equivalently its family tree is dominated by a subcritical Galton–Watson total progeny. The Catalan/path multiplicity is included in this comparison, which gives

$$\mathbb{P}\{\text{at least } j \text{ creation blocks before return}\} \leq Ce^{-cj}.$$

Reaching $M = \rho N$, or changing Q by a fixed fraction of N without first returning to $M = 0$, requires order N creation blocks. The resulting bound is $Ce^{-cN} = O(N^{-k})$ for every fixed k . This supplies the claimed geometric-factorial comparison rather than inferring it merely from a pointwise negative drift.

The implication for the two Q -boundaries is exact. Before return to zero, let I_0 count extra 0-immigrations, U_T unary-to-quadratic events, D_T quadratic-to-unary events, and D_0 top particle absorptions. Then

$$M = 1 + I_0 + U_T - D_T - D_0, \text{quad}Q - N = -U_T + D_T.$$

Thus $Q \leq N/2$ forces $U_T \geq N/2$, while $Q \geq 2N$, together with $M \geq 0$, forces $I_0 \geq N - 1$. Either boundary therefore requires order N of the slow creation blocks controlled above.

Together with the phase-type L_0 killing blocks, this gives, for all sufficiently large N ,

$$\sup_N \mathbb{E}_{z_N} \tau_N^p < \infty, \quad \sup_N \mathbb{E}_{z_N} \left(1 + \frac{M_{\tau_N} + C_{\tau_N}}{N}\right)^p < \infty \quad (4.3)$$

for every fixed p . A direct construction is to expose consecutive single-particle killing blocks. During one block a unary birth has probability $O(N^{-1})$; after such a birth the same estimate restarts from at most two particles. Quadratic and degree-preserving reactions may change the particle labels, but do not increase M , and every label has a finite top path to 0. Iteration up to $M = \rho N$ gives a subcritical geometric bound. Reaching that boundary requires order N successful immigration or unary-birth races against order- N top clocks. Hence, for every $k < \infty$,

$$\mathbb{P}_{z_N}\{M_{\tau_N} \geq \rho N \text{ or } Q_{\tau_N} \notin (N/2, 2N)\} = O(N^{-k}). \quad (4.4)$$

The boundary overshoot is bounded because every population jump is bounded, and (4.3) makes every fixed polynomial endpoint cost on (4.4) negligible. This argument retains the physical quadratic clocks; they only insert label changes or deaths and never increase M .

The probability of two or more T -reactions in the same busy excursion is $O(N^{-2})$. The probability that the first degree-changing T -reaction is quadratic-to-unary is also $O(N^{-2})$, by (4.1). Kac's cycle identity applied to (3.3) gives the first-order coefficient of the unary-to-quadratic event. Since $\pi_N(z_N) = 1 + O(N^{-1})$,

$$\begin{aligned}\mathbb{E}_{z_N}(Q_{\tau_N} - N) &= -\frac{a_-}{\Lambda N} + O(N^{-2}), \\ \mathbb{E}_{z_N}(Q_{\tau_N} - N)^2 &= \frac{a_-}{\Lambda N} + O(N^{-2}).\end{aligned}\tag{4.5}$$

For completeness, if σ_N denotes the unperturbed positive L_0 -return cycle, continuous-time Kac gives

$$\mathbb{E}_{z_N} \int_0^{\sigma_N} f(X_t) dt = \frac{\mathbb{E}_{\pi_N} f}{\pi_N(z_N) \Lambda}.\tag{4.5a}$$

Applied to the aggregate unary-to-quadratic propensity, (3.2) makes the right side $a_-/(\Lambda N) + O(N^{-2})$. Total unary T -occupation is $O(N^{-1})$, total quadratic occupation is $O(N^{-2})$, and after one unary event the remaining busy occupation is again $O(N^{-1})$. Compensation therefore makes two T -events an $O(N^{-2})$ event. Coupling the full and unperturbed cycles until their first T -event changes the Kac coefficient only by $O(N^{-2})$, which proves both lines of (4.5).

These are full-reaction stopped-cycle expansions; (4.4) is included in their remainders. A unary conversion followed by a degree-changing reaction, a reversal after a unary birth, and a boundary carrier launched near the return time all contain a second lower-during- fast contest and are part of the $O(N^{-2})$ remainder.

17.5 5. Strict quadratic trace drift

At an ordinary base endpoint $M = 0$, one has $Q = C \geq 0$; the boundary endpoints in (4.4) have negligible quadratic contribution. Combining the two lines of (4.5),

$$\begin{aligned}\mathbb{E}_{z_N}[Q_{\tau_N}^2 - N^2] &= 2N \mathbb{E}_{z_N}(Q_{\tau_N} - N) + \mathbb{E}_{z_N}(Q_{\tau_N} - N)^2 \\ &= -\frac{2a_-}{\Lambda} + O(N^{-1}).\end{aligned}\tag{5.1}$$

Therefore there are $N_0 < \infty$ and $\delta > 0$, depending on the oriented network and rates, such that

$$\mathbb{E}_{z_N}[Q_{\tau_N}^2 - N^2 + \delta \tau_N] \leq -\delta, \quad N \geq N_0.\tag{5.2}$$

This is the required positive- Q physical-time sign. In particular, the equality of the two canonical surplus-word exponents cannot produce a null or transient reflected level on these rows.

17.6 6. Relation to reflected debt

On a fast L_0 carrier, entries add one to both D and M , top exits at positive debt subtract one from both, and top conversions preserve both. Thus $D - M$ is unchanged by the carrier dynamics until a surplus exit. A unary-to-quadratic T -reaction raises M by one and hence lowers positive base debt by one after the carrier is drained. It occurs at one slow-before-fast contest. A quadratic-to-unary reaction can create one unit of base debt, but first requires two simultaneous cofactors and is one order rarer. The canonical depth-two surplus word from $D = 1$ is just two consecutive depth-one debt reductions.

17.7 7. A single proper potential for all eighty-three rows

The quadratic trace need not be glued to the rest of the atlas as a separate potential. Fix any network-dependent linear corrector $\ell \in \mathbb{R}^3$, add a constant K , and put

$$F_\ell(x) = K + \sum_i \log(x_i!) + \ell \cdot x, \quad V(x) = F_\ell(x)^2,\tag{7.1}$$

where K is large enough that $F_\ell \geq 1$. This V is proper. On the base z_N ,

$$F_\ell(z_N) = N \log N + O(N), \quad F_\ell(z_{N-1}) - F_\ell(z_N) = -\log N + O(1). \quad (7.2)$$

The event $Q_{\tau_N} = N - 1$ has probability $a_-/(\Lambda N) + O(N^{-2})$; every nonzero event other than this leading one has total probability $O(N^{-2})$, with the uniform endpoint moments from (4.3)–(4.4). Therefore

$$\mathbb{E}_{z_N}\{V(X_{\tau_N}) - V(z_N)\} = -\frac{2a_-}{\Lambda}(\log N)^2 + O(\log N). \quad (7.3)$$

In particular, the same proper V , rather than Q^2 , pays the physical duration with a diverging margin. Beginning at any fixed inactive population on the same positive- Q shell, first run the full physical cleanup to $M = 0$. The no- T cleanup lowers F_ℓ by $M \log N + O_M(1)$; a T -interruption has probability $O_M(N^{-1})$ and has all fixed moments. Appending the base cycle preserves (7.3), with constants allowed to depend on that fixed inactive state.

The dependence on the inactive state is not a hidden finite-box assumption. In a bad-sequence proof, a genuinely one-active sequence has eventually fixed inactive coordinates. If either inactive coordinate is unbounded, the exact source-rate subsequence has at least two active coordinates and is routed to the corresponding multi-active descriptor.

The fifteen pairs have eighty-three feasible failures. The seventy-five rows treated above are the five capped active- C rows on each pair. The remaining eight rows consist of

- six direct-enabled rows with no positive-debt base; and
- two zero-source seed rows with zero-contest service and strictly deeper debt creation.

After normalizing the active species to C and exchanging inactive A/B where needed, the executable payload gives exactly the following four templates; every row has cap $(A, B) = (0, 0)$:

$L^{(1)}$	$L^{(2)}$	multiplicity/access
$\{0, AB, AC\}$	$\{B, 2B, BC\}$	2 zero-source
$\{0, AB, AC\}$	$\{C, 2B, BC\}$	2 direct
$\{0, AB, AC\}$	$\{B, C, 2B, BC\}$	2 direct
$\{0, AB, AC\}$	$\{B, C, BC\}$	2 direct.

(7.4a)

For a direct row, mark an enabled active-degree-one source and follow a shortest top path to its first lower target. Strong connectivity makes the fast path phase type. A completed path has

$$\Delta F_\ell = -\log X + O(1), \quad (7.4)$$

whereas a lower-during-fast interruption has probability $O(X^{-1})$ and positive F_ℓ -cost $O(\log X)$. Endpoint and duration moments are uniform from the fixed inactive start. Hence

$$\mathbb{E}\Delta V = 2F_\ell \mathbb{E}\Delta F_\ell + \mathbb{E}(\Delta F_\ell)^2 \leq -cF_\ell \log X \quad (7.5)$$

for all sufficiently large X .

More explicitly, in each direct template the initially enabled source is C . A C -edge to B or $2B$ is already a strict active exit. A C -edge to BC launches the stripped open phase on $\{0, B\}$: on the X -clock, $C \rightarrow BC$ is immigration and $BC \rightarrow C$ is linear death. It is countable, not a finite carrier. Kill this immigration–death phase at the first C - or BC -source edge to B or $2B$. Strong connectivity makes that killing accessible from its unique closed fast class, so the killed phase has a uniform exponential absorption time and exponential B -endpoint moments. All competitors from the first linkage or from $B, 2B$ have order at most polynomial in this exponentially controlled B and act for physical time $O(X^{-1})$. Their aggregate interruption probability and propensity-times-log cost are therefore $O(X^{-1})$. The successful endpoint loses one C , so (7.4)–(7.5) follow with all clocks retained.

For either zero-source row, start at the actual target of the 0-edge and run the finite lower/top return prefix with all clocks retained. A failed seed attempt returns to the exact capped face; the number of

attempts is geometric. At the first enabled unpaired top exit, (7.4) holds. Positive interrupted-carrier cost is again $O(\log X/X)$. Thus (7.5) holds for these two rows as well.

For clarity, the two possible seed targets are AB and AC . A seed at AB immediately enables the order- X BC carrier in the second linkage. A seed at AC either returns to 0, an exact failed reset, or reaches AB through the first strongly connected linkage and then launches that same BC carrier. Strong connectivity makes the number of resets geometric with a rate-dependent positive success parameter. During an AC or BC carrier, every noncarrier clock is order one, so the total exceptional probability is $O(X^{-1})$; its positive squared-factorial cost is $O(F_\ell \log X/X)$, negligible beside the successful $-cF_\ell \log X$ term. This verifies the two seed rows rather than appealing to the withdrawn uniform old-debt lemma.

Finally, along every feasible descriptor of these pairs which is not one of the eighty-three failed rows, the Anderson–Kim top-source argument gives the usual strict factorial drift for F_ℓ . For bounded jumps,

$$\mathcal{L}V = 2F_\ell \mathcal{L}F_\ell + \sum_r \lambda_r (\Delta_r F_\ell)^2. \quad (7.6)$$

On a passing source-rate sequence, the second term is $o(F_\ell |\mathcal{L}F_\ell|)$: every $\Delta_r F_\ell = O(\log(2 + |x|))$, while $F_\ell \asymp |x| \log(2 + |x|)$, and the strict top-source logarithmic ratio diverges. Explicitly, if α_n is a maximal source propensity and $g_n \rightarrow \infty$ the certified logarithmic exit gap, the ordinary tier estimate gives $\mathcal{L}F_\ell \leq -c\alpha_n g_n$, whereas

$$\sum_r \lambda_r (\Delta_r F_\ell)^2 \leq C\alpha_n \log^2(2 + |x_n|).$$

The ratio to $F_\ell \alpha_n g_n$ is at most $C \log(2 + |x_n|) / \{|x_n| g_n\} = o(1)$. Thus the passing-region drift also remains strict for the same V .

At a boundary in (4.2), the exact descriptor has at least two active coordinates, has already lost a fixed fraction of Q , or is a super-polynomially rare larger positive- Q restart. In the first case every affine-feasible descriptor of these fifteen pairs passes the top-source test: by construction they are a subset of the exact 1,227-pair selector whose nonempty feasible failure set is entirely one-active. In the last case the same critical episode restarts at the new level. The squared-factorial estimate (7.6) charges the promoted endpoint, while (4.4) makes the upward-restart V -cost summable.

These estimates give the following independently audited exact composition statement.

Theorem 7.1. Give each linkage of any of the fifteen selected support pairs an arbitrary strongly connected orientation and arbitrary positive channel rates. Then every closed irreducible class is positive recurrent.

Indeed, if no finite exceptional set admitted a random-time V +duration inequality, choose a bad state sequence in one closed class and pass to its exact D-tier/source-rate descriptor. An infeasible descriptor cannot occur in that affine class. A passing descriptor is excluded by (7.6); one of the seventy-five equality rows is excluded by (7.3); and one of the eight companion rows is excluded by (7.5). This exhausts the finite descriptor table. The episode endpoints have the moments used above, so the statewise selector gives

$$\mathbb{E}_x[V(X_\tau) - V(x) + \tau] \leq -1 \quad (7.7)$$

outside a finite classwise set. Only source-molecularity-zero or source-molecularity-one reactions can increase total population, at an affine rate, so the chain is nonexplosive. Random-time Foster then gives finite mean return to the finite set.

17.8 8. Audit status and universal interface

The argument above explains what a universal 1,227-pair theorem would have to prove. For every arbitrary strong orientation, classify reflected creation and debt-reduction histories by their first nonzero slow-before-fast coefficient. Strict kinetic separation plugs directly into the common squared-factorial potential V . Equality cannot be settled by a Hamilton-cycle exponent table; for the exact fifteen-pair equality family it is resolved by the invariant Q and the shell trace (4.3). Any other equality family would require its own coefficient identity.

No assertion in this note promotes any pair outside these exact fifteen. The canonical regression is not an arbitrary-orientation theorem, a box exit is not automatically a promotion, and no count is inferred from kinetic depth alone.

Two independent audits checked (4.3)–(4.5a), the super-polynomial boundary restart, all four companion templates, the squared-potential gluing estimate (7.6), and the exact prior-branch disjointness. The certified update is

$$(1835, 187) \mapsto (1820, 187).$$

The analytic and pair-level flags are true only for these fifteen pairs. The global T3-2 flag remains false.

18 The exact 1,227-pair fourth-power composition

18.1 1. Scope and frozen inputs

This note composes the exact one-active selector with the physical-time interface proved in `research_notes/universal_fourth_power_one_active_interface.md`. It makes no claim about the 795 support pairs left outside this selector and no global T3-2 claim. An independent audit replayed the composition below at pre-promotion note SHA-256 652e41ccd7ae36183862a798fcd3bd5acf92ab2528bb356816d14df003b09a and returned **PASS** (confidence 0.91).

The inputs are:

1. the affine-feasibility and exact-tier selector in `src/one_active_phase_shape.py`, which gives 1,227 support pairs and 3,297 affine-feasible failing incidences, every one of them one-active;
2. the arbitrary-strong-orientation resistance theorem in `research_notes/one_active_arbitrary_orientation_graph_theorem.md`;
3. the all-23 killed-resolvent, random-overshoot, moving-boundary, and common-fourth-power theorem in `research_notes/universal_fourth_power_one_active_interface.md`; and
4. the all-species reflected target and physical-time gluing theorem in `research_notes/all_species_reflected_debt_target.md`.

Fix one support pair, an arbitrary strongly connected directed graph on each nontrivial linkage support, arbitrary strictly positive rate constants, and one closed irreducible population class Γ . No constant below is required to be uniform over different rate vectors or different classes.

Use the single population potential

$$\mathcal{F}(x) = K + \sum_{i=1}^3 \log(x_i!), \quad G = 1 + \mathcal{F} \geq 1, \quad W(x) = G(x)^4. \quad (1.1)$$

Thus the linear correction is $\ell = 0$ in every chart. The function W is proper on $\mathbb{N}_0^3$.

18.2 2. Fixed-class descriptor dichotomy

Lemma 2.1 (classwise bad-sequence extraction). Let the support pair belong to the 1,227-pair selector. Every divergent population sequence $x_n \in \Gamma$ has a subsequence satisfying exactly one of the following alternatives.

1. It realizes an Anderson–Kim passing descriptor, and $\mathcal{L}W(x_n) \rightarrow -\infty$.
2. It realizes one of the 3,297 selected one-active failures. After relabelling, its active coordinate $X_n \rightarrow \infty$, while the two inactive integer coordinates are fixed on a further subsequence.

18.2.1 Proof

The exact source/D-tier extraction has finitely many descriptors. Pass to a subsequence realizing one of them. A state sequence contained in the affine class of Γ can realize only an affine-feasible descriptor; this is the necessity direction of the stoichiometric gate theorem. By definition of the 1,227-pair selector, every nonempty affine-feasible failure is one-active. Any other descriptor passes the universal top-source/top-D criterion, and Section 2 of the universal fourth-power note gives $\mathcal{LW}(x_n) \rightarrow -\infty$.

For a one-active exact tier, atlas identity (8.6) says that a coordinate belongs to the active mask exactly when its population diverges. Thus any diverging nominally inactive coordinate makes the exact descriptor at-least-two-active and hence passing. The remaining inactive integer coordinates are bounded and therefore constant after one more subsequence. This is a state-sequence fixed-width tube, not a claim that a tight distribution has finite support. $\square$

The usual contradiction now gives a finite-exception generator bound on the passing part: otherwise choose a divergent sequence with $\mathcal{LW} > -1$ and apply Lemma 2.1. In fact the stronger bound $\mathcal{LW} \leq -cW^{3/4}$ is available on every passing subsequence, so the concave transform from the universal note controls its physical exit time as well as its endpoint W -moment.

The orientation quantifier is already included. On a fixed class, every exact descriptor is either affine-infeasible (and therefore impossible), universally passing (and handled by the common W), or one of the selected affine-feasible one-active failures. For any actual orientation, a descending row may use the generator drift directly; every remaining failing orientation is covered by the graph theorem, which quantifies over all strong orientations of the selected supports. Positive rate constants change only the classwise constants, not the resistance or tier alternatives.

There is also a genuinely finite statewise bad-tube decomposition on the fixed class. Otherwise, for every M and every finite set one could choose a non- W -good state whose second-largest coordinate exceeds M . Such a sequence has two diverging coordinates; Lemma 2.1 extracts an at-least-two-active passing subsequence, a contradiction. Hence there are $M_\Gamma < \infty$ and a finite set K_0 such that every non- W -good state outside K_0 lies in one of the finitely many tubes

$$G = \{x \in \Gamma : \mathcal{LW}(x) \leq -1\}, \quad B_{X,e} = \{x \in \Gamma \setminus (K_0 \cup G) : (x_i)_{i \neq X} = e\}, \quad e \in \{0, \dots, M_\Gamma\}^2. \quad (2.1)$$

Taking a minimum over these finitely many tubes makes the local episode constants uniform on Γ . This finite family is derived from the state-sequence contradiction; it is not inferred from tight occupation marginals.

18.3 3. Reachable marked bad tubes

Fix a reference $x^\circ \in \Gamma$. On every physical reaction $x \mapsto x + \zeta$, update all three reflected marks by

$$D_i^+ = (D_i + \zeta_i)^+, \quad H_i = X_i - D_i. \quad (3.1)$$

Then $0 \leq D_i \leq X_i$ and $H_i \leq x_i^\circ$ pathwise. Consider a one-active bad-tube subsequence from Lemma 2.1, with active coordinate X .

If $D_X = 0$, then

$$X = H_X \leq x_X^\circ. \quad (3.2)$$

Together with the fixed inactive coordinates, this contradicts divergence. Equivalently, the $D_X = 0$ part of every fixed-width tube is a finite class-dependent exception. Thus every divergent reachable marked bad tube has $D_X > 0$.

The graph theorem separates the remaining rows into a strict old-debt reduction alternative and a frozen/no-history alternative. A no-history base is, by definition, incompatible with a physical history carrying $D_X > 0$. A completely frozen population is an absorbing singleton; if Γ is nontrivial it is absent, and if Γ is that singleton positive recurrence is immediate. Consequently every divergent marked bad-tube sequence which actually occurs uses one of the service alternatives.

Family II causes no hidden uniform-box requirement. Its apparent spectator is an exact invariant a_Γ on the fixed class. The cap label 2 means $a_\Gamma \geq 2$, not $a_\Gamma = 2$. The killed Green and carrier constants may depend on this fixed value. Families I and III have the origin base, direct-active rows use their killed exponential phase, and the 222 wholly-top rows use the audited open Poisson block.

18.4 4. The common-potential episode

Lemma 4.1 (one-active physical episode). Outside a finite subset of the reachable marked class, every one-active bad-tube state with $D_X > 0$ admits an all-reaction physical stopping time $\tau > 0$ such that

$$\mathbb{E}_{x,d}[W(X_\tau) - W(x) + \tau] \leq -1. \quad (4.1)$$

Its endpoint and duration satisfy the uniform-integrability conditions required for localization on that fixed class.

18.4.1 Proof

The arbitrary-orientation theorem gives aggregate down resistance $m \leq 2$, a strictly positive leading down coefficient, and no same-base up coefficient through resistance m . In the wholly-top open family, the separate Poisson regeneration theorem replaces word resistance by its effective aggregate value $m = 0$. Proposition 4.2 and Theorem 4.1 of the universal fourth-power note turn these inputs into the repeated physical kernel.

Neutral base endpoints telescope before the first nonneutral return. A nonboundary up endpoint has all conditional moments of its random active overshoot and inactive factorial cost; no deterministic overshoot bound is used. A moving-boundary endpoint, including a simultaneous down/up tie, is charged by the endpoint-weighted three-interruption estimate and lands in an at-least-two-active passing region. The successful endpoint strictly reduces the already-positive reflected debt. These estimates give (4.1) for all sufficiently large active levels in each extracted fixed-width tube.

If no finite exception made (4.1) valid throughout the reachable bad-tube set, choose a divergent sequence where it failed. Lemma 2.1 fixes a template and the inactive base data on a subsequence; the classwise local theorem then gives (4.1), a contradiction. This also proves the required endpoint and duration uniform integrability by the same bad-sequence argument. $\square$

18.5 5. Pair-level theorem

Theorem 5.1 (the exact 1,227-pair branch). For every support pair in the one-active selector, every strongly connected orientation of its linkage supports, every positive rate vector, and every closed irreducible population class, the physical mass-action CTMC is positive recurrent.

18.5.1 Proof

Work on the reachable all-species marked class and use W from (1.1). Lemma 2.1 gives $\mathcal{L}W \leq -1$ outside a finite set and the one-active bad tubes. Section 3 and Lemma 4.1 show that every reachable bad-tube state outside a finite set has positive selected debt and the physical episode (4.1). These are exactly the hypotheses of the all-debt finite-target gluing theorem. The marked chain therefore hits a finite marked set in finite mean physical time. Its physical projection is finite and has a finite mean positive return; irreducibility promotes this to positive recurrence of every state in Γ .

It remains only to justify nonexplosion. Every population-increasing reaction has source molecularity zero or one: a binary source already has the maximal allowed target molecularity and cannot increase total population. Hence the total positive population drift rate is at most $C(1 + \|x\|_1)$. Stopping at total-population levels and applying Gronwall gives nonexplosion. Quadratic reactions which do not increase total population cannot generate infinitely many jumps inside a finite population sublevel, because that sublevel contains finitely many states with bounded total rate. $\square$

18.6 6. Exact disjoint contribution

The executable selector in `src/one_active_prospective_composition.py` gives the following frozen arithmetic:

	positive	signed	total
candidate one-active set	1076	151	1227
already certified overlap	15	0	15
new disjoint contribution	1061	151	1212
remainder after contribution	759	36	795.

(6.1)

The overlap is exactly the already-certified critical one-active 15-pair branch; the candidate set has zero overlap with every other certified branch. The frozen pair fingerprints are

```
candidate_1227      3ab28358663c45a089a5bdf4144c28573718b0c4f8b05472a0af208ca919fcf8
new_disjoint_1212   a7784a1f98da2fbadd70a62bc97fe852393cb410a24e666a6d6c246998f0f579
remainder_after_795 6a1327e6c38bfca30d334691415ba457e84d45d1dfe53d81df4c02aad868123
```

The audit confirmed both Theorem 5.1 and this disjoint arithmetic. The 1,227-pair theorem is therefore certified, with a net new contribution of 1,212 pairs after removing the exact 15-pair prior overlap. Global T3-2 remains uncertified.

19 Audited common-factorial theorem for the prospective 26 pairs

19.1 1. Scope and claim status

This note supplies the proof requested by the claim-neutral selector in `src/prospective_no_promotion_26.py`. It proves two local extensions and then composes them on a fixed irreducible class:

1. the audited arbitrary-orientation one-active graph/kernel theorem applies to the selector’s 30 one-active incidences, with the same arbitrary fixed linear correction used below;
2. the curvature-safe reversible two-node all-active estimate lifts to the **discrete** rate-adjusted factorial and to its fourth power; and
3. one common potential closes the one-active and all-active descriptions by the existing all-species reflected-debt gluing theorem.

The resulting pair statement has passed an independent proof audit. The finite scope certificate is `src/prospective_26_candidate_theorem.py`, and its focused tests are in `tests/test_prospective_26_candidate_theorem.py`. Its two scoped analytic flags and its 26-pair recurrence flag are certified. The global T3-2 flag remains false.

There is one correction to the older selector note. Proposition 5.2 of `three_active_shell_gluing_gate.md` is literally written for the continuous entropy

$$U_{\theta}(x) = \sum_i \left[x_i \left(\log \frac{x_i}{\theta_i} - 1 \right) + \theta_i \right],$$

not for $\sum_i \log(x_i!)$. The two functions do not differ by an affine term. Section 4 below proves the missing discrete comparison and does not identify the two potentials by assertion.

19.2 2. Exact selector and the common correction

The selector contains 26 positive-invariant ordered support pairs, with pair fingerprint

```
393474671be0bf095868e66cbcbf3164d941b99191517f172a41f157e20b21af
```

Their 124 affine-feasible failed incidences split as

$$30 \text{ one-active,} \quad 0 \text{ two-active,} \quad 94 \text{ all-active.} \quad (2.1)$$

The selector is disjoint from every prior-certified branch and lies inside the authoritative claim-neutral prospective 795. The frozen arithmetic is

claim-neutral set	positive	signed	total
prospective parent	759	36	795
selected candidate, prior-certified overlap zero	26	0	26
prospective remainder after candidate	733	36	769

The 769-pair remainder fingerprint is

4e645eca1ba23849680f2f983e1fb8c9465001c5b0e8c0090a31a339bf18ec06

Before the independent verdict this was prospective arithmetic only. The audit has now promoted exactly these 26 pairs, with zero overlap, and changes the authoritative remainder from (759, 36) to (733, 36).

Fix one pair, arbitrary strongly connected orientations of both linkage supports, arbitrary positive rates, and one closed irreducible population class Γ . The whole-top linkage in every failed all-active descriptor of this pair is one fixed support $T = \{y, z\}$. It is either

$$\{2A, B + C\} \quad \text{or} \quad \{A + C, B + C\}. \quad (2.2)$$

A strongly connected orientation on two vertices contains both directions. After aggregating any parallel labels, write their positive rates as κ_{yz} and κ_{zy} , and put $d = z - y$. Choose $\theta \in (0, \infty)^3$ so that

$$\kappa_{yz}\theta^y = \kappa_{zy}\theta^z, \quad \ell_i = -\log \theta_i. \quad (2.3)$$

Only the scalar equation

$$\ell \cdot d = -\log(\kappa_{yz}/\kappa_{zy}) \quad (2.4)$$

is prescribed, so a solution always exists. Because T is fixed for the pair, the same ℓ works on all 94-menu cones belonging to that pair. Choose K large enough that

$$F_\ell(x) = K + \sum_{i=1}^3 \log(x_i!) + \ell \cdot x, \quad G = 1 + F_\ell \geq 1, \quad W = G^4 \quad (2.5)$$

on $\mathbb{N}_0^3$. Such a K exists because the factorial term dominates every fixed linear term. The function W is proper and

$$G(x) \asymp R \log R, \quad qquad R = 2 + \|x\|_1, \quad (2.6)$$

along every divergent sequence.

19.3 3. The 30 one-active rows

19.3.1 3.1 Exact support audit

Relabel the active species as $X = C$ and the inactive species as A, B , as in the audited one-active graph theorem. The 30 physical rows collapse to 15 exact normalized support profiles. Every row has capped inactive state $(A, B) = (0, 0)$, and both linkages are mixed. Their exact partition is

arbitrary-orientation route	incidences
a mixed linkage contains the pure active source X	20
Family III, resistance-zero service from the origin	8

arbitrary-orientation route	incidences
Family III, frozen origin/no positive-debt history	2

The full 30-row and 15-profile hashes are

one-active rows: 091dc59dc83b6445d36135a2d6b1e6c1bfb9565db88328fc1725c39f78bbc7e4
normalized profiles: 36b6eb1c1beadc939681fc7515e2adf3dfb9e16d116da4b8d2bd73ac77aabe12

The certificate checks the following support facts row by row.

- In each of the 20 direct rows, one mixed linkage has stripped top source 0, hence physical source X , and also has a lower complex.
- In each of the ten remaining rows the two stripped top sources are exactly A and B , one in each mixed linkage. The no-fast set is therefore the origin.
- In the eight service rows, a zero-containing linkage has another lower vertex. In the other two rows neither lower support contains 0.

These are exactly the hypotheses used in Sections 3 and 6 of the audited arbitrary-orientation graph proof. They do not depend on a Hamilton-cycle orientation or a bounded CEGAR box.

19.3.2 Lemma 3.1 (scope extension of the graph ordering)

For every strong orientation and every positive rate vector, every historically consistent positive-debt row among these 30 incidences has

$$m_- = 0 < m_+, \quad (3.1)$$

where resistance counts a lower-source firing while an active-degree-one source is enabled. The two remaining rows are frozen at the displayed origin and admit no history carrying positive reflected X -debt.

Proof In a direct row, positive old debt implies $X \geq 1$, so the source X is enabled. In its strong mixed linkage take a simple directed complex path from X to a lower complex. Before the first lower target every source has active degree one. The top-to-top edges have active reward zero and the first top-to-lower edge has reward -1 . Consecutive edges are physically enabled because the target complex of one firing supplies the source of the next. This is a resistance-zero strict decrease.

In a Family-III service row, the only no-fast base is the origin. Follow a simple path in the zero-containing strong linkage from 0 to a nonzero lower vertex. If the path enters an active complex first, follow its first exit; the surviving nonzero cofactor enables the other active source or a second exit. In either case the path supplies one more exit than entry and strictly lowers the relative active level without firing a lower source against an enabled fast clock. Hence again $m_- = 0$.

A primitive positive return to the origin cannot have resistance zero. Its only free launch occurs while no fast source is enabled. Thereafter, in a zero-resistance word, every reaction is active-sourced until the origin is reached. Each entry contributes $+1$, each exit contributes -1 , and return of the inactive population requires at least as many exits as free entries. Its active reward is therefore nonpositive. Thus $m_+ \geq 1$.

Finally, in either no-history row the origin has no enabled source: the two top sources require A and B , and every lower source is nonzero. Moreover no linkage contains target 0, so this origin cannot be reached from a nonorigin state. It is an absorbing initial singleton and cannot carry positive reflected debt. This proves the lemma. $\square$

19.3.3 Lemma 3.2 (aggregate kernel and arbitrary fixed ℓ)

For every nonfrozen row of Lemma 3.1, the complete physical one-active block of the audited fourth-power kernel theorem satisfies, uniformly over the finite base marks of the fixed class,

$$p_D(n) \geq a, \quad p_U(n) \leq \frac{b}{n}, \quad \mathbb{E}\sigma_n \leq T, \quad (3.2)$$

for all sufficiently large active population n . Every fixed endpoint-polynomial moment is bounded in the sense of the kernel theorem, and its stopped three-interruption moving-boundary remainder is $O((L_n^2/n)^3)$ with $L_n = n^{1/8}$. The conclusion remains true for the common W in (2.5), with the arbitrary fixed ℓ selected in (2.3).

Proof The analytic proof of the resistance-to-kernel proposition uses only the support properties just checked, binary molecularity, strong connectivity, and fixed positive rates.

In a direct row, divide time by the active population. Before service, the stripped active-source phase is an immigration–conversion–death process on complexes contained in $\{0, A, B\}$, killed on its first active exit. Every transient top component has a directed route to the lower part because the original mixed linkage is strong. The exponential killed-Green estimate and its polynomially size-biased versions therefore apply exactly as in the audited direct-block proof. The simple path in Lemma 3.1 has a strictly positive product of rate fractions, which supplies the order-one down coefficient.

In a Family-III service row the regenerative no-fast set is the single origin. Between visits to it, the same killed unimolecular Green estimate applies. The resistance-zero witness supplies the positive order-one down coefficient. A positive origin return requires at least one suppressed lower clock by Lemma 3.1, so the ordered-compensation expansion makes its aggregate probability $O(n^{-1})$, after all neutral loops have been summed. Binary molecularity gives the same carrier estimate and the same endpoint-weighted three-interruption bound as in the audited proof. There is no unlisted countable no-fast phase in these rows.

This proves (3.2), including duration and endpoint moments, rather than only the probability of one selected word. The repeated-base fourth-power theorem now applies with $m = 0$.

It remains to check ℓ . A bounded reaction changes $\ell \cdot x$ by the fixed number $\ell \cdot \zeta_r$. Thus it changes neither resistance, the killed generator, nor any power of n in the compensation expansion. At a down endpoint,

$$F_\ell(X_\tau) - F_\ell(X_0) = -\log n + O(1 + Z), \quad (3.3)$$

where Z is the inactive endpoint cost; at an up endpoint the analogous positive bound differs from the $\ell = 0$ bound by at most a fixed multiple of the endpoint population. The already proved exponential/factorial endpoint moments absorb that linear term. At the moving boundary the term is $O(L_n + J)$, not merely $O(L_n)$: nested active entries can make the active overshoot as large as a constant multiple of the counted number J of suppressed interruptions. The audited boundary estimate is endpoint-weighted and controls every fixed J -moment, in particular an order strictly above eight, on the three-interruption remainder. Applying that estimate first to the leading linear endpoint term and then with moment order $q > 8$ to the fourth-power tail gives the same $O(n^{2+7/8}(\log n)^4) = o(n^3(\log n)^4)$ completed-episode boundary cost as for $\ell = 0$. Hence the fourth-power endpoint calculation and the promotion-access estimate are unchanged. $\square$

19.4 4. Discrete all-active fourth-power lift

Fix one of the 94 all-active rows and a tier sequence x_n realizing it. All three coordinates diverge. Hence, after deleting a finite prefix, both top sources are enabled, both neighboring states $x_n \pm (z - y)$ lie in the population lattice whenever they occur below, and all factorial quotients are literal. At boundary states the usual convention is $x^y = 0$ when $y \not\leq x$; no boundary state remains in this all-active tail. Let $T = \{y, z\}$, $d = z - y$, and write

$$A(x) = \kappa_{yz}x^y, \quad B(x) = \kappa_{zy}x^z, \quad h(x) = \log \frac{A(x)}{B(x)}. \quad (4.1)$$

The two top source monomials occupy one D-tier, so along a further tier subsequence

$$0 < c \leq A/B \leq C < \infty. \quad (4.2)$$

Let R now denote the other linkage, let β_n be its maximal source propensity, and put

$$\mathcal{D}_n = (A_n - B_n) \log(A_n/B_n) \geq 0. \quad (4.3)$$

19.4.1 Lemma 4.1 (discrete reversible decomposition)

For the top linkage,

$$\mathcal{L}_T F_\ell(x_n) \leq -\mathcal{D}_n + C\beta_n, \quad (4.4)$$

and, for $k = 2, 3, 4$,

$$\sum_{r \in T} \lambda_r(x_n) |\Delta_r F_\ell(x_n)|^k \leq C(\mathcal{D}_n + \beta_n). \quad (4.5)$$

Proof For the forward jump, the factorial identity is exact:

$$\prod_i \frac{(x_i + d_i)!}{x_i!} = \frac{(x + d)^{\underline{z}}}{x^{\underline{y}}}. \quad (4.6)$$

Together with (2.4), it gives

$$e^{F_\ell(x+d) - F_\ell(x)} = \frac{\kappa_{zy}(x + d)^{\underline{z}}}{\kappa_{yz}x^{\underline{y}}}, \quad (4.7)$$

the exact discrete endpoint detailed-balance ratio. To compare both top clocks at the same state, bounded falling-factorial expansion gives

$$\Delta_{yz} F_\ell = -h + \varepsilon_+, \quad qquad \Delta_{zy} F_\ell = h + \varepsilon_-, \quad (4.8)$$

where, with $I = \{i : d_i \neq 0\}$,

$$|\varepsilon_+| + |\varepsilon_-| \leq C s_n, \quad qquad s_n = \sum_{i \in I} \frac{1}{x_{n,i}}. \quad (4.9)$$

Indeed each logarithm in the quotient of two shifted factorials is a sum of finitely many terms $\log(1 + j/x_i)$. Consequently

$$\mathcal{L}_T F_\ell = -(A - B)h + A\varepsilon_+ + B\varepsilon_-. \quad (4.10)$$

The curvature-safe premise is exactly

$$\frac{A_n + B_n}{x_{n,i}} \leq C\beta_n \quad (i \in I). \quad (4.11)$$

To see this, choose a top endpoint e with $e_i > 0$. Top flatness makes $A + B \asymp x_n^e$, while $x_n^e/x_{n,i} = x_n^{e - \mathbf{e}_i}$; the certificate checks that this curvature cofactor lies at or below the maximal R -source D-tier in all 94 rows. Equations (4.9)–(4.11) prove (4.4).

On the compact ratio range (4.2),

$$\mathcal{D}_n \asymp (A_n + B_n)h_n^2. \quad (4.12)$$

Equations (4.8)–(4.9), boundedness of h_n , and $(A + B)s_n^2 \leq C\beta_n$ therefore yield, for $2 \leq k \leq 4$,

$$(A + B)(|\Delta_{yz} F_\ell|^k + |\Delta_{zy} F_\ell|^k) \leq C\{(A + B)h^2 + (A + B)s_n^2\} \leq C(\mathcal{D}_n + \beta_n),$$

which is (4.5). $\square$

19.4.2 Lemma 4.2 (lower-linkage drift and jump moments)

There is a sequence $a_n \rightarrow \infty$ such that

$$\mathcal{L}_R F_\ell(x_n) \leq -\beta_n a_n, \quad (4.13)$$

and, for $k = 2, 3, 4$,

$$\sum_{r \in R} \lambda_r(x_n) |\Delta_r F_\ell(x_n)|^k \leq C \beta_n (1 + \log^k R_n), \quad R_n = 2 + \|x_n\|_1. \quad (4.14)$$

Proof The lower-linkage part of Proposition 5.2 proves $\mathcal{L}_R U_\theta / \beta_n \rightarrow -\infty$. Here

$$F_\ell - U_\theta = C_0 + \sum_i \{\log(x_i!) - x_i \log x_i + x_i\}. \quad (4.15)$$

For every bounded reaction vector, the jump of the right side is $O(\sum_i x_i^{-1}) = o(1)$ on an all-active sequence, by the discrete Stirling difference (or directly by factorial ratios). The total R -reaction rate is $O(\beta_n)$. Hence

$$\mathcal{L}_R(F_\ell - U_\theta) = o(\beta_n), \quad (4.16)$$

which proves (4.13). Finally, every bounded reaction has $|\Delta F_\ell| \leq C(1 + \log R_n)$, and every R -source propensity is at most a fixed multiple of β_n on the tier. The reaction menu is finite, proving (4.14). $\square$

19.4.3 Proposition 4.3 (powered all-active drift)

Along every selected failed all-active sequence,

$$\mathcal{L}(1 + F_\ell)^4(x_n) \longrightarrow -\infty. \quad (4.17)$$

The conclusion is uniform over the finite failed-cone menu of a fixed pair in the finite-exception sense.

Proof Absorb the $C\beta_n$ term of (4.4) into (4.13). After decreasing a_n , still with $a_n \rightarrow \infty$, Lemmas 4.1–4.2 give

$$\mathcal{L} F_\ell \leq -\mathcal{D}_n - \beta_n a_n \quad (4.18)$$

and, for $k = 2, 3, 4$,

$$M_{k,n} := \sum_r \lambda_r |\Delta_r F_\ell|^k \leq C \{\mathcal{D}_n + \beta_n (1 + \log^k R_n)\}. \quad (4.19)$$

The exact discrete binomial, or carré/Taylor, identity is

$$\begin{aligned} \mathcal{L} W &= 4G^3 \mathcal{L} F_\ell + 6G^2 \sum_r \lambda_r (\Delta_r F_\ell)^2 \\ &\quad + 4G \sum_r \lambda_r (\Delta_r F_\ell)^3 + \sum_r \lambda_r (\Delta_r F_\ell)^4. \end{aligned} \quad (4.20)$$

The third moment is signed, so bounding it by its absolute moment can only increase the right side. Substituting (4.18)–(4.19) gives

$$\begin{aligned} \mathcal{L} W &\leq -4G^3(\mathcal{D}_n + \beta_n a_n) \\ &\quad + C(G^2 + G + 1)\mathcal{D}_n \\ &\quad + C\beta_n \{G^2(1 + \log^2 R_n) + G(1 + \log^3 R_n) + 1 + \log^4 R_n\}. \end{aligned} \quad (4.21)$$

Because $G \rightarrow \infty$, the second line is absorbed by $-G^3\mathcal{D}_n$. By (2.6), the ratio of the largest term in the last line to $G^3\beta_n a_n$ is

$$\frac{C(1 + \log^2 R_n)}{Ga_n} \rightarrow 0, \quad (4.22)$$

and the other two ratios are smaller. Thus eventually

$$\mathcal{L}W(x_n) \leq -2G_n^3(\mathcal{D}_n + \beta_n a_n) \rightarrow -\infty. \quad (4.23)$$

This controls both the reversible top carré term near detailed balance and the Taylor remainder away from it. It does not infer convex-transform drift merely from $\mathcal{L}F_\ell < 0$.

Equivalently, if the harmless finite shifts in (4.8) are temporarily deleted and $q = \Delta_{yz}F_\ell = -h$, the two top directions pair as

$$\begin{aligned} (G+q)^4 - G^4 + e^q\{(G-q)^4 - G^4\} \\ = 4G^3q(1-e^q) + 6G^2q^2(1+e^q) + 4Gq^3(1-e^q) + q^4(1+e^q). \end{aligned} \quad (4.24)$$

On the bounded flat-top ratio range, $q(1-e^q) \leq -cq^2$, while the cubic term is nonpositive; hence this core is nonpositive for large G . Restoring the shifts costs $O((A+B)s_n G^3) = O(\beta_n G^3)$ by (4.11), which is absorbed by the $-\beta_n a_n G^3$ lower exit. This paired form is the same domination as (4.18)–(4.23), and records explicitly that the two enabled top directions are paired before estimating their discrete curvature.

There are finitely many failed all-active descriptors for a fixed pair. If no common finite exception existed, a violating divergent sequence would have a subsequence in one descriptor and contradict (4.23). This is the claimed finite-exception uniformity. $\square$

19.5 5. Fixed-class common- W composition

19.5.1 Theorem 5.1 (audited prospective 26-pair branch)

For every selected support pair, every strongly connected orientation, every positive rate vector, and every closed irreducible population class Γ , the physical mass-action CTMC is positive recurrent.

Proof Use the one function W in (2.5) on the whole class. Take any divergent sequence in Γ and extract an exact source/D-tier subsequence. An affine-infeasible descriptor cannot occur in the fixed affine class. Every affine-feasible descriptor is therefore one of the following.

1. It passes the ordinary top-S/top-D criterion. The quantitative Anderson–Kim factorial estimate, with an arbitrary fixed linear correction, and the standard fourth-power remainder calculation give $\mathcal{L}W \rightarrow -\infty$.
2. It is one of the 94 all-active failures. Proposition 4.3 gives $\mathcal{L}W \rightarrow -\infty$.
3. It is one of the 30 one-active failures. There is no two-active failed descriptor by (2.1).

The usual bad-sequence contradiction therefore gives a finite set K_0 such that $\mathcal{L}W \leq -1$ outside K_0 and finitely many fixed-width one-active tubes. A moving-boundary exit from such a tube has two active coordinates and hence is in the generator-good region; no promotion kernel or return to the old axis is needed.

On the all-species reflected marked chain update

$$D_i^+ = (D_i + \zeta_i)^+, \quad H_i = X_i - D_i. \quad (5.1)$$

Then $0 \leq D_i \leq X_i$ and H_i is pathwise bounded above by its initial value. In a divergent one-active bad tube, $D_X = 0$ would imply $X = H_X$ bounded, a contradiction. Thus every reachable divergent bad tube has positive old debt. The two no-history rows of Lemma 3.1 are incompatible with that marked history.

Lemma 3.2 and the repeated-base fourth-power theorem give, outside another finite set, an all-reaction physical stopping time $\tau > 0$ with

$$\mathbb{E}_{x,d}[W(X_\tau) - W(x) + \tau] \leq -1, \quad (5.2)$$

whose successful endpoint strictly reduces old D_X . It uses the same ℓ and the same W as the generator-good region.

Generator steps and the episodes (5.2) are exactly the hypotheses of the existing all-debt finite-target gluing theorem. Alternating them and localizing at finite W -sublevels gives a nonnegative supermartingale; bounded-index telescoping and Fatou yield a finite mean hit of the finite marked target where every debt is zero. Its physical projection is finite. The finite trace has a positive recurrent state, and irreducibility of Γ promotes positive recurrence to every state in the class.

Finally the process is nonexplosive. In a binary network, a reaction with a bimolecular source cannot increase total population because targets have molecularity at most two. Every population-increasing propensity is therefore constant or linear. Stopping at population levels and applying Gronwall controls the total population, while every finite population sublevel has finitely many states and bounded total jump rate. $\square$

19.6 6. Reproduction and frozen hashes

The finite certificate freezes the following hashes:

```
one-active rows:      091dc59dc83b6445d36135a2d6b1e6c1bfb9565db88328fc1725c39f78bbc7e4
normalized profiles: 36b6eb1c1beadc939681fc7515e2adf3dfb9e16d116da4b8d2bd73ac77aabe12
all-active rows:     55669e3dc9b3fbb7ee84895ea158b3b596d339265cde1a23189bdf312a314bea
pair-fixed tops:     d1de16c1c45eaab3bce35a17db75bd947a8398a049722f100d96dd227b7e21d5
certificate payload: a1f528caffb63729d889e792ce8831b899865e70c1564b2a0a33d46f295a39da
prospective remainder: 4e645eca1ba23849680f2f983e1fb8c9465001c5b0e8c0090a31a339bf18ec06
```

Run

```
PYTHONPATH=src python3 -B src/prospective_26_candidate_theorem.py
PYTHONPATH=src python3 -B -m unittest \
  tests/test_prospective_26_candidate_theorem.py -v
```

The machine checks exact supports, caps, categories, top reaction vectors, cofactor tiers, and the discrete endpoint factorial identity. It does not replace the analytic estimates above. The independent audit replayed the one-active arbitrary-orientation scope extension, the discrete powered-top remainder, the common- W gluing, nonexplosion, and the exact zero-overlap arithmetic, and returned **PASS**. Accordingly the scoped flags are

```
analytic_one_active_scope_extension_certified = true
analytic_powered_all_active_lift_certified = true
candidate_26_pair_recurrence_certified = true
global_t3_2_certified = false
```

20 The easy 943 promotion incidences and the certified 416-pair theorem

20.1 1. Scope and certification boundary

This note separates a local analytic theorem from a larger pair-level candidate theorem. The finite selectors are frozen in `src/two_active_easy_common_w.py`.

The post-exact-26 remainder has 769 support pairs. Its two-active promotion table has 1,350 incidences on 749 pairs and splits as follows.

physical mechanism	incidences	pair union
enabled top seed	929	550
dormant rank-two access word (the nominal Poisson rows)	8	8
dormant finite rank-one shell	6	6
dormant, no whole-top linkage	407	333

The first three rows are the **easy 943**. Their pair union has 555 pairs (527 positive and 28 signed). The exact pair fingerprints are

easy-943 union: 19420077a2e54b88498f8f791fbe05a967c26bef882a06c5e76a48973df676d5

hard-407 union: d3c9dad6e8510a81efee6c56873de0f1f2cf6f24d3f50b46d4cf22abb2ad9484

Exactly 416 pairs have **every** promotion incidence in the easy family. They consist of 414 positive pairs and two signed pairs and have fingerprint

8c3325983568c53772f024080c0b95d37873cfe0a149386ec9829d1d9323e186

This note proves the easy-943 stopped estimate and composes every remaining interface on those 416 pairs. Independent audit replayed the stochastic endpoint estimates, all arbitrary-orientation cuts, the common correction, and the marked fixed-class argument. The exact 416-pair recurrence flag is therefore certified. Global T3-2 remains false.

There is one important qualification. The 937 bounded access-word rows work for any fixed linear correction ℓ . The six finite rank-one shell rows do **not**. They require the usual rate-adjusted correction on their fixed whole shell. Section 5 gives an exact counterexample to the stronger arbitrary- ℓ assertion.

20.2 2. The common fourth-power potential

For a fixed network and a fixed closed population class, put

$$F_\ell(x) = K + \sum_{i=1}^3 \log(x_i!) + \ell \cdot x, \quad G = 1 + F_\ell, \quad W = G^4, \quad (2.1)$$

where K is chosen so that $F_\ell \geq 1$. A fixed linear correction does not destroy properness: the factorial term dominates every linear term at infinity.

The correction menu on the 416-pair selector is exact and disjoint.

common correction	pairs	fingerprint
arbitrary fixed ℓ , chosen as 0 if desired	371	77b42a7079d38c1b83b322b1e46f a4ea61dc4a7203a33760dd856b0a 883ebca4
reversible-shell rate adjustment	29	d5026791e3166315e347a6d4b938 f1d9f8315a898f785d1ace920262 86437392
directed-triple rate adjustment	16	3d1a90f712f00ed4d343a97dcfc1 0b218a779a26640789dec232799d 727f2712

For a reversible top $y \rightleftharpoons z$, the constraint is

$$\ell \cdot (z - y) = \log \frac{\kappa_{zy}}{\kappa_{yz}}. \quad (2.2)$$

For $\{2X, X + Y, 2Y\}$, the correction is the rate-dependent one from the directed fluid polynomial in the certified rank-one triple theorem. The 23 reversible and 16 directed-triple all-active pairs use exactly the same top mask in all their two-active closed-rank-one failures. The six finite promotion-shell pairs are disjoint from those 39 pairs. Thus (2.2), or its directed-triple analogue, imposes at most one correction on each pair.

20.2.1 2.1 A powered shell-overshoot lemma

The fourth power needs more than the first-moment endpoint estimate quoted in the earlier corrected-factorial theorem. The following strengthening is the precise input used below.

Lemma 2.1 (positive shell overshoot). On any rank-one top shell used in Sections 5 and 7, let

$$Z_N = F_\ell - \min_{\text{shell}} F_\ell. \quad (2.3)$$

Start in an exact-tier compact interior, allowing a fixed number of bounded lower perturbations. If S is an independent exponential time, or if σ is one of the killed top-window endpoints in the carrier construction, then, for every fixed $r < \infty$,

$$\sup_N \mathbb{E} [\{(F_\ell(X_S) - F_\ell(X_0))^+\}^r] < \infty, \quad (2.4)$$

and the same bound holds with S replaced by σ .

Use the one-dimensional shell coordinate from the audited rank-one endpoint theorem. Put $s_N = 1$ for $A \rightleftharpoons B$, and $s_N = N$ for $B \rightleftharpoons 2A$ and for a homogeneous quadratic shell. The exact paired finite differences, strict inward shell drift, and discrete convexity give, on the compact interior,

$$\begin{aligned} \mathcal{L}_* Z_N &\leq s_N(C - cZ_N), \\ \sum_e \lambda_e (\Delta_e Z_N)^2 e^{\theta|\Delta_e Z_N|} &\leq C s_N(1 + Z_N) \end{aligned} \quad (2.5)$$

for every sufficiently small fixed $\theta > 0$. For $A \rightleftharpoons B$, insert $\Delta_- Z_N = \log\{\kappa_{BA}(B+1)/(\kappa_{AB}A)\}$ and its reverse. For $B \rightleftharpoons 2A$, the two logarithms in the exact generator formula give dissipation $N^2(r - r_*)^2$, while $1 + Z_N \asymp 1 + N(r - r_*)^2$. For an arbitrary directed homogeneous triple, the audited $p(r)\Phi(r)$ calculation gives the first line of (2.5), and bounded top jumps give the second. Taylor's formula, after reducing θ once, therefore yields

$$\mathcal{L}_* e^{\theta Z_N} \leq -c s_N e^{\theta Z_N} + C s_N. \quad (2.6)$$

Consequently the stopped semigroup satisfies

$$\mathbb{E}_x e^{\theta Z_N(X_t)} \leq e^{-c s_N t} e^{\theta Z_N(x)} + C. \quad (2.7)$$

The compensation formula gives the same estimate at a killed endpoint: on the descriptor compact set its normalized killing density is a fixed degree polynomial, which is absorbed after decreasing θ . Hence

$$\mathbb{P}_x\{Z_N(X_\rho) - Z_N(x) > u\} \leq C e^{-cu} \quad (2.8)$$

for every endpoint ρ above, up to exit from the enlarged compact tube. The existing exponential-barrier estimate makes that exit super-polynomially unlikely on the required horizons. The full shell factorial oscillation is polynomial in N , so choosing the barrier power after r removes the stop. Integrating (2.8) proves (2.4); a fixed number of bounded perturbations changes only the constant.

20.3 3. A bounded top-access word

The reusable local statement is the following.

Lemma 3.1 (all-reaction bounded access word). Let x_n realize an exact descriptor, fix ℓ , and let $A_n \rightarrow \infty$ be the common source scale of its global top D-tier. Suppose a directed physical word $y_0 \rightarrow y_1 \rightarrow \cdots \rightarrow y_m$ has bounded length and satisfies:

1. y_0 is enabled at x_n ;

2. every prescribed source along the actual-target word has propensity between cA_n and CA_n ;
3. every preterminal target stays in the exact top D-tier; and
4. the terminal edge has factorial gap $g_n \rightarrow \infty$ from its top source to its lower target.

Stop at word completion or at the first reaction not prescribed by the word. Then all reactions are retained, the word succeeds with probability at least $p > 0$, and for every fixed $q > 0$

$$\begin{aligned} \mathbb{E}\tau_n^q &= O(A_n^{-q}), \\ \sup_n \mathbb{E}[(\Delta F_\ell)^+]^q &< \infty, \\ \mathbb{E}\Delta W &\leq -cG_n^3 g_n \end{aligned} \tag{3.1}$$

for all sufficiently large n . The population displacement at the endpoint is bounded by a word-dependent constant.

20.3.1 Proof

At each word state, the desired edge has rate at least cA_n . Because the displayed tier is the **global** top D-tier, every physical source has propensity at most CA_n . There are finitely many channels, so the total all-reaction competitor rate is at most CA_n . A lower competitor with source rate $b_n \leq CA_n$ can have positive factorial cost only of order

$$1 + \log^+(A_n/b_n). \tag{3.2}$$

Successive exponential races therefore give a fixed success probability. They also give the endpoint-weighted bound

$$\frac{b_n}{A_n} \left\{ 1 + \log \frac{A_n}{b_n} \right\}^q \leq C_q, \quad 0 < b_n \leq A_n, \tag{3.3}$$

with a harmless constant change when b_n/A_n stays in a compact range. There are finitely many competitors and finitely many word positions, so (3.3) proves every fixed positive endpoint moment. The total hazard before each stop is at least cA_n , which proves the duration estimate. Notice that a competitor is not deleted: its physical firing is precisely an endpoint of the stop.

On word success the exact factorial-ratio identity gives

$$\Delta F_\ell = -g_n + O(1). \tag{3.4}$$

The fixed correction changes each bounded jump by only $O(1)$. Also $g_n \leq C \log(2 + \|x_n\|_1) = o(G_n)$. On the compact part of the success event,

$$(G_n - g_n + O(1))^4 - G_n^4 \leq -cG_n^3 g_n. \tag{3.5}$$

The complement and every failed race cost only $O(G_n^3)$ in expectation by (3.3), with the quadratic, cubic, and quartic terms smaller still. Since $g_n \rightarrow \infty$, (3.5) dominates. This proves (3.1). In particular, the statement supplies the endpoint orders $q > 8$ used by the global gluing interface. $\square$

20.4 4. Applying the word lemma to 937 rows

20.4.1 4.1 The 929 enabled rows

In every enabled row, a vertex of the proper top subset is physically enabled. Starting there, strong connectivity of that linkage supplies a simple directed path to its complement. Stop the path at its first target outside the global top D-tier. Every intermediate source is an actual previous target; hence it is enabled, including when it needs the newly created inactive species. Exact D-tier equivalence makes all prescribed source rates comparable to A_n . Lemma 3.1 applies for arbitrary strong orientations, arbitrary positive rates, and any fixed ℓ .

This argument includes the four enabled rows that also display a whole-top finite shell. The access word takes $O(A_n^{-1})$ physical time, so the whole linkage is simply one of the retained top competitors; it does not need to be averaged or deleted.

20.4.2 4.2 The eight nominal Poisson rows

These rows do not need a Poisson corrector. Their supports, after the displayed labelling, are

$$\{A, B, AC\} \text{ paired with one of } \begin{cases} \{0, 2C, BC\}, \\ \{0, C, 2C, BC\}, \\ \{0, C, BC\}, \\ \{C, 2C, BC\}, \end{cases} \quad (4.1)$$

or the four copies obtained by exchanging A and B , hence exchanging AC and BC . The proper top intersection is the singleton BC in the first four rows and AC in the other four.

At $C = 0$, the whole linkage has an enabled vertex A or B . Strong connectivity gives a simple whole-linkage path to AC , respectively BC , and that actual target creates C . The proper top singleton is then enabled, and strong connectivity of the other linkage gives a path from it to a lower target. Concatenating the two simple paths gives the word in Lemma 3.1. Every source lies in the same global top D-tier. Thus all eight rows again work for arbitrary orientations, rates, and fixed ℓ , with all reactions retained.

20.5 5. The six finite shells and why ℓ is load-bearing

The six rank-one dormant supports are exactly

whole shell	proper linkage
$\{A, B\}$	$\{0, C, AC, BC\}$
$\{B, 2A\}$	$\{0, A, C, BC\}$
$\{B, 2A\}$	$\{0, A, AC, BC\}$
$\{B, 2A\}$	$\{0, C, AC, BC\}$
$\{B, 2A\}$	$\{A, C, AC, BC\}$
$\{B, 2A\}$	$\{0, A, C, AC, BC\}$.

(5.1)

Use the rate-adjusted F_ℓ from (2.2). The earlier dormant-priority theorem was stated for supports containing $2C$, whereas none of the six supports in (5.1) contains that vertex. The following explicit cut lemma is the needed extension.

On the $\{A, B\}$ shell, reflected debt is at most one. Before service, the dominant reset graph is contained in $\{0, AC, BC\}$; the remaining proper vertex C is a service vertex. A closed service-free reset class would therefore be a proper closed subset of $\{0, C, AC, BC\}$, contradicting strong connectivity.

On the $\{B, 2A\}$ shell, reflected debt is at most two. Whenever $C > 0$, the unique BC top source has priority. Every neutral dominant reset component is contained in $\{0, A, BC\}$, while each of the five displayed proper supports contains C or AC outside that set. If a dominant reset component had no service exit, it would again project to a proper closed subset of the strongly connected proper linkage. Thus a cut edge leaves it. All choices from its source have the same source scale, so the exit has a fixed positive conditional probability. Lower-priority clocks are retained as stopping interruptions. Internal equal-scale $AC \leftrightarrow BC$ moves, when present, have a geometric count because the aggregate exit rate is comparable to their total rate.

This proves, for arbitrary strong orientations and positive rates, that after a fixed number K of retained physical macrotransitions,

$$\mathbb{E}\Delta H_w \leq -\delta, \quad |\Delta H_w| \leq C_K. \quad (5.2)$$

Lemma 2.1 gives uniformly bounded positive F_ℓ -endpoint moments of every fixed order at the independent zero-source waits and the killed priority windows. Let $Y_n \geq 0$ be the sum of those positive shell costs. For every fixed q ,

$$\sup_n \mathbb{E} Y_n^q < \infty, \quad \Delta F_\ell \leq \Delta H_w \log N_n + C + Y_n. \quad (5.3)$$

The right side of (5.3), call it Z_n , is an upper endpoint surrogate, has $\mathbb{E} Z_n \leq -\delta \log N_n + O(1)$, and has q -moments $O(\log^q N_n)$. Monotonicity of $u \mapsto u^4$ and the exact binomial identity give

$$\begin{aligned} \mathbb{E} \Delta W &\leq 4G_n^3 \mathbb{E} Z_n + 6G_n^2 \mathbb{E} Z_n^2 + 4G_n \mathbb{E} |Z_n|^3 + \mathbb{E} Z_n^4 \\ &\leq -cG_n^3 \log N_n, \end{aligned} \quad (5.4)$$

because $\log N_n = o(G_n)$. The physical duration has every fixed moment; the inactive endpoint is bounded by K , and the scaled active endpoint has every fixed moment. Thus (5.4) has the required $q > 8$ endpoint version and retains the whole-shell motion throughout.

20.5.1 5.1 Exact counterexample to arbitrary ℓ

Consider the first row of (5.1). Orient the whole shell reversibly with

$$A \xrightarrow{2} B, \quad B \xrightarrow{1} A, \quad (5.5)$$

and orient the proper linkage as the unit-rate directed cycle

$$0 \longrightarrow AC \longrightarrow C \longrightarrow BC \longrightarrow 0. \quad (5.6)$$

Start from $(A, B, C) = (N, N, 0)$, and take the uncorrected choice $\ell = 0$. Let $T \sim \text{Exp}(1)$ be the first $0 \rightarrow AC$ clock. It is independent of the whole-shell chain. Since total mass $A + B = 2N$,

$$\mathbb{E} B_t = \frac{4N}{3} - \frac{N}{3} e^{-3t}, \quad \mathbb{E} B_T = \frac{5N}{4}. \quad (5.7)$$

On $1 \leq T \leq 2$, particle independence and a Chernoff bound imply $B_T \geq 1.25N$ with probability $1 - e^{-cN}$. Convexity and Stirling's formula then give

$$\log(A_T!) + \log(B_T!) - 2 \log(N!) \geq c_0 N - O(\log N), \quad (5.8)$$

where one may take any

$$c_0 < 0.75 \log 0.75 + 1.25 \log 1.25.$$

After activation, two consecutive $AC \rightarrow C$ firings have probability bounded below independently of N ; the second is surplus service. They take $O(N^{-1})$ time and change the factorial potential by only $O(\log N)$. Hence the positive $\Theta(N)$ shell cost in (5.8) overwhelms the $-\Theta(\log N)$ service reward, and its fourth-power cost is larger by the same leading G^3 factor.

The required correction here is

$$\ell_B - \ell_A = -\log 2, \quad (5.9)$$

which centers the factorial-linear potential at the actual shell balance. Thus “arbitrary common ℓ ” is false for the finite-six block. This is an exact rate/orientation obstruction, not a numerical simulation and not a counterexample to recurrence with the rate-adjusted potential.

20.6 6. The one-active structural extension on all 416 pairs

The 416 pairs have 1,455 one-active failed incidences. Their exact classification under the existing graph-theorem predicates is

structural family	incidences	pair union
direct physical C -source	1,356	415
Family I	65	13
exact Family II	0	0
Family III	10	10
open whole-top phase	24	8
generalized Family II	0	0

The pair-union fingerprints, in table order after the direct row, are

```

direct:      f0ca0f316e4796aef91c46c68bc24e780c808890ac8eb631c023739e4b8d5943
Family I:    e9cd9bdfdc6032dca816b851a83c9dcb46dc31365e2b229b5c6ba0efa5f0564d
Family III:  12cd9a7ca2abcc3508e691b7730d22ff2993e6516061ca588b0a74d39b131ecc
open whole:  a1c3277feba8d795479840d300772a9031d2ae5e616d4c19369bb25073807ae0
empty set:   4f53cda18c2baa0c0354bb5f9a3ecbe5ed12ab4d8e11ba873c2f11161202b945

```

More finely, all 65 Family I rows are `family_i_origin_down_0`; the Family III split is six `origin_down_0` plus four `origin_down_1`; and all 24 open rows are `open_wholly_top_down_1`. Every existing predicate ran without an assertion failure. There are 736 distinct normalized support/cap/phase profiles.

This is not exact-signature reuse: none of the 1,455 normalized support/cap signatures is literally one of the previously frozen 1,227-pair selector signatures. The extension is structural. The proof of the one-active fourth-power theorem uses only the following predicate-level facts:

1. a direct mixed C -source has resistance-zero physical access;
2. Family I has one lower-only and one two-cofactor mixed phase, with the certified origin-service cut;
3. Family III has two singleton mixed phases and down resistance at most two strictly below same-base up resistance; and
4. the open whole-top row is the same immigration-death phase with its factorial maximum tail.

The finite classification checks exactly those facts here. Extra lower vertices are retained as interruptions; the compensation estimate uses $r\{1 + \log(1/r)\}^q$, so it is insensitive to adding finitely many such channels. The moving-cutoff endpoint estimate and the ordered three-interruption bound therefore give every moment $q > 8$, for the same fixed ℓ selected in Section 2. No generalized Family II spectator axis occurs. Consequently all 416 pairs—not merely a proper subset—are the immediate structural composition target.

Independent replay checked that this predicate-level extension uses no selector-specific finite-box hypothesis.

20.7 7. Powering the other two-active rank-one episodes

Of the 416 pairs, 377 have failures in active dimensions $\{1, 2\}$ only. The other 39 have profiles $\{1, 2, 3\}$. On those 39 pairs there are 117 additional two-active closed-rank-one rows: 115 have an enabled maximal lower source and two use lower-layer activation. Every pair contributes three rows.

The already audited corrected-factorial endpoint theorem gives, for a direct row,

$$\Delta F_\ell \leq -I_n g_n + Y_n, \quad \mathbb{P}(I_n = 1) \geq p > 0, \quad g_n \rightarrow \infty, \quad (7.1)$$

where Y_n^+ has uniformly bounded moments of every fixed order. This is now an explicit consequence of Lemma 2.1 on the homogeneous and $\{B, 2A\}$ shells, the exponential cofactor bound in the exceptional template, and

$$r\{1 + \log(1/r)\}^q \rightarrow 0 \quad (7.2)$$

for every submaximal interruption. Thus (7.1) is stronger than a bare first-moment statement. Since $g_n = O(\log R_n) = o(G_n)$, the same binomial calculation as Lemma 3.1 gives

$$\mathbb{E}\Delta W \leq -cG_n^3 g_n. \quad (7.3)$$

The two lower-activation rows have top $\{2A, A + B, 2B\}$. Their existing reflected-debt endpoint has $\mathbb{E}\Delta H_w \leq -\delta$, bounded debt, and every fixed positive shell-endpoint moment by Lemma 2.1. Equations (5.3)–(5.4) apply verbatim and give $-cG^3 \log N$. Therefore all 117 closed-rank-one rows have the powered physical-time endpoint estimate with the same correction used in the all-active phase.

The load-bearing point in this section is the positive q -moment strengthening, not merely $\mathbb{E}\Delta F_\ell \rightarrow -\infty$; Lemma 2.1 and the independent replay certify that strengthening.

20.8 8. The 117 all-active rows

Exactly the same 39 pairs have all-active failures, again three per pair. The exact split is

fixed whole-top shape	incidences	pairs
curvature-safe reversible two-node rank one	69	23
arbitrary directed triple $\{2X, X + Y, 2Y\}$	48	16

Every two-node row satisfies the discrete curvature-cofactor premise. The audited reversible decomposition and fourth-power proof from the exact-26 theorem therefore applies without change: its proof is premise-based, not selector-specific.

It remains to record the directed-triple power estimate. Write $N = X + Y$ on the flat top and let β_n be the maximal propensity in the other linkage. The audited directed-triple theorem gives a fixed rate-dependent correction such that

$$\mathcal{L}_T F_\ell \leq KN, \quad \mathcal{L}_R F_\ell \leq -\beta_n a_n, \quad \beta_n \geq cN, \quad a_n \rightarrow \infty. \quad (8.1)$$

Flatness gives $X \asymp Y$, so every top jump has $|\Delta_T F_\ell| = O(1)$, even though the total top rate is $O(N^2)$. Every other reaction has $|\Delta_R F_\ell| \leq C \log R_n$. Hence, for $k = 2, 3, 4$,

$$M_{k,n} := \sum_r \lambda_r |\Delta_r F_\ell|^k \leq C \{N^2 + \beta_n (1 + \log^k R_n)\}. \quad (8.2)$$

After absorbing KN into $-\beta_n a_n$, substitute (8.1)–(8.2) into

$$\mathcal{L}W = 4G^3 \mathcal{L}F_\ell + 6G^2 M_2 + 4G \sum_r \lambda_r (\Delta_r F_\ell)^3 + M_4. \quad (8.3)$$

The lower-linkage remainder divided by the leading negative term is at most $C(1 + \log^2 R_n)/(G_n a_n) \rightarrow 0$. The largest top remainder has ratio

$$\frac{N^2}{G_n \beta_n a_n} \leq \frac{C}{a_n \log(N+1)} \rightarrow 0, \quad (8.4)$$

because $G_n \geq cN \log(N+1)$. The cubic and quartic ratios are smaller. Thus

$$\mathcal{L}W(x_n) \leq -cG_n^3 \beta_n a_n \rightarrow -\infty. \quad (8.5)$$

This closes the directed-triple carré seam; the independent audit replayed the ratio in (8.4).

20.9 9. The 416-pair composition theorem

Theorem 9.1. Choose any one of the 416 support pairs, give both linkages arbitrary strongly connected orientations and arbitrary positive rates, and choose ℓ from the exact menu in Section 2. Then the common potential $W = (1 + F_\ell)^4$ supplies a physical-time Foster episode outside a finite set in every closed irreducible class. Consequently every such class is positive recurrent.

The proof is the following finite composition:

1. all 1,455 one-active failures use Section 6;
2. the 762 promotion failures on the selector use Sections 3–5;
3. the additional 117 closed-rank-one two-active failures use Section 7;
4. the 117 all-active failures use Section 8; and
5. every passing descriptor uses the standard powered descending-source generator estimate with the same fixed W .

All stopped episodes retain every reaction and include physical duration. Their endpoints have moments above order eight. Here is the fixed-class gluing argument, including the moving-boundary case rather than importing it from the earlier 1,227-pair selector.

Fix a closed irreducible class Γ , a reference population $x^\circ \in \Gamma$, and mark all three species with reflected debts

$$D_i^+ = (D_i + \Delta_i)^+. \quad (9.1)$$

Then $X_i - D_i$ is pathwise nonincreasing and is at most x_i° . In a fixed-width one-active tube, a state with selected debt $D_X = 0$ therefore has $X \leq x_X^\circ$; the two inactive coordinates and all their marks are tube-bounded. These states form a finite class-dependent exception. Every divergent reachable one-active tube sequence consequently has $D_X > 0$ and uses the Section 6 episode.

That episode already includes in its expected W -increment the jump which reaches its moving boundary. If the endpoint is an at-least-two-active **passing** state, the usual generator-good rule applies next. If it is a two-active **failed** state, no uncharged continuation is inferred: at the next macrostep it uses one of the 762 promotion episodes or one of the 117 closed-rank-one episodes. An all-active failed endpoint uses Section 8. Thus the union of the one- and two-active bad sets is closed under the stopping rule without deleting a reaction or discarding an endpoint cost.

For completeness, suppose the required generator/episode alternatives did not hold outside a finite subset of the reachable marked class. Choose a divergent countersequence and extract a fixed exact-tier descriptor, fixed tube widths, and fixed invariant caps. An affine-infeasible descriptor cannot occur. A descriptor which passes for the actual orientation is generator-good for the common W . A remaining one-active failure is covered by Section 6; a remaining two-active failure is in Sections 3–5 or 7; and an all-active failure is in Section 8. The corresponding local estimate has constants depending only on the fixed class, rates, orientation, and template, and contradicts the selected countersequence. This is the standard bad-sequence finite-exception lemma, now applied to the complete mixed-profile list rather than to the old one-active list.

The all-species marked common-potential theorem therefore gives finite mean hitting of a finite marked target. One ordinary physical jump from that finite target, followed by the same estimate from its finitely many successors, gives finite mean positive return. Projecting the target and using irreducibility yields positive recurrence of Γ .

Binary mass action is nonexplosive because every population-increasing reaction has source molecularity at most one; its positive total-population drift is bounded by $C(1 + |x|)$. Finite mean return and local finiteness then imply positive recurrence.

The independent audit replayed:

1. the all-competitor race moment in Lemma 3.1 for all 937 access rows;
2. the exponential shell-overshoot calculation in Lemma 2.1;
3. the six-support priority cut in Section 5;
4. the predicate-only one-active extension in Section 6;
5. the powered closed-rank-one endpoint in Section 7;
6. the directed-triple top-carré ratio (8.4); and
7. the marked fixed-class gluing argument in Section 9.

All seven obligations passed. The 416 pairs contain 414 positive-invariant and two signed members and have zero overlap with the previously certified set. Thus the ordered certified remainder changes by

$$(733, 36) \mapsto (319, 34). \quad (9.2)$$

The resulting 353-pair fingerprint is

9868f965cc8af951fd7545f8832ed0275a8d60bab70b2593b7424654cba7d8ec

20.10 10. The separated hard 407

Nothing in this note treats the 407 dormant promotion rows with no whole top linkage. They lie on 333 pairs and have row/pair split disjoint from the easy mechanism. Every such row has one proper linkage whose global-top intersection is exactly one complex of the form inactive-plus-active, while the other linkage is disjoint from the global top. Their normalized workload profiles are

$$(1, 3) \text{ (333 rows)}, \quad (1, 2) \text{ (37 rows)}, \quad (4, 5) \text{ (37 rows)}. \quad (10.1)$$

The bounded priority-word regression passes all 407 rows, but that finite search is not an arbitrary-rate/orientation analytic proof. These rows remain the separate dormant-resolvent problem and are not silently included in Candidate Theorem 9.1.

20.11 11. Reproduction and frozen hashes

Run

```
PYTHONPATH=src python3 -B src/two_active_easy_common_w.py
PYTHONPATH=src python3 -B -m unittest tests/test_two_active_easy_common_w.py -v
```

The canonical incidence hashes are

```
all promotion rows:      e03857257080c80b0426c400f45781f6e291634ee32ab2757f46564bfab41e86
easy 943:                d7cace7ff05356f6fd899ee622b6718413643d00b4da87ec61cf29941224a20a
hard 407:                ddd4c217b0236d7a44aa684873e6f6a9d5356c6741dea0d8575703e6263b7567
promotion rows on 416:   318a861e2fae514680a1e42cfc74e5e2cdfa3d2f6cc8b5d3b117b51352f09333
one-active classified:   d835320fd024e14d2e3a3198b4546d7bd83889fa23ffb7d1238ca91ffd7534f9
one-active profiles:     3652ab6d5d7e660fffbfbbee85ac48d440411f604bc39b534e6ed0c7f3c0ea55c
closed rank-one rows:    515ed4fbf3603d2c3489b73c7d9f26dd23ad160d5ec6bc48589f697c24a76124
all-active rows:         658010646b5ac720a9acc1fbf14fd08691620df7698042d9e0c9af7370a9c2a7
certificate payload:     40547e6856855ce5b128cf944a4e81aa44e1db77a35e29ea1d099e8b26ca3097
```

The certificate ends with the scoped certification boundary

```
independent_audit_passed = true
analytic_easy_943_common_w_certified = true
analytic_one_active_structural_extension_certified = true
analytic_closed_rank_one_power_lift_certified = true
analytic_directed_triple_power_lift_certified = true
exact_416_pair_recurrence_certified = true
global_t3_2_certified = false
```

21 A common polynomial scalar for the thirteen rank-two switch pairs

21.1 1. Scope and status

This note gives the independently audited common-potential proof for the thirteen pairs isolated in *rank_two_linear_switch_13.md*. It fixes the seam identified there: the linear workload is positive on some passing boundary states, whereas factorial entropy can be positive in the failed all-active interior.

The proof uses one state function on the whole population class. Independent replay passed every orientation, rate, scaling, endpoint, and classwise obligation. The exact thirteen-pair recurrence flag is true; global T3-2 remains false.

21.2 2. The scalar and its exact exponents

Fix one pair, arbitrary strongly connected orientations, positive rates, and a closed irreducible class. Put

$$F(x) = K + \sum_{i=A,B,C} \log(x_i!), \quad G = 1 + F, \text{quad}W = G^4, \quad (2.1)$$

where K makes $G \geq 1$. Let w be the positive rank-two workload and $H = 1 + w \cdot x$. For any fixed $\eta > 0$, define

$$V(x) = W(x) + \eta H(x)^q, \quad (2.2)$$

with

$$q = \begin{cases} 6, & w = (1, 1, 1), \\ 5, & w = (2, 1, 1) \text{ or } w = (1, 2, 1). \end{cases} \quad (2.3)$$

The finite certificate finds eleven rows of the first kind and two of the second. Every top complex has w -level two and the top difference space has rank two. Thus the top generator annihilates every function of H . The lower linkage is $0 \rightleftharpoons C$, so exactly

$$\mathcal{L}H^q = \kappa_{0C}\{(H + w_C)^q - H^q\} + \kappa_{C0}C\{(H - w_C)^q - H^q\}. \quad (2.4)$$

Consequently, as $H \rightarrow \infty$,

$$\mathcal{L}H^q \leq K_0 H^{q-1} - c_0 C H^{q-1}. \quad (2.5)$$

In particular (2.2) is nonnegative and proper.

21.3 3. Failed all-active sequences

Take a failed all-active exact-tier sequence. Every top source is enabled and all top source monomials are mutually comparable. Since the top difference space has rank two, its orthogonal complement is exactly $\text{span}\{w\}$. Flatness says that every top-complex scalar product with $\log x$ differs by $O(1)$. Hence $\log x = \lambda w + O(1)$ for some $\lambda \rightarrow \infty$. Writing $N = e^\lambda$ and $m = \max_i w_i$ gives

$$x_i = \Theta(N^{w_i}), \quad H = \Theta(N^m), \quad \beta_T = \Theta(N^2) = \Theta(H^{2/m}). \quad (3.1)$$

Here the constants may change along a compact-ratio subsequence, which is all the sequence argument needs. The exact factorial ratio across a top jump is bounded on this flat tier.

The lower birth must be retained separately. Its exact entropy increment is

$$d_+(x) = F(x + e_C) - F(x) = \log(C + 1). \quad (3.2)$$

The fourth-power binomial identity therefore gives

$$\begin{aligned} \mathcal{L}_{0C}W &\leq K_1\{G^3 d_+ + G^2 d_+^2 + G d_+^3 + d_+^4\} \\ &= O\{H^3(\log H)^4\}. \end{aligned} \quad (3.3)$$

This is strictly lower order than the top bound in both workload classes: the latter is $H^5(\log H)^3$ in the homogeneous rows and $H^4(\log H)^3$ in the weighted rows. Thus, since $G = \Theta(H \log H)$, the complete positive part satisfies

$$\mathcal{L}W \leq CH^{3+2/m}(\log H)^3. \quad (3.4)$$

The lower death term in (2.5), on the other hand, has polynomial order

$$CH^{q-1} = \Theta(H^{q-1+w_C/m}). \quad (3.5)$$

For the eleven homogeneous rows, (3.4) has exponent 5, while (3.5) has exponent 6. For either weighted row, the exponents are 4 and 9/2. Thus the strict power gaps are respectively 1 and 1/2, and

$$\mathcal{L}V(x_n) \longrightarrow -\infty. \quad (3.6)$$

This is why the exponent must change on the two unary-top supports. A uniform sixth power is unnecessarily large at their unary boundary axes; the fifth power has exactly the required half-power all-active margin.

21.4 4. Passing sequences

Every affine-feasible descriptor other than the unique all-active failure passes the ordinary top-S/top-D condition. The quantitative powered Anderson–Kim estimate therefore supplies, with A_n the largest enabled source propensity and $g_n \rightarrow \infty$ the descending factorial gap,

$$\mathcal{L}W(x_n) \leq -cG_n^3 A_n g_n \quad (4.1)$$

after absorbing its lower-order fourth-power remainders.

If $C_n \rightarrow \infty$, (2.4) is eventually negative. Both summands in (2.2) are then generator-good, so the desired drift follows immediately.

It remains to consider C_n bounded. Then $H \asymp A + B$. The exact support certificate gives the following pure sources in the top linkage:

w	q	coercive sources
(1, 1, 1)	6	$2A, 2B$
(2, 1, 1)	5	$A, 2B$
(1, 2, 1)	5	$2A, B$.

(4.2)

Hence $A_n \geq cH^{q-4}$. Since $G \asymp H \log H$, (4.1) has magnitude at least

$$cH^{q-1}(\log H)^3 g_n, \quad (4.3)$$

whereas the possibly positive term in (2.5) is only $O(H^{q-1})$. The logarithmic factor and $g_n \rightarrow \infty$ close the boundary seam:

$$\mathcal{L}V(x_n) \longrightarrow -\infty \quad (4.4)$$

on every passing C -bounded sequence as well.

21.5 5. Classwise recurrence theorem

Every divergent sequence in a fixed class has an exact-tier subsequence. An affine-infeasible descriptor cannot occur in that class. Sections 3–4 therefore show that every divergent sequence has a subsequence along which $\mathcal{L}V \rightarrow -\infty$. The usual bad-sequence contradiction makes

$$\mathcal{L}V \leq -1 \quad (5.1)$$

outside a finite subset of the class. Nonexplosion follows from the standard binary-network population bound, and localized Dynkin would then give positive recurrence on every closed irreducible class.

Independent replay checked (3.1)–(3.5), the exact discrete fourth-power bound in (3.3)–(3.4), the quantitative passing estimate (4.1), properness, nonexplosion, and localized Dynkin. It found no counterorientation or rate exception. Therefore all thirteen pairs are classwise positive recurrent.

They are disjoint from the previously certified 416-pair branch and contain thirteen positive-invariant and zero signed pairs. The certified remainder changes exactly as

$$(319, 34) \mapsto (306, 34). \quad (5.2)$$

The global T3-2 theorem is not claimed.

21.6 6. Reproduction

```
PYTHONPATH=src python3 -B src/rank_two_linear_switch_13_common_scalar.py
PYTHONPATH=src python3 -B -m unittest \
  tests/test_rank_two_linear_switch_13_common_scalar.py -v
```

The executable freezes all thirteen supports, the $11 + 2$ power split, the all-active strict exponent gaps, and the C -bounded coercive source menu.

22 A stopped-service common scalar for the final rank-two seven

22.1 1. Scope and audit status

This note gives the independently audited recurrence proof for exactly the seven pairs selected by *rank_two_o_mixed_profile_7.py*. It composes:

1. a localized Perron–Frobenius activation trial at every dormant nonservice vertex;
2. a full-chain density-dependent service window after activation; and
3. one common scalar on the stopped wedges and their pointwise-good complement.

The exact seven-pair fingerprint is

93717536ce82eceeef6909c62568afab31e06695dada8b69defb93335d576957

The final proof replaces an invalid geometric-tail shortcut for the total seed count, distinguishes every birth and death count in the episode, uses random-time-change exponential estimates for the service clock, and restores all physical reactions through a deterministic endpoint envelope. Independent replay passed the stopping, endpoint, ODE, moment, gluing, and arbitrary-orientation obligations. The exact seven-pair analytic and recurrence flags are true; global T3-2 remains false.

22.2 2. Exact geometry and the common scalar

Every pair has lower linkage

$$0 \rightleftharpoons C \quad (2.1)$$

and a strongly connected top linkage of molecularity two and stoichiometric rank two. The exact service-zero geometry is

T	sources on $C = 0$	dormant nonservice vertices
$\{2A, 2C, AB, AC, BC\}$	$\{2A, AB\}$	B
$\{2A, 2C, AB, BC\}$	$\{2A, AB\}$	B
$\{2A, AB, AC, BC\}$	$\{2A, AB\}$	B
$\{2B, 2C, AB, AC\}$	$\{2B, AB\}$	A
$\{2B, 2C, AB, AC, BC\}$	$\{2B, AB\}$	A
$\{2B, AB, AC, BC\}$	$\{2B, AB\}$	A
$\{2C, AB, AC, BC\}$	$\{AB\}$	A, B .

(2.2)

There are eight dormant rows. Every pair contains AB , has no two-active failed descriptor, and has one all-active failed descriptor with workload $(1, 1, 1)$.

Put

$$\begin{aligned}
n &= A + B + C, & h &= 1 + n, \\
F(x) &= K_F + \sum_{i=A,B,C} \log(x_i!), & G &= 1 + F, \\
V(x) &= G(x)^4 + \lambda h(x)^6, & \lambda &> 0,
\end{aligned} \tag{2.3}$$

where K_F makes $G \geq 1$. This is the single proper scalar used at every gluing endpoint.

22.3 3. Pointwise drift away from dormant wedges

Write $\alpha > 0$ and $\delta > 0$ for the rates of $0 \rightarrow C$ and $C \rightarrow 0$. The top linkage preserves n , and hence

$$\begin{aligned}
\mathcal{L}h^6 &= \alpha\{(h+1)^6 - h^6\} + \delta C\{(h-1)^6 - h^6\} \\
&\leq K_0 h^5 - c_0 C h^5.
\end{aligned} \tag{3.1}$$

At a dormant vertex d , relabel d as X , the service species C as A , and the remaining species as B . Section 4 constructs $R_d = v_A A + v_B B$, with positive coefficients. Choose a common sufficiently small η , after the fixed pair, orientation, and rates are given, and define

$$\mathcal{W}_d = \{R_d \leq \eta m\}, \quad \mathcal{W} = \bigcup_{d \in D(\text{pair})} \mathcal{W}_d. \tag{3.2}$$

The one or two wedges are disjoint above a finite population sublevel. Every divergent sequence outside $\mathcal{W}$ has a subsequence on which

$$\mathcal{L}V \longrightarrow -\infty. \tag{3.3}$$

The exact-tier cases are as follows.

22.3.1 3.1 All-active failure

Here $A, B, C = \Theta(N)$, $h = \Theta(N)$, and the top source rates are $\Theta(N^2)$. The exact discrete fourth-power estimate from the audited homogeneous rank-two scalar theorem gives

$$(\mathcal{L}G^4)^+ \leq K N^5 (\log N)^3. \tag{3.4}$$

The service term in (3.1) has negative order N^6 , which is a strict polynomial margin.

22.3.2 3.2 Passing cones with $C \rightarrow \infty$

The powered exact-tier estimate gives

$$\mathcal{L}G^4 \leq -cG^3 A_N g_N, \quad g_N \longrightarrow \infty, \tag{3.5}$$

after its lower-order binomial terms are absorbed. Here A_N is the largest enabled source propensity. At the same time, (3.1) is eventually negative. Both summands of V are generator-good.

22.3.3 3.3 Passing cones with C bounded

Now $h \asymp A + B$. On the service-zero edge, (2.2) gives the coercive source explicitly. A non-dormant pure endpoint has a $2A$ or $2B$ source of order h^2 . If both A and B have positive limiting fractions, the universal AB source has order h^2 . The only omitted pure endpoints are the wedges in (3.2). Compactness of their complement therefore gives

$$A_N \geq ch^2. \tag{3.6}$$

Since $G \asymp h \log h$, the negative magnitude in (3.5) is at least

$$ch^5(\log h)^3 g_N, \quad (3.7)$$

which absorbs the $O(h^5)$ birth term in (3.1).

There are no two-active failures. Every one-active failure enters one of the dormant wedges, and an affine-infeasible descriptor cannot occur in a fixed population class. The bad-sequence contradiction turns (3.3) into

$$\mathcal{L}V \leq -1 \quad (3.8)$$

outside $\mathcal{W}$ and one finite set.

22.4 4. Localized PF trials and the exact activation counts

Fix a dormant chart. Its normalized top support is one of

$$\{2A, 2B, AB, AX, BX\}, \quad \{2A, 2B, AX, BX\}, \quad \{2A, AB, AX, BX\}, \quad \{2B, AB, AX, BX\}. \quad (4.1)$$

Killing each carrier edge which enters an inactive quadratic gives a transient two-state generator Q . Let

$$q = (-Q)^{-1} \mathbf{1} > 0, \quad v_i = 1 - \varepsilon q_i > 0. \quad (4.2)$$

For sufficiently small ε , the rate-weighted reward at both carrier sources is positive. Thus, exactly for falling-factorial propensities,

$$R = v_A A + v_B B, \quad \mathcal{L}_T R \geq cXR - KR^2. \quad (4.3)$$

The physical birth increases R ; the death contributes at worst $-KR$. After reducing η , for $0 < R \leq \eta n$ and all large n ,

$$\mathcal{L}R \geq c_1 nR, \quad \Gamma R := \sum_r \lambda_r (\Delta_r R)^2 \leq C_1 nR. \quad (4.4)$$

All jumps of R are bounded. For a sufficiently small $\theta > 0$,

$$\mathcal{L}e^{-\theta R} \leq -c_2 \theta n R e^{-\theta R}. \quad (4.5)$$

Starting with one physical $0 \rightarrow C$ seed and stopping at $R = 0$ or $R \geq \eta n$, optional stopping gives a success probability

$$p_d \geq 1 - e^{-\theta v_A} + o(1) > 0. \quad (4.6)$$

This probability statement alone does not imply a geometric tail for the total number of births. The required repair is a localized trial estimate. Choose r_0 larger than twice the maximum R -jump. On $r_0 \leq R \leq \eta n$, Taylor's formula yields

$$\mathcal{L} \log R \geq c_3 n. \quad (4.7)$$

Below r_0 , only finitely many inactive phases occur and every enabled carrier race has order n . The killed-carrier cut excludes a closed nonabsorbing finite phase. Dynkin applied to (4.7), followed by the strong Markov property over dyadic bands and the finite lower phase, gives constants $a, b, C > 0$ such that the fast part ρ_n of a trial obeys

$$\mathbb{P} \left\{ \rho_n > u \frac{C \log n}{n} \right\} \leq a e^{-bu}, \quad u \geq 1, \quad (4.8)$$

until activation, return to a dormant vertex, or a population-shell exit. This is a physical-time estimate, not a fast-jump count.

Let J be the number of extra $0 \rightarrow C$ births in the fast part. Since ρ_n depends on the birth clock, J is not asserted to be Poisson conditional on ρ_n . Instead, the stopped counting-process exponential martingale gives

$$\mathbb{E} \exp\{sJ - \alpha(e^s - 1)\rho_n\} \leq 1. \quad (4.9)$$

Apply Cauchy–Schwarz to the martingale with parameter $2s$. Equations (4.8)–(4.9) then give a uniform exponential moment for J , and that moment tends to one as $n \rightarrow \infty$. At a dormant vertex, one additional seed is charged before the next fast trial. The number of localized trials is dominated by a geometric variable with success parameter

$$p_* = \min_{d \in D(\text{pair})} p_d > 0, \quad (4.10)$$

where the minimum is taken only over the one or two dormant charts of the fixed pair after its rates are fixed. A completed macroepisode may be re-marked at the other dormant vertex; the next application uses the same finite minimum. No constant is claimed uniformly over rate vectors.

Let N_0 be the population at the beginning of the episode, K the total number of activation-stage births, and D_{pre} the total number of activation-stage deaths. The compound-geometric calculation from (4.9)–(4.10), including an arbitrary initial point in a wedge, gives some $s_0 > 0$ with

$$\sup_{N_0 \geq N_*} \mathbb{E} e^{s_0 K} < \infty, \quad \sup_{N_0 \geq N_*} \mathbb{E} \tau_{\text{act}}^m < \infty \quad (m \geq 1). \quad (4.11)$$

Localize the episode at $n \leq N_0/2$ or $n \geq 2N_0$. A downward exit is favorable. An upward exit requires at least N_0 births, so (4.11) implies

$$\mathbb{P}\{\text{upward exit}\} \leq C e^{-cN_0}. \quad (4.12)$$

A downward exit requires order- N_0 service deaths during a compound-geometric sum of fast trials. The localized trial-time estimate (4.8), the death counting-process exponential martingale, and a split at $\varepsilon N_0 / \log N_0$ trials give the bound. Above the split, the geometric trial tail is $e^{-cN_0 / \log N_0}$. Below it, the summed compensator is at most $C\varepsilon N_0$ outside an exponentially small trial-duration event; choose ε first and apply the counting martingale. Thus, for every m ,

$$\mathbb{P}\{\text{activation-stage downward exit}\} = o(N_0^{-m}). \quad (4.13)$$

This tail estimate is used only to remove the branch from the regular service calculation. Its actual scalar contribution is favorable and is retained separately in Section 7.

On the regular activation branch, with activation population N_a , the exact identity is

$$N_a = N_0 + K - D_{\text{pre}}, \quad \frac{N_0}{2} \leq N_a \leq 2N_0. \quad (4.14)$$

No lower clock has been removed in (4.8)–(4.14).

22.5 5. Infinite top-ODE service integral

On the unit simplex, let $z = (a, b, c)$ solve

$$\frac{dz}{d\tau} = \sum_{y \rightarrow y' \in T} \kappa_{yy'} z^y (y' - y), \quad a + b + c = 1. \quad (5.1)$$

Let $\mathcal{A}$ be the finite union of normalized activation shells. It is compact and separated from every dormant vertex.

Lemma 5.1. For every $z(0) \in \mathcal{A}$,

$$\int_0^\infty c(\tau) d\tau = \infty. \quad (5.2)$$

If the integral were finite, boundedness of the polynomial vector field would make c uniformly continuous, and Barbalat's argument would give $c(\tau) \rightarrow 0$. The omega-limit would be a nonempty compact invariant subset of $c = 0$.

The directed cut in (2.2) identifies the largest invariant subset of that face. If $a, b > 0$, every present service-free source is enabled, and strong connectivity gives a cut edge to a C -containing node, so $\dot{c} > 0$. At the pure A vertex, an existing $2A$ source either creates C or first creates B , after which the same cut applies. Thus the vertex remains in the face only when $2A$ is absent. The corresponding statement holds at the pure B vertex. Hence the largest invariant subset consists exactly of the dormant vertices in (2.2).

An omega-limit contained in this finite set is one vertex. In its chart, the fluid form of (4.3) gives

$$\dot{R} \geq cR - KR^2. \quad (5.3)$$

For small positive R , the right side is positive, so an orbit which has left the vertex cannot approach it. This contradiction proves (5.2).

For fixed T , the map

$$I_T(z) = \int_0^T c_z(\tau) d\tau \quad (5.4)$$

is continuous. The increasing open sets $\{z : I_T(z) > M\}$ cover the compact set $\mathcal{A}$ as $T \rightarrow \infty$. Therefore, for every prescribed M , there is one finite $T = T(M)$ such that

$$\inf_{z \in \mathcal{A}} I_T(z) > M. \quad (5.5)$$

No stationary or fast-mixing assumption is used.

22.6 6. Lattice-uniform full-chain service

On the regular activation branch, run every physical clock for time T/N_a . For the fluid estimate, localize at $N_t \notin [N_a/2, 2N_a]$; the endpoint analysis below retains either localized exit and does not replace the physical chain. Let

$$Z_{N_a}(\tau) = N_a^{-1} X(\tau_{\text{act}} + \tau/N_a), \quad 0 \leq \tau \leq T. \quad (6.1)$$

Top clocks have rates $N_a^2 \kappa z^y + O(N_a)$, so their compensated density martingales have quadratic variation $O(N_a^{-1})$. Let B_{win} and D_{win} be the birth and death counts in this window. For the unlocalized deterministic horizon, the birth count is Poisson with mean $\alpha T/N_a$; the stopped count is bounded by that same underlying clock. The death compensator is

$$\Lambda_{\text{win}} = \delta \int_0^{T/N_a} C_t dt = \delta \int_0^T (Z_{N_a}(\tau))_C d\tau. \quad (6.2)$$

Since deaths only reduce population,

$$\Lambda_{\text{win}} \leq \delta T + \frac{\delta T}{N_a} B_{\text{win}}. \quad (6.3)$$

The exponential counting-process martingale gives, after localization,

$$\mathbb{E} \exp\{s D_{\text{win}} - (e^s - 1) \Lambda_{\text{win}}\} \leq 1. \quad (6.4)$$

Apply Cauchy–Schwarz to (6.4) with parameter $2s$, and then use (6.3). Together with the Poisson exponential moment of B_{win} , this gives uniform exponential moments for both window counts. Thus the death clock is controlled through its random compensator; it is not replaced by a fixed pathwise Poisson clock.

Random time changes, high-moment Burkholder estimates, and Gronwall now give, uniformly over every lattice point whose normalized value is within $O(N_a^{-1})$ of $\mathcal{A}$,

$$\mathbb{E} \sup_{0 \leq \tau \leq T} \|Z_{N_a}(\tau) - z(\tau)\|^m \longrightarrow 0 \quad (6.5)$$

for each fixed m . The lower counts have $O(1)$ moments and therefore move density by $o(1)$. By the compensator identity and (5.5),

$$\inf_{\text{activation lattice states}} \mathbb{E} D_{\text{win}} \geq \delta \inf_{z \in \mathcal{A}} I_T(z) - o(1). \quad (6.6)$$

Let

$$\mu_K = \sup_{N_0 \geq N_*} \mathbb{E} K < \infty.$$

Choose M , hence T , so that $\delta M > \mu_K + 4$. Since $\mathbb{E} B_{\text{win}} = o(1)$, define

$$\overline{\Delta} := K + B_{\text{win}} - D_{\text{win}}. \quad (6.7)$$

Let $\mathcal{R}_{N_0}$ denote the regular branch on which neither the activation stage nor the service window exits its displayed population shell. Upward activation exits have probability $O(e^{-cN_0})$ by (4.11)–(4.12), while (4.13) makes the removed activation-down branch super-polynomially small. Upward service exits require order- N_0 births and have an exponential bound. A service-window downward exit requires order- N_0 deaths; (6.3)–(6.4) make both its probability and its truncated count moments exponentially small. Each downward branch is also retained as a separate favorable contribution in Section 7; its negative scalar contribution is not used to improve any conditional count estimate.

On $\mathcal{R}_{N_0}$, use the event-weighted inequalities

$$\mathbb{E}[K; \mathcal{R}_{N_0}] \leq \mathbb{E} K \leq \mu_K, \quad \mathbb{E}[B_{\text{win}}; \mathcal{R}_{N_0}] = o(1). \quad (6.8)$$

Uniformity of (6.6), followed by the exponential count bounds for the removed service-shell event, gives

$$\mathbb{E}[D_{\text{win}}; \mathcal{R}_{N_0}] \geq \delta M - o(1). \quad (6.9)$$

No estimate of the form $\mathbb{E}[K \mid \mathcal{R}_{N_0}] \leq \mu_K$ is asserted or needed. Therefore, for all sufficiently large N_0 ,

$$\mathbb{E}[\overline{\Delta}; \mathcal{R}_{N_0}] \leq -3, \quad \sup_{N_0} \mathbb{E} e^{s_1 |\overline{\Delta}|} < \infty \quad (6.10)$$

for some $s_1 > 0$. The second, unconditional, statement uses (4.11) and (6.3)–(6.4). The exact physical endpoint identity, with no count discarded or counted twice, is

$$\begin{aligned} N_a &= N_0 + K - D_{\text{pre}}, \\ N_{\text{end}} &= N_a + B_{\text{win}} - D_{\text{win}} \\ &= N_0 + K - D_{\text{pre}} + B_{\text{win}} - D_{\text{win}} = N_0 + \overline{\Delta} - D_{\text{pre}}. \end{aligned} \quad (6.11)$$

In particular,

$$N_{\text{end}} \leq N_0 + \overline{\Delta}. \quad (6.12)$$

For every $q > 8$, (4.11), (6.10), and the shell localization give

$$\begin{aligned} \sup_{N_0} \mathbb{E} \tau^q &< \infty, \\ \mathbb{E}(1 + N_{\text{end}})^q &\leq C_q (1 + N_0)^q, \\ \sup_{N_0} \mathbb{E} \{(N_{\text{end}} - N_0)^+\}^q &< \infty. \end{aligned} \quad (6.13)$$

These are the endpoint and duration uniform-integrability estimates used in the scalar calculation.

22.7 7. Deterministic endpoint envelope and stopped scalar drift

Define the monotone deterministic envelope

$$\mathcal{P}(m) = \{1 + K_F + \log(m!)\}^4 + \lambda(1 + m)^6, \quad m \geq 0. \quad (7.1)$$

For every state y of total population m ,

$$V(y) \leq \mathcal{P}(m). \quad (7.2)$$

Also, the exact multinomial inequality gives

$$0 \leq \log(N_0!) - \sum_i \log(x_i!) = \log \frac{N_0!}{A!B!C!} \leq N_0 \log 3. \quad (7.3)$$

Consequently,

$$\mathcal{P}(N_0) - V(x) \leq CN_0^4(\log N_0)^3. \quad (7.4)$$

On the regular branch, (6.12) and monotonicity of $\mathcal{P}$ give the pathwise deterministic inequality

$$V(X_\tau) - V(x) \leq \mathcal{P}(N_0 + \bar{\Delta}) - \mathcal{P}(N_0) + CN_0^4(\log N_0)^3. \quad (7.5)$$

This does not delete D_{pre} from the process or compare two reaction systems. Every pre-service death remains in the exact identity (6.11); it only improves the total-population envelope (7.5). All top reactions are included through the factorial maximum in $\mathcal{P}$.

The exponential moment in (6.10) permits an exact Taylor expansion with a uniform remainder. The sixth-power part satisfies

$$\mathbb{E}[\{(1 + N_0 + \bar{\Delta})^6 - (1 + N_0)^6\}; \mathcal{R}_{N_0}] = 6(1 + N_0)^5 \mathbb{E}[\bar{\Delta}; \mathcal{R}_{N_0}] + O(N_0^4) \leq -cN_0^5. \quad (7.6)$$

The positive change of the factorial-maximum part is $O(N_0^3(\log N_0)^4)$, while (7.4) costs $O(N_0^4(\log N_0)^3)$. Both are strictly below (7.6). Hence

$$\mathbb{E}_x[\{V(X_\tau) - V(x) + \tau\}; \mathcal{R}_{N_0}] \leq -c_1 N_0^5 \quad (7.7)$$

for every sufficiently large wedge state. This is an event-weighted regular contribution, not a conditional expectation.

At an activation-stage downward shell exit, the endpoint population is at most $N_0/2 + 1$. Equations (7.1)–(7.4) give a negative order- N_0^6 increment. At an upward exit, the first crossing has population at most $2N_0 + 1$, so its positive envelope is $O(N_0^6)$; (4.12) makes its expected contribution $O(N_0^6 e^{-cN_0})$.

For a service-window shell exit, first split on $K \leq N_0/4$. Then $N_a \leq N_0 + K \leq 5N_0/4$, so a downward crossing of $N_a/2$, combined with the deterministic envelope, is favorable at order N_0^6 , while an upward crossing of $2N_a$ requires order- N_0 window births. The complement $K > N_0/4$ has exponentially small probability by (4.11), and its endpoint is charged by the same exponential moment and deterministic envelope. Equations (6.3)–(6.4) give the corresponding exponential window tail. Combining all branches yields

$$\mathbb{E}_x\{V(X_\tau) - V(x) + \tau\} \leq -1 \quad (7.8)$$

outside a finite wedge sublevel.

22.8 8. Fixed-class gluing and nonexplosion

The sole population-increasing clock has constant rate α . Hence total population is pathwise bounded above by its initial value plus a rate- α Poisson process. On every finite population shell, the top state space and all reaction rates are finite. This proves nonexplosion and the finite-time moments used in localization.

Use the partition

$$E = K \cup (E \setminus \mathcal{W}) \cup (\mathcal{W} \setminus K), \quad (8.1)$$

where K contains the finite exceptions in (3.8) and (7.8). Equation (3.8) is the generator-good hypothesis of the common-entropy physical-time gluing theorem; (7.8) is its exceptional-episode hypothesis for the same globally proper V . The stopping rules are state-based and strong-Markov compatible. The theorem gives finite mean hitting of K . After restricting to a closed irreducible population class, the finite mixture of return bounds from K gives positive recurrence.

This proves recurrence for the seven pairs, for each fixed strong top orientation and positive rate vector. The independent audit replayed Sections 3–8 and found no load-bearing gap.

22.9 9. Reproduction and independent audit

Run

```
PYTHONPATH=src python3 -B src/rank_two_mixed_profile_7_stopped_service.py
PYTHONPATH=src python3 -B -m unittest \
  tests/test_rank_two_mixed_profile_7_stopped_service.py -v
```

The independent replay verified:

1. localized PF trial survival, duration, and compound-birth exponential moments from arbitrary wedge starts;
2. exact identities (4.14) and (6.11), including both shell exits;
3. the directed-cut/Barbalat proof of infinite integrated service;
4. lattice-uniform fluid convergence and random-compensator exponential estimates;
5. $q > 8$ endpoint and duration uniform integrability;
6. the C -bounded passing estimate; and
7. the deterministic envelope and fixed-class gluing.

All seven scoped recurrence flags are promoted. Global T3-2 remains uncertified because the disjoint hard 333-pair family is still open.

23 The hard-333 common-potential fixed-class theorem

Proof-first composition theorem, repaired 2026-08-12 PDT. The local inputs are proved analytically; finite computation is used only for descriptor exhaustion and compatibility of one pair-fixed correction. The canonical one-active 1,104 theorem is now audited at the exact bytes pinned in Section 8. This hard-333 theorem remains at candidate status only until its own fresh composition replay is complete. No argument below infers a stochastic estimate from a search over orientations, histories, or population states.

23.1 1. Statement

Fix one of the final 333 support pairs, give each linkage an arbitrary strongly connected orientation and arbitrary positive rate constants, and let X be its stochastic mass-action chain. For every closed irreducible population class Γ , there is a vector ℓ , fixed for the whole pair, such that

$$G_\ell(x) = K_\ell + \sum_{i=1}^3 \log(x_i!) + \ell \cdot x \geq 1, \quad W_\ell(x) = G_\ell(x)^4 \quad (1.1)$$

is a common population potential for every active-dimensional chart. The choice menu is

correction family	pairs
arbitrary fixed ℓ	291
reversible-top detailed balance	34
directed-triple correction	8.

(1.2)

For an arbitrary- ℓ pair we choose any fixed admissible value, and in particular choose $\ell = 0$ on the four H_w pairs. The constant K_ℓ is chosen once, independently of invariant shells and stopping episodes.

Theorem 1.1. Every closed irreducible class Γ of every hard support pair is positive recurrent.

The proof occupies Sections 2–7. Section 8 records exactly where the finite atlas enters.

23.2 2. Properness, nonexplosion, and the marked lift

The potential is proper. If $N = |x|_1$, one coordinate is at least $N/3$, so Stirling’s formula gives a factorial contribution of order $N \log N$, whereas the negative part of $\ell \cdot x$ is only linear.

The chain is nonexplosive independently of recurrence. Any reaction which increases total population has source molecularity at most one, since every target complex has molecularity at most two. The sum of all population-increasing propensities is therefore bounded by $C(1 + |x|_1)$. Localization and Gronwall bound the total population on finite time intervals. Within a fixed population sublevel there are only finitely many states and all total rates are bounded. Hence infinitely many jumps cannot accumulate in finite physical time.

Fix $x^\circ \in \Gamma$. On the reachable lift starting from $(x^\circ, 0)$, update three reflected debt marks after a physical jump $x \mapsto x + \zeta$ by

$$D_i^+ = (D_i + \zeta_i)^+, \quad H_i = X_i - D_i. \quad (2.1)$$

Then, pathwise,

$$0 \leq D_i \leq X_i, \quad H_i(t) \leq H_i(0) = x_i^\circ. \quad (2.2)$$

Indeed, H_i is unchanged when reflection is inactive and decreases when reflection occurs. The set with all debts zero is consequently finite. The lifted potential $\widehat{W}(x, d) = W_\ell(x)$ is proper because a fixed population admits only finitely many marks.

The mark is used only to establish eligibility of a one-active old-species service block. If a divergent one-active sequence selects active species V but has $D_V = 0$, then $V = H_V \leq x_V^\circ$, a contradiction. Thus every genuinely divergent reachable one-active bad tube has positive selected debt after descriptor reclassification. A structural “no-history” face, on which V cannot have changed, can never carry that positive debt. This replaces every false assertion that an arbitrary unmarked one-active start must admit service.

23.3 3. The local common-potential contract

Every stopped lemma used below is a physical strong-Markov rule. It retains all enabled clocks, includes the reaction causing its terminal event, evolves the marks along the actual path, and stops at the actual endpoint. For a chart scale $s \rightarrow \infty$ it supplies some $h_s \rightarrow \infty$ and some fixed moment order $q > 8$,

$$\begin{aligned} \mathbb{E}_x[W_\ell(X_\tau) - W_\ell(x) + \tau] &\leq -cG_\ell(x)^3 h_s, \\ \mathbb{E}_x|G_\ell(X_\tau) - G_\ell(x)|^q &< \infty, \quad \mathbb{E}_x \tau^q < \infty. \end{aligned} \quad (3.1)$$

The exact displayed orders differ between charts, but all dominate a fixed negative constant outside a finite class-dependent set. Moving-boundary events are path-labelled. Only a cutoff reached at an outer no-fast return is a promotion endpoint P ; a cutoff first reached while an excursion is open remains an auxiliary endpoint B , even if later cleanup reaches the same population face. Both are already charged in (3.1). Reclassification at X_τ therefore has zero comparison toll because the next rule uses the identical W_ℓ . The inputs often prove every fixed moment, but the composition below uses only the existence of one fixed $q > 8$.

23.3.1 3.1 One active coordinate

The 1,104 failed incidences split into

$$951 + 99 + 44 + 4 + 6. \quad (3.2)$$

The first group uses the generalized 146-template theorem: exact-cloud, mixed Schur, separated regenerative, and invariant no-history alternatives. The 99 direct physical-source rows use the audited multi-service block with $K = 1 + u$, where u is the initial inactive mass. A tagged Harris path gives a background-uniform service clock; compensated active-free-source hazards and the inactive endpoint entropy remain polynomial in $1 + u + K$. Forty-four translated exact Family-II rows satisfy the abstract mixed fast-Schur hypotheses, and four zero-inactive faces cannot carry reachable positive debt. The six open rows use the audited all-clock multi-service block. Its stopping rule is autonomous in (B, C) , so the conditional A -endpoint remains exactly binomial plus Poisson. Its designated zero-source launch at $B = 0$ is allowed; only a competing active-free-source reaction outside that launch is a defect.

In both repaired routes, D_1 is the internal first net service. It lowers incoming reflected debt but is not the atlas endpoint. The included K -th service D_K is the clean terminal endpoint; services between D_1 and D_K after reflection reaches zero are surplus services. The $K \log n$ active factorial reward amortizes the entire arbitrary subpower inactive endpoint entropy. Together with the other three routes, the audited canonical 1,104 theorem proves (3.1) for the same arbitrary fixed ℓ , with some endpoint and physical-time moment order $q > 8$.

23.3.2 3.2 Two active coordinates

There are

$$407 + 181 + 114 = 702 \quad (3.3)$$

failed incidences.

The 407 dormant incidences normalize, with physical multiplicity, to 188 macroscopic carrier templates. Their disjoint analytic split is 19 exact carriers and 169 nonexact carriers. Exact product-form carrier occupation, sourcewise ordered Green bounds, and a killed exponential overshoot estimate give (3.1) on the exact group. The nonexact Schur theorem contracts only physical self returns and uses a strong cut to obtain the same estimate. Both the endpoint and all elapsed raw time are retained. The composition $188 = 19 + 169$, representing all 407 physical incidences, has received an independent proof replay.

The 181 enabled incidences are literal members of the analytic bounded-access lemma. A simple path in the exact top D-tier ends with an edge leaving that tier. All-reaction race estimates, including every competitor, yield a fixed success probability and an endpoint moment of some fixed order $q > 8$. This is an analytic graph-and-clock argument; the finite certificate only verifies membership in the lemma's premise set.

The remaining 114 incidences are closed rank-one shells. The powered corrected-factorial endpoint theorem gives an integrable positive overshoot on reversible two-node and directed homogeneous-triple shells, including the four activation rows. There are 90 reversible and 24 directed-triple rows. Their shell mask agrees with the all-active top of the same pair, so their correction in (1.1) is literally the same vector. The local 114-row theorem has received a strict independent proof audit. The final finite replay checks all 114 rows against the pair-fixed correction and finds zero mismatches.

On each of the twelve H_b switch pairs, both the all-active chart and the two-active rank-one chart have one top mask, and those masks are identical. Each of the four H_w switch pairs has one all-active rank-two mask and no two-active rank-one mask. These are support identities, not stochastic estimates.

23.3.3 3.3 Three active coordinates

The 154 failed incidences split as

$$110 + 24 + 16 + 4. \quad (3.4)$$

For 110 curvature-safe reversible two-node rows, paired top reactions give the powered corrected-factorial generator descent. Twenty-four homogeneous directed-triple rows use the fixed directed-fluid correction. On the flat top, jumps of corrected entropy are $O(1)$, total top rate is $O(N^2)$, the lower negative rate is at least order N , and its tier gap diverges; the exact fourth-power carré ratios vanish.

The sixteen H_b curvature incidences on twelve pairs use the guard-free same-state Kac shell resolvent. Polynomial source tilts of the reversible factorial shell law control the first lower reaction and its actual jump. The top relaxation-to-killing ratio tends to infinity, and the stationary lower entropy reward diverges negatively. The pointwise complement is the union of separately negative top-flux and lower high-cut branches; no false single-margin estimate is used. This gives (3.1) for the same reversible correction used in every lower-dimensional chart.

The four H_w rows use $\ell = 0$. Their weighted transverse coordinate has a proof-level dyadic activation theorem: source-balance and an adaptive Chernoff bound give positive escape from the dormant region, deterministic service supplies a negative-increment exponential comparator, and the fractional total-population return yields a common factorial fourth-power drop. This theorem begins from every all-active start and includes failures, upper exits, overshoots, and duration.

The finite compatibility replay compares the correction attached to every one of these 154 all-active incidences with its pair-fixed choice in (1.2) and finds zero mismatches: 110 safe and sixteen H_b rows use the reversible correction, 24 triples use the directed-triple correction, and the four H_w rows use the arbitrary family specialized to $\ell = 0$.

23.4 4. Passing descriptors

It remains to treat descriptors which pass the actual oriented source-tier criterion. Let A_s be the largest enabled source propensity and $g_s \rightarrow \infty$ the logarithmic gap of a certified top source to its lower target. For $d_r = G_\ell(x + \zeta_r) - G_\ell(x)$, the quantitative descending-source estimate is

$$\mathcal{L}G_\ell \leq -cA_s g_s, \quad \sum_r a_r \leq CA_s, \quad |d_r| \leq C \log(1 + |x|_1). \quad (4.1)$$

The exact binomial identity

$$\mathcal{L}W_\ell = 4G_\ell^3 \mathcal{L}G_\ell + 6G_\ell^2 \sum_r a_r d_r^2 + 4G_\ell \sum_r a_r d_r^3 + \sum_r a_r d_r^4 \quad (4.2)$$

and $G_\ell \asymp |x|_1 \log(1 + |x|_1)$ show that every remainder in (4.2) is lower order than $-G_\ell^3 A_s g_s$. Hence

$$\mathcal{L}W_\ell \longrightarrow -\infty \quad (4.3)$$

along every divergent passing descriptor.

23.5 5. Descriptor exhaustion is only a compactness step

Every divergent population sequence has a subsequence with a fixed exact source/D-tier order, availability pattern, and active-coordinate set. An affine-infeasible descriptor cannot occur in Γ . A realizable passing descriptor satisfies (4.3). Every realizable failure is assigned by Sections 3.1–3.3. The finite proposition used here is exactly

active coordinates	failed incidences	
1	1104	$1104 + 702 + 154 = 1960.$
2	702	
3	154,	

(5.1)

No orientation or stochastic path was enumerated to prove a local estimate. The table checks only that the analytic predicates are disjoint and exhaustive and that the correction menu is (1.2).

If infinitely many reachable marked states were neither generator-good nor episode-good, properness would give a divergent sequence among them. Exact descriptor extraction would produce a subsequence covered by Section 3 or 4, a contradiction. Thus the uncovered set is finite. Because the chart library is

finite and every local negative magnitude diverges along its escaping sequences, after enlarging that finite set there are constants $a, \eta, \delta > 0$ such that

$$\mathcal{L}W_\ell \leq -a \quad (5.2)$$

on the generator-good set and

$$\mathbb{E}_z[W_\ell(X_{\tau_z}) - W_\ell(x) + \eta\tau_z] \leq -\delta \quad (5.3)$$

on its complement outside the finite target. Endpoint integrability is a part of every local theorem, not a consequence of the finite exhaustion.

23.6 6. Random-time Foster theorem

For completeness we prove the gluing statement used here. Let K be the finite target from Section 5. From a generator-good state, run until first entrance into the episode-good set or K . Localized Dynkin applied to (5.2), followed by Fatou, gives

$$\mathbb{E}[W_\ell(X_\sigma) - W_\ell(x) + \eta\sigma] \leq 0. \quad (6.1)$$

At an episode-good state append its physical stopping time from (5.3), then repeat from the actual endpoint. Stop the construction when K is first visited, although a containing local episode may be completed for drift accounting. If N is the number of completed exceptional episodes and S_m accumulated physical time, conditional iteration gives

$$\delta \mathbb{E}(N \wedge m) + \eta \mathbb{E}S_{N \wedge m} \leq W_\ell(x) + \delta. \quad (6.2)$$

Monotone convergence yields finite mean time to K from every reachable marked state. Exact physical self returns may have been erased inside a local resolvent only analytically: their holding times, internal cutoffs, and actual mark evolution were already restored in (3.1), so (6.2) is a physical time statement.

23.7 7. From finite mean hitting to positive recurrence

If Γ is an absorbing singleton there is nothing to prove. Otherwise take one ordinary physical jump from the finite marked target and apply the hitting estimate to each of its finitely many successors. The successive positive returns to the target define a finite trace chain. Pick a recurrent state of that trace. Its discrete return count has finite mean, and each physical target-to-target segment has uniformly finite mean, so the marked process has a finite mean positive return to that state. Projecting to the physical population can only shorten the return. Irreducibility of Γ therefore makes every state positive recurrent. This proves Theorem 1.1.

23.8 8. Frozen finite certificate and audit boundary

The companion `src/hard333_final_descriptor_coverage.py` verifies only:

1. the 333-pair correction histogram (1.2);
2. the exact local-route histogram in (3.2)–(3.4);
3. the physical normalization of 407 dormant rows to the audited 188 templates;
4. one all-active mask on every switch pair, the identical unique rank-one mask on each of the twelve H_b overlaps, and absence of a rank-one mask on the four H_w pairs;
5. zero pair-correction mismatch on the 114 rank-one and 154 all-active incidences;
6. the frozen audited one-active input bytes; and
7. the exhaustion (5.1).

The audited canonical one-active input is pinned as

```
theorem: d2d2eff482190abb78ae7d5b0a83440881931ff8387f35435d2066fb8b20bca8
source: 03f61eb180792201cf5bbdcdb1ac85b198cd02bcd8b85e7e46771d9fdd4c47cc
tests: 29856f4c64c30bb376527389e4c90d06e79c4bd5647aff511c19428df1d141db
rows: 66d931827f2b118306cd10b732d368b50d77f929dafae3c36b1bfae47a40d1b4
payload: 8a5c7f6f6c55fbc3ddb9fa1f438677212c15075d0294f706958c44cc6b91eef3
```

Its current canonical hashes are

```
pair rows: a1765765a7bc7f9eafd2f9ca0744f082a61a83ad7df9f8efc01a3f38aaa9de08
failure rows: b96325940e9d4178e9ef3dcd7f221fcd0e758b2f1c30b4904e2b3b0fc3fb7ca9
payload: 9537f53f0aa3af10ac612b01da85081a3bd44b0382f689149da5f5917e353cc1
```

Pair and global promotion remain false until an independent replay pins the complete hard-333 composition.

Part IV

Completed mixed orbit

24 Recurrence of the 27,462-pair completed mixed orbit

Proof-first pair theorem, 2026-08-12 PDT. This note isolates the part of the inherited mixed support atlas whose seed-level analytic branches were already complete. The 110 seeds formerly routed only by an invariant positive on two chart-active coordinates are excluded before taking any symmetry orbit. Consequently this theorem does not inherit the invalid claim that such an invariant controls an all-three-active escape.

Finite computation is used only for support, tier, affine, and set identities. No orientation, rate vector, population state, reaction history, or communicating class is enumerated.

24.1 1. Exact seed partition

The finite certificate is

```
src/other_mixed_orbit_27462_certificate.py
SHA-256 57d8904dd86cd0bf626e344dbfd7b7f248b239cdeaace48489796058c6875f08
```

```
tests/test_other_mixed_orbit_27462_certificate.py
SHA-256 e708c52f6cbc1bbcd4abf33f246d72379dff58c74cc97a38fdd8076ac3ae7d13
```

The inherited positive-shield and signed-shield tables contain 5,169 ordered support pairs. Exactly 110 were assigned the label `common_active_invariant`. Remove those seeds *before* applying any species permutation or linkage reversal. The remaining 5,059 seeds have the exact disjoint partition

$$5059 = 187 + 974 + 9 + 1378 + 2511, \quad (1.1)$$

with the following literal meanings.

branch	seed pairs	analytic conclusion
strictly positive invariant	187	every fixed class is finite
full deficiency zero	974	classwise product form and recurrence
three literal physical seams	9	standalone physical-time theorems
corrected S-superlevel tier pass	1378	statewise entropy Foster
corrected residual union	2511	fourteen standalone pair theorems

The eligible-seed fingerprint is

```
c45f67990ff841e1ba7b7d5d8a2795539f495f2434d68135ad3b2483d2fda44f.
```

The branch manifest fingerprint is
22f10cf6ea09a7b36650df174a866fd15470c770b3da6079728cfe5301f61c76.

24.2 2. Analytic seed branches

Fix one eligible ordered seed pair, arbitrary strongly connected labelled orientations on its two supports, arbitrary positive rates, and a closed irreducible population class Γ .

24.2.1 2.1 Strict invariant and deficiency zero

If $h \in \mathbb{R}_{>0}^3$ annihilates every reaction vector, then $h \cdot x$ is fixed on Γ . Its nonnegative integer level set is finite, so Γ is finite.

For a full deficiency-zero seed, the entire two-linkage network is weakly reversible and deficiency zero. The stochastic complex-balanced product form for arbitrary positive rates is normalizable on every closed class by restriction of the full product-Poisson sum. Together with the binary nonexplosion argument in Section 4, this gives positive recurrence.

24.2.2 2.2 Literal physical seams

The nine seeds in this branch are exactly six seven-support pairs, two signed-service pairs, and one residual-pair core. Their standalone arbitrary-orientation physical-time theorems are frozen at

```
research_notes/certified_exact_shielded_seam.md
d8e037a1bc2f396928011404513b5a55d0fae0c28d55cf46de5a4ed391fd9e3d
```

```
research_notes/signed_service_seam_full_proof.md
4ec0ae7007184f2c5bda82bd55df5707d2c3570c7fdf2683ad87b97f75930738
```

```
research_notes/residual_pair_full_proof.md
dcca51ed7ed30523cfdce1db74e24b9b3a59aafbeb0cb8b7de72ac9b254fb7db
```

and their independent exact-scope audit has SHA-256
e7e76b76cd1371f98d19da0a1f5362ab4a0696548fba62028b29ccd2950617c9.

The signed-service support condition is not used monotonically: only the literal certified supports occur in this branch.

24.2.3 2.3 Corrected tier-pass seeds

For each of these 1,378 seeds, every one of the exact 259 tier descriptors satisfies the corrected S-tier-superlevel cut. Let E be the global top S-tier and let U_L be the D-superlevel in linkage L at the D-level of E . The exact graph theorem says that every pair of strong orientations has an E -sourced D-descending edge precisely when

$$\emptyset \neq U_L \subsetneq L, \quad U_L \subseteq E \quad (2.1)$$

for some linkage. The corrected theorem and finite descriptor proof are frozen at

```
research_notes/s_tier_superlevel_cut_and_affine151_corrected.md
d91f369d34cadfb28ddb872df8fb9f6d17799ec207da29933037f55ae95f0407
```

```
src/s_tier_superlevel_interface.py
1a4e27fcf40af76cac6281f8830b7644bf086b3c05d97a963ce9f5bac736ad57
```

Suppose the entropy generator were not negative outside any finite subset of Γ . Properness gives an escaping bad sequence in Γ ; after passing to one of the exact tier subsequences, (2.1) supplies the top-S descending reaction. The proof of Anderson–Kim (2018, journal Theorem 9) then forces the entropy drift to minus infinity, a contradiction. Thus the entropy has a statewise Foster drift outside a finite classwise set. This is not a drift-or-chart-exit argument.

24.2.4 2.4 The 2,511 residual seeds

The publication-safe residual certificate is

```
src/corrected_t3_2_two_linkage_union.py  
501d96c4cea2de33ed34db2c31702d3104e8ed80c1abb8cf15e895c56201593f
```

```
tests/test_corrected_t3_2_two_linkage_union.py  
2976ea43e18f0cb024fd3a5f30bac039c38329767253d022986659e4f609216b
```

Its exact dependency manifest rehashes fourteen pair-theorem scopes and their independent audits. Those scopes are pairwise disjoint and their union is the exact 2,511-pair residual, whose fingerprint is

```
0c57f530eb44a688520cc1706f830afa18063f4d08d24e5006f47a5666edd0b3.
```

Each of the fourteen analytic dependencies is already a standalone positive-recurrence theorem for its fixed support pair, with all clocks, actual endpoints, and physical time retained. The finite union therefore does not splice potentials along one trajectory: the support pair is fixed and belongs to exactly one theorem scope.

24.3 3. Symmetry transfer and exact orbit

Species permutations conjugate population states, propensities, and the generator. Reversing the order in which the two linkage supports are listed changes nothing in the physical network. Hence every symmetry image of an eligible seed inherits its seed theorem with the relabelled constants.

The orbit of the 5,059 eligible seeds has exactly 27,462 distinct ordered support pairs and fingerprint

```
1bf337cf143c6eb4cee5088827bb9e9b9ce704f01a1b1f57bde6aed856d2812.
```

It is disjoint from the 432-pair exclusive active-invariant orbit gap. The 282 active-invariant orbit points which also lie here are legitimate: each has a *different eligible seed representation* and is charged to that completed seed theorem, never to active-only invariance.

24.4 4. Nonexplosion and conclusion

A binary reaction with a degree-two source cannot increase total molecular population. Every population-increasing channel therefore has source degree at most one, bounded jump size, and aggregate intensity $O(1 + |x|_1)$. Linear pure-birth comparison prevents population escape in finite time. Each bounded-population sublevel is finite with bounded total rate, so neutral quadratic clocks cannot accumulate there. The CTMC is nonexplosive.

Theorem 4.1. Every ordered support pair in the exact 27,462-pair completed mixed orbit, under every strongly connected orientation and every positive labelled rate vector, is nonexplosive and positive recurrent on each closed irreducible population class.

The conclusion is obtained pairwise from the seed theorem and generator conjugacy. No orientation, population box, rate vector, or reaction history is enumerated, and no result is claimed for the separately repaired 432-pair orbit gap.

Part V

Orbit gap and level-set residual

25 Population-Foster closure of the 432-pair active-invariant orbit gap

Proof-first standalone theorem, 2026-08-12 PDT. This note closes the support-pair orbit which had been removed by an invariant positive on only two coordinates before species permutations were taken. The

original active-coordinate invariant is not used as a globally proper Foster function. Instead, the exact orbit splits into deficiency-zero pairs, pairs with a statewise factorial fourth-power drift, and two explicit rank-two exceptional families with a direct population-level service proof.

Finite computation is used only for support-orbit, deficiency, invariant, tier, affine-feasibility, and orbit-representative identities. No orientation, rate vector, population box, or stochastic history is enumerated.

25.1 1. The family, process, and one population potential

Begin with the inherited ordered shielded/available seed pairs. Among them, 110 were assigned to the common-active-invariant prebranch. Close these pairs under all species permutations and linkage reversal, then subtract the orbit of every other inherited seed pair. Denote the resulting exclusive family by $\mathcal{G}$.

Fix $P \in \mathcal{G}$, an arbitrary strongly connected directed graph on each of its two linkage supports, arbitrary positive labelled rate constants, and a closed irreducible population class Γ . Let X_t be the ordinary stochastic mass-action chain. Choose $K > 1$ and put

$$G(x) = K + \sum_{i=A,B,C} \log(x_i!), \quad W(x) = G(x)^4. \quad (1.1)$$

This is one population function: there is no reaction mark, chart-dependent linear correction, or entrance recharge. It is nonnegative and proper on $\mathbb{N}_0^3$.

The chain is nonexplosive for every pair considered here. A quadratic source cannot increase total molecularity because every target has molecularity at most two. Every population-increasing reaction therefore has source degree zero or one, total increasing intensity $O(1 + |x|_1)$, and bounded jump size. A linear pure-birth comparison bounds total population on finite time intervals; each population sublevel is finite and has bounded total hazard.

25.2 2. Exact finite decomposition

The authoritative finite certificate verifies all of the following.

1. The 110 seeds have a 714-pair orbit. Removing the orbit of the other 5,059 inherited seeds leaves exactly 432 pairs in $\mathcal{G}$.
2. Every pair has stoichiometric rank two. Its primitive normal has exactly one zero coefficient and two coefficients of the same strict sign. Each choice of zero coordinate occurs 144 times.
3. Exactly 174 pairs have full-network deficiency zero.
4. Of the remaining 258 pairs, 234 have no affine-feasible failure of the corrected S-tier-superlevel cut.
5. The last 24 pairs form two disjoint species-permutation/linkage-reversal orbits, each of size 12, represented by

$$\begin{array}{c|c|c} & L_1 & L_2 \\ \hline \text{I} & \{A, A + B\} & \{2A, 2C, A + C\} \\ \text{II} & \{A, A + B\} & \{C, 2A, B + C\}. \end{array} \quad (2.1)$$

Their primitive nonnegative invariants are respectively $A + C$ and $A + 2C$. Thus the only unbounded coordinate on a fixed class is B , the zero coordinate of the invariant.

6. The 192 affine-feasible corrected-cut failures include 48 deficiency-zero pairs and the 24 pairs in (2.1). In every failed row, the invariant-zero coordinate is exactly the unique active coordinate.

The principal exact fingerprints are

```
all 432 pairs
5516d6071b2b9d07b0e4e02613b9caee217ba3ebb0082e21f2bc664e6247ea36
deficiency-zero 174
d894c296de7c24e3c855f116228b852ef940b15ac913556ffea9c87e6655a974
non-DZ / no feasible failure 234
```

```

4c6420ad47c2ee57736484b213595dde69c3734ce20d19ff4f97aeb31ab77d6a
exceptional 24
051a641f3987ec93b129ad044d96292a97e536e9ef3d2724234dc4af9bdfef69
type-I orbit 12
50804d58a48bb1a2683014442a436761ec0ba0df34af3ef3ac0d1939fe850886
type-II orbit 12
cc76d3c7bcc7942f956f1efc8cf0f718ad1e6e08d911f8f8127ebc3cf8c5002f
invariant manifest
9dec8108276e9d439c18aacda1ec35d9bac08e097f8833e3446c50b40d8148ca
failure/invariant alignment
a9368dd934b7ac6135c3df4866e2322700d3a607dd58f8327aeb709065880ab2

```

These are support identities only. The analytic implications are proved in Sections 4–6.
The authoritative certificate bytes are

```

src/active_invariant_orbit_gap_432_certificate.py
31fa24a20e18546e9c623d3aaf6d3b845c1708d5782f86333c02417fa366cd53

tests/test_active_invariant_orbit_gap_432_certificate.py
09c434fc162ff33f51e331e298ec2e35407a8e0501f1a3b8d771bf33c2fe708b

```

Its nine dedicated tests and the combined 22-test replay pass.

25.3 3. Elementary fourth-power estimates

On any fixed conserved (A, C) -phase, as $B \rightarrow \infty$,

$$G(A, B, C) = B \log B + O(B). \quad (3.1)$$

If a bounded reaction changes G by u , then exactly

$$(G + u)^4 - G^4 = 4G^3u + 6G^2u^2 + 4Gu^3 + u^4. \quad (3.2)$$

Consequently, uniformly over a finite (A, C) -phase, for all sufficiently large B :

$$\begin{aligned} u &= -\log B + O(1) &\implies \Delta W &\leq -cG^3 \log B, \\ u &= \log(B + 1) + O(1) &\implies (\Delta W)^+ &\leq CG^3 \log B, \\ |u| &\leq C &\implies |\Delta W| &\leq CG^3. \end{aligned} \quad (3.3)$$

All constants in (3.1)–(3.3) may depend on the fixed population class and rate vector. No uniformity across different classes or rates is needed.

We will also use the following elementary random-time criterion.

Lemma 3.1 (bounded-depth population Foster rule). Let a nonexplosive CTMC have a proper nonnegative function V and a finite set K_0 . Suppose that from every state outside K_0 there is an all-clock stopping time $\tau > 0$, ending after between one and $d < \infty$ actual jumps, such that

$$\mathbb{E}_x[V(X_\tau) - V(x) + \tau] \leq -1, \quad (3.4)$$

and $V(X_\tau) + \tau$ is integrable. Then the mean hitting time of K_0 is finite. On a closed irreducible class the chain is positive recurrent.

25.3.1 Proof

Let $\sigma_0 = 0$, and recursively complete the selected episode from its actual endpoint. Define N to be the first positive episode count j for which the path segment from σ_{j-1} through σ_j visits K_0 ; complete that episode only for accounting. Conditional summation of (3.4) through $N \wedge n$ gives

$$\mathbb{E}V(X_{\sigma_{N \wedge n}}) + \mathbb{E}\sigma_{N \wedge n} + \mathbb{E}(N \wedge n) \leq V(x). \quad (3.5)$$

Fatou and monotone convergence give $\mathbb{E}N < \infty$ and finite mean accounting time. Indeed, on $\{N = \infty\}$, the episode endpoints include an infinite subsequence of actual jump times because every episode contains at least one jump. Nonexplosion forces $\sigma_n \rightarrow \infty$, contradicting the finite expectation obtained from (3.5). Thus $N < \infty$ almost surely, and the actual hit of K_0 occurs no later than σ_N .

For the recurrence promotion, fix $o \in K_0 \cap \Gamma$, which is nonempty by the hitting result. From each of the finitely many $c \in K_0 \cap \Gamma$, irreducibility gives a finite actual labelled path to o . On the event that each prescribed label fires before every competitor, that path is completed. Across the finitely many path states, this event has a positive uniform probability and the attempt has finite uniform mean duration. The possible first-competitor endpoints form a finite set; the preceding hitting bound gives a finite uniform mean return from them to $K_0 \cap \Gamma$. Geometric repetition gives a finite mean hit of o . A singleton class is already positive recurrent. Otherwise, after the first jump from o , the same bound gives a finite mean return to o . Hence the closed irreducible class is positive recurrent. $\square$

25.4 4. The 174 deficiency-zero pairs

Each directed linkage graph is strongly connected, so the full network is weakly reversible. For every positive rate vector, weak reversibility and full deficiency zero give a positive complex-balanced vector $c \in (0, \infty)^3$. On a closed irreducible class Γ , the stochastic product form is

$$\pi_\Gamma(x) = Z_\Gamma^{-1} \prod_{i=A,B,C} \frac{c_i^{x_i}}{x_i!}, \quad x \in \Gamma. \quad (4.1)$$

It is normalizable because its mass is bounded by $\exp(c_A + c_B + c_C)$, the full-lattice product-Poisson mass before the constant factors. Nonexplosion was proved in Section 1. Thus every closed irreducible class is positive recurrent. This is the class-restricted Anderson–Craciun–Kurtz deficiency-zero theorem, applied to the full two-linkage network, not separately to its linkages.

25.5 5. The 234 pairs with no feasible corrected-cut failure

We prove a direct generator theorem under the population potential (1.1). Suppose, to the contrary, that no finite set supports $\mathcal{L}W \leq -1$ on a fixed class Γ . Properness supplies a divergent sequence $x_n \in \Gamma$ with $\mathcal{L}W(x_n) > -1$. Extract one of the finitely many exact tier/cap descriptors.

Because the sequence lies in one affine stoichiometric class, the necessity direction of the exact affine-feasibility theorem makes this descriptor feasible. By the defining 234-pair identity it therefore satisfies the corrected S-tier-superlevel cut. For every strong orientation, that cut gives a D-descending reaction sourced in the literal global top S-tier.

Let A_n be the largest enabled source propensity and let $g_n \rightarrow \infty$ be its certified logarithmic D-gap. The exact factorial calculation gives

$$\mathcal{L}G(x_n) \leq -cA_n g_n, \quad \sum_r a_r(x_n) \leq CA_n, \quad |\Delta_r G(x_n)| \leq C \log(2 + |x_n|_1). \quad (5.1)$$

Using (3.2) reaction by reaction,

$$\begin{aligned} \mathcal{L}W &= 4G^3 \mathcal{L}G + 6G^2 \sum_r a_r (\Delta_r G)^2 \\ &\quad + 4G \sum_r a_r (\Delta_r G)^3 + \sum_r a_r (\Delta_r G)^4. \end{aligned} \quad (5.2)$$

Since $G(x_n) \asymp |x_n|_1 \log(2 + |x_n|_1)$, every remainder in (5.2) is lower order than $-G^3 A_n g_n$. Hence $\mathcal{L}W(x_n) \rightarrow -\infty$, a contradiction. Therefore

$$\mathcal{L}W \leq -1 \quad (5.3)$$

outside a finite subset of Γ . Continuous-time Foster, localized at finite W -sublevels and followed by Fatou, gives finite mean hit of that finite set and positive recurrence. This branch has no chart exits and no random-potential seam.

25.6 6. The two exceptional orbits

Relabel an exceptional pair as in (2.1). The invariant makes A, C range over a finite set on every fixed class, while B is the only coordinate that can diverge.

25.6.1 6.1 Type II: pointwise service

Fix the invariant value

$$M = A + 2C. \quad (6.1)$$

If $M = 0$, all sources are disabled and each population state is an absorbing singleton. Assume $M > 0$. Since both linkage graphs are strong, the source $A + B$ has a positive total rate to A , and the source $B + C$ has a positive total rate to the B-free complexes C or $2A$. Thus the aggregate rate of B-removing reactions is

$$B(\delta_1 A + \delta_2 C) \geq c_M B. \quad (6.2)$$

Every such reaction changes G by $-\log B + O_M(1)$. All B-creating sources are among $A, C, 2A$; their aggregate propensity is bounded by a classwise constant. Reactions which do not change B also have bounded propensity and bounded G -increment. Equations (3.3) and (6.2) give

$$\mathcal{L}W \leq -c_M B G^3 \log B + C_M G^3 \log B + C_M G^3 \longrightarrow -\infty. \quad (6.3)$$

Hence $\mathcal{L}W \leq -1$ outside a finite set, and ordinary continuous-time Foster proves positive recurrence.

25.6.2 6.2 Type I: activation followed by service

Now fix

$$M = A + C. \quad (6.4)$$

The first linkage necessarily contains both arrows $A \rightleftharpoons A + B$. Whenever $A > 0$, the B-death rate is δAB , whereas its B-birth rate is $\beta A = O_M(1)$. The second linkage has no B-coordinate and has total rate $O_M(1)$. Therefore, at a state $z = (A, B, C)$ with $A > 0$,

$$\mathcal{L}W(z) \leq -c_M B G(z)^3 \log B + C_M G(z)^3 \log B. \quad (6.5)$$

Its total hazard satisfies

$$c_M B \leq \lambda(z) \leq C_M(1 + B). \quad (6.6)$$

If J is the next actual jump and T_J its holding time, then (6.5)–(6.6) imply, for large B ,

$$\mathbb{E}_z[W(X_J) - W(z) + T_J] = \frac{\mathcal{L}W(z) + 1}{\lambda(z)} \leq -c'_M G(z)^3 \log B. \quad (6.7)$$

It remains to cover $A = 0$. If $M = 0$, or if $M = 1$ and $C = 1$, no complex in either linkage is enabled; the state is an absorbing singleton. If $M \geq 2$, the only enabled source is $2C$. Every outgoing target from $2C$ is either $2A$ or $A + C$, because its linkage is strong. Thus the first actual jump occurs at a fixed positive classwise rate, leaves B unchanged, and lands at a state z with $A > 0$. Its G -increment is bounded by a classwise constant, and hence its W -increment is at most $C_M G^3$. Append the next actual jump from z . By (6.7),

$$\mathbb{E}_{(0,B,M)}[W(X_\tau) - W(0, B, M) + \tau] \leq C_M G^3 - c'_M G^3 \log B + C_M \leq -1 \quad (6.8)$$

for all sufficiently large B . From $A > 0$, define τ to be one actual jump; from $A = 0, M \geq 2$, define it to be the two jumps in (6.8). Every clock is retained. The endpoint changes B by at most one, A, C stay in their finite invariant phase, and the duration has finite mean bounded uniformly in B. Thus the endpoint and duration in (6.8) are integrable.

The complement of the large-B region is finite on the fixed class. Lemma 3.1 applied to (6.7)–(6.8) proves positive recurrence of every non-singleton type-I class.

25.7 7. Pair theorem and seam disposition

Theorem 7.1 (the exclusive 432-pair orbit). For every support pair in $\mathcal{G}$, every strongly connected orientation on its two linkage supports, every fixed positive labelled rate vector, and every closed irreducible population class, the stochastic mass-action chain is nonexplosive and positive recurrent.

25.7.1 Proof

The exact decomposition in Section 2 is disjoint and exhaustive. Section 4 handles 174 pairs, Section 5 handles 234, and Section 6 handles the final 24. The counts sum to 432. $\square$

This proof intentionally supersedes the earlier marked- W drift-or-chart-exit draft. A terminal strongly connected component can cycle entirely through chart exits, so common marked potential plus unweighted exit circulation is not sufficient. Here the 234 passing pairs have statewise population-generator drift, type II has statewise direct service, and type I has an unconditional one- or two-jump population Foster episode from every large state. No chart exit, cap promotion, marked recharge, or weighted terminal seam remains.

25.8 8. Frozen analytic dependencies and claim boundary

The passing calculation uses the exact analytic statements in:

```
research_notes/s_tier_superlevel_cut_and_affine151_corrected.md
d91f369d34cadfb28ddb872df8fb9f6d17799ec207da29933037f55ae95f0407
```

```
research_notes/stoichiometric_gate_feasibility.md
27b40b61903ae6c2e223d007ec08323ec9aec10e9198deb99d2d7c60d878d007
```

```
research_notes/universal_fourth_power_one_active_interface.md
9d4239f4fc6b45a9522b94b09523c9f98ac7a3b089c919bd9594f12409c78cc2
```

The deficiency-zero publication scope is independently checked in the primary-literature audit.

The theorem is pair-level. A final global union must pin the authoritative 432 certificate and prove that this orbit and the other certified branches cover the intended global support universe with the stated precedence.

26 The workload-only physical-time Foster lemma for level-set pairs

Proof-first conditional composition lemma, 2026-08-12 PDT. This note isolates the exact stochastic statement needed to promote a level-set pair from local all-active drift to classwise positive recurrence. It uses the single linear workload preserved by the upper linkage. There is no marked factorial, chart potential, embedded-chain margin, or weighted seam.

The theorem below is conditional only on one explicit direct-death occupation macro. Finite computation is neither used nor needed.

26.1 1. Level-set network and exact workload identity

Fix a closed irreducible population class Γ of a reduced binary stochastic mass-action network. Constant coordinates have already been deleted. Suppose the two active linkage supports are T, R , with

$$h \cdot y = 2s \quad (y \in T), \quad R = \{0\} \cup U, \quad h \cdot u = s \quad (u \in U), \quad (1.1)$$

where $h \in (0, \infty)^d$, $s > 0$, and U consists of distinct unary complexes. Give both linkages arbitrary fixed strongly connected labelled graphs and arbitrary fixed positive rate constants. Put

$$H(x) = h \cdot x. \quad (1.2)$$

Parallel labelled channels are retained. Projected zero-displacement labels have already been discarded, so an ordinary jump below is an off-diagonal physical transition. The conditional Foster argument uses only (1.1); in the exact 336 application $d = 3$ and $\text{rank span}(T - T) = 2$.

This is proper on the reduced lattice. For a lower unary vertex u , set

$$\delta_u = \sum_{u \rightarrow 0} \kappa_{u0}, \quad D = \{u \in U : \delta_u > 0\}, \quad \beta = \sum_{0 \rightarrow u} \kappa_{0u}. \quad (1.3)$$

Strong connectivity of R gives $D \neq \emptyset$: the last edge of a directed path from a unary vertex to zero is sourced at an element of D . Every top reaction and every lower unary transfer preserves H . Thus the physical generator has the exact global identity

$$\mathcal{L}H(x) = s\beta - s \sum_{u \in D} \delta_u x_u. \quad (1.4)$$

No upper clock is estimated or averaged in (1.4), regardless of its rate.

26.2 2. Direct-death boundary and macro hypothesis

Fix an integer $k \geq 1$, and define

$$\mathcal{D}_k = \{x \in \Gamma : x_u < k \text{ for every } u \in D\}. \quad (2.1)$$

Let $C_R = \{x \in \Gamma : H(x) \leq R\}$. It is finite because H is proper. Choose R large enough that $C_R \neq \emptyset$. The load-bearing hypothesis is the following literal physical-time statement.

Occupation macro $\mathbf{M}(k, R, \eta)$. There is $\eta > 0$ such that, for every $x \in \mathcal{D}_k$ with $x \notin C_R$, one may select an all-clock stopping time τ_x satisfying:

1. τ_x contains at least one ordinary off-diagonal physical jump;
2. $\tau_x < \infty$ almost surely and $\mathbb{E}_x \tau_x < \infty$;
3. its endpoint is the actual physical endpoint, with $\mathbb{E}_x H(X_{\tau_x}) < \infty$; and
4. the exact stopped workload inequality is

$$\mathbb{E}_x [H(X_{\tau_x}) - H(x) + \eta \tau_x] \leq 0. \quad (2.2)$$

The selection is a function of the current state and the fixed network data. No condition is imposed on a future activation or reaction label.

A uniform negative constant on the right of (2.2) is stronger than needed. The positive physical-time term is the coercive quantity.

Here is the stopped-integrability point in full. For any almost surely finite all-clock stopping time τ with $\mathbb{E}_x \tau < \infty$, let B_τ be the aggregate number of zero-source births and let $N_{u,\tau}$ count the labels $u \rightarrow 0$. Nonexplosion makes these counts finite pathwise, and the exact workload balance is

$$H(X_\tau) - H(x) = s \left(B_\tau - \sum_{u \in D} N_{u,\tau} \right). \quad (2.3)$$

The aggregate birth clock has constant intensity β . The stopped counting-process compensation formula, obtained first under bounded time-and-count localization and then by monotone convergence, gives

$$\mathbb{E}_x B_\tau = \beta \mathbb{E}_x \tau. \quad (2.4)$$

Since $H(X_\tau) \geq 0$, (2.3) gives the pathwise domination

$$\sum_{u \in D} N_{u,\tau} \leq H(x)/s + B_\tau. \quad (2.5)$$

Thus every direct-death count is integrable. Applying the same localized compensation formula to those counts now gives

$$\mathbb{E}_x \sum_{u \in D} N_{u,\tau} = \mathbb{E}_x \int_0^\tau \sum_{u \in D} \delta_u X_u(t) dt < \infty. \quad (2.6)$$

Equations (2.3)–(2.6) also prove $\mathbb{E}_x H(X_\tau) < \infty$; endpoint integrability is a consequence of duration integrability, not an additional moment demand. Taking expectations in (2.3) yields the exact stopped Dynkin identity

$$\mathbb{E}_x[H(X_\tau) - H(x)] = s\beta\mathbb{E}_x\tau - s\mathbb{E}_x \int_0^\tau \sum_{u \in D} \delta_u X_u(t) dt. \quad (2.7)$$

Consequently the precise minimal occupation form of (2.2) is exactly

$$\mathbb{E}_x \int_0^{\tau_x} \sum_{u \in D} \delta_u X_u(t) dt \geq \left(\beta + \frac{\eta}{s}\right) \mathbb{E}_x \tau_x. \quad (2.8)$$

Equivalently, with $D_\tau = \sum_{u \in D} N_{u,\tau}$,

$$s\mathbb{E}_x(D_\tau - B_\tau) \geq \eta\mathbb{E}_x\tau_x. \quad (2.9)$$

Adding any fixed $a > 0$ to the right of (2.9), or $-a$ to the right of (2.2), is useful slack but is not part of the minimal theorem.

26.3 3. Pointwise complement needs no macro

Let

$$\delta_* = \min_{u \in D} \delta_u, \quad c_k = s(\delta_* k - \beta). \quad (3.1)$$

Increase k , if necessary, so $c_k > 0$, and put

$$\eta_0 = \min\{\eta, c_k/2\} > 0. \quad (3.2)$$

At $x \notin \mathcal{D}_k$, (1.4) gives

$$\mathcal{L}H(x) \leq -c_k. \quad (3.3)$$

Let T_1 be the next ordinary off-diagonal jump time. In a nontrivial irreducible class its rate $q(x)$ is finite and positive. The elementary holding-time identity gives

$$\mathbb{E}_x\{H(X_{T_1}) - H(x) + \eta_0 T_1\} = \frac{\mathcal{L}H(x) + \eta_0}{q(x)} \leq 0. \quad (3.4)$$

The margin in (3.4) may vanish as $q(x) \rightarrow \infty$. This is harmless: the proof below sums physical time, not embedded episode count.

26.4 4. Positive-duration episode tiling

For $x \in C_R$, set $N = 0$ and $S_N = 0$. Starting from $x \notin C_R$, define stopping times S_n recursively. At S_n :

- if $X_{S_n} \in \mathcal{D}_k$, append the macro of Section 2;
- otherwise append the one-jump episode of Section 3.

Stop the recursion when an episode endpoint lies in C_R . Every nonterminal episode contains at least one actual off-diagonal jump. Chart classification, active-set relabelling, or an algebraic declaration is not an episode and consumes no index; all zero-time declarations are folded into the rule which takes the next physical jump.

By the strong Markov property and (2.2), (3.4),

$$\mathbb{E}_x [H(X_{S_{(n+1) \wedge N}}) - H(X_{S_{n \wedge N}}) + \eta_0(S_{(n+1) \wedge N} - S_{n \wedge N}) | \mathcal{F}_{S_{n \wedge N}}] \leq 0, \quad (4.1)$$

where N is the first endpoint index in C_R . Summing gives

$$\mathbb{E}_x H(X_{S_{n \wedge N}}) + \eta_0 \mathbb{E}_x S_{n \wedge N} \leq H(x). \quad (4.2)$$

The induction behind (4.2) is integrable, not merely formal. At every nonterminal state, either (2.2) or (3.4) rearranges to

$$\mathbb{E}[H(\text{next endpoint}) + \eta_0(\text{duration}) | \text{current state}] \leq H(\text{current state}). \quad (4.3)$$

Both terms on the left are nonnegative. Starting from deterministic finite $H(x)$, (4.3) therefore supplies all first moments needed at the next induction step. The stopped count calculation (2.3)–(2.7) proves the same fact directly from episode-duration integrability. No polynomial or exponential endpoint moment is used.

26.5 5. Conditional Foster theorem

Theorem 5.1 (workload-only physical-time Foster theorem). Assume the reduced level-set network (1.1) is nonexplosive and satisfies the occupation macro $\mathbf{M}(k, R, \eta)$ for some k, R, η , with k large enough that the quantity in (3.1) is positive. Then

$$\mathbb{E}_x S_N \leq \frac{H(x)}{\eta_0} < \infty, \quad (5.1)$$

so every state has finite mean physical hitting time of the finite set C_R . The closed irreducible class Γ is positive recurrent.

26.5.1 Proof

Let

$$S_\infty = \lim_{n \rightarrow \infty} S_{n \wedge N}.$$

Since $H \geq 0$, monotone convergence for the times in (4.2) gives

$$\eta_0 \mathbb{E} S_\infty \leq H(x). \quad (5.2)$$

On $\{N = \infty\}$, the episode endpoints contain an infinite subsequence of ordinary physical jump times. Nonexplosion forces their limit to be infinite. Hence (5.2) implies $\mathbb{P}(N = \infty) = 0$, and (5.1) follows.

For completeness, fix $o \in C_R$. The set C_R is finite. For each $c \in C_R$, irreducibility supplies a finite actual labelled population path from c to o . An attempt follows that path until either its next prescribed label fires or a competing label fires, in which case the attempt stops immediately. Across the finitely many chosen path states, the probability of completing the prescribed path has a positive uniform lower bound p , and the mean duration of one attempt has a finite uniform upper bound a . The possible first competing endpoints form a finite set E . By (5.1),

$$m = \begin{cases} \max_{e \in E} \mathbb{E}_e S_N, & E \neq \emptyset, \\ 0, & E = \emptyset \end{cases} < \infty. \quad (5.3)$$

After a failed attempt, wait for this endpoint return to C_R , then use the path selected from the new state. At every restart the conditional success probability is at least p , while the conditional mean cost of the failed attempt and return is at most $a + m$. Thus the number of attempts is dominated by a geometric variable and the mean hit of o is finite.

If Γ is a singleton there is nothing more to prove. Otherwise the first ordinary jump from o has finite mean holding time and one of finitely many endpoints. Applying the preceding finite-mean hit bound from each such endpoint gives a finite mean return time to o . Thus Γ is positive recurrent. $\square$

26.6 6. Nonexplosion and fixed-class exceptions

For a binary network, nonexplosion is automatic here. A degree-two source cannot increase total molecule count because every target also has degree at most two. Every population-increasing channel therefore has source degree zero or one. The aggregate positive population drift is bounded by $C(1 + |x|_1)$, and all jumps are bounded. Localization followed by a Yule comparison prevents population escape in finite time. On a fixed population sublevel there are finitely many states and bounded total rates, so population-preserving upper reactions cannot accumulate infinitely many jumps.

For the exact at-most-three-species $h = (1, 1, 1)$ and permuted $h = (1, 1, 2)$ families, there are only two genuine reduction exceptions.

1. If the reduced class has no active reaction, irreducibility makes it a singleton.
2. On an exceptional common-catalyst face, every upper source contains a species Y , the face has $Y = 0$, and Y is absent from the lower unary support. Then Y is an exact full-network invariant. It is fixed before the theorem is invoked, the upper linkage is inactive on that class, and the remaining network is an open unary linkage.

The latter reduction is positive recurrent directly. Kill the unary complex graph on first hitting zero. Its subgenerator Q is transient by strong connectivity, so $v = (-Q)^{-1} \mathbf{1} > 0$. The linear function $v \cdot x$ has generator equal to a constant immigration term minus $\sum_i x_i$, up to the fixed labelled-rate convention. It is a standard linear Foster function. Equivalently, the open unary linkage is weakly reversible and deficiency zero.

No other closed no-service face is hidden in the theorem. If an upper source is active on a proper face, a strong upper path leaks from that face; if the upper linkage is dead, strong connectivity of the open lower linkage gives finite physical access to a direct-death unary unless the required coordinate is the invariant just described.

26.7 7. Exact remaining macro for the two weights

For either weight, the hard analysis has exactly the same target. Fix the network, orientation, rates, class, k , and a state $x \in \mathcal{D}_k$ with $H(x)$ large. Construct one all-clock stopping time satisfying

$$\mathbb{E}_x \int_0^{\tau_x} \sum_{u \in D} \delta_u X_u(t) dt \geq \left(\beta + \frac{\eta}{s} \right) \mathbb{E}_x \tau_x, \quad (7.1)$$

with $0 < \mathbb{E}_x \tau_x < \infty$. That is the precise minimal macro. It need not bound the number of neutral upper reactions, reach a prescribed complex by a fixed word, mix before the first lower event, or provide any moment beyond first physical time.

For $h = (1, 1, 1)$, a symbolic proof may use only the three exhaustive upper activation geometries:

- a killed two-carrier Perron–Frobenius wedge;
- the dyadic two-minimum source-balance kernel; and

- the exact common-catalyst linear-chain kernel.

For completeness, fix a dormant pure- X ray and write the other species as Y, Z . Dormancy excludes $2X$. Rank two forces at least one of XY, XZ . Up to swapping Y, Z : if both occur, one has the two-carrier kernel; if only XY occurs and $2Z$ occurs, one has the dyadic two-minimum kernel; if only XY occurs and $2Z$ is absent, rank two forces the exact support $\{XY, YZ, 2Y\}$, the common- Y -catalyst kernel. This proves the symbolic exhaustion without listing orientations.

For $h = (1, 1, 2)$, the exact upper supports are

$$\{C, 2A, 2B\}, \quad \{C, 2A, A + B\}, \quad \{C, 2B, A + B\}, \quad \{C, 2A, 2B, A + B\}. \quad (7.2)$$

Indeed the level-two shell is $\{C, 2A, A + B, 2B\}$; its three quadratic vertices are collinear, so rank two is equivalent to including C and at least two of those vertices.

Supports containing the opposite pure double use the two-reservoir source-balance kernel for propensities C and $B(B - 1)$. The sole no-opposite-double support $\{C, A + B, 2A\}$ uses the killed-complex linear ascent to an A -occupied region, followed by an occupation/service handoff. A false uniform reaction-depth or $O(H^{-1/2})$ entrance claim is unnecessary: (7.1) charges all service accumulated during the complete physical macro.

Once (7.1) is proved in these kernels, Theorem 5.1 performs the entire global classwise composition under the single function H .

27 Homogeneous two-carrier and dyadic activation with every lower support

Proof-first standalone theorem, 2026-08-12 PDT. This note proves the all-clock activation-or-ledger macro for the first two symbolic kernels at a top-dead homogeneous pure vertex. It covers every optional top support compatible with the rank-two hypothesis, every lower support of two or three unaries, every strong labelled orientation, and every fixed positive rate vector. It does not enumerate orientations, paths, populations, or rate regimes.

After relabelling the dead bulk vertex as X , write Y, Z for the two transverse species. The upper support T is a strongly connected subset of the homogeneous quadratic shell, $2X \notin T$, and

$$\dim \text{span}(T - T) = 2. \quad (0.1)$$

The two kernels are:

1. **two carrier:** $X + Y, X + Z \in T$;
2. **dyadic:** $T = \{X + Y, 2Z\} \cup Q$, where $\emptyset \neq Q \subseteq \{Y + Z, 2Y\}$, and $X + Z \notin T$.

Thus the carrier kernel has at least one vertex outside $\{X + Y, X + Z\}$, and the three possible dyadic supports are exactly $\{X + Y, 2Z, Y + Z\}$, $\{X + Y, 2Z, 2Y\}$, and their union. The rank-one support $\{X + Y, 2Z\}$ is not in the theorem.

The lower linkage is $R = \{0\} \cup U$, where U consists of two or three unaries and the labelled graph on R is strongly connected.

27.1 1. Exact theorem, disjoint wedges, and the workload ledger

Put $H = X + Y + Z$. Let B_t count all zero-source births and D_t all labelled direct lower deaths after the beginning of the episode. Every top reaction and every nonzero lower transfer preserves H , so

$$H(X_t) - H(X_0) = B_t - D_t. \quad (1.1)$$

There are only finitely many top-dead pure vertices. At each such vertex d , choose the positive transverse linear coordinate S_d constructed in Section 3 or Section 4, after relabelling d as X . On the normalized simplex Δ , S_d vanishes only at d . Hence a fixed sufficiently small $\varepsilon_0 > 0$ makes the normalized wedge neighborhoods

$$\widehat{\mathcal{W}}_d = \{u \in \Delta : S_d(u) < \varepsilon_0\} \quad (1.2)$$

have pairwise disjoint closures. Let $\mathcal{W}_d$ be the population lift $\{x : x/H(x) \in \widehat{\mathcal{W}}_d\}$. Define the single global activated region

$$\mathcal{A} = \Delta \cap \left(\bigcup_d \widehat{\mathcal{W}}_d \right)^c. \quad (1.3)$$

It is compact. Reaction jumps are uniformly bounded. Consequently, after enlarging the finite set once more, the first actual jump out of $\mathcal{W}_X$ cannot land in another $\mathcal{W}_d$; its normalized endpoint is literally in $\mathcal{A}$.

Theorem 1.1 (activation or direct-death ledger). For each fixed network in either kernel there are $R, C < \infty$, independent of the integer $L \geq 1$, such that the following holds. From every x with $H_0 := H(x) \geq R$ in $\mathcal{W}_X$, there is an all-clock stopping time $\sigma_{x,L}$ satisfying

$$\mathbb{E}_x B_{\sigma_{x,L}} \leq C, \quad \mathbb{E}_x \sigma_{x,L} \leq C, \quad (1.4)$$

and one of the following *labelled* alternatives holds at its actual endpoint. Labels are assigned after the terminal jump in the priority order $F > D > I$:

$$\begin{aligned} F : \quad & H(X_{\sigma_{x,L}}) \leq H_0/2, \\ D : \quad & H(X_{\sigma_{x,L}}) > H_0/2, \quad D_{\sigma_{x,L}} \geq L, \\ I : \quad & H(X_{\sigma_{x,L}}) > H_0/2, \quad D_{\sigma_{x,L}} < L, \quad X_{\sigma_{x,L}}/H(X_{\sigma_{x,L}}) \in \mathcal{A}. \end{aligned} \quad (1.5)$$

Every episode contains at least one ordinary physical jump. Every clock, including every direct death, remains present throughout the construction.

The independence of C from L is essential. We first construct a base time σ_∞ , stopping only at F or at the first exit from $\mathcal{W}_X$, with

$$\sup_{H_0 \geq R} \mathbb{E}_x \sigma_\infty \leq C, \quad \sup_{H_0 \geq R} \mathbb{E}_x B_{\sigma_\infty} \leq C. \quad (1.6)$$

Deaths are not absorbing in this base construction. If ρ_L is the actual time of the L -th direct death, set

$$\sigma_{x,L} = \sigma_\infty \wedge \rho_L. \quad (1.7)$$

Then (1.4) follows by monotonicity, and (1.5) is exhaustive with the stated priority. Thus there is no circular choice of L against an L -dependent prelude debt.

27.2 2. Lower-linkage sign and literal finite establishment

Let δ_X be the total rate coefficient of direct arrows $X \rightarrow 0$. Inside $\mathcal{W}_X$, $X \geq (1 - C\varepsilon_0)H$. If $\delta_X > 0$, then, up to the exit of the wedge or F ,

$$\mathcal{L}H \leq \beta_0 - cH, \quad (2.1)$$

where β_0 is the total constant birth rate. Stopped Dynkin applied to H , followed by localization, gives a uniform mean time to the exit or F , and the birth compensator gives the second bound in (1.6). This proves the theorem in that branch. We may therefore assume in the hard branch that

$$\delta_X = 0. \quad (2.2)$$

If $X \in U$, every lower arrow sourced at X is then a transfer to Y or Z . It strictly increases the transverse height S used below. The only lower arrows that can decrease a transverse coordinate are sourced in Y, Z , with aggregate rate

$$\lambda_{\text{adv}} \leq C(1 + Y + Z). \quad (2.3)$$

Thus an order- H X -source clock is favorable and is never hidden in a bounded error term.

Fix a sufficiently large integer K . We now prove, without a prescribed slow reaction word, the uniform establishment estimate on $0 \leq S < K$.

Two carriers. Use the height $S = v_Y Y + v_Z Z$ of Section 3. At $S = 0$ the top linkage is disabled. If $X \in U$, (2.2) and strong connectivity give an $X \rightarrow Y$ or $X \rightarrow Z$ arrow of intensity at least cX . If $X \notin U$, then $U = \{Y, Z\}$, and an arrow out of 0 gives a transverse birth at a fixed positive rate. Thus a state with $0 < S \leq C_0$ is reached in uniformly bounded mean time and birth count.

On $0 < S < K$, the exact killed-resolvent estimate (3.2), the favorable sign of every X -source lower transfer, and (2.3) give, for large H_0 and before F or wedge exit,

$$\mathcal{L}S \geq cH_0S, \quad \Gamma S \leq CH_0S. \quad (2.4)$$

Indeed $H \geq H_0/2$, $X \geq cH$, and S is bounded. Stop a seeded trial at $S = 0$, $S \geq K$, F , or wedge exit. For a fixed small $\theta > 0$, Taylor's formula makes $e^{-\theta S}$ a supermartingale on the open interval. Since a nonzero population has $S \geq s_* := \min(v_Y, v_Z) > 0$, optional stopping gives

$$\inf \mathbb{P}\{S \geq K, F, \text{ or wedge exit before } S = 0\} \geq 1 - e^{-\theta s_*} =: p_K > 0. \quad (2.5)$$

Stopped Dynkin applied to S , whose endpoint is bounded by K plus one jump, gives mean seeded-trial time $O(H_0^{-1})$. After a return to $S = 0$, run the same all-clock seed mechanism again. Geometric repetition of (2.5) has uniformly bounded mean duration and birth count. All fast carrier firings and all deaths remain present.

Dyadic kernel. Here $S = 2Y + Z$. First suppose $X \in U$. By (2.2) some $X \rightarrow Y$ or $X \rightarrow Z$ arrow has intensity at least cX , and every such firing raises S . While $S < K$, all clocks capable of lowering S are sourced in Y, Z or at an optional high top complex, and their total intensity is $O_K(1)$. Neutral minimum-source firings do not affect this race. Before F or wedge exit, $X \geq cH_0$. Consequently the probability that K good X -source firings occur before the first adverse firing is bounded below uniformly, and their mean physical time is $O(H_0^{-1})$. This reaches $S \geq K$; an adverse firing merely restarts the trial at its actual finite transverse state.

It remains to treat $X \notin U$, hence $U = \{Y, Z\}$. On $S < K$, the pair (Y, Z) ranges over a finite set. We quotient only the order- H firings sourced at $B = X + Y$. A $B \rightarrow A = 2Z$ firing is S -neutral and lowers Y by one; a B -firing to $Y + Z$ or $2Y$ raises S . Thus at most K consecutive fast neutral firings can occur before either strict progress or a state with $Y = 0$.

At $S = 0$, a lower birth creates Y or one Z at fixed positive rate. A lone Z either moves to Y , receives a second Z birth, or is lost and restarts; the second-arrival-or- Y event has fixed positive probability. Once $Z \geq 2$, the source $A = 2Z$ has fixed positive intensity. Its firing either makes a strict cut or reaches B . In the latter case the order- H B -clock fires before the bounded slow competitors with probability tending to one, and it either makes a strict cut or returns to A . The pair $\{A, B\}$ cannot be closed in the strong top graph because $Q \neq \emptyset$. Hence at least one of A, B has a strict outgoing label, with a fixed positive same-source label probability.

This gives a finite quotient chain with no closed unsuccessful class: from every $S < K$ phase, a bounded number of slow A /lower steps and contracted fast B -steps has a fixed positive probability of raising S , reaching F , or exiting the wedge. Concatenating at most K such strict increments gives one uniform positive probability of reaching $S \geq K$. Every wait for a slow step has total progress-or-reset intensity bounded below by a fixed constant; fast steps only shorten it. Geometric repetition from the actual endpoint therefore has uniformly bounded mean physical duration. This explicitly includes the lone- Z resistance-two phase and the neutral $A \leftrightarrow B$ loop.

In all cases the constant-rate birth compensator now proves

$$\sup_{H_0 \geq R} \mathbb{E}_x \tau_K \leq C_K, \quad \sup_{H_0 \geq R} \mathbb{E}_x B_{\tau_K} \leq \beta_0 C_K. \quad (2.6)$$

where τ_K stops at $S \geq K$, wedge exit, or F , but not at a death. This is the required literal establishment. It uses no population-dependent slow top word and no conditional-Poisson shortcut.

27.3 3. Two carriers: killed resolvent and multiplicative ascent

Kill the strong top graph when it first exits the carrier set $\{X + Y, X + Z\}$ to any other top complex. The carrier subset cannot be closed: otherwise strong connectivity would make it the whole top graph, whose stoichiometric rank would be one rather than two. Its killed two-state subgenerator Q is therefore transient. Put

$$g = (-Q)^{-1} \mathbf{1} > 0, \quad v_i = 1 - \epsilon g_i > 0 \quad (i = Y, Z), \quad S = v_Y Y + v_Z Z. \quad (3.1)$$

For sufficiently small fixed $\epsilon > 0$, the rate-weighted S -reward at each carrier source is strictly positive. Every noncarrier source has propensity $O((Y + Z)^2)$ and bounded S -jump. Exactly for falling-factorial propensities,

$$\mathcal{L}_T S \geq cXS - CS^2. \quad (3.2)$$

Every X -sourced lower transfer is favorable by Section 2. Direct deaths of Y, Z and all other adverse lower terms contribute only $O(1 + S)$. Choose ϵ_0 small and K large. On $K \leq S < \epsilon_0 H$, before F ,

$$\mathcal{L}S \geq cHS, \quad \Gamma S := \sum_e \lambda_e (\Delta_e S)^2 \leq CHS + CS^2 \leq C'HS. \quad (3.3)$$

The last inequality uses $S \leq \epsilon_0 H$; the fixed and order- H lower variances are absorbed because $S \geq K$. All S -jumps are bounded.

Here are the stopped estimates used in the ascent. Enter a band at $S \in [r, r + C_0]$ and stop at $S \leq r/2$, $S \geq 2r$, wedge exit, or F . Give wedge exit and F cemetery value zero. Taylor's formula and (3.3), with a fixed sufficiently small $\theta > 0$, give on the open band

$$\mathcal{L}^\dagger e^{-\theta S} \leq -cHSe^{-\theta S}. \quad (3.4)$$

Optional stopping of this nonnegative supermartingale yields

$$\mathbb{P}\{S \leq r/2 \text{ before the upper or favorable exits}\} \leq Ce^{-cr}. \quad (3.5)$$

For the time estimate, do not use the killed function. Let J bound the absolute S -jump and take $K > 2J$. Define a bounded function f_r that equals $\log S$ throughout the open band and at every state reachable from it in one jump, and clamp it arbitrarily outside that one-jump enlargement. Thus $\mathcal{L}f_r = \mathcal{L} \log S$ before the stopping time. The bounded-jump Taylor formula gives

$$\mathcal{L}f_r \geq \frac{\mathcal{L}S}{S} - C \frac{\Gamma S}{S^2} \geq cH \quad (3.6)$$

after increasing K . Every actual stopped endpoint has $r/2 - J \leq S \leq 2r + J$, so its f_r -value differs from the starting value by at most $\log 4 + O(r^{-1})$. Before the wedge exit and F , one has $H \geq cr$. Dynkin's formula therefore gives

$$\mathbb{E}t_r \leq \frac{C}{r}. \quad (3.7)$$

No exponential holding-time claim is needed.

Sum (3.5)–(3.7) over the dyadic bands $K, 2K, 4K, \dots$. The error sum is at most Ce^{-cK} , the physical-time sum is at most C/K , and this remains true if births move the terminal wedge boundary outward. Choosing K large gives a fixed positive chance that one establishment-plus-ascent attempt exits $\mathcal{W}_X$ or reaches F . A downward return simply restarts at its actual state. Geometric repetition with (2.6) proves (1.6) in the two-carrier kernel. Direct deaths have remained active throughout.

27.4 4. Dyadic kernel: aggregate minimum clocks and exact source balance

Put

$$S = 2Y + Z. \quad (4.1)$$

The minimum-height sources are

$$A = 2Z, \quad B = X + Y, \quad (4.2)$$

both of S -height two. Optional sources $Y + Z, 2Y$ have heights three and four. In a band

$$r/2 < S < 2r, \quad r \leq \varepsilon_0 H, \quad (4.3)$$

we have $X \geq H - S \geq (1 - \varepsilon_0)H$, and for large $K \leq r$

$$\lambda_{\min} = a_A Z(Z - 1) + a_B XY \geq c\{Z(Z - 1) + HY\} \geq cr^2. \quad (4.4)$$

Call optional high-source reactions, adverse Y/Z -sourced lower events, and births **bad**. Direct Y/Z deaths are included, not stopped. Since the optional-source propensities are $O(YS)$, their aggregate intensity satisfies

$$\frac{\lambda_{\text{bad}}}{\lambda_{\min}} \leq C(\varepsilon_0 + r^{-1}). \quad (4.5)$$

Lower transfers sourced at X are **good**: each increases S by one or two. Their intensity may be order H . They are retained in the actual prefix and are never placed in (4.5).

Consider a reaction prefix that remains in the band. Let m_A, m_B be the numbers of minimum-source firings, e the number of bad firings, g the number of good X -source firings, and c_* the number of minimum-source firings whose target has strictly larger S -height. Every good firing and every strict cut increases S by at least one; a firing $A \leftrightarrow B$ has zero S -reward; only a bad firing can decrease S , by a bounded amount. Hence the exact height ledger gives

$$g + c_* \leq Cr + Ce. \quad (4.6)$$

Strong connectivity supplies a direct-cut source without enumerating orientations. If B has no strict outgoing cut, every B -firing goes to A , consumes one Y , and therefore

$$m_B \leq Y_0 + 2m_A + C(e + g). \quad (4.7)$$

Then A must have a strict outgoing cut. If A has no strict outgoing cut, every A -firing goes to B , consumes two Z , and

$$2m_A \leq Z_0 + 2m_B + C(e + g). \quad (4.8)$$

Then B must have a strict outgoing cut. If both have strict cuts, every minimum firing is already from a direct-cut source. Thus in all cases the number q_* of firings from direct-cut sources obeys

$$q_* \geq c(m_A + m_B) - Cr - C(e + g). \quad (4.9)$$

At any fixed source, all outgoing labels share the same source monomial. Consequently a strict label has a fixed conditional probability $p_* > 0$ at every direct-cut opportunity.

We now give the complete prefix estimate. Inspect $M = L_0 r$ successive actual nonterminal reactions, padding only after an exit for the probability calculation. At each pre-exit state the conditional chance of a bad reaction is at most $q = C(\varepsilon_0 + r^{-1})$. Choose L_0 large, then ε_0 small and K large enough that qL_0 is smaller than all fixed fractions below. Adaptive Chernoff gives, outside an event of probability Ce^{-cr} ,

$$e \leq 2qM. \quad (4.10)$$

If the prefix stays in the band, (4.6) first gives $g \leq Cr + Ce$. Hence all but $O(r + e)$ of the M reactions are minimum-source firings. Equations (4.9) and (4.6) then imply that a fixed fraction of M are direct-cut opportunities, whereas the height ledger permits only $O(r + e)$ successful cuts. A second adaptive Chernoff bound for the fixed label probability p_* contradicts this when L_0 is large. A lower exit before M requires at least cr bad negative reward and is covered by the same first Chernoff estimate. Therefore

$$\mathbb{P}\{\text{lower exit, or no exit by } M\} \leq Ce^{-cr}. \quad (4.11)$$

Restart M -reaction prefixes after a nonexit. Their number is geometrically dominated, so the expected reaction count before a band exit is $O(r)$. While the prefix remains in the band, (4.4) bounds every holding time by an exponential clock of rate cr^2 . Thus

$$\mathbb{E}t_r \leq \frac{C}{r}. \quad (4.12)$$

Equations (4.11)–(4.12), summed over dyadic bands, give a fixed positive chance and uniformly bounded mean time for one establishment-plus-ascent attempt to exit $\mathcal{W}_X$ or reach F . The sum $\sum_r C/r$ is finite over doubling r , even if births move the wedge boundary outward. Failed attempts restart from their actual state. The small-phase bound (2.6) and geometric repetition prove (1.6) in the dyadic kernel. In particular, direct deaths remain live; no prefix is restarted merely because a death occurred.

27.5 5. Truncation, integrability, and exact composition scope

The preceding constructions tile actual physical paths by establishment and band stopping times. Each tile contains a jump unless it ends at the same terminal jump as the previous tile; such zero-length classifier handoffs are concatenated and never counted as episodes. The geometric attempt bounds and (2.6), (3.7), or (4.12) give $\mathbb{E}\sigma_\infty < \infty$. The chain is nonexplosive under binary mass action: pathwise, $H_t \leq H_0 + N_t$, where N is the constant-rate zero-source birth Poisson process; on every bounded interval this bounds all binary hazards. Localization and monotone convergence therefore justify all stopped Dynkin and compensator identities. In particular,

$$\mathbb{E}B_{\sigma_\infty} = \beta_0 \mathbb{E}\sigma_\infty \leq C, \quad (5.1)$$

with the same C for every later truncation $\sigma_{x,L}$. This proves Theorem 1.1.

The residual support certificate identifies the homogeneous dead-ray incidences as

$$360 = 168 + 144 + 48, \quad (5.2)$$

where the first two summands are respectively the two-carrier and dyadic kernels proved here, and the final 48 are common-catalyst kernels. The common-catalyst family, including all four relative lower supports, is proved separately at exact SHA

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The present theorem deliberately stops at activation-or-ledger. A common deterministic top-service and stochastic compensator window can be appended at an I -endpoint. By (1.1), an F -endpoint already has $D - B \geq H_0/2$, while a D -endpoint has $D \geq L$. Because the birth debt C is independent of L , one may choose $L > C$ only after the theorem is applied. This is the exact interface required by the workload-only Foster composition.

The support identity is finite geometry only. No finite stochastic orientation, history, or population enumeration enters Sections 2–4.

28 Common catalyst with every residual lower support

Proof-first standalone theorem, 2026-08-12 PDT. This note proves the all-clock workload macro for the homogeneous common-catalyst top support

$$T = \{X + Y, Y + Z, 2Y\} \quad (1.1)$$

paired with every lower support occurring in the residual 336 family:

$$R = \{0\} \cup U, \quad U \in \{\{X, Y\}, \{X, Z\}, \{Y, Z\}, \{X, Y, Z\}\}. \quad (1.2)$$

Both supports carry arbitrary strongly connected labelled directed graphs and arbitrary fixed positive rates. The argument is symbolic and retains every clock. In particular, order- H lower activity sourced at X is not contracted, conditioned away globally, or treated as a bounded phase.

28.1 1. Exact workload theorem

Put $H = X + Y + Z$. Let B_t count all zero-source births and let D_t count all labelled direct lower deaths $u \rightarrow 0$. Every top reaction and every nonzero lower transfer preserves H , so pathwise

$$H(X_t) - H(X_0) = B_t - D_t. \quad (1.3)$$

Theorem 1.1 (general common-catalyst macro). Fix a network with supports (1.1)–(1.2). On every closed irreducible class which is not a catalyst-free invariant reduction, there are constants $R, C < \infty$, $a > 0$, and $\eta > 0$ such that, from every state x with $H(x) \geq R$, an all-clock stopping time τ_x can be selected with an actual physical endpoint and at least one ordinary jump, satisfying

$$0 < \mathbb{E}_x \tau_x \leq C, \quad \mathbb{E}_x (D_{\tau_x} - B_{\tau_x}) \geq a, \quad (1.4)$$

and therefore

$$\mathbb{E}_x \{H(X_{\tau_x}) - H(x) + \eta \tau_x\} \leq 0. \quad (1.5)$$

Moreover $\mathbb{E}_x H(X_{\tau_x}) < \infty$. If $Y \notin U$ and a class lies in $Y = 0$, that face is invariant, the top linkage is dead, and the class is an open-unary reduction; it is positive recurrent by a linear Foster function.

The theorem supplies exactly the occupation macro in the workload-only physical-time Foster theorem. It makes no bound on the number of neutral top reactions.

28.2 2. Labelled top particles and protected paths

Assume $Y \geq 1$, remove one persistent catalyst, and set

$$z_X = X, \quad z_Z = Z, \quad z_Y = Y - 1. \quad (2.1)$$

The three source propensities factor exactly:

$$(x)_{X+Y} = Y z_X, \quad (x)_{Y+Z} = Y z_Z, \quad (x)_{2Y} = Y z_Y. \quad (2.2)$$

After aggregating labelled top rates, let Q be the irreducible three-state generator on $\{X, Z, Y\}$. With

$$A(t) = \int_0^t Y(s) ds, \quad (2.3)$$

the $H - 1$ free labels move independently with generator Q in operational time:

$$\mathcal{L}_T = Y \mathcal{L}_Q. \quad (2.4)$$

Fix an operational block length $a_0 > 0$. For each initial type i , choose a simple Q -path to Y , of length at most two. For one initial free label define a protected event as follows.

1. In prescribed disjoint operational subintervals, its chosen top arrows occur and its competing top arrows do not.
2. Once it first reaches Y , every later top arrow out of Y is absent through operational time a_0 .
3. Until that first Y -hit, no lower unary clock sourced at X or Z acts on this label.

To realize the last requirement without conditioning on the block duration, use the standard per-molecule graphical construction for unary mass action: attach to each label an independent physical-time Poisson clock of rate K_* , where K_* dominates the total lower rate of one X - or Z -molecule, and require no mark in the first a_0 units of physical time. Thinning realizes every actual lower label clock. The top and lower graphical clocks are independent between distinct labels. On the global event of Section 3, $Y \geq 1$, so the physical block duration is at most a_0 . Hence there is a fixed

$$p_* \geq p_Q e^{-K_* a_0} > 0 \quad (2.5)$$

such that every initial free label is protected with probability at least p_* , independently over labels.

The per-label lower-clock requirement prevents a protected X - or Z -label from being transferred or killed before its terminal hit. Once it is at Y , the global clean event in Section 3 suppresses its lower Y -clock and the protected top event suppresses its top Y -out arrows. Arbitrary lower events involving other labels remain active and do not invalidate a protected success. One could allow a helpful $Z \rightarrow Y$ transfer as an earlier terminal hit, but the stronger no-lower-mark event is used here so that the protected indicators are literally independent.

Let S be the number of protected initial labels. Uniform binomial Chernoff gives constants $c > 0$ and n_0 such that, for every initial free-particle profile,

$$\mathbb{P}\{S \geq p_*(H - 1)/2\} \geq 1 - e^{-cH} \quad (H \geq n_0). \quad (2.6)$$

28.3 3. The global clean event

Let K_Y^R be the total lower rate per Y -molecule, with $K_Y^R = 0$ when $Y \notin U$, and let $\beta > 0$ be the aggregate zero-source birth rate. From $Y \geq 1$, run every clock until the first of

- an increment a_0 of A ;
- a lower event sourced at Y ;
- a zero-source birth; or
- the fractional-return stop defined in Section 5.

Call the block **unspoiled** when neither listed adverse clock occurs before the earlier of the operational endpoint and the fractional-return stop. Conditional on all top and all X/Z -sourced lower graphical histories, the top linkage preserves at least one catalyst before a Y -sourced lower event and lower X/Z -events cannot remove it. Hence $Y \geq 1$, and an operational interval of length a_0 takes at most a_0 physical time. The integrated adverse hazards before the earlier endpoint are bounded by

$$K_Y^R \int Y(t) dt + \beta t \leq (K_Y^R + \beta)a_0. \quad (3.1)$$

Therefore the conditional probability of the global unspoiled event is at least

$$q_0 = e^{-(K_Y^R + \beta)a_0} > 0. \quad (3.2)$$

The conditioning order is important: first reveal the protected-label events and every allowed lower history, and then suppress the disjoint birth and Y -source clocks. Combining (2.6) with (3.2) gives

$$\mathbb{P}\{F \text{ occurs, or the block reaches the shell}\} \geq q_0(1 - e^{-cH}). \quad (3.3)$$

On the fractional-return branch the macro has already succeeded. Otherwise the unspoiled block reaches its operational endpoint, and every protected label is still at Y . No birth occurred, so $H_{\text{end}} \leq H_{\text{start}}$. With $\rho = p_*/4$, for all large block-start workloads,

$$Y_{\text{end}} \geq \rho H_{\text{start}} \geq \rho H_{\text{end}}. \quad (3.4)$$

Thus one active block reaches a catalyst shell with a fixed probability, uniformly over every allowed lower support and all large initial profiles. All nonsuppressed lower deaths remain in D and are favorable.

28.4 4. Uniform seeding of the catalyst

Suppose $Y = 0$ and $Y \in U$. The top linkage is dead. Choose a simple directed lower path

$$0 = u_0 \longrightarrow u_1 \longrightarrow \cdots \longrightarrow u_m = Y, \quad (4.1)$$

with $m \leq 3$. Wait for the selected constant label $0 \rightarrow u_1$, tag that molecule, and require its successive selected unary labels. Before the final step, $Y = 0$, so no top clock can act. The tagged molecule's total unary rate is fixed, its selected branch probabilities are positive, and a deviation simply starts a new trial from the actual endpoint. Trials are geometrically dominated. Other molecules do not delay the selected constant clock or the tagged unary lifetimes; if another event creates Y , stop successfully, and count every direct death as favorable.

Consequently the seed time σ_s and seed birth count have uniform finite first moments:

$$\sup_{x:Y(x)=0} \mathbb{E}_x \sigma_s < \infty, \quad \sup_{x:Y(x)=0} \mathbb{E}_x B_{\sigma_s} < \infty, \quad Y(X_{\sigma_s}) \geq 1. \quad (4.2)$$

If $Y \notin U$, the coordinate face $Y = 0$ is invariant for the full network. On a class in this face the top linkage is dead and the remaining system is an open unary linkage. Killing its finite unary graph on reaching zero gives a transient subgenerator Q_R ; the positive vector $(-Q_R)^{-1} \mathbf{1}$ supplies a linear Foster function. Such a class is already positive recurrent. On any other class, $Y \geq 1$ and no seed is needed.

28.5 5. Geometric prelude

Fix a start x and put $H_0 = H(x)$. If already $Y \geq \rho H$, stop the prelude at time zero and proceed to service. Otherwise repeat complete attempts:

1. when $Y = 0$, use the seed of Section 4;
2. from $Y \geq 1$, use the active block of Section 3.

Stop the prelude immediately after any physical jump for which

$$H \leq H_0/2. \quad (5.1)$$

Also stop on the catalyst shell (3.4). Until (5.1), every active block starts above $H_0/2$; after enlarging the fixed workload threshold, (3.3) gives one conditional success probability at every attempt. The number of complete attempts is geometrically dominated. Each active block has physical duration at most a_0 and stops at its first adverse birth, so it adds at most one birth on failure. Each seed has the uniform bounds (4.2). Therefore constants independent of H_0 satisfy

$$\mathbb{E} \tau_{\text{pre}} \leq C_{\text{pre}}, \quad \mathbb{E} B_{\tau_{\text{pre}}} \leq C_B. \quad (5.2)$$

The endpoint alternatives are disjoint:

$$\begin{aligned} F : \quad & H \leq H_0/2, \quad D_{\text{pre}} - B_{\text{pre}} \geq H_0/2; \\ I : \quad & Y \geq \rho H, \quad H > H_0/2. \end{aligned} \quad (5.3)$$

All deaths in failed attempts are retained. No first-death restart or local ledger threshold is used.

28.6 6. Uniform service on the catalyst shell

Let

$$\delta_i = \sum_{i \rightarrow 0} \kappa_{i0}, \quad i \in \{X, Y, Z\}. \quad (6.1)$$

At least one δ_i is positive by strong connectivity of R . On the unit simplex, the top ODE is exactly

$$\frac{dq}{ds} = q_Y q_Q, \quad q(\theta) = q(0)e^{Q\theta}, \quad \frac{d\theta}{ds} = q_Y. \quad (6.2)$$

Uniformly over $q_Y(0) \geq \rho$, bounded operational intervals obey $q_Y(\theta) \geq \rho e^{-K\theta}$, so every fixed operational time is reached in uniformly finite physical time. Since Q is irreducible, $q(0)e^{Q\theta}$ converges uniformly over the simplex to its strictly positive invariant vector. Hence, after a uniform finite physical time, every coordinate is bounded below by one fixed positive constant. For every prescribed D_0 , a finite physical fluid horizon $T(D_0)$ therefore satisfies

$$\inf_{q_Y(0) \geq \rho} \int_0^{T(D_0)} \sum_i \delta_i q_i(s) ds \geq 2D_0. \quad (6.3)$$

Start the full chain at a shell endpoint of workload n , and run all clocks for physical time $T(D_0)/n$. On the rescaled interval the top density martingale has quadratic variation $O(n^{-1})$, the top drift is uniformly Lipschitz on the simplex, and lower unary and constant reactions make only $O(1)$ jumps and hence an $o(1)$ density perturbation. Localization and Gronwall give uniform L^1 convergence to (6.2). The direct-death compensator and (6.3) imply, for all sufficiently large n ,

$$\inf_{\text{shell}} \mathbb{E}D_{\text{serv}} \geq D_0, \quad \mathbb{E}B_{\text{serv}} = \frac{\beta T(D_0)}{n}. \quad (6.4)$$

Append one final ordinary jump. The total hazard is at least $\beta > 0$, so its holding time has mean at most $1/\beta$, and it adds at most one birth. This guarantees a genuine jump even when the macro starts on the shell and the deterministic service window contains no reaction.

28.7 7. Expected ledger closure

Choose

$$D_0 = C_B + 3. \quad (7.1)$$

Then enlarge R so that $R/2 \geq D_0$, the graphical block estimate is uniform above $R/2$, (6.4) holds, and the service-window expected birth count is at most one. On F , (5.3) gives net ledger at least D_0 . On I , discard all favorable prelude deaths, use conditional service at least D_0 , and subtract the prelude, service, and final-jump births. Thus

$$\begin{aligned} \mathbb{E}(D_\tau - B_\tau) &\geq D_0 \mathbb{P}(F) + D_0 \mathbb{P}(I) - (C_B + 2) \\ &\geq 1. \end{aligned} \quad (7.2)$$

This is an expectation-level ledger; no successful block is asked to repay the realized births in preceding failures. Equations (5.2), (6.4), and the final holding bound give $\mathbb{E}\tau \leq C < \infty$. Since $H(X_\tau) \leq H(x) + B_\tau$, endpoint workload is integrable. With $a = 1$ and $\eta = 1/(2C)$, (1.3) and (7.2) prove (1.4)–(1.5).

28.8 8. Classwise conclusion and scope

Population-increasing channels have at most linear propensity; quadratic top clocks preserve the proper workload and cannot accumulate in a finite sublevel. Thus the chain is nonexplosive. Every failure restarts at its actual endpoint by the strong Markov property, every clock remains active, and the complete macro contains a physical jump.

The theorem covers all four relative lower-support patterns in (1.2). It therefore repairs precisely the scope restriction of the two-unary $R = \{0, Y, Z\}$ operational theorem while preserving its physical-time ledger interface. Combined with the workload-only physical-time Foster theorem, it proves positive recurrence on every nonreduced fixed class; the only excluded face is an exact catalyst-free open-unary reduction, which is positive recurrent directly.

29 Homogeneous 336 workload occupation theorem

Proof-first standalone theorem, 2026-08-12 PDT. This note proves the missing global boundary estimate for the 312 homogeneous incidences in the residual level-set family. The proof is symbolic. It does not enumerate a reaction orientation, a rate vector, a population box, or a reaction path.

After relabelling, the two supports have the form

$$T \subseteq \{2A, 2B, 2C, A + B, A + C, B + C\}, \quad R = \{0\} \cup U, \quad (1.1)$$

where T has internal rank two, U consists of two or three distinct unaries, and both directed labelled graphs are strongly connected with arbitrary fixed positive rates. Put

$$H = A + B + C. \quad (1.2)$$

The theorem supplies the literal direct-death occupation macro required by the workload-only physical-time Foster theorem. The upper linkage may make arbitrarily many neutral reactions inside one episode.

29.1 1. Exact workload ledger and statement

For $u \in U$, write

$$\delta_u = \sum_{u \rightarrow 0} \kappa_{u0}, \quad D = \{u \in U : \delta_u > 0\}, \quad \beta = \sum_{0 \rightarrow u} \kappa_{0u}. \quad (1.3)$$

Strong connectivity gives $D \neq \emptyset$ and $\beta > 0$. If B_t is the number of zero-source births and D_t the number of labelled $u \rightarrow 0$ deaths up to time t , then every top reaction and every lower unary transfer preserves H , and hence, pathwise,

$$H(X_t) - H(X_0) = B_t - D_t. \quad (1.4)$$

Fix k and let

$$\mathcal{D}_k = \{x : x_u < k \text{ for every } u \in D\}. \quad (1.5)$$

Theorem 1.1 (homogeneous workload macro). For every fixed network in (1.1), on every closed irreducible class not contained in a catalyst-free invariant reduction, one can choose $k, R < \infty$ and $\eta > 0$ so that, from every $x \in \mathcal{D}_k$ in that class with $H(x) > R$, there is an all-clock stopping time τ_x which contains an actual reaction, has its actual physical endpoint, and satisfies

$$0 < \mathbb{E}_x \tau_x < \infty, \quad \mathbb{E}_x H(X_{\tau_x}) < \infty,$$

together with

$$\mathbb{E}_x \{H(X_{\tau_x}) - H(x) + \eta \tau_x\} \leq 0. \quad (1.6)$$

Equivalently,

$$\mathbb{E}_x (D_{\tau_x} - B_{\tau_x}) \geq \eta \mathbb{E}_x \tau_x. \quad (1.7)$$

Every catalyst-free invariant reduction is an open-unary network and is positive recurrent directly. Consequently every closed irreducible population class of (1.1) is positive recurrent.

The last conclusion follows from the workload-only physical-time Foster theorem and its independent strict audit, pinned respectively at

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9d8fc8b5e15178e7a8305422ba7fd08e6875e851c37951207815d5d84babcc67

Sections 2–7 prove the unconditional macro (1.6).

29.2 2. Dead faces and the finite state-selected partition

Normalize a nonzero population by $z = x/H(x)$ on the unit simplex. A coordinate face is **top-dead** if no source in T is supported on it. There are only two kinds of proper top-dead faces.

- A pure X -vertex is dead exactly when $2X \notin T$.
- A two-species face, say $Y = 0$, is dead only when every top complex contains Y . Rank two then forces

$$T = \{X + Y, Y + Z, 2Y\}. \quad (2.1)$$

Indeed, if the $Y = 0$ face is dead, the only available quadratic complexes are the three displayed in (2.1); rank two forces all three. Conversely they all contain Y . A singleton assertion is immediate.

If a proper face is not dead, an enabled source lies on it. A directed path in the strong top graph eventually leaves the face; its first crossing edge has strictly positive production of a missing coordinate. Repeating this argument enters the relative interior.

If (2.1) does not occur, the maximal top-dead sets are finitely many pure vertices. Choose small normalized wedges $\mathcal{W}_d$ around them with pairwise disjoint closures, and put

$$\mathcal{A} = \Delta \cap \left(\bigcup_d \mathcal{W}_d \right)^c, \quad \Delta = \{z \geq 0 : \sum_i z_i = 1\}. \quad (2.2)$$

The wedge thresholds are chosen so that exiting a wedge means crossing a fixed positive transverse-height level. Since all reaction jumps are bounded and distinct wedge closures have positive normalized separation, a large-population jump which first exits one wedge cannot land in another. Its normalized endpoint lies literally in $\mathcal{A}$. Every point of $\mathcal{A}$ has an enabled top source, and the first-crossing argument enters the relative interior.

When (2.1) occurs, its face $Y = 0$ is the unique maximal top-dead set and contains both dead pure endpoints. It is not covered by overlapping pure wedges. Section 5 invokes the separate all-lower-support catalyst theorem on the whole face neighborhood. Thus the state-selected partition has no ambiguous chart handoff and no zero-time episode.

There is one exact invariant reduction. In (2.1), if the catalyst Y is absent from U , the face $Y = 0$ is invariant for the full network. On a closed class in that face the top linkage is inactive and the remaining network is an open unary linkage. Killing its finite unary graph on first hitting zero gives a transient subgenerator Q_R , and $(-Q_R)^{-1}\mathbf{1} > 0$ gives a linear Foster function. Such a class is already positive recurrent. Hence below, a live catalyst face always has physical lower access to its catalyst.

29.3 3. Symbolic exhaustion and exact finite scope

Fix a dead pure X -vertex and call the other species Y, Z . Thus $2X \notin T$. Rank two forces at least one carrier $X + Y, X + Z$, since otherwise $T \subseteq \{2Y, Y + Z, 2Z\}$ has affine rank at most one. After interchanging Y, Z , precisely one of the following holds.

1. **Two carriers:** $X + Y, X + Z \in T$.
2. **Dyadic carrier:** $X + Y \in T$, $X + Z \notin T$, and $2Z \in T$.
3. **Common catalyst:** $X + Y \in T$, while $X + Z, 2Z \notin T$. The only remaining available quadratic vertices are $2Y, Y + Z$; rank two forces both and hence the exact support (2.1).

This is a symbolic trichotomy, not an orientation list. The lower support need not omit X . Every X -source lower transfer is retained below; if it is not a death, it creates transverse mass and is favorable.

The exact residual support certificate freezes the finite set identity at

```
src/all_active_residual_levelset_336_certificate.py
SHA-256 4149b682d1222bd3327548b0eb95921f7aae20663816b345b48285239c12f93d
```

```
tests/test_all_active_residual_levelset_336_certificate.py
SHA-256 6f5802976d4de479a0728648248a2291f5d518e04de29b9b7053802eb7f1b9c2
```


Its six focused tests give 312 homogeneous selected incidences and 360 dead-ray incidences, split exactly as

$$360 = 168 \text{ two-carrier} + 144 \text{ dyadic} + 48 \text{ common-catalyst.} \quad (3.1)$$

The dead-ray fingerprint is

c968fadc060af8225121efc84aa17380e11c41e677ed107d2d078c63d0f241fe

Exactly 270 dead rays have $X \in U$. Among the 48 catalyst rays the relative lower patterns $\{X, Y\}, \{X, Z\}, \{Y, Z\}, \{X, Y, Z\}$ occur 12 times each. These are support identities only. No orientation, rate, population, or stochastic history is part of the certificate.

29.4 4. The two-carrier and dyadic activation dependency

For a noncatalyst top support, choose the disjoint wedges in (2.2). The standalone carrier/dyadic theorem and its two exact-byte strict audits (first, independent) are frozen at

f4d8cc40ccea1c6d9e0df9302f75c8cc1d58dd7c89669fd19ad48fc4bca735b0
30d780d5853a956bd1502fae8517483caac9f0bd77606f6f6bc13caccb56d783
219a21a59f57a839d52c4fa0c7cdac3df3dacdac47e16655db957f0a8a026c62

They prove the following stronger statement. There is one constant $C_{\text{pre}} < \infty$, independent of the integer L , and an all-clock stopping time σ_L from every sufficiently large state in a dead pure wedge such that

$$\mathbb{E}B_{\sigma_L} \leq C_{\text{pre}}, \quad \mathbb{E}\sigma_L \leq C_{\text{pre}}, \quad (4.1)$$

and, with the priority order fractional return, then death ledger, then activation, its endpoint has exactly one label

$$\begin{aligned} F : & \quad H(X_{\sigma_L}) \leq H(x)/2, \\ D : & \quad H(X_{\sigma_L}) > H(x)/2, \quad D_{\sigma_L} \geq L, \\ I : & \quad H(X_{\sigma_L}) > H(x)/2, \quad D_{\sigma_L} < L, \quad X_{\sigma_L}/H(X_{\sigma_L}) \in \mathcal{A}. \end{aligned} \quad (4.2)$$

The construction first runs a base activation-or-fractional episode with every direct-death clock live, and only then stops it earlier at the L -th death. Thus (4.1) is independent of L ; there is no circular choice of a death threshold beyond an L -dependent birth debt. If the fixed support has more than one dead pure vertex, take the maximum of the finitely many C_{pre} 's and the maximum finite cutoff.

The dependency proves the sourcewise killed-carrier and dyadic source-balance estimates, the literal single- Z resistance-two establishment phase, the disjoint-wedge actual endpoint, and the L -independent birth/time debt. It assumes exactly $\dim \text{span}(T - T) = 2$, so it excludes the rank-one support $\{X + Y, 2Z\}$, which is absent from the certified residual family. No finite orientation or population selector is used.

29.5 5. The common-catalyst dependency

When (2.1) holds, the whole top-dead face must be treated at once. The standalone all-lower-support theorem, frozen at

81a48c007e092570cd500d8f124c0546538d44f7e62599100ecf00480f401496

proves Theorem 1.1 directly for all four lower patterns. Its two independent exact-byte strict audits (first, independent) are frozen at

5a85a422345a2dd8d640dcd2986f31cd035d293d8bfbb9735941915e547c39f8
4fbf8fd6384940e5a8919270e4da6a2cbe2885fb465b68957ec51ce592f979f2

The proof uses the exact factorization

$$(x)_{X+Y} = Yz_X, \quad (x)_{Y+Z} = Yz_Z, \quad (x)_{2Y} = Yz_Y, \quad \mathcal{L}_T = Y\mathcal{L}_Q. \quad (5.1)$$

Protected label events use independent operational top clocks and independent dominating physical lower clocks. A separate clean event suppresses only births and Y -sourced lower events until the earlier of the

operational endpoint and fractional return. The repaired success union is therefore fractional return or a macroscopic catalyst shell. Geometric complete attempts have uniform mean birth debt and duration, and the shell service window closes the expected death-minus-birth ledger. If $Y \notin U$, a closed class in $Y = 0$ is an exact open-unary invariant reduction and is positive recurrent directly.

The catalyst theorem already includes its own shell service and expected ledger closure. Catalyst states therefore terminate through that theorem; they are not sent through the noncatalyst service window below.

29.6 6. Uniform physical service after noncatalyst activation

Let $\mathcal{A}$ be the compact activated set from Section 2 for a noncatalyst support. For every direct-death species $d \in D$ and every $M < \infty$, there is a finite top-fluid horizon $T = T(M)$ such that

$$\inf_{z(0) \in \mathcal{A}} \int_0^T z_d(s) ds > M. \quad (6.1)$$

Here is the boundary and compact-uniform argument in full. Write $\phi_t(z)$ for the normalized top ODE on the simplex $\Delta = \{z \geq 0 : \sum_i z_i = 1\}$. For each coordinate i , split its equation as

$$\dot{\phi}_i = P_i(\phi) - C_i(\phi), \quad P_i \geq 0, \quad C_i(w) \leq K_i w_i \quad (w \in \Delta). \quad (6.2)$$

The last inequality is sourcewise: a reaction can consume i only from a quadratic source containing i , and on Δ its monomial is at most w_i . Hence a positive coordinate stays positive by variation of constants. If a source complex y_0 is enabled at a boundary point, choose a directed walk in the strong complex graph which starts at y_0 and visits every complex. Along its first edge, every new target coordinate has strictly positive production. Equation (6.2) makes that coordinate positive and keeps it positive. Induction along the walk enables every complex. Rank two and conservation of total degree imply that all three species occur, so the trajectory enters Δ° in finite time. By construction, every point of $\mathcal{A}$ has such an enabled source.

The top system is autonomous, weakly reversible, and has one linkage class. Its stoichiometric subspace is the full plane $\{v : \sum_i v_i = 0\}$, because its rank is two. Thus its normalized positive stoichiometric class is exactly Δ° . Theorem 4.2 of Boros and Hofbauer, *Permanence of Weakly Reversible Mass-Action Systems with a Single Linkage Class*, applies with no deficiency restriction: every positive trajectory enters a compact subset of Δ° in finite time. Consequently

$$\int_0^\infty \phi_d(s; z) ds = \infty \quad (z \in \mathcal{A}). \quad (6.3)$$

Fix M . For each $z \in \mathcal{A}$, choose $T_z < \infty$ so that the integral through T_z is larger than $M + 1$. Finite-time continuous dependence of the polynomial ODE makes the same integral larger than M on a relative neighborhood U_z of z in Δ . Choose a finite subcover $U_{z_1}, \dots, U_{z_m}$ of $\mathcal{A}$ and put $T = \max_j T_{z_j}$. Since the integrand is nonnegative, this proves (6.1). This finite-subcover step, rather than pointwise permanence alone, is what makes the service horizon uniform up to the boundary.

Start the full stochastic chain at an activated endpoint of population N , put $Z^N(s) = N^{-1}X(s/N)$, and stop the fluid clock at

$$\zeta_N = T \wedge \inf\{s : H(X(s/N)) \notin [N/2, 2N]\}. \quad (6.4)$$

Before ζ_N , the total lower unary intensity is at most CN in physical time. Its count during $[0, T/N]$ is therefore dominated by a counting process with compensator CT ; divided by N , its density perturbation tends to zero uniformly. The top compensated density martingale has predictable quadratic variation $O(N^{-1})$, and replacing falling factorials by their quadratic monomials contributes $O(N^{-1})$ to the fluid drift. Doob's inequality and Gronwall's lemma consequently give

$$\sup_{z \in \mathcal{A} \cap N^{-1}\mathbb{Z}^3} \mathbb{E}_z \sup_{s \leq \zeta_N} |Z^N(s) - \phi_s(z)| \longrightarrow 0. \quad (6.5)$$

Leaving the localization interval requires at least $N/2$ population-changing lower events, whereas their stopped compensator is bounded by a constant depending only on T . The exponential counting martingale therefore gives

$$\sup_{z \in \mathcal{A} \cap N^{-1}\mathbb{Z}^3} \mathbb{P}_z\{\zeta_N < T\} \longrightarrow 0. \quad (6.6)$$

Let $D_{\text{win},d}$ count only the labelled $d \rightarrow 0$ reactions up to physical time ζ_N/N . Its exact compensator, (6.1), and (6.5)–(6.6) give

$$\begin{aligned} \inf_{\text{activated starts}} \mathbb{E}D_{\text{win},d} &= \delta_d \inf_{\text{activated starts}} \mathbb{E} \int_0^{\zeta_N} Z_d^N(s) ds \\ &\geq \delta_d \inf_{z \in \mathcal{A}} \int_0^T \phi_d(s; z) ds - o(1). \end{aligned} \quad (6.7)$$

The right side can be made arbitrarily large by choosing M . Meanwhile

$$\mathbb{E}B_{\text{win}} \leq \frac{\beta T}{N}, \quad \mathbb{E}\tau_{\text{win}} \leq \frac{T}{N}. \quad (6.8)$$

The stopped lower count has a uniformly bounded compensator, so its counting-process exponential martingale gives uniform exponential moments for $D_{\text{win},d}$, B_{win} , and every other lower-window count. The localization time is an actual physical endpoint when an exit occurs. To ensure a genuine reaction even when the deterministic window contains none, append the first ordinary all-clock jump after ζ_N/N . Its holding time is dominated by an exponential variable of rate $\beta > 0$, it adds at most one birth, and any death it adds is favorable. Thus the complete service block has an actual reaction, uniform finite mean duration and birth debt at most $1 + o(1)$, while its mean death service is still bounded below by (6.7). No top clock or lower clock has been suppressed.

29.7 7. Assembly of the occupation macro

From a low-direct-death state, use the following finite state-selected menu.

- For a common-catalyst support, apply the complete theorem of Section 5. It already gives (1.6), including its own shell service. On a catalyst-free invariant face use the open-unary reduction of Section 2.
- For a noncatalyst support, begin the service window of Section 6 directly on $\mathcal{A}$. In a dead pure-vertex wedge first use the carrier/dyadic episode of Section 4.

It remains only to assemble the noncatalyst branch. Let σ be its prelude endpoint, with disjoint labels F, D, I from (4.2); a start already in $\mathcal{A}$ has label I and zero prelude. The preceding sections give constants $C_{\text{pre}}, C_\tau < \infty$, independent of L , such that

$$\mathbb{E}B_\sigma \leq C_{\text{pre}}, \quad \mathbb{E}\sigma \leq C_\tau. \quad (7.1)$$

On F , the exact ledger (1.4) gives $D_\sigma - B_\sigma \geq H(x)/2$. Enlarge the workload cutoff so that this is at least L . On D , $D_\sigma \geq L$, while all expected prelude births are charged globally by C_{pre} . On I , fix any $d \in D$, append the service block, and choose its mean d -death threshold D_0 . Set $C_B = C_{\text{pre}} + 1$, charging the possible birth on the final actual jump of Section 6. By (6.7)–(6.8), for large starting workload,

$$\mathbb{E}(D_\tau - B_\tau) \geq L\{\mathbb{P}(F) + \mathbb{P}(D)\} + D_0\mathbb{P}(I) - C_B - o(1) \geq \min\{L, D_0\} - C_B - o(1). \quad (7.2)$$

First choose $L, D_0 > C_B + 4$, and then enlarge the finite workload cutoff so that the error in (7.2) is below one. The complete episode has a finite uniform mean duration, say at most C_* . Taking

$$\eta = \frac{1}{C_* + 1} \quad (7.3)$$

and, if necessary, increasing L, D_0 once more, gives

$$\mathbb{E}(D_\tau - B_\tau) \geq \eta \mathbb{E}\tau. \quad (7.4)$$

Every episode contains a physical jump. Its birth count and duration are integrable, so (1.4) gives endpoint integrability. Equations (1.4) and (7.4) are exactly (1.6). The pointwise one-jump rule outside $\mathcal{D}_k$, followed by the positive-duration episode tiling of the pinned workload theorem, gives finite mean hitting of a finite set and positive recurrence on every closed irreducible class. $\square$

29.8 8. Scope boundary

The theorem uses only the homogeneous rank-two shell, the open-unary lower support, strong connectivity, and fixed positive rates. Its three activation kernels are forced by rank two. The common-catalyst argument retains repeated deaths and reseeding, so it does not assume a uniformly positive probability of reaching the interior before the first death. It also makes no bound on neutral reaction depth.

The 24 anisotropic incidences are outside this note and are covered by the quotient-entropy physical-time theorem and its exact-byte audit, frozen at

9206aa2b07aa802e4d06a769b3b60d520b2dbd12752312497aa5b41156780d48
992448ad8b6520f014e783adb26a4f9b393b0e6a5f38c3a6262dd9b2fa0c1764

30 Anisotropic 336: quotient entropy and physical-time Foster

Proof-first standalone theorem, 2026-08-12 PDT. This note closes the global boundary problem for the 24 anisotropic incidences in the residual 336 family. After a coordinate permutation their supports are

$$\begin{aligned} h &= (1, 1, 2), & R &= \{0, A, B\}, \\ T &= \{C\} \cup Q, & Q &\subseteq \{2A, A + B, 2B\}, & |Q| &\geq 2. \end{aligned} \quad (1.1)$$

Each support carries an arbitrary strongly connected directed graph and arbitrary fixed positive labelled rates. No orientation, rate vector, reaction history, or population box is enumerated below. The proof uses one global quotient of the marked factorial entropy and a physical-time Foster lemma which permits the expected decrement per jump to vanish at rate H^{-1} .

30.1 1. Statement

Fix a closed irreducible population class of the stochastic mass-action chain generated by (1.1), and suppose the class is infinite. Carry the actual target t of the last physical reaction. Put

$$\begin{aligned} H &= A + B + 2C, & F(x, t) &= \sum_{i=A,B,C} \log((x_i - t_i)!), \\ V(x, t) &= H(x) + \epsilon \frac{F(x, t)}{H(x) + 1}. \end{aligned} \quad (1.2)$$

The first ordinary reaction supplies the initial mark; an absorbing state is already recurrent. The result proved here is the following.

Theorem 1.1 (global anisotropic level-set theorem). For sufficiently small fixed $\epsilon > 0$, there are a finite marked set K , a constant $\eta > 0$, and a finite menu of all-clock, state-selected physical episodes which tile the trajectory until K , contain at least one actual jump each, retain every competing clock, and satisfy

$$\mathbb{E}[V(Y_{S_{n+1}}) - V(Y_{S_n}) + \eta(S_{n+1} - S_n) \mid \mathcal{F}_{S_n}] \leq 0. \quad (1.3)$$

Every endpoint is the actual population and target mark of the last included reaction. Consequently $\mathbb{E}_y \tau_K \leq V(y)/\eta$, and the physical chain is nonexplosive and positive recurrent on its fixed irreducible class.

The proof is global across all coordinate faces. It does not assume that a compact $A, B = \Theta(\sqrt{H})$ shell is reached on a prescribed time scale, and it uses no stationary averaging of the neutral linkage.

30.2 2. Exact marked identities and global hazard

For a complex y , let

$$K_y = \sum_{e:s(e)=y} \kappa_e, \quad \lambda_y(x) = K_y(x)_y, \quad \Lambda(x) = \sum_y \lambda_y(x), \quad p_y = \frac{\lambda_y}{\Lambda}. \quad (2.1)$$

If the next labelled reaction is $e : y \rightarrow u$, with population increment $\zeta_e = u - y$, factorial cancellation gives

$$F(x + \zeta_e, u) - F(x, t) = \log \frac{(x)_t}{(x)_y} =: r_e(x, t). \quad (2.2)$$

Thus one ordinary all-clock jump has mean reward

$$D(x, t) = \sum_e \frac{\lambda_e(x)}{\Lambda(x)} r_e(x, t) = \log p_t - \sum_y p_y \log p_y - \log K_t + \sum_y p_y \log K_y, \quad (2.3)$$

and hence

$$D(x, t) \leq \log p_t + C, \quad \sup_{x,t} \sum_e \frac{\lambda_e(x)}{\Lambda(x)} [r_e(x, t)]_+^q < \infty \quad (2.4)$$

for every fixed $q < \infty$.

Every T -reaction preserves H . The lower linkage changes H only through zero-source births and unary-to-zero deaths. If

$$\beta = \sum_{0 \rightarrow u} \kappa_{0u}, \quad \delta_A = \sum_{A \rightarrow 0} \kappa_{A0}, \quad \delta_B = \sum_{B \rightarrow 0} \kappa_{B0}, \quad (2.5)$$

then

$$\mathcal{L}H = \beta - \delta_A A - \delta_B B. \quad (2.6)$$

At least one of δ_A, δ_B is positive. More importantly, the total hazard has the global lower bound

$$\Lambda(x) \geq c_0(H(x) + 1). \quad (2.7)$$

Indeed the strong graph on R gives a positive total outgoing rate at each of the sources A, B , while $C \in T$ gives one at C . Their propensities bound $A + B + C \geq H/2$. The constant source covers $H = 0$.

The quotient in (1.2) is proper because $V \geq H$. Since t is binary,

$$0 \leq F(x, t) \leq C(H + 1) \log(H + 2). \quad (2.8)$$

30.3 3. Exact quotient increment

Write $d_e = H(x + \zeta_e) - H(x)$. Here $d_e = 0$ on T and on lower unary transfers, $d_e = 1$ on lower births, and $d_e = -1$ on lower deaths. Combining (2.2) with (1.2) gives the exact identity

$$\begin{aligned} \Delta_e V &= d_e + \epsilon \left\{ \frac{F + r_e}{H + d_e + 1} - \frac{F}{H + 1} \right\} \\ &= d_e + \frac{\epsilon r_e}{H + 1} - \frac{\epsilon d_e (F + r_e)}{(H + 1)(H + d_e + 1)}. \end{aligned} \quad (3.1)$$

There is a useful sign which prevents a logarithmic denominator toll. Write $F' = F + r_e \geq 0$ for the endpoint factorial value. On a birth,

$$1 + \epsilon \left\{ \frac{F'}{H + 2} - \frac{F}{H + 1} \right\} \leq 1 + \frac{\epsilon r_e}{H + 1}, \quad (3.2)$$

because the omitted term is $-\epsilon F'/((H+1)(H+2))$. On a death,

$$-1 + \epsilon \left\{ \frac{F'}{H} - \frac{F}{H+1} \right\} = -1 + \frac{\epsilon r_e}{H+1} + \frac{\epsilon F'}{H(H+1)}. \quad (3.3)$$

By (2.8), the last term in (3.3) is $o(1)$, uniformly over one bounded-displacement jump. It can therefore be absorbed into half of the favorable workload death outside a finite set.

A birth has probability at most $\beta/\Lambda = O(H^{-1})$. Its workload cost has expected size

$$O(H^{-1}). \quad (3.4)$$

Every other reaction has $d_e = 0$, so its quotient reward is exactly $\epsilon r_e/(H+1)$. These estimates remain valid, with changed fixed constants, along every bounded-depth prescribed path: populations and H move by $O(1)$, and (2.7) continues to hold.

30.4 4. Rare-source Bellman episodes

Fix one linkage $L \in \{T, R\}$, an actual target $t \in L$, and a terminal complex $c \in L$. Choose a simple directed path

$$t = y_0 \longrightarrow \cdots \longrightarrow y_m = c. \quad (4.1)$$

Retain all clocks. Continue only on the designated label, stop at the actual endpoint of the first competitor, and after designated arrival at c take one final ordinary jump. Let $J(x, t; c)$ denote its expected total F -reward. If the source probability of c , at the deterministic success endpoint, tends to zero along an escaping sequence, the exact Bellman recursion

$$J_i = D_i + a_i J_{i+1} \quad (4.2)$$

and the first-vanishing-source argument give

$$J(x, t; c) \longrightarrow -\infty. \quad (4.3)$$

All positive F -reward moments and all jump counts are uniformly bounded. Equation (2.7) gives

$$\mathbb{E}\tau \leq \frac{C}{H+1}. \quad (4.4)$$

At every stage, use the sign in (3.2)–(3.3), discard the remaining favorable death contribution, and charge the expected births by (3.4). The stage workload differs from its initial value by $O(1)$. Moreover, whenever the current target and next source are enabled binary complexes,

$$|r_e(x, t)| \leq C \log(H+2). \quad (4.5)$$

Replacing each denominator $H_i + 1$ by the initial $H + 1$ therefore costs only $O(\log(H+2)/H^2)$ over the bounded episode. Consequently

$$\mathbb{E}\Delta V \leq \frac{\epsilon J(x, t; c) + C}{H+1} + \frac{C \log(H+2)}{(H+1)^2}. \quad (4.6)$$

Hence, once $-J$ is larger than a fixed constant, (4.4) and (4.6) imply (1.3), with one common sufficiently small $\eta > 0$. Sequential compactness over the finite menu makes this uniform outside a finite set. No fixed embedded-jump decrement is asserted or needed.

For the lower linkage use $c = 0$. Its propensity is constant, while $\Lambda \geq c_0 H$, so

$$p_0 \leq \frac{C}{H+1} \longrightarrow 0. \quad (4.7)$$

Thus every mark in R starts a rare-source episode.

30.5 5. Exhaustive top-source dichotomy

It remains to treat marks in T . For the actual target t and each $c \in T$, fix a simple path from t to c . Its deterministic success population is

$$z_c = x - t + c. \quad (5.1)$$

Because all members of T have the same H -level, $H(z_c) = H(x)$.

The quantitative Bellman recursion behind (4.3) has the following finite- menu consequence. There is a decreasing function $A(\delta) \uparrow \infty$ as $\delta \downarrow 0$ such that

$$p_c(z_c) \leq \delta \implies J(x, t; c) \leq -A(\delta). \quad (5.2)$$

Otherwise a sequence with $\delta \downarrow 0$ would contradict the first-vanishing-source recursion (4.2). Here is a direct quantitative version. Let $\alpha_* > 0$ be the smallest conditional branching probability of a labelled edge used by the finite path menu, and let C_0 be the constant in $D_i \leq \log p_{y_i} + C_0$. Set

$$B_0(\delta) = \log \delta + C_0, \quad B_{k+1}(\delta) = C_0 + \sup_{0 < p \leq 1} \{\log p + \alpha_* p B_k(\delta)\}. \quad (5.3)$$

Once $B_k < 0$, replacing an actual edge branching probability $\alpha \geq \alpha_*$ by α_* only increases the upper bound. Induction in (4.2) therefore gives $J \leq B_m(\delta)$ for a path of length m . If $B_k = -L$, the supremum in (5.3) is at most $-\log(\alpha_* L) - 1$ for large L . Thus every $B_k(\delta) \rightarrow -\infty$. Taking the maximum over the finitely many path lengths proves (5.2).

Choose $\delta > 0$ so small that (4.4),(4.6) imply (1.3) whenever (5.2) applies.

Suppose instead that

$$p_c(z_c) > \delta \quad (c \in T). \quad (5.4)$$

Apply this first to $c = C$. The global lower bound (2.7) at z_C and $\lambda_C(z_C) \geq \delta \Lambda(z_C)$ give

$$C(x) \geq cH, \quad \Lambda(z_C) \leq CH. \quad (5.5)$$

The states z_c , z_C , and x differ by bounded vectors. For any quadratic $q \in Q$, (5.4) and (2.7) at z_q give

$$(z_q)_q \geq cH, \quad (5.6)$$

while the upper bound in (5.5), transported across bounded displacement, gives

$$(z_q)_q \leq CH. \quad (5.7)$$

Hazard transport across the bounded displacement must retain a linear remainder, because the displacement can activate a mixed source on a face. Sourcewise, for every fixed bounded integer vector b for which both populations are nonnegative,

$$(w + b)_i \leq (w)_i + C_b, \quad (w + b)_{2i} \leq C_b\{(w)_{2i} + w_i + 1\},$$

and

$$(w + b)_{i+j} \leq C_b\{(w)_{i+j} + w_i + w_j + 1\}. \quad (5.8)$$

After multiplication by the fixed source rates and summation over the finite source menu, these inequalities give the correct global comparison

$$\Lambda(w + b) \leq C_b\{\Lambda(w) + H(w) + 1\}. \quad (5.9)$$

Since $\Lambda(z_C) = O(H)$, equation (5.9) gives $\Lambda(z_q) = O(H)$, including when the bounded displacement activates a previously disabled mixed source.

Since Q contains at least two of $2A, A + B, 2B$, (5.6)–(5.7) imply

$$c\sqrt{H} \leq A \leq C\sqrt{H}, \quad c\sqrt{H} \leq B \leq C\sqrt{H}. \quad (5.10)$$

Indeed, the pair $\{2A, 2B\}$ is immediate. For $\{2A, A+B\}$, the $2A$ bounds give $A = \Theta(\sqrt{H})$, and the $A+B$ bounds then give $B = \Theta(\sqrt{H})$; the remaining pair is symmetric. Extra vertices do not alter the conclusion. Equations (5.5),(5.10) also give at the current state

$$cH \leq \Lambda(x) \leq CH. \quad (5.11)$$

Let $d \in \{A, B\}$ be a species with $\delta_d > 0$. Equations (2.6) and (5.10) give

$$\mathcal{L}H \leq -C_1\sqrt{H}. \quad (5.12)$$

Take one ordinary all-clock jump. Its expected duration is $1/\Lambda \leq C/H$ by (2.7). Combine a death's denominator correction with half of its favorable workload decrement as in (3.3). Birth denominator corrections are favorable. The remaining expected quotient reward is at most $\epsilon D/(H+1) \leq C\epsilon/H$ by (2.3)–(2.4). After this absorption, the workload part is bounded above by

$$\frac{\beta - \frac{1}{2}\delta_A A - \frac{1}{2}\delta_B B}{\Lambda}.$$

Thus (5.11)–(5.12) give

$$\mathbb{E}\Delta V \leq -\frac{c}{\sqrt{H}} + \frac{C}{H}. \quad (5.13)$$

The duration term is at most $\eta C/H$, so the one-jump rule satisfies (1.3) for all sufficiently large H . The alternatives (5.2) and (5.4) are pointwise and exhaustive. There are only finitely many targets and paths, so they define a finite state-selected menu without a compactness gap.

30.6 6. The episode-time Foster lemma

For completeness, here is the exact random-time interface used above.

Lemma 6.1 (vanishing-margin physical-time Foster lemma). Let Y be a nonexplosive CTMC, K a finite set, and $V \geq 0$ one proper function. Until K , suppose state-selected physical episodes tile the trajectory at strict stopping times

$$0 = S_0 < S_1 < S_2 < \dots, \quad (6.1)$$

every episode contains at least one actual jump, and (1.3) holds for one fixed $\eta > 0$. Suppose the stopped conditional sums are justified by localization; it is sufficient that, for every finite V -sublevel, the positive endpoint increment of an episode started in that sublevel is integrable. Then

$$\mathbb{E}_y \tau_K \leq \frac{V(y)}{\eta}. \quad (6.2)$$

Stop the conditional sum of (1.3) at the first **episode endpoint** in K , at the first endpoint outside a finite V -sublevel, and after n episodes. Localization and conditional summation give

$$\eta \mathbb{E} S_{n \wedge N_K \wedge \sigma_R} \leq V(y). \quad (6.3)$$

Let $R \rightarrow \infty$ and then $n \rightarrow \infty$. Nonnegativity of V and Fatou handle the endpoint term. On non-hitting, the endpoints contain an infinite subsequence of genuine jump times, so nonexplosion forces $S_n \rightarrow \infty$. Equation (6.3) rules out non-hitting with positive probability, and monotone convergence gives (6.2).

In the present network, every rule in Sections 4–5 contains at least one ordinary jump. The actual endpoint mark chooses the next rule immediately; there is no zero-duration classifier episode and hence no artificial zero-time cycle. A localized V -sublevel contains only finitely many marked starts by properness. From any such start an episode has bounded jump count and bounded population displacement, so its positive endpoint increment is bounded on that localized start set. This verifies the exact integrability hypothesis. Globally, (2.4) also supplies all fixed source-weighted logarithmic positive moments.

30.7 7. Recurrence and scope

Binaryity gives nonexplosion: population-increasing reactions have degree at most one, so their aggregate rate is at most $C(1 + |x|_1)$, while neutral quadratic clocks cannot accumulate inside a finite population sublevel.

Lemma 6.1 gives finite mean hitting of K . One physical jump from this finite set has finitely many successors, each with the same bound, so K has finite mean return. Irreducibility normalizes its finite occupation measure to the unique stationary probability; forgetting the target mark proves positive recurrence of the physical class.

Thus the theorem covers exactly the 24 residual anisotropic permutations of (1.1); the homogeneous $h = (1, 1, 1)$ family has different dormant faces and lies outside its scope.

31 Exact residual 336 pair recurrence theorem

Proof-first composition theorem, 2026-08-12 PDT. This note composes the frozen finite support certificate with the two frozen global stochastic theorems for the homogeneous and anisotropic level-set families. It proves positive recurrence for every fixed pair in the exact residual 336 set, for every strongly connected labelled orientation and every fixed positive rate vector. It does not enumerate orientations, rates, population states, or reaction histories.

31.1 1. Certified pair set

Let $\mathfrak{P}_{336}$ be the projection onto ordered support pairs of the rows returned by the certificate function `selected_incidences`, in the frozen files

```
src/all_active_residual_levelset_336_certificate.py
SHA-256 4149b682d1222bd3327548b0eb95921f7aae20663816b345b48285239c12f93d
```

```
tests/test_all_active_residual_levelset_336_certificate.py
SHA-256 6f5802976d4de479a0728648248a2291f5d518e04de29b9b7053802eb7f1b9c2
```

An incidence is an ordered disjoint support pair together with an all-active tier descriptor. The certificate proves

$$\#\{\text{incidences}\} = \#\{\text{distinct ordered pairs}\} = 336, \quad (1.1)$$

and that the selected set equals the geometric level-set set. For every row, the certificate function `levelset_geometry` identifies a unique side T , the other side R , a positive weight h , and a scale $s > 0$ such that

$$\dim \text{span}(T - T) = 2, \quad h \cdot y = 2s \ (y \in T), \quad R = \{0\} \cup U, \quad h \cdot u = s \ (u \in U), \quad (1.2)$$

where U consists of two or three distinct unary complexes. The two independent row fingerprints are

```
d0c31db81db2400e0ead6e4a1a86b237fbf3b8bbb597340856a2756e9f6c884d
2bd4025f29d20ea4af467d46704c598652c9332ac4e32df18669cb7eb75c75a0
```

The exact weight split is

$$336 = 312[h = (1, 1, 1)] + 8[h = (1, 1, 2)] + 8[h = (1, 2, 1)] + 8[h = (2, 1, 1)]. \quad (1.3)$$

The same certificate proves that the 312 homogeneous top supports are entirely quadratic, whereas all 24 anisotropic top supports contain a unary complex.

The six focused certificate tests were replayed with the project's modern Python runtime and pass. This finite computation certifies support identity and normal form only; it contains no stochastic orientation or population search.

31.2 2. Network realizations and theorem

Fix a pair in $\mathfrak{P}_{336}$. On each of its two supports choose an arbitrary finite strongly connected labelled directed graph, with arbitrary fixed positive rate constants. The union of the two linkage graphs defines the stochastic mass-action chain on $\mathbb{Z}_{\geq 0}^3$. The side T and lower side R are the unique sides selected by (1.2); their position in the ordered pair is immaterial.

Theorem 2.1 (exact 336 pair theorem). Every such stochastic mass-action chain is non-explosive. Every closed irreducible population class is positive recurrent. Equivalently, every infinite closed class has finite mean hitting time to a finite subset and hence a unique stationary probability; finite closed classes are positive recurrent trivially.

This assertion is global on each fixed population class. It is not a chart-local drift statement and requires no additional chart-switching or terminal-exit lemma.

31.3 3. Homogeneous branch: 312 pairs

For a row with $h = (1, 1, 1)$, equation (1.2) and binaryity give

$$T \subseteq \{2A, 2B, 2C, A + B, A + C, B + C\}, \quad R = \{0\} \cup U, \quad (3.1)$$

where T has internal rank two and U contains two or three unaries. This is exactly the scope of the frozen global homogeneous theorem

research_notes/proof_first_336_h111_workload_occupation_theorem.md
SHA-256 e3c484cdbda44949ba070dae6c911a2c7de465064857b61b5d9883e9dd03bdf

research_notes/proof_first_336_h111_workload_occupation_exact_byte_audit.md
SHA-256 740d929dfa460818df2cd134fc6beba70d015c7409f3c74ce0979316b8d4af89

That theorem is already global across every boundary face. It symbolically exhausts every top-dead pure ray into the two-carrier, dyadic, or common-catalyst kernel; invokes exact all-clock physical-time macros for those kernels; proves uniform top-service exposure on the compact activated region; and applies one workload-only random-time Foster theorem on each fixed class. Catalyst-free invariant faces are open-unary reductions and are proved positive recurrent directly. Thus Theorem 2.1 holds for all 312 homogeneous pairs, for arbitrary strong orientations and positive rates.

31.4 4. Anisotropic branch: 24 pairs

For an anisotropic row, permute the species so that $h = (1, 1, 2)$. Write the weight-one species as A, B and the weight-two species as C . Equation (1.2) forces

$$R = \{0, A, B\}. \quad (4.1)$$

The unique unary complex on top level $2s$ is C , and the certificate says every anisotropic top support contains a unary. Therefore

$$T = \{C\} \cup Q, \quad Q \subseteq \{2A, A + B, 2B\}. \quad (4.2)$$

Rank two forces $|Q| \geq 2$. Hence (4.1)–(4.2) are exactly the scope of the frozen global anisotropic theorem

research_notes/proof_first_336_h112_quotient_foster_theorem.md
SHA-256 9206aa2b07aa802e4d06a769b3b60d520b2dbd12752312497aa5b41156780d48

research_notes/proof_first_336_h112_quotient_foster_exact_byte_audit.md
SHA-256 992448ad8b6520f014e783adb26a4f9b393b0e6a5f38c3a6262dd9b2fa0c1764

That theorem uses the single proper marked potential

$$V(x, t) = H(x) + \epsilon \frac{F(x, t)}{H(x) + 1}, \quad H = A + B + 2C, \quad (4.3)$$

and a finite all-clock episode menu which tiles every face of the trajectory. Its physical-time Foster inequality gives finite mean hitting of a finite marked set; forgetting the mark proves positive recurrence of the physical class. Coordinate permutation preserves explosion times, communicating classes, and recurrence. Thus Theorem 2.1 holds for all $8 + 8 + 8 = 24$ anisotropic pairs.

31.5 5. Exhaustion and classwise conclusion

The alternatives in (1.3) are disjoint and exhaustive. Sections 3 and 4 therefore cover all

$$312 + 24 = 336 \quad (5.1)$$

certified fixed pairs. Both analytic dependencies retain every physical clock, include actual reaction endpoints, prove nonexplosion, and conclude positive recurrence on each fixed irreducible class. No inference from orientation enumeration, bounded reaction depth, stationary averaging, or a drift-or-exit terminal condition enters the composition. $\square$

Part VI

Outside-mixed population and marked theorems

32 Population-Foster closure of the 11,842 no-failure pairs

Proof-first standalone theorem, 2026-08-12 PDT. This note closes the 11,842 support pairs in the exact 18,496-pair outside-mixed remainder which have no affine-feasible failure of the corrected S-tier-superlevel cut. The proof is global on each fixed population class. It uses one population function and a statewise generator inequality; it has no reaction mark, chart exit, terminal strongly connected component, or potential switch.

Finite computation is used only for the exact support/descriptor set identity. No orientation, rate vector, population state, stochastic path, or communicating class is enumerated.

32.1 1. Exact finite input

Let $\mathcal{C}_2$ be the ten binary complexes on three species. Begin with all 46,872 ordered disjoint pairs of nontrivial supports, remove the 27,894-pair orbit of the inherited mixed atlas, remove the 146 pairs having a strictly positive linear invariant, and then remove the frozen 336 level-set pairs. The exact remaining universe has 18,496 pairs.

For each remaining pair, the authoritative certificate checks all 259 exact tier/cap descriptors. A row is called a failure when

1. its rational affine flag is feasible in the pair's stoichiometric subspace; and
2. the exact S-tier-superlevel condition fails.

Exactly 11,842 pairs have no such row. Denote this set by $\mathcal{N}_{11842}$. Its exact fingerprint is
b425db9040d0836462f4240a4a3acf51d067d356eb4f2bfe4ce2cf648e42db26

The containing 18,496-pair fingerprint is
eb7db151e42eb9562b1a1d519ea7dad212df52c6df368ffa08edbf79410db4ad

The authoritative finite bytes are

```
src/outside_mixed_remaining_18496_certificate.py
314f378664052cabe23910e118c9a43acf99884ccb5c63b61daf014a206e4c63
```

```
tests/test_outside_mixed_remaining_18496_certificate.py
28d3cf0087bcd77e24d6dbfa280b226b34d3d026c35e743bc10487c829667769
```

```
research_notes/proof_first_outside_mixed_remaining_18496_certificate_exact_byte_audit.md
2539d8eee4d1d584ed5566f7494c7262843676dc368078647db12481d6a5822f
```

The five dedicated tests pass. These are set identities only. Their analytic implication is proved below.

For the elementary population fourth-power estimate used in Section 4, one additional exact identity is load-bearing. Among all 3,010,738 affine-feasible descriptors of the 11,842 pairs, the global top S-tier is nonempty and is contained in the literal global top D-tier. There are zero exceptions. The canonical incidence fingerprint is

```
a965d56c3b116a603ae147ad9bf22450c5cec9fb81477a3f99366920a0482ec8
```

and the independent summary fingerprint is

```
8f83a44a578f45597ea968c551d5dcdbba5a529833ba9eea01e14f1179af6bf5
```

The finite certificate and its five passing tests are

```
src/no_failure_11842_top_s_top_d_certificate.py
e72da319fd49df6df0c34a9a3090d95f2baf30199c0464966f9081990e347b4a
```

```
tests/test_no_failure_11842_top_s_top_d_certificate.py
ea8d042875c0c4f5c47bdabdaddc913e70138db7becedddafbe72c171ea1b572
```

This stronger identity is used for the direct global-top-D domination below. It is not asserted to be a necessary hypothesis of the more general Anderson–Kim tier theorem.

32.2 2. Process, nonexplosion, and one population function

Fix $P \in \mathcal{N}_{11842}$. Give each of its two linkage supports an arbitrary strongly connected directed reaction graph and arbitrary positive labelled rate constants. Fix a closed irreducible population class Γ , and let X_t be the stochastic mass-action chain restricted to Γ .

The chain is nonexplosive. A reaction with a quadratic source cannot increase total molecularity because every target is binary. Thus every population-increasing reaction has source degree zero or one, its total increasing intensity is $O(1 + |x|_1)$, and every jump is bounded. A linear pure-birth comparison bounds total population on finite time intervals. A fixed population sublevel is finite and has bounded total hazard, so neutral quadratic clocks cannot accumulate there.

Use the population entropy

$$V(x) = \sum_{i=A,B,C} \{x_i(\log x_i - 1) + 1\}, \quad 0 \log 0 = 0. \quad (2.1)$$

It is nonnegative and proper on $\mathbb{N}_0^3$, hence on Γ . We will prove that a finite $K_\Gamma \subset \Gamma$ satisfies

$$\mathcal{L}V(x) \leq -1 \quad (x \in \Gamma, x \notin K_\Gamma). \quad (2.2)$$

This is a statewise population-generator assertion, not a chart-local negative-or-exit alternative.

32.3 3. Every escaping fixed-class sequence has a physical descent

Suppose (2.2) is false. Properness gives a sequence $x_n \in \Gamma$, $|x_n|_1 \rightarrow \infty$, with

$$\mathcal{L}V(x_n) > -1. \quad (3.1)$$

Pass to a tier subsequence. Completeness of the rational binary arrangement assigns one of the 259 exact descriptors: it records the full D-preorder, the active-coordinate set, and the exact eventual caps $0, 1, \geq 2$ of bounded coordinates.

Because x_n lies in one affine stoichiometric class, the necessity part of the exact affine flag theorem makes this descriptor feasible for the pair. Indeed, at every positive weight level the normalized increments of the sequence supply a stoichiometric direction which is zero below that level and strictly positive on its new coordinate block. Equivalently, failure of such a direction would give by Gordan’s alternative an affine invariant whose constant value contradicts the prescribed divergence.

By definition of $\mathcal{N}_{11842}$, this feasible descriptor cannot fail the corrected S-tier-superlevel condition. The exact identity pinned in Section 1 additionally says that its literal global top S-tier E is contained in the literal global top D-tier. Let r be that global top D-level. For some linkage L ,

$$\emptyset \neq U_L(r) \subsetneq L, \quad U_L(r) \subseteq E, \quad (3.2)$$

where $U_L(r)$ consists of the complexes of L at or above level r . Strong connectivity supplies a directed path from $U_L(r)$ to its complement. The first exiting edge

$$y \longrightarrow z \quad (3.3)$$

has its source $y \in E$, hence simultaneously in the global top S- and top D-tiers, and its target strictly below D-level r . Thus every orientation has a literal physical descending reaction sourced in both global top tiers. No orientation search is involved in this cut argument.

32.4 4. Entropy drift contradiction

Put $\bar{x}_i = x_i \vee 1$. For a reaction $y \rightarrow z$, the bounded binary displacement and the factorial/entropy first-difference estimate give, along the tier sequence,

$$V(x_n - y + z) - V(x_n) = \log \frac{\bar{x}_n^z}{\bar{x}_n^y} + O(1). \quad (4.1)$$

The bars are load-bearing at a bounded-zero coordinate: the entropy jump is finite when a reaction creates the first molecule. They are also exactly the coordinates used to define the D-tier preorder. Since every source under discussion is enabled, replacing its positive bounded coordinates by their barred values changes only the uniform $O(1)$ term.

Every source in E is enabled and has propensity comparable, up to fixed positive rate constants, with the largest enabled source propensity A_n . For (3.3), the strict D-tier gap gives

$$g_n := \log \frac{\bar{x}_n^y}{\bar{x}_n^z} \longrightarrow \infty. \quad (4.2)$$

The standard tier summation groups reactions by their source tiers. The edge (3.3) contributes at most $-cA_n g_n$. Reactions from lower S-tiers have propensities negligible relative to the first higher tier at which they can contribute positively, while reactions within one D-tier have bounded logarithmic increment. Since the reaction menu is finite, the complete sum satisfies

$$\mathcal{L}V(x_n) \longrightarrow -\infty. \quad (4.3)$$

This is the elementary global-top-D form of the class-local Anderson–Kim tier contradiction. The broader Anderson–Kim theorem only requires an appropriate top-S-sourced D-descent; here the stronger exact identity from Section 1 puts the selected source literally in the global top D-tier and makes the displayed domination immediate. Equation (4.3) contradicts (3.1), and hence proves (2.2).

For completeness, the same conclusion may be expressed with the discrete factorial fourth-power function

$$G(x) = K + \sum_i \log(x_i!) \geq 1, \quad W(x) = G(x)^4. \quad (4.4)$$

On the sequence above, the exact factorial calculation gives $\mathcal{L}G \leq -cA_n g_n$, while

$$\sum_e a_e \leq C A_n, \quad |\Delta_e G| \leq C \log(2 + |x_n|_1), \quad G \asymp |x_n|_1 \log(2 + |x_n|_1). \quad (4.5)$$

Expanding $(G + \Delta_e G)^4 - G^4$ reaction by reaction shows that every quadratic, cubic, and quartic remainder is lower order than $-G^3 A_n g_n$. Consequently $\mathcal{L}W(x_n) \rightarrow -\infty$ as well. Thus either (2.1) or (4.4) is one global population Foster function for the whole fixed class.

32.5 5. Classwise recurrence theorem

Theorem 5.1 (the 11,842 no-failure pairs). For every support pair $P \in \mathcal{N}_{11842}$, every strongly connected orientation on its two linkage supports, every fixed positive labelled rate vector, and every closed irreducible population class, the stochastic mass-action chain is nonexplosive and positive recurrent.

32.5.1 Proof

Nonexplosion is Section 2. Equation (2.2) gives finite mean hit of the finite set K_Γ by continuous-time Foster, localized first on finite population sublevels and then passed to the limit by Fatou. From the finite set, irreducibility and finitely many fixed labelled paths give finite mean return to one reference state by geometric repetition. Hence every closed irreducible class is positive recurrent. A finite or singleton class is already covered. $\square$

The proof never follows a structural exit. It never switches between marked and unmarked potentials, and it does not invoke a terminal SCC.

32.6 6. Frozen analytic dependencies and exact claim boundary

The symbolic cut, affine necessity, and fourth-power lift used above are proved in the following frozen dependencies:

```
research_notes/s_tier_superlevel_cut_and_affine151_corrected.md
d91f369d34cadfb28ddb872df8fb9f6d17799ec207da29933037f55ae95f0407
research_notes/stoichiometric_gate_feasibility.md
27b40b61903ae6c2e223d007ec08323ec9aec10e9198deb99d2d7c60d878d007
research_notes/universal_fourth_power_one_active_interface.md
9d4239f4fc6b45a9522b94b09523c9f98ac7a3b089c919bd9594f12409c78cc2
```

The exact 11,842-specific top-S/top-D identity used to specialize these general inputs to the elementary global-top-D calculation is pinned in Section 1.

The identical pair-independent population-Foster argument was independently audited inside the frozen 432-pair theorem:

```
research_notes/proof_first_active_invariant_orbit_gap_432_common_w_theorem.md
7edab78daabf7e492851efe5326ccc228adfc57f02cd5ff55eaa7056e034c8
research_notes/proof_first_active_invariant_orbit_gap_432_common_w_exact_byte_audit.md
1110efc0760ed8714fc4bf203739152820f6f9a18cbdc0e92716638a707140fd
```

The remaining 6,654 pairs have feasible failures only of type B/B, B/F0, or AA. They require a separate global theorem; local drift-or-chart-exit statements do not suffice.

33 Unconditional two-active AA contraction and the cap-SCC obstruction

Proof-first candidate, 2026-08-12 PDT. This note separates two claims which must not be conflated.

1. The proposed finite graph on cap descriptors is not a valid Markov reward quotient. A cap records source availability, not the exact bounded population, and a directed support graph carries neither transition probabilities nor reward occupation weights.

2. No such graph is needed for the two-active part of the exact 18,496-pair remainder. Every affine-feasible two-active descriptor is available/available (AA), and the available-target Bellman episode can be run **without stopping at a cap or tier change**. It is therefore an unconditional common-marked-potential episode.

The finite identity used in the second claim is frozen at

```
src/remaining_18496_all_feasible_two_active_aa_certificate.py
SHA-256 25e5c2fce812d2e3d3f02c0c9377533cf16c68313b77bc0d1e7a485052bd68ee
```

```
tests/test_remaining_18496_all_feasible_two_active_aa_certificate.py
SHA-256 2445edcbcf66ff2204c821baec455b4f8a416180360a39a6d757712af5fc22e0
```

It enumerates support/descriptor incidences only. It does not enumerate orientations, rate vectors, populations, paths, or stochastic histories.

33.1 1. Why a directed SCC is not yet a Foster corrector

Consider first a genuine finite Markov-additive kernel. Let I be a finite phase set, $P = (P_{ij})$ a stochastic matrix, and r_i the expected reward of one episode begun in phase i . A bounded phase corrector $a : I \rightarrow \mathbb{R}$ with

$$r_i + \sum_j P_{ij} a_j - a_i \leq -\delta \quad (1.1)$$

can exist only if every stationary law π of every closed recurrent class satisfies

$$\sum_i \pi_i r_i \leq -\delta. \quad (1.2)$$

This is immediate after multiplying (1.1) by π_i and summing. The converse, with a possibly smaller $\delta > 0$, is the finite Poisson/Farkas alternative: solve the centered Poisson equation on each recurrent class and then extend the corrector backwards over the transient condensation graph.

Thus the load-bearing finite datum is the **rate-weighted reward kernel**, not its directed support. The assertion that a closed SCC merely contains a coercive node is insufficient unless one also proves all of the following.

- The quotient phase after an episode is determined by the node and the episode endpoint.
- Every edge used to reach the coercive node has a uniformly positive transition probability on the relevant escaping family.
- The expected positive reward accumulated before that hit is uniformly integrable.
- The negative reward at the coercive hit dominates that complete prefix.

For a state-dependent family of kernels the exact replacement for (1.2) is negativity against every invariant occupation measure of every limiting rate/reward kernel. A graph-only SCC test discards precisely this measure.

33.2 2. The cap quotient is not Markov

The descriptor cap 2 means only “at least two,” because binary sources cannot distinguish 2 from a larger population for enablement. It does not mean that the bounded coordinate equals two.

This failure occurs inside the exact 18,496-pair family. Use the ordered supports

$$L_b = \{0, A, 2A, A + B\}, \quad L_0 = \{C, 2C, A + C\}, \quad (2.1)$$

and the one-active descriptor with active species B , weight $(0, 1, 0)$, and A -cap zero. The exact remainder certificate contains its B/F0 failure rows at each C -cap $0, 1, 2$. Choose any strong orientation of L_0 containing the label $2C \rightarrow C$, for example the cycle



At populations

$$(A, B, C) = (0, N, k), \quad k \geq 2, \quad (2.3)$$

the same descriptor node records C -cap 2. If the label $2C \rightarrow C$ fires, then

$$k = 2 \implies \text{cap } 1, \quad k \geq 3 \implies \text{cap } 2. \quad (2.4)$$

Hence the endpoint cap is not a function of the cap node and the reaction label. Moreover, within the same cap node the $2C$ -source propensity is proportional to $k(k-1)$, whereas the C -source propensity is proportional to k . The all-clock transition probabilities are not functions of the finite cap node either.

Refining cap 2 to exact k repairs Markovianity only by producing a countable phase. Sending $k \rightarrow \infty$ is not another bounded-cap node: it is promotion of C to the active set and requires a new tier compactification. There is no fixed population threshold at which this sequence-defined promotion occurs.

The kinetic obstruction is real rather than cosmetic. An accelerated immigration–death cofactor can leave every fixed box before a slow service clock with probability tending to one, even though countable-phase averaging gives a uniform physical-time service probability. Thus finite box exits may form an apparent exit-only circulation while the full countable phase is recurrent. A cap-SCC calculation cannot replace the needed killed countable-phase resolvent or Poisson corrector.

We therefore record:

Proposition 2.1 (finite cap-SCC proposal: fail). A node recording only the active mask, tier partition, caps, and mark does not define a finite Markov reward quotient. Consequently a certificate that every closed directed SCC contains a locally coercive node would not, by itself, prove a bounded Foster corrector or an unconditional macro.

This proposition does not say that the networks are transient. It says that the proposed finite object forgets the rate-weighted occupation needed to prove recurrence.

3.3.3 3. The exact all-feasible-two-active AA identity

For the exact outside-mixed remainder, the frozen finite certificate gives

quantity	count
remainder pairs	18,496
pairs with a feasible two-active descriptor	18,322
affine-feasible two-active incidences	1,140,984
corrected-cut passes	1,137,900
corrected-cut failures	3,084
incidences with an S-classified linkage	0.

(3.1)

The unordered available-kind histogram is

kind	Q/Q	Q/U	C/Q	C/U	U/U	C/C
count	639,468	321,858	165,870	9,360	4,212	216.

(3.2)

The incidence and identity fingerprints are respectively

```
a4c4aa42dadabe7e73d46a690f29fe1d457bf57c5bece422b8c0a3fafa72eb99
bc9195d09dc8717381486ae114fcd59e22786c729d957fe209ceac432deaaac8
```

The analytic use of (3.1) is only that both linkage supports satisfy the symbolic Q/U/C available-target construction. The counts are not a stochastic proof.

33.4 4. Remove the structural-exit test from an AA path

Fix a strongly connected orientation and positive labelled rates. Let (x_n, t) be an escaping marked sequence with active coordinates i, j and bounded coordinate b . Thus

$$x_{n,i}, x_{n,j} \longrightarrow \infty, \quad \sup_n x_{n,b} < \infty, \quad x_n \geq t. \quad (4.1)$$

After a subsequence, fix its complete D-tier type. Suppose the linkage containing t is Q-, U-, or C-classified. The symbolic bridge supplies complexes q, c in that linkage such that

$$q_b \leq c_b, \quad q \succ_D c. \quad (4.2)$$

Here $q \succ_D c$ means that the enabled falling-factorial ratio of c to q tends to zero on the fixed tier sequence. Strong connectivity supplies a simple directed path

$$t = y_0 \longrightarrow y_1 \longrightarrow \cdots \longrightarrow y_m = c. \quad (4.3)$$

Run the following episode with every physical clock retained.

1. At y_r , take the next ordinary all-clock jump.
2. Continue only if the designated label $y_r \rightarrow y_{r+1}$ fires; otherwise stop at the competitor's actual endpoint and actual target.
3. On reaching c , take one final ordinary all-clock jump and stop at its actual endpoint and target.

There is **no cap, enabled-support, active-set, or tier structural-exit test** in this rule.

Indeed, after a successful prefix the exact population is

$$x_{r,n} = x_n - t + y_r. \quad (4.4)$$

It dominates y_r , so every designated source is physically enabled by the preceding actual target. At the pre-final endpoint

$$z_n = x_n - t + c \geq c. \quad (4.5)$$

the active coordinates still diverge. Equation (4.2) and (4.5) also give $z_{n,b} \geq c_b \geq q_b$, so q is enabled even if the bounded cap has changed. Bounded displacement preserves every strict active-coordinate D-comparison. Therefore

$$p_c(z_n) \leq \frac{K_c(z_n)_c}{K_q(z_n)_q} \longrightarrow 0. \quad (4.6)$$

This is the decisive point: a cap change along the successful word does not remove the comparison source. Intermediate cap and tier labels were proof bookkeeping, not a physical obstruction.

33.5 5. Exact all-clock reward

Carry the actual target after every reaction and put

$$F(x, t) = \sum_{r=1}^3 \log((x_r - t_r)!), \quad W(x, t) = 1 + F(x, t). \quad (5.1)$$

For a source y , let

$$\lambda_y(x) = K_y(x)_y, \quad \Lambda(x) = \sum_y \lambda_y(x), \quad p_y(x) = \frac{\lambda_y(x)}{\Lambda(x)}. \quad (5.2)$$

The one-jump marked identity is

$$D(x, t) := \mathbb{E}_{x,t} \Delta F = \log p_t - \sum_y p_y \log p_y - \log K_t + \sum_y p_y \log K_y \leq \log p_t + C_K. \quad (5.3)$$

Let $D_{r,n} = D(x_{r,n}, y_r)$, let $a_{r,n}$ be the probability of the designated label at stage r , and let $J_{r,n}$ be the expected remaining stopped reward. The literal all-clock recursion is

$$J_{m,n} = D_{m,n}, \quad J_{r,n} = D_{r,n} + a_{r,n} J_{r+1,n}. \quad (5.4)$$

All competing clocks are already in $D_{r,n}$. At the first prefix source whose probability tends to zero, (5.3) tends to minus infinity. Every earlier designated probability stays bounded below after source-ratio compactification, while the positive tail after the first rare source is multiplied by $O(p_{y_r}) \rightarrow 0$. If there is no earlier rare source, use c and (4.6). Iterating (5.4) gives

$$J_{0,n} \longrightarrow -\infty. \quad (5.5)$$

The episode contains at most ten jumps. Source entropy gives, for every fixed $s < \infty$,

$$\sup_n \mathbb{E}[(\Delta F)^+]^s < \infty. \quad (5.6)$$

At each stage the carried target is enabled, so total hazard is bounded below by the minimum positive labelled out-rate. All fixed moments of the physical duration are therefore uniformly bounded.

33.6 6. Finite-menu state selection

For a fixed support pair and fixed orientation, put in one finite menu every simple target-following path associated with every two-active tier type, available linkage, actual target, and admissible terminal c . At a marked state select the path with least exact expected stopped reward; break ties deterministically.

If uniform coercivity failed on an AA two-active escaping family, a violating sequence would have a subsequence with fixed active pair, tier type, actual target, available terminal, and simple path. The corresponding menu score tends to minus infinity by (5.5), and the selected minimum score is no larger. This is a contradiction. Hence, outside a finite subset of every AA two-active escaping family,

$$\mathbb{E}_{x,t}[W(X_\tau, T_\tau) - W(x, t) + \eta\tau] \leq -1 \quad (6.1)$$

for a fixed sufficiently small $\eta > 0$. Every episode contains at least one actual jump, has an actual integrable endpoint, and carries the same proper W . Episodes concatenate without an entrance toll or a structural-exit handoff. Since a bounded episode cannot destroy either of two diverging active coordinates along the fixed escaping sequence, the corresponding endpoint sequence remains in the same two-active asymptotic regime. This is a sequential statement, not pointwise invariance of a cap node.

Theorem 6.1 (unconditional two-active AA episode). Fix any of the 18,496 outside-mixed remainder pairs, any strongly connected orientations, any positive labelled rates, and any closed population class. Along every affine-feasible two-active escaping sequence, the finite-menu all-clock rule above satisfies the unconditional common-marked- W estimate (6.1). This includes all 3,084 corrected-cut failure incidences. No cap graph, descriptor-exit circulation, weighted seam, or orientation enumeration is used.

This is a complete **local two-active contract**. A global pair-recurrence deduction still requires every complementary all-active and one-active escaping sequence to be covered by compatible rules. In particular, this theorem does not claim that the one-active complement has already been closed.

33.7 7. Exact residual after the repair

The theorem removes the complete two-active part of the proposed SCC problem. All-active feasible descriptors pass the corrected cut and have no bounded cap. The remaining cap issue is therefore one-active, and in particular the B/F0 phase begun at a Flat0 mark. There the inactive process may be a genuine countable recurrent phase; Proposition 2.1 shows why it cannot be replaced by the three availability caps.

A valid completion must use one of the following stronger objects.

1. an unconditional finite killed completion that appends the newly enabled B-service before stopping;
2. a countable-phase killed resolvent/Poisson corrector, with promotion to Theorem 6.1 when an inactive coordinate diverges; or
3. a population potential giving direct one-active generator descent.

What is no longer needed is a finite cap-SCC theorem for two-active charts.

34 The repaired one-active Bellman/Flat0 all-clock prelude

Proof-first seam theorem, 2026-08-12 PDT. This note replaces, rather than edits, the earlier draft of the Bellman/Flat0 prelude. It closes the one-active interface in which one linkage has an available lower terminal and the other linkage is flat of active degree zero. It enumerates no orientation, rate vector, reaction history, or population box. Its only finite object is the inactive phase of one already fixed terminal chart.

The marked all-clock theorem used below is frozen at SHA-256

157e94cd035dec9a41947129dfcbbab0ebc6e72c01abde6bcf6626052954f1ed,

and its Q/U/C bridge is frozen at SHA-256

014a317602b60c765dc9a9eb98f0921ba3fd8f779221e271e0dd7f53e245f54c.

The proof below does not invoke the two-active wording of that theorem. It repeats the dimension-free marked identity and the one-active endpoint argument explicitly.

34.1 1. Setting and the marked identity

Write the active species as X , and the two bounded species as U, V . Fix a closed irreducible population class, arbitrary strongly connected orientations of two disjoint linkage supports, and arbitrary positive labelled rates. In the one-active chart,

$$X = n \longrightarrow \infty, \quad (U, V) \in B, \quad (1.1)$$

where B is the fixed finite padded inactive phase. A reaction that leaves the padded phase or changes any declared active-set, enabled-source, source-order, lattice, or shell datum is retained as a physical structural exit, including its actual endpoint and target.

Let L_0 be flat of active degree zero,

$$y_X = 0 \quad (y \in L_0), \quad (1.2)$$

and let L_b contain complexes q, c such that

$$q_X = 1, \quad c_X = 0, \quad q_U \leq c_U, \quad q_V \leq c_V. \quad (1.3)$$

After each physical reaction carry its actual target t , so $x \geq t$, and put

$$F(x, t) = \sum_{i=X,U,V} \log((x_i - t_i)!), \quad W(x, t) = 1 + F(x, t). \quad (1.4)$$

For a complex y , let K_y be the sum of labelled rates out of y , $\lambda_y(x) = K_y(x)_y$, $\Lambda = \sum_y \lambda_y$, and $p_y = \lambda_y/\Lambda$. If the next reaction is $y \rightarrow z$, exact factorial cancellation gives

$$F(x - y + z, z) - F(x, t) = \log \frac{(x)_t}{(x)_y}. \quad (1.5)$$

Consequently the expected reward of one ordinary all-clock jump is

$$D(x, t) = \log p_t - \sum_y p_y \log p_y - \log K_t + \sum_y p_y \log K_y \leq \log p_t + C_K. \quad (1.6)$$

The current target is enabled. Finiteness of the binary source menu also gives, for every fixed $r < \infty$,

$$\sup_{x,t} \sum_y p_y(x) \left[\log \frac{(x)_t}{(x)_y} \right]_+^r < \infty. \quad (1.7)$$

This is the source-entropy bound $s(1 + \log(1/s))^r \leq C_r$.

34.2 2. A Bellman mark needs no prelude

Suppose the current target t belongs to L_b . Choose a simple directed path, supplied by strong connectivity,

$$t = y_0 \longrightarrow y_1 \longrightarrow \cdots \longrightarrow y_m = c. \quad (2.1)$$

Retain all clocks, continue only when the designated label fires, stop at the actual endpoint of the first competitor or structural exit, and on designated success take one final ordinary jump at c . On success the pre-final population is

$$z = x - t + c \geq c. \quad (2.2)$$

Thus (1.3) enables q at z , and in the fixed inactive phase

$$p_c(z) \leq \frac{K_c(z)_c}{K_q(z)_q} \leq \frac{C}{z_X} \rightarrow 0. \quad (2.3)$$

Let D_i be (1.6) at a success-prefix state, a_i the probability of the next designated label, and J_i the expected remaining stopped reward. The literal all-clock recursion is

$$J_m = D_m, \quad J_i = D_i + a_i J_{i+1}. \quad (2.4)$$

At the first prefix source whose probability tends to zero, its D_i tends to minus infinity. Earlier designated probabilities stay bounded below after source-ratio compactification, and a later positive tail is multiplied by $a_i = O(p_{y_i}) \rightarrow 0$. Hence

$$J_0 \longrightarrow -\infty, \quad (2.5)$$

unless a fixed prefix reaches a structural exit with probability bounded below. The episode has bounded depth, all fixed positive endpoint-reward moments by (1.7), and all fixed physical-duration moments because the carried target supplies a fixed positive lower bound on the total hazard.

This proof uses only one diverging coordinate. It is the exact one-active corollary of the frozen marked theorem.

34.3 3. The finite phase before Bellman access

Now suppose the carried target lies in L_0 . If any X -degree-one source of L_b is enabled already, declare **top access** immediately. Otherwise record the phase

$$e = (U, V, t, \text{fixed chart flags}). \quad (3.1)$$

The phase set E is finite. Until absorption, the only in-chart population transitions are L_0 reactions. Degree-zero L_b reactions and structural exits remain as competing absorbing hazards. Every such rate depends on e and the fixed labelled rates, but not on n .

Use four disjoint absorbing sections, with structural exit given first priority. Thus any reaction that causes a declared exit is classified in E_{out} , even if its endpoint would also enable a top source or the reaction is itself a Bellman launch.

1. E_{out} , structural exit: include the exit-causing reaction and its actual endpoint.
2. A , top access: after a **nonexit** in-chart flat jump, a positive- X -degree L_b source is enabled. Stop the finite phase before taking the next jump.
3. B , Bellman launch: a degree-zero-source L_b reaction fires and does not cause a declared exit. Include that reaction and retain its actual target.
4. C , closed no-access class: enter a closed communicating class of flat phases from which none of the first three sections is possible.

Declare every such closed class C absorbing. The remaining embedded kernel Q is a finite transient substochastic matrix. Therefore, unconditionally on which absorbing section is reachable,

$$\rho(Q) < 1, \quad \mathbb{P}_e(N > k) \leq C_0 \rho_0^k \quad (3.2)$$

for some $\rho_0 < 1$, uniformly over the finite set of starting phases. This corrects the false formulation that required no closed class to be reachable: closed classes have already been made absorbing.

The current flat target is enabled and has an outgoing positive labelled rate. Its falling factorial is a positive integer. Thus the total hazard is uniformly bounded below, and (3.2) gives

$$\sup_{n,e} \mathbb{E}_e N^r < \infty, \quad \sup_{n,e} \mathbb{E}_e \sigma^r < \infty \quad (3.3)$$

for every fixed $r < \infty$, where σ is the finite-phase absorption time.

The flat reward does not accumulate per attempt. It telescopes. Before absorption the active population is n , the initial and final inactive phases lie in a finite set, and both the carried mark and every flat source have X -degree zero. Hence

$$|F(X_\sigma, T_\sigma) - F(x, t)| \leq C_B \quad (3.4)$$

on entry to A or C . The same bound holds after a degree-zero Bellman launch or structural-exit jump: in (1.5) the initial mark and the source both have active degree zero, irrespective of the target. Thus the prelude has uniformly bounded positive endpoint reward.

34.4 4. The two coercive absorbing sections

At A , the current flat target has bounded propensity, while an enabled degree-one source $r \in L_b$ has propensity at least $c_B n$. Take one ordinary all-clock jump and stop at its actual endpoint. No selected reaction is awaited. Equations (1.6)–(1.7) give

$$\mathbb{E}[\Delta F \mid A] \leq -\log n + C_B, \quad (4.1)$$

with uniformly bounded positive moments. A fast top reaction may preempt an exit, but this only creates the coercive unconditional charge (4.1); no uniform exit probability is asserted at an A -phase.

At B , the included launch reaction is the terminal jump of the finite prelude. Its actual target initializes the path in Section 2. It is not counted again. The bounded prefix (3.4) plus (2.5) tends to minus infinity, unless the appended physical path has positive structural-exit probability.

Let a_e, b_e, c_e be the probabilities, from a fixed finite starting phase, of absorption in $A \cup B$, E_{out} , and C , respectively. Before A , all competing hazards have degree zero, so these are constants independent of n , with

$$a_e + b_e + c_e = 1. \quad (4.2)$$

The exhaustive split is as follows. If $c_e > 0$, Section 5 shows that the fixed irreducible population class is finite. Otherwise $c_e = 0$. If $a_e > 0$, the corresponding finite mixture of (4.1) and (2.5) is coercively negative, unless one of the finitely many Bellman continuations has positive structural-exit probability. All complementary rewards have bounded positive part by (1.7) and (3.4), so they cannot offset that divergence. Finally, if $c_e = 0 = a_e$, then $b_e = 1$: the rule records a physical exit. These three cases exhaust (4.2). This phasewise dichotomy does not infer a uniform exit probability along phases at which a degree-one hazard grows like n .

34.5 5. Closed no-access phases are finite physical classes

In a closed section C , no L_b source is enabled: a degree-one source would be top access, and an enabled degree-zero source has a positive Bellman-launch hazard. No structural exit is enabled, and every enabled reaction belongs to L_0 , stays in C , and preserves X .

For the entrance value $X = n$, the populations represented by C form a finite closed physical subset. If it is reachable in a closed irreducible population class Γ , irreducibility forces $\Gamma = C$. Thus Γ is finite and positive recurrent. In particular, in an infinite class supporting an alleged one-active escape, $c_e = 0$ for every reachable phase. This justifies the use of the first two cases after (4.2) without deleting a merely rare linkage.

34.6 6. Repaired prelude theorem

Theorem 6.1 (Bellman/Flat0 prelude). Fix a terminal one-active chart of a reduced binary two-linkage network. Suppose one linkage satisfies (1.3) and the other satisfies (1.2). For every strongly connected orientation, every fixed positive labelled rate vector, and every actual carried target, there is a finite-menu all-clock stopping rule using the single proper marked potential (1.4) such that, outside a finite set, at least one of the following applies:

1. for some $\eta > 0$,
$$\mathbb{E}_{x,t}[W(X_\tau, T_\tau) - W(x, t) + \eta\tau] \leq -1; \quad (6.1)$$
2. the rule records a physical structural-exit reaction with a positive phasewise probability; or
3. the fixed irreducible population class is finite.

The endpoint is always the actual endpoint and target of the last included reaction. For every fixed $r < \infty$, the duration and the positive endpoint increment have uniformly bounded r -th moments.

Uniformity in (6.1) follows by contradiction over the finite phase, target, and simple-path menu. Once the expected reward is below -2 , (3.3) and the bounded Bellman depth permit a common sufficiently small $\eta > 0$. On a terminal Green trace the second alternative has zero normalized flux, and the third cannot carry escape; hence the negative alternative excludes an escaping Bellman/Flat0 occupation.

Every physical jump is counted exactly once. In particular, a Bellman activation is the terminal jump of the preceding finite prelude, and its actual target begins the next Bellman episode. It is never also counted as the first jump of that episode. This nonoverlap is why the failed conditional-activation counterexample does not apply.

35 One-active quadratic charts with the common marked factorial potential

Proof-first seam-avoidance theorem, 2026-08-12 PDT. This note replaces the separate unmarked generator potential previously used for the one-active quadratic branch. The quadratic branch admits the same marked factorial potential as the both-available, Bellman/Bellman, and repaired Bellman/Flat0 branches. The physical episode has at most two ordinary jumps, retains every competing clock, and stops at an actual population and target mark.

No orientation, rate vector, reaction history, or population box is enumerated. The proof is one application of the exact marked factorial identity and the fact that $2X$ is the unique binary source with X -degree two.

35.1 1. Scope and the common marked chain

Fix a reduced binary stochastic mass-action network with at most three dynamic species. Write them as X, U, V , allowing one or both of U, V to be absent. Fix a closed irreducible population class, arbitrary strongly connected linkage orientations, and arbitrary fixed positive labelled rate constants. Work in one terminal one-active chart

$$x_X \longrightarrow \infty, \quad (x_U, x_V) \in B, \quad (1.1)$$

where B is a fixed finite padded phase. Every reaction which changes a declared active set, inactive box, enabled-source flag, support, shell, or ratio cell is retained as a named structural exit.

Assume one active linkage contains

$$q = 2X. \quad (1.2)$$

After every ordinary population-changing labelled reaction, mark the physical target t . At a marked state (x, t) , necessarily $x \geq t$. Put

$$F(x, t) = \sum_i \log((x_i - t_i)!), \quad W(x, t) = 1 + F(x, t). \quad (1.3)$$

This is exactly the potential used in the current-target Bellman theorem. It is nonnegative and proper on the reachable marked state space: the mark ranges over a finite binary complex set, so $|x|_1 \rightarrow \infty$ forces some $x_i - t_i \rightarrow \infty$.

For a source complex y , let

$$K_y = \sum_{e:s(e)=y} \kappa_e, \quad \lambda_y(x) = K_y(x)_y, \quad p_y(x) = \frac{\lambda_y(x)}{\sum_z \lambda_z(x)}. \quad (1.4)$$

If the next labelled reaction is $e : y \rightarrow u$, its marked endpoint is $(x - y + u, u)$, and cancellation is exact:

$$F(x - y + u, u) - F(x, t) = \log \frac{(x)_t}{(x)_y}. \quad (1.5)$$

Consequently the expected increment of one ordinary all-clock jump is

$$D(x, t) = \log p_t(x) - \sum_y p_y(x) \log p_y(x) - \log K_t + \sum_y p_y(x) \log K_y, \quad (1.6)$$

and finiteness of the source set gives

$$D(x, t) \leq \log p_t(x) + C_K. \quad (1.7)$$

For every fixed $r < \infty$, the positive one-jump reward also satisfies

$$\sup_{x,t} \sum_y p_y(x) \left[\log \frac{(x)_t}{(x)_y} \right]_+^r < \infty. \quad (1.8)$$

Indeed, sourcewise (1.6) bounds the positive logarithm by a fixed constant plus $\log(1/p_y)$, and $s(1 + \log(1/s))^r$ is bounded on $[0, 1]$.

35.2 2. The quadratic dominance estimate

The binary universe has only one complex with X -degree two, namely $q = 2X$. Uniformly for (x_U, x_V) in the finite padded phase,

$$\lambda_q(x) = K_q x_X (x_X - 1) = \Theta(x_X^2), \quad \lambda_y(x) = O(x_X) \quad (y \neq q). \quad (2.1)$$

Therefore

$$p_q(x) \longrightarrow 1, \quad p_t(x) = O(x_X^{-1}) \quad (t \neq q). \quad (2.2)$$

If the carried target is $t \neq q$, take one ordinary all-clock jump and stop. Equations (1.7) and (2.2) give

$$\mathbb{E}_{x,t} \Delta F = D(x, t) \longrightarrow -\infty. \quad (2.3)$$

It remains only to treat the nonrare mark $t = q$.

35.3 3. The two-jump rule from the mark q

Strong connectivity of the linkage containing q , whose support has at least two distinct vertices, supplies one fixed nonzero labelled edge

$$e : q \longrightarrow u, \quad u \neq q. \quad (3.1)$$

Starting from (x, q) , take the next ordinary all-clock jump.

1. If a competing label fires, stop at its actual endpoint and target mark.
2. If e fires and causes a named structural exit, include it, record the exit, and stop at its actual endpoint and mark.
3. If e fires without an exit, take one further ordinary all-clock jump from its marked endpoint and stop there, including any exit caused by that second jump.

Thus every episode contains one or two ordinary jumps, and every physical clock competes at both stages. Put

$$x' = x - q + u, \quad a_e(x) = \frac{\kappa_e(x)_q}{\sum_z \lambda_z(x)}. \quad (3.2)$$

If case 2 persists along an escaping chart sequence, then by (2.2)

$$a_e(x) = \frac{\kappa_e}{K_q} p_q(x) \longrightarrow \frac{\kappa_e}{K_q} > 0, \quad (3.3)$$

so the episode records a structural exit with uniformly positive probability.

Otherwise the endpoint x' remains in the fixed padded chart. Since $u \neq q$, it has X -degree at most one. Also $x'_X = x_X - 2 + u_X \rightarrow \infty$, so the quadratic source remains enabled and

$$p_u(x') = O(x_X^{-1}). \quad (3.4)$$

Let $J(x, q)$ be the expected total F -increment of the rule. The first ordinary jump contributes its full all-clock expectation $D(x, q)$. Only the literal event that e fires launches the second jump. Hence the Markov property gives the exact Bellman identity

$$J(x, q) = D(x, q) + a_e(x) D(x', u). \quad (3.5)$$

By (1.7), $D(x, q) \leq C_K$. By (1.7), (3.3), and (3.4), the second term in (3.5) tends to minus infinity. Therefore

$$J(x, q) \longrightarrow -\infty. \quad (3.6)$$

Together, (2.3) and (3.6) prove a uniform finite-menu alternative: outside a finite subset of the fixed chart, the selected one- or two-jump rule either records a named physical exit with a fixed positive probability, or

$$\mathbb{E}_{x,t}[W(X_\tau, T_\tau) - W(x, t)] \leq -2. \quad (3.7)$$

Uniformity follows without a quantitative compactness guess: if it failed, a violating sequence has a fixed mark and fixed padded phase along a subsequence, and the corresponding limit (2.3) or (3.6) contradicts it.

35.4 4. Endpoints, moments, duration, and nonoverlap

The endpoint in every case is the actual endpoint of the last included physical reaction, with that reaction's actual target mark. Its population displacement from the start is bounded by two binary reaction increments.

Applying (1.8) at the first and possible second stage and using $(a + b)^r \leq 2^{r-1}(a^r + b^r)$ shows, for every fixed $1 \leq r < \infty$,

$$\sup \mathbb{E}_{x,t} ([W(X_\tau, T_\tau) - W(x, t)]^+)^r < \infty. \quad (4.1)$$

At every stage the carried target is enabled. Its falling factorial is a positive integer, and its linkage has a positive outgoing labelled rate. If $\kappa_* > 0$ is the minimum positive labelled rate, the total hazard is at least κ_* . A sum of at most two such holding times has all fixed moments, so

$$\sup \mathbb{E}_{x,t} \tau^r < \infty. \quad (4.2)$$

Choose $\eta > 0$ smaller than the reciprocal of the uniform mean-duration bound. Shrinking the exterior finite set if necessary, (3.7) gives the physical-time inequality

$$\mathbb{E}_{x,t}[W(X_\tau, T_\tau) - W(x, t) + \eta\tau] \leq -1. \quad (4.3)$$

When episodes are concatenated, the last reaction of the current episode is never counted as the first reaction of the next. Its actual target merely selects the next rule after the endpoint is reached. A finite surrounding chart cover may therefore reclassify that endpoint under the same function W , without a comparison toll. This is the exact common-potential interface shared by the other marked Bellman branches.

35.5 5. The seam-avoidance theorem

Theorem 5.1 (quadratic branch with common marked W). In every fixed one-active chart (1.1) whose reduced support contains $2X$, for every strongly connected orientation and every fixed positive labelled rate vector, the common marked factorial potential (1.3) admits a finite menu of one- or two-jump all-clock physical episodes satisfying the following alternative outside a finite set:

1. the episode obeys the physical-time drift inequality (4.3); or
2. it records a named structural-exit reaction with uniformly positive probability.

Every episode ends at its actual physical endpoint and mark, has bounded depth, all fixed positive reward moments, and all fixed duration moments. Episodes concatenate without overlap. Hence the one-active quadratic branch may be composed with the both-available, Bellman/Bellman, and Bellman/Flat0 branches using the identical proper marked potential W ; no terminal-chart potential switch or weighted seam estimate occurs among those branches.

The theorem is local to the quadratic one-active branch. It does not claim that a direct linear workload used on a different all-active family agrees with W ; that separate interface must still be closed or avoided.

36 The repaired one-active no-mixed exhaustion

Proof-first composition lemma, 2026-08-12 PDT. This new note replaces the corrupted combined draft. It proves a symbolic exhaustion of every one-active chart whose fixed support pair has no raw mixed two-active occurrence, and then invokes the separately repaired all-clock prelude for the only stochastic interface not covered by direct drift, the marked Bellman recursion, or a fixed-class invariant.

No orientation, rate vector, reaction history, or population box is enumerated. The two short truth tables below are literal evaluations of the ordered Q/U/C/S support classifier on the ten-complex binary universe.

The dependencies are frozen at the following SHA-256 values.

Both-available marked theorem:

157e94cd035dec9a41947129dfcbbab0ebc6e72c01abde6bcf6626052954f1ed

Exact Q/U/C bridge:

014a317602b60c765dc9a9eb98f0921ba3fd8f779221e271e0dd7f53e245f54c

Repaired Bellman/Flat0 prelude:

f8ad11189d41fc5f1d09d0cf306c90d77a9b2b4b18cd00fe3dc06918d762c19b

36.1 1. Exact universe and hypothesis

Write the active species as X , and the bounded species as U, V . The literal binary complex universe is

$$\mathcal{C}_2 = \{0, X, U, V, 2X, 2U, 2V, X + U, X + V, U + V\}. \quad (1.1)$$

A support is a subset of (1.1) with at least two complexes. The two linkage supports are disjoint. Each has an arbitrary strongly connected orientation and arbitrary fixed positive labelled rates.

For each of the three choices of active coordinate pair, apply the exact ordered classifier of the frozen bridge: flat top; active quadratic top; one-active-particle-flat obstruction; unary top; shared bounded cofactor; otherwise shielded. Use the four primitive workload representatives

$$(1, 1), \quad (2, 3), \quad (1, 2), \quad (1, 3), \quad (1.2)$$

and their coordinate exchanges. The pair is **globally nonmixed** if its two supports have the same available/shielded output in every such literal cell. This support-level hypothesis is stronger than merely being nonmixed in the present one-active chart: any pair that is mixed anywhere has already been routed to the completed pair-specific theorem, which covers all charts of that fixed pair.

Fix a terminal one-active chart in one closed irreducible population class:

$$X \longrightarrow \infty, \quad (U, V) \in B. \quad (1.3)$$

The inactive padded phase B is finite. Any reaction changing a declared active-set, source-order, enabled-source, lattice, box, or shell datum is retained as a physical structural exit.

36.2 2. One-linkage symbolic categories

For a linkage support L , define mutually exclusive categories in the following order.

1. **Q**: $2X \in L$.
2. **F0**: every complex has X -degree zero.
3. **F1**: every complex has X -degree one.
4. **B**: none of Q, F0, F1 holds, and there are $q, c \in L$ with

$$q_X = 1, \quad c_X = 0, \quad q_U \leq c_U, \quad q_V \leq c_V. \quad (2.1)$$

5. **D**: none of the preceding alternatives holds.

After Q fails, the degree-one menu is

$$X, \quad X + U, \quad X + V, \quad (2.2)$$

and the degree-zero menu is

$$0, U, V, 2U, U + V, 2V. \quad (2.3)$$

If a nonflat linkage contains X , (2.1) holds with every lower complex. If it contains $X + U$, failure of (2.1) forbids every lower complex containing U ; the statement for $X + V$ is symmetric. Hence every D support has exactly one of

degree-one block	nonempty degree-zero block
$\{X + U\}$	$S \subseteq \{0, V, 2V\}$
$\{X + V\}$	$S \subseteq \{0, U, 2U\}$
$\{X + U, X + V\}$	$\{0\}$.

(2.4)

This is a symbolic consequence of binaryity and coordinatewise dominance.

36.3 3. The single-token dormant rows are necessarily mixed

By U, V symmetry, consider

$$L = \{X + U\} \cup S, \quad \emptyset \neq S \subseteq \{0, V, 2V\}. \quad (3.1)$$

Write A and S for available and shielded, and write $(XV; a, b)$ for the ordered classifier with active coordinates X, V and weights a, b . The following table evaluates all seven possible lower blocks.

S	d_1	$L(d_1)$	d_2	$L(d_2)$
$\{0\}$	$(XV; 3, 2)$	S	$(UV; 3, 2)$	S
$\{V\}$	$(XV; 3, 2)$	S	$(XV; 2, 3)$	A
$\{2V\}$	$(XV; 3, 1)$	S	$(XV; 3, 2)$	A
$\{0, V\}$	$(XV; 3, 2)$	S	$(XV; 2, 3)$	A
$\{0, 2V\}$	$(XV; 3, 1)$	S	$(XV; 3, 2)$	A
$\{V, 2V\}$	$(XV; 3, 1)$	S	$(XV; 3, 2)$	A
$\{0, V, 2V\}$	$(XV; 3, 1)$	S	$(XV; 3, 2)$	A .

(3.2)

The complete disjoint survivor lists after d_1 , in the row order of (3.2), are:

1. $\{U, 2U\}, \{V, U + V\}$;
2. $\{0, U\}, \{0, 2U\}, \{U, 2U\}, \{0, U, 2U\}, \{0, U + V\}$;
3. the preceding five supports together with $\{V, U + V\}$;
4. $\{U, 2U\}$;
5. $\{U, 2U\}, \{V, U + V\}$;
6. the five supports in item 2; and
7. $\{U, 2U\}$.

In the first row both surviving partners are available at d_2 . In every other row all surviving partners are shielded at d_2 . Thus every disjoint partner disagrees with L at one of the two displayed tests. The table and list are exhaustive because the lower block has seven nonempty subsets and the list is the literal remaining support universe after the first ordered classifier test. No stochastic inference enters (3.2).

Therefore a D support in a globally nonmixed pair must, up to exchanging U, V , be

$$L_D = \{0, X + U, X + V\}. \quad (3.3)$$

36.4 4. The exact five dormant partners and their invariant

The support (3.3) is available in every XU and XV cell. In the UV chart it is shielded on the equality wall and available in either open chamber. A disjoint support has the same complete signature exactly when it is one of

$$L_F \in \{\{U, V\}, \{2U, 2V\}, \{2U, U + V\}, \{2V, U + V\}, \{2U, 2V, U + V\}\}. \quad (4.1)$$

Here is the literal ordered-classifier proof. After deleting the three vertices of L_D , the partner universe is

$$\{X, U, V, 2X, 2U, U + V, 2V\}. \quad (4.2)$$

On the UV equality wall, every member of $\{2U, U + V, 2V\}$ has top degree two. If the partner contains one of them and also a lower-shell vertex, the active-quadratic test returns A, contrary to the required S. It must therefore lie wholly in that quadratic shell. If it contains no quadratic-shell vertex but contains U or V together with a lower X or $2X$, the unary-top test returns A. Therefore an S partner on the equality wall lies wholly in exactly one of

$$\{U, V\}, \quad \{2U, U + V, 2V\}, \quad \{X, 2X\}. \quad (4.3)$$

A nontrivial support in $\{X, 2X\}$ remains flat and shielded in both open UV chambers, so it cannot match L_D . The unary shell has the single nontrivial support $\{U, V\}$, which is available in both open chambers. The quadratic shell has three two-vertex subsets and its full three-vertex set; every one has an active-quadratic top in both open chambers. These are exactly the other four supports in (4.1). No other classifier branch or binary shell remains. This proves (4.1) without a support scan.

Every complex in L_D has value zero under

$$H = X - U - V, \quad (4.4)$$

and every support in (4.1) lies in one level set of $U + V$. Hence every physical reaction of either linkage preserves H , independently of orientation and rates. In a fixed class and the one-active chart, $X = H + U + V$ is bounded. The D/F0 residue cannot escape.

36.5 5. Flat degree one cannot partner a Bellman support

An F1 support is a nontrivial subset of $\{X, X + U, X + V\}$. There are four possibilities up to U, V exchange. The table lists a cell where it is shielded and shows that every disjoint B support is available there. In the third row one equality-wall survivor $\{0, X\}$ remains, and the second displayed cell separates it.

L_{F1}	first cell	L_{F1} status	B partners with same status
$\{X, X + U\}$	$(XV; 3, 2)$	S	$\emptyset$
$\{X, X + V\}$	$(XU; 3, 2)$	S	$\emptyset$
$\{X + U, X + V\}$	$(UV; 1, 1)$	S	$\{0, X\}$
$\{X, X + U, X + V\}$	—	—	$\emptyset$.

(5.1)

For the sole survivor in row three,

$$L_{F1}(UV; 3, 2) = A, \quad \{0, X\}(UV; 3, 2) = S. \quad (5.2)$$

The fourth row has no disjoint B support because it uses the entire degree-one menu. Thus a globally nonmixed B/F pair is necessarily B/F0.

36.6 6. The complete symbolic alternative

If either linkage is Q, the source $2X$ is enabled for all large X . It is the unique binary complex of X -degree two, so every nonzero outgoing reaction lowers active degree. For

$$V(x) = \sum_i \log(x_i!), \quad (6.1)$$

one such strong-connectivity exit contributes $-\Theta(X^2 \log X)$. Every other source contributes at worst $O(X \log X)$, uniformly in the inactive box. Hence

$$\mathcal{L}V(x) \longrightarrow -\infty. \quad (6.2)$$

If both linkages are flat, every reaction preserves X exactly, even when their constant active degrees differ. A fixed class cannot then have $X \rightarrow \infty$.

If Q and D are absent and the pair is not flat/flat, it is B/B or B/F0 by Section 5. For B/B, every actual target belongs to a linkage satisfying (2.1). The dimension-free marked identity and Bellman recursion of the frozen theorem apply verbatim: a same-linkage path reaches c , the endpoint $x - t + c$ enables q , and

$$p_c(x - t + c) \leq \frac{K_c(x - t + c)_c}{K_q(x - t + c)_q} = O(X^{-1}) \rightarrow 0. \quad (6.3)$$

Thus the common proper marked potential

$$W(x, t) = 1 + \sum_i \log((x_i - t_i)!) \quad (6.4)$$

has coercive all-clock reward or records a physical structural exit, with actual endpoints, bounded episode depth, all fixed positive reward moments, and all fixed physical-duration moments. This is also proved explicitly in Section 2 of the repaired prelude.

For B/F0, invoke the repaired prelude whose frozen SHA-256 is
f8ad11189d41fc5f1d09d0cf306c90d77a9b2b4b18cd00fe3dc06918d762c19b

It stops before every fast-source phase, gives exit-causing reactions first priority, includes a degree-zero Bellman launch exactly once, and proves the exhaustive alternative: common- W negative drift, positive physical exit, or a finite fixed irreducible class.

Combining Sections 2–6 proves the following.

Theorem 6.1 (one-active globally nonmixed exhaustion). Every terminal one-active chart of a globally nonmixed disjoint binary two-linkage pair belongs to exactly one structural category:

1. Q descent;
2. flat/flat active-coordinate invariance;
3. B/B marked Bellman drift-or-exit;
4. B/F0 repaired prelude drift-or-exit-or-finite-class; or
5. one of the five D/F0 pairs (4.1), with common invariant $X - U - V$.

Hence no such chart carries an escaping fixed-class terminal occupation. Every activation and structural-exit jump is retained once at its actual endpoint; no conclusion conditions on future activation.

36.7 7. Exact 9,489 regression profile

The analytic proof is Sections 2–6. As a non-load-bearing exact check, assign each nontrivial binary support its first category in the order Q, F0, F1, B, D; take unordered disjoint pairs; and retain exactly the globally nonmixed pairs of Section 1. For fixed active species X , the literal profile is

unordered category pair	support pairs
Q/B	6,050
$Q/F0$	1,352
$Q/F1$	54
B/B	1,224
$B/F0$	731
$F0/F0$	19
$F0/F1$	54
$D/F0$	5
total	9,489.

(7.1)

The universe and selector are explicit above, so (7.1) is reproducible without an undocumented filter. The five D/F0 rows agree exactly with (4.1), and the absence of D/B, D/D, and B/F1 rows independently checks the three symbolic cuts above. No count in (7.1) supplies a probability, a drift estimate, or a recurrence claim.

36.8 8. Composition scope

On a terminal Green trace, a positive structural-exit flux contradicts terminality. Q or marked negative reward contradicts the corresponding nonnegative terminal workload/entropy balance, while the flat and signed invariants and the finite no-access phase contradict fixed-class escape. The proof fixes one terminal chart and does not follow an exit into a chart using a different potential; hence it does not claim an unsupported global potential-switching Foster theorem.

Pairs with any raw mixed two-active occurrence are outside this note and must be prioritized globally to their completed pair-specific theorem. The theorem here covers precisely the remaining one-active charts. It does not enlarge the scope of either the mixed atlas or the two-active both-available theorem.

37 The cap-free one-active B/F0 killed-resolvent theorem

Proof-first candidate, 2026-08-12 PDT. This note removes the finite inactive-cap stop from the one-active Bellman/Flat0 argument. It proves a literal all-clock theorem on the countable inactive phase. No orientation, reaction word, population box, or stochastic history is enumerated.

The result is classwise. If the killed inactive phase has a closed no-kill component, that component is proved positive recurrent directly. Otherwise its exact killed resolvent is used, and the first access or Bellman-linkage firing is completed before the episode stops.

37.1 1. Statement

Fix two disjoint binary linkage supports L_B, L_0 on species X, U, V , arbitrary strongly connected orientations, arbitrary positive labelled rates, and a closed irreducible population class Γ . Assume that L_0 is **Flat0** in X :

$$y_X = 0 \quad (y \in L_0). \quad (1.1)$$

Assume that L_B has a Bellman witness $q, c \in L_B$:

$$q_X = 1, \quad c_X = 0, \quad q_U \leq c_U, \quad q_V \leq c_V. \quad (1.2)$$

Let (x_n, t_n) be an escaping sequence of physical marked states in Γ , with $t_n \in L_0$, for which X is the unique active coordinate. Thus

$$X_n \longrightarrow \infty, \quad \log(1 + U_n) + \log(1 + V_n) = o(\log X_n), \quad x_n \geq t_n. \quad (1.3)$$

Carry the actual reaction target as the mark and use

$$F(x, t) = \sum_{i \in \{X, U, V\}} \log((x_i - t_i)!), \quad W(x, t) = 1 + F(x, t). \quad (1.4)$$

Theorem 1.1 (cap-free B/F0 completion). Under (1.1)–(1.3), one of the following two classwise alternatives holds.

1. The full physical class Γ is a fixed- X , fixed- U one-species binary class and is positive recurrent (a singleton is allowed).
2. A finite support-dependent menu selects, at every sufficiently large start in (1.3), an almost surely finite all-clock stopping time $\tau_n > 0$, with its actual population endpoint and actual target, such that, for constants $\eta > 0$ and $C_m < \infty$,

$$\mathbb{E}_{x_n, t_n}[W(X_{\tau_n}, T_{\tau_n}) - W(x_n, t_n) + \eta\tau_n] \longrightarrow -\infty, \quad (1.5)$$

$$\mathbb{E}_{x_n, t_n}[(F(X_{\tau_n}, T_{\tau_n}) - F(x_n, t_n))^+]^m \leq C_m. \quad (1.6)$$

The killed Flat0 prefix has a uniform exponential relative marked factorial bound, and its physical duration satisfies the sharp bound

$$\mathbb{E}\sigma^m \leq C_m \{1 + \log(2 + V_n)\}^m. \quad (1.7)$$

The rule never stops merely because a cap, tier, active-set label, or enabled-source flag changes. A Flat0 access jump and a Bellman-linkage launch are terminal events only after their physical reward completion has been included.

The logarithm in (1.7) cannot be replaced by a constant uniform in the inactive start. For the pure-death phase $V \rightarrow 0$, followed by a launch available only at zero, the mean time from $V = v$ is $\sum_{k=1}^v (\kappa k)^{-1} = \kappa^{-1} \log v + O(1)$.

37.2 2. The unconditional B episode

Let $t \in L_B$ be any actual target. Strong connectivity gives a simple directed path

$$t = y_0 \longrightarrow y_1 \longrightarrow \cdots \longrightarrow y_m = c. \quad (2.1)$$

Follow its designated labels with every physical clock retained, stop on the first competitor, and, on reaching c , take one final ordinary all-clock jump. There is no structural-exit test. On designated success the pre-final population is

$$z = x - t + c \geq c. \quad (2.2)$$

Equation (1.2) and divergence of X make q enabled at z . Since the inactive coordinates differ from their start values by a bounded amount,

$$p_c(z) \leq \frac{K_c(z)_c}{K_q(z)_q} \leq \frac{C(1 + U + V)^2}{X}. \quad (2.3)$$

For one ordinary marked jump put $D(x, t) = \mathbb{E}_{x, t} \Delta F$. Exact factorial cancellation gives

$$D(x, t) = \log p_t - \sum_y p_y \log p_y - \log K_t + \sum_y p_y \log K_y \leq \log p_t + C. \quad (2.4)$$

If a_i is the probability of the designated label at stage i , the literal all-clock Bellman recursion is

$$J_m = D_m, \quad J_i = D_i + a_i J_{i+1}. \quad (2.5)$$

The first source probability tending to zero gives the negative term; all earlier designated probabilities stay bounded below, and the later positive tail is multiplied by the rare designated probability. Equations (2.3)–(2.5) give $J_0 \rightarrow -\infty$.

This already proves the qualitative unconditional B episode whenever $\log(1 + U) + \log(1 + V) = o(\log X)$, including the immediate $q = X$ case and the B/B case with two subpower inactive coordinates.

We need one quantitative refinement only for a B episode **appended after the prefix absorption of Section 3**. Before absorption $U = 0$, so the absorbing target leaves U bounded by two; V , denoted v , is the only possibly large inactive coordinate. Along (2.1), every enabled source propensity is, up to a fixed factor,

$$X^a(1+v)^b, \quad a \in \{0, 1\}, \quad b \in \{0, 1, 2\}, \quad a+b \leq 2. \quad (2.6)$$

Here is the complete first-rare product argument. Expand (2.5):

$$J_0 = D_0 + a_0 D_1 + \cdots + (a_0 \cdots a_{m-1}) D_m. \quad (2.6a)$$

After a subsequence, either v is bounded or $v \rightarrow \infty$, and every enabled source in every one of the finitely many bounded-displacement states has a fixed exponent pair (a, b) from (2.6). The total hazard is comparable with the lexicographically largest enabled pair, where the X -exponent is compared first because $v = X^{o(1)}$. Consequently a source probability is either bounded below, at most C/v , or at most Cv^2/X .

Let $j < m$ be the first path source whose probability is not bounded below. Then $a_0 \cdots a_{j-1} \geq b_* > 0$, while (2.4) gives

$$D_j \leq \begin{cases} -\log v + C, & p_{y_j} \leq C/v, \\ -\log X + 2\log v + C, & p_{y_j} \leq Cv^2/X. \end{cases} \quad (2.6b)$$

Every $D_i \leq C_0$, and the complete positive tail after j is multiplied by $a_j \leq Cp_{y_j}$. Hence it is $O(p_{y_j})$, while the negative term in (2.6b) is multiplied only by the fixed b_* . If no such j exists, all preterminal a_i 's are bounded below and the terminal estimate (2.3) supplies the second line of (2.6b). Since $\log(1+v) = o(\log X)$, this proves constants $a, C > 0$, uniform over the finite target/path menu, such that

$$\begin{cases} J_0 \leq -a \log X + C, & v \leq M, \\ J_0 \leq -a \log(2+v) + C, & v > M \end{cases} \quad (2.7)$$

for a fixed sufficiently large M . This is only the integer-exponent comparison (2.6); no population ranges are enumerated. Source entropy gives every fixed positive reward moment, and the episode has at most ten physical jumps with uniformly bounded duration moments.

37.3 3. Exact reduction before access

If the witness has a cofactor, binaryity and relabelling give

$$q = X + U. \quad (3.1)$$

If $q = X$, or if $q = X + U$ and $U > 0$, the witness is already enabled. The actual mark is still in L_0 , so one does **not** start the same-linkage path (2.1) from that mark. Instead take one ordinary all-clock payoff jump. The marked source has rate at most $C(1+U+V)^2$, whereas q has rate at least cX . Thus (2.4) gives

$$\mathbb{E}\Delta F \leq -\log X + 2\log(2+U+V) + C. \quad (3.2)$$

It remains to have $q = X + U$ and start with $U = 0$. Stop the **prefix** at the first of:

1. a reaction of L_B ; or
2. a reaction of L_0 whose target contains U .

Before that event, X and U are constant, and every enabled source and target lies in

$$E = \{0, V, 2V\}. \quad (3.3)$$

The actual mark also lies in E . The nonabsorbed marked phase is

$$\mathcal{E} = \{(v, s) : v \in \mathbb{N}_0, s \in \{0, 1, 2\}, v \geq s\}, \quad (3.4)$$

where s is the V -degree of the actual target. No cap replaces v .

At a type-1 event, include the causing reaction and start Section 2 from its actual B target. At a type-2 event, include the causing reaction and then take the access jump in (3.2), unless the causing reaction is itself a B reaction, in which case type 1 has priority. These are the disjoint absorbing payoff kernels.

37.4 4. The marked killed resolvent

Let P be the exact substochastic embedded kernel of nonabsorbed transitions in (3.4), and let S be its physical absorption kernel, including the causing reaction. For $0 < \theta < 1/4$, put

$$w_\theta(v, s) = ((v - s)!)^\theta. \quad (4.1)$$

On a terminal state reached with source y , write $v = V$ for its pre-jump inactive count and define

$$\widehat{w}_\theta(y; X, v) = \left\{ \frac{(X - y_X)!(v - y_V)!}{X!} \right\}^\theta. \quad (4.2)$$

Every enabled source before absorption has $y_U = 0$. Thus (4.2) is exactly the terminal marked factorial divided by the constant prefix factor $(X!)^\theta$; the actual target cancels. A B source of X -degree one supplies the favorable factor $X^{-\theta}$.

Lemma 4.1 (uniform relative marked-factorial resolvent). If the phase has no reachable closed no-absorption class, then

$$(I - P)^{-1} S \widehat{w}_\theta(\cdot; X, \cdot)(v, s) \leq C_\theta w_\theta(v, s) \quad (4.3)$$

for all $X \geq 1$ and $(v, s) \in \mathcal{E}$. Equivalently, at the physical prefix endpoint σ ,

$$\mathbb{E}_{x,t} \exp\{\theta(F(X_\sigma, T_\sigma) - F(x, t))\} \leq C_\theta. \quad (4.4)$$

37.4.1 Proof

Delete zero-vector self labels. Let d be the largest V -degree of an enabled degree-zero source occurring in either linkage. There is only one one-species complex of degree d . If it belongs only to L_B , every firing from it is absorbed and the continuation is contractive. Otherwise it is the maximal Flat0 source dV .

For a nonabsorbed jump with source yV , factorial cancellation gives

$$\frac{w_\theta(v - y + u, u)}{w_\theta(v, s)} = \left(\frac{(v - y)!}{(v - s)!} \right)^\theta. \quad (4.5)$$

The target u cancels. When $y < d$, divide its source rate by the order- $(v)_d$ maximal-source rate. If $y < s$,

$$\frac{K_y(v)_y}{K_d(v)_d} \left(\frac{(v - y)!}{(v - s)!} \right)^\theta \leq C v^{-(d-s)-(1-\theta)(s-y)}. \quad (4.6)$$

If $s \leq y < d$, the exponent is $-(d - s) + (1 - \theta)(y - s) < 0$. Hence every lower-source continuation is $O(v^{-\eta})$, unless $s < d$, in which case the maximal-source jump itself contracts by $O(v^{-\theta(d-s)})$.

Only $s = d$ permits one weight-neutral maximal-source jump. Strong connectivity supplies an outgoing nonself label from dV . It is either absorbed or has target $u < d$; in the latter case the successor mark is strictly lower, so the next maximal-source jump contracts. Consequently, for a kernel Q retaining maximal-source nonabsorbed jumps, there are $M < \infty$, $\rho < 1$, and $\eta > 0$ such that

$$\|Q^2\|_{w_\theta, \{v > M\}} \leq \rho, \quad \|P - Q\|_{w_\theta, \{v > M\}} \leq CM^{-\eta}. \quad (4.7)$$

Choose M so the second bound is smaller than $(1 - \rho)/4$.

Absorbing degree-one B sources cannot spoil the estimate. If r is their propensity divided by the propensity of the enabled marked source, their source probability is at most $Cr/(1 + r)$, while the exact marked factorial ratio is at most $Cr^{-\theta}$. Their weighted terminal contribution is at most

$$C \frac{r^{1-\theta}}{1+r} \leq C. \quad (4.8)$$

The identical sourcewise estimate handles every degree-zero absorbing label. Thus the terminal operator from w_θ to $\widehat{w}_\theta$ has finite weighted norm.

On the finite marked core $v \leq M + 4$, absence of a reachable closed no-absorption class says that the substochastic kernel is transient. For a fixed core phase, write the total Flat0 continuation rate as a , the total absorbing degree-zero rate as b , and the total coefficient of absorbing degree-one B rates as c . Its nonabsorbed row is the fixed Flat0 numerator divided by $a + b + cX$, hence is entrywise nonincreasing in X . The entrywise largest core kernel occurs at $X = 1$, and its finite Green matrix is bounded. Combining that inverse with (4.7), or using the Schur complement at first core return, gives

$$\|(I - P)^{-1}\|_{w_\theta} < \infty. \quad (4.9)$$

Equations (4.8)–(4.9) prove (4.3), and exact factorial cancellation turns (4.3) into (4.4). $\square$

For every fixed m , (4.4) implies

$$\sup_{X,v,s} \mathbb{E}[((F_\sigma - F_0)^+)^m] < \infty. \quad (4.10)$$

This uniform relative statement is what permits a Bellman decrement which may diverge arbitrarily slowly.

We also need the continuation maximum before absorption, without the favorable $X^{-\theta}$ factor at a degree-one B terminal. For $M > v + 4$, let ρ_M be the first nonabsorbed phase with $V \geq M$. Stop the Green series at $\rho_M \wedge \sigma$. The same outer contraction (4.7) and finite-core inverse (4.9), now with boundary payoff w_θ at ρ_M and zero at absorption, give

$$\mathbb{E}_{v,s}[w_\theta(V_{\rho_M}, S_{\rho_M}); \rho_M < \sigma] \leq C_\theta w_\theta(v, s). \quad (4.11)$$

Since jumps are bounded and $w_\theta(M - O(1), 2) \geq \exp\{cM \log(2 + M)\}$, (4.11) implies

$$\mathbb{P}_{v,s}\left\{\sup_{r < \sigma} V_r \geq M\right\} \leq C \exp\{\theta v \log(2 + v) - cM \log(2 + M)\}. \quad (4.12)$$

This is the required survival-before-absorption tail. It is stronger than the terminal relative bound (4.4) in precisely the case where a terminal B source contributes $X^{-\theta}$.

37.5 5. Physical duration

Lemma 5.1 (sharp logarithmic duration). Under Lemma 4.1, for every fixed $m \geq 1$,

$$\mathbb{E}_{v,s} \sigma^m \leq C_m \{1 + \log(2 + v)\}^m. \quad (5.1)$$

37.5.1 Proof

Put $L(v) = 1 + \log(v + e)$, and give an absorbed state value zero. If the largest source degree is one, its nonabsorbed strict jump lowers v by one and has rate at least cv ; every upward jump has constant rate. If the largest degree is two, a maximal-source nonabsorbed strict jump lowers v by one or two and has rate at least cv^2 , whereas every upward jump has source degree at most one. A maximal-source absorbing jump adds a negative killed term. Bounded-jump Taylor estimates give, outside a fixed core,

$$\mathcal{K}L(v) \leq -c, \quad (5.2)$$

where $\mathcal{K}$ is the physical killed generator. In degree zero, either absorption has a fixed positive hazard or the phase is closed. Finite-core killed transience supplies (5.2) globally after adding a bounded finite-core corrector.

Dynkin gives (5.1) for $m = 1$. Applying the calculation to L^j and inducting with the strong Markov property gives $\mathcal{K}L^j \leq -c_j L^{j-1} + C_j 1_{\{v \leq M\}}$, hence all integer moments. Other fixed moments follow by domination. $\square$

37.6 6. The closed no-kill alternative

Suppose the nonabsorbed phase has a reachable closed communicating class. There every physical reaction preserves X , preserves $U = 0$, and remains in (3.3). It is a closed physical subset of Γ . Irreducibility forces it to be the whole class.

The remaining coordinate is a one-species binary mass-action chain. If its largest nontrivial source degree is one, strong connectivity supplies a strict $V \rightarrow 0$ edge of linear rate and every upward edge has constant rate. If it is two, strong connectivity supplies a strict edge from $2V$ to a lower complex of quadratic rate, while every upward edge has source degree at most one. Therefore, for $0 < \theta < 1/2$,

$$\Psi_\theta(v) = \exp\{\theta v \log(v + e)\}$$

satisfies

$$\mathcal{L}\Psi_\theta(v) \leq C - c(1 + v)^d \Psi_\theta(v), \quad d \in \{1, 2\}, \quad (6.1)$$

outside a finite set. Degree zero is finite. Ordinary Foster theory proves nonexplosion and positive recurrence. This is alternative 1 of Theorem 1.1, with no episode handoff.

37.7 7. Completion and the physical-time ledger

Assume the transient alternative. At type 1, append Section 2 from the actual B target. At type 2, append the ordinary access jump (3.2). Call the resulting stop τ . Every jump is counted once: the absorption-causing jump ends the killed prefix, and the appended rule starts only at its actual endpoint.

Let $r_0 = V_0 - (t_0)_V$, and let r_σ be the residual inactive population at the prefix endpoint. On $r_\sigma < r_0/2$, exact factorial telescoping gives

$$F_\sigma - F_0 \leq -cr_0 \log(2 + r_0) + C. \quad (7.1)$$

The appended B/access rule has expected reward bounded above by a fixed constant, because every one-jump $D \leq C$ and its Bellman recursion has finite depth. Thus the full completed reward obeys the same estimate on this event, after changing C .

On

$$r_0/2 \leq r_\sigma \leq 2r_0 + 4, \quad (7.2a)$$

(2.7) or (3.2) gives conditional appended reward at most

$$-a \log(2 + r_0) + C \quad (7.2)$$

when $r_0 \rightarrow \infty$. On $r_\sigma > 2r_0 + 4$, the universal upper bound $D \leq C$, the finite Bellman recursion, and source entropy bound the appended expected reward above by a constant. Moreover (4.12), with $M = 2r_0 + O(1)$, gives

$$\mathbb{P}\{r_\sigma > 2r_0 + 4\} \leq C \exp\{-cr_0 \log(2 + r_0)\}. \quad (7.2b)$$

The positive prefix overshoot is uniformly integrable by (4.10). The low, middle, and high events form a partition; (7.2b) makes the high probability vanish. Therefore

$$\mathbb{E}[F(X_\tau, T_\tau) - F(x, t)] \leq -a_0 \log(2 + V_0) + C. \quad (7.3)$$

If V_0 is bounded, (4.12) lets us choose a fixed M such that the prefix endpoint lies below M with probability at least $3/4$, uniformly in n . On that event (2.7) or (3.2) contributes $-a \log X + C$. On the complement, the positive part of the complete reward is uniformly integrable by (4.10) and source entropy. Thus

$$\mathbb{E}[F(X_\tau, T_\tau) - F(x, t)] \leq -a_1 \log X + C. \quad (7.4)$$

Equations (5.1), (7.3), and (7.4) permit one fixed sufficiently small $\eta > 0$ and give (1.5). Equation (4.10), the finite-jump source-entropy bound, and Hölder give (1.6). Every endpoint is physical and the physical duration is integrable.

The menu is finite: choose the witness, the simple B path, and tie ordering from the finite support. If uniform constants failed, extract a one-active subsequence with fixed actual mark, witness, path, and bounded reaction offsets. Sections 2–7 contradict the failure. This proves Theorem 1.1.

37.8 8. Exact scope and the remaining global interface

Theorem 1.1 is the cap-free one-active B/F0 contract. The qualitative part of Section 2 also proves the unconditional B/B contract with two subpower inactive coordinates: every actual target belongs to a B linkage, and (2.3) tends to zero. The two-active Q/U/C bridge and the same cap-free path argument prove the unconditional AA contract.

This note does not alone assert recurrence of every 6,654 failure pair. A final pair theorem must use one proper potential on complementary passing and all-active sequences, or prove an episode-level population boundary estimate lifting these marked episodes to that common potential. No terminal-chart exit, descriptor SCC, or unweighted drift-or-exit gluing is claimed here.

38 Recurrence of the 6,654 failure pairs under one common marked factorial

Proof-first composition candidate, 2026-08-12 PDT. This note proves a standalone pair theorem for the 6,654 support pairs in the exact outside-mixed 18,496-pair remainder which have at least one affine-feasible corrected-cut failure. A single proper marked factorial is used on every noninvariant scale. No descriptor exit is discarded, no terminal-chart circulation is invoked, and no finite cap graph is used.

Finite computation below certifies only support, tier, affine-feasibility, and set identities. Orientations, rate vectors, population states, and reaction histories are not enumerated.

38.1 1. Scope and exact finite inputs

Fix an ordered pair of disjoint nontrivial binary linkage supports from the 6,654-pair failed set in

```
src/outside_mixed_remaining_18496_certificate.py
SHA-256 314f378664052cabe23910e118c9a43acf99884ccb5c63b61daf014a206e4c63
```

```
tests/test_outside_mixed_remaining_18496_certificate.py
SHA-256 28d3cf0087bcd77e24d6dbfa280b226b34d3d026c35e743bc10487c829667769
```

The independent support-only audit has SHA-256

```
2539d8eee4d1d584ed5566f7494c7262843676dc368078647db12481d6a5822f.
```

The exact failure data are

pair set	pairs	
outside-mixed remainder	18,496	(1.1)
no affine-feasible cut failure	11,842	
at least one feasible failure	6,654	

and the 21,906 failed incidences split as

$$15,204 \text{ B/F0} \cup 3,618 \text{ B/B} \cup 3,084 \text{ AA.} \quad (1.2)$$

There are no failed all-active descriptors. The AA rows are exactly the two-active rows; B/F0 and B/B are exactly the one-active rows.

The literal global-nonmixing definition used by the one-active symbolic theorem has been replayed independently on all 18,496 pairs. The focused support-only certificate is

```
src/remaining_18496_globally_nonmixed_certificate.py
SHA-256 54e0a5c96c2fe5e0e54c48bbb91f0f2eccbd140d62bce39ed55df54dd5a486fb
```

```
tests/test_remaining_18496_globally_nonmixed_certificate.py
SHA-256 9228597561617ef92b93f6387a43cd954f534796e7de1b51f542613bfce06060
```

It checks all three active-coordinate pairs and all seven ordered primitive workload representatives. The exact result is 18,496 globally nonmixed pairs and zero violations. Its pair manifest is

```
eb7db151e42eb9562b1a1d519ea7dad212df52c6df368ffa08edbf79410db4ad
```

and its support-signature fingerprint is

```
216a07ec3265cd9c072ff6975235198b839ab44b9dfd832c51ff70dd304459c1.
```

Its independent exact-byte audit has SHA-256

```
b2e54902d95f4fa52bb3857acc2b4d5e7fa247138c21a1064aa06b40db140fc2.
```

The exact all-feasible-two-active AA identity is frozen at source/test SHA-256 values

```
25e5c2fce812d2e3d3f02c0c9377533cf16c68313b77bc0d1e7a485052bd68ee
2445edcbcf66ff2204c821baec455b4f8a416180360a39a6d757712af5fc22e0,
```

with exact-byte audit SHA-256

```
27bfc707721fab03d829d14ca764505694556dbf0d2fd325ea25aa8573b2beef.
```

It gives 1,140,984 affine-feasible two-active incidences, all AA, including all 3,084 failures.

38.2 2. The common marked chain

Fix arbitrary strongly connected orientations, arbitrary positive labelled rates, and one closed irreducible population class Γ . After every physical reaction carry its actual target t as a mark. Then $x \geq t$. Put

$$F(x, t) = \sum_{i=1}^3 \log((x_i - t_i)!), \quad W(x, t) = 1 + F(x, t). \quad (2.1)$$

Marks range over the finite binary complex set, so $W \geq 1$ is proper on the reachable marked state space. For a source y , write

$$K_y = \sum_{e:s(e)=y} \kappa_e, \quad p_y(x) = \frac{K_y(x)_y}{\sum_z K_z(x)_z}. \quad (2.2)$$

If the next reaction is $y \rightarrow u$, factorial cancellation gives

$$F(x - y + u, u) - F(x, t) = \log \frac{(x)_t}{(x)_y}. \quad (2.3)$$

Thus one ordinary all-clock jump has expected reward

$$D(x, t) = \log p_t - \sum_y p_y \log p_y - \log K_t + \sum_y p_y \log K_y \leq \log p_t + C_K. \quad (2.4)$$

For every fixed $m < \infty$, its positive reward has a uniform m -th moment. Indeed, on a positive sourcewise term with $R = (x)_t / (x)_y \geq 1$, one has $p_y \leq C/R$, and $R^{-1}(\log R)^m$ is bounded.

38.3 3. All-active sequences: unconditional corrected-cut reward

Take any all-active escaping marked sequence in Γ . Its descriptor is affine feasible. By (1.2), it passes the corrected S-superlevel cut. Let E be the global top S-tier, and choose the certified edge

$$e : y \longrightarrow z \quad (3.1)$$

whose source $y \in E$ and whose target is in a strictly lower D-tier. After a source-ratio subsequence,

$$p_y(x_n) \geq b > 0. \quad (3.2)$$

Take the next ordinary all-clock jump. Continue only if the labelled edge e fires; on that event, take one final ordinary jump from the actual marked endpoint $(x_n - y + z, z)$. Stop on a competitor at its actual endpoint. There is no structural-exit test.

Every coordinate diverges on the sequence. A bounded displacement cannot disable any source, and it preserves every strict tier comparison. Hence, for any $q \in E$,

$$p_z(x_n - y + z) \leq \frac{K_z(x_n - y + z)_z}{K_q(x_n - y + z)_q} \longrightarrow 0. \quad (3.3)$$

If a_e is the physical probability of the labelled edge, first-step conditioning and (2.4) give

$$J_n = D(x_n, t) + a_e(x_n)D(x_n - y + z, z) \longrightarrow -\infty. \quad (3.4)$$

This is the depth-two corrected-tier theorem with its former boundary-exit alternative removed for the all-active domain, where no such source loss is possible.

38.4 4. Two-active sequences: unconditional AA reward

Every affine-feasible two-active descriptor is Q/U/C available on both linkages by the exact certificate in Section 1. The cap-free actual-target AA theorem is frozen at

research_notes/proof_first_remaining_18496_two_active_unconditional_aa_and_cap_scc_obstruction.md
SHA-256 fd55879fe932e9389b6b468a00c046b59ec3c214ad08afdd0eb696b3f42844e3

with independent exact-byte audit SHA-256

2de8466793c774eaccfde4d0160c39dbafe49f581b68154a6a39419e5fd74a8a.

For completeness, its decisive endpoint identity is short. If t is the actual mark, the symbolic Q/U/C bridge supplies a same-linkage simple path from t to c and a comparison source q such that

$$q_b \leq c_b, \quad q \succ_D c, \quad (4.1)$$

where b is the bounded coordinate. At successful endpoint $x - t + c$, both active coordinates still diverge and $q_b \leq c_b$ makes q literally enabled. Thus $p_c(x - t + c) \rightarrow 0$. The exact finite Bellman recursion along the simple path makes the total common- W reward tend to minus infinity. All clocks compete, and cap changes are retained inside the same episode rather than treated as exits.

38.5 5. One-active sequences: the globally-nonmixed exhaustion

Let X be the unique active coordinate, so, after the exact logarithmic compactification,

$$X_n \longrightarrow \infty, \quad \log(1 + U_n) + \log(1 + V_n) = o(\log X_n). \quad (5.1)$$

Because the pair is globally nonmixed, the audited symbolic exhaustion in

research_notes/proof_first_one_active_no_mixed_exhaustion_repaired.md
SHA-256 9fb1828f5660ffae83e6e1a08a0cb33ce8bd2813d7394a90187d9bccc64895c4

applies. Its exact-byte audit SHA-256 is

6f7619e7696d1dfe3ab332746eea0be54899416dff3b3488f5ef53f43672e682.

It gives the following exhaustive support alternatives.

38.5.1 5.1 Quadratic branch

If a linkage contains $q = 2X$, it is the unique source with active degree two. Every other binary source has propensity $X^{1+o(1)}$, while $\lambda_q = \Theta(X^2)$. Hence $p_q \rightarrow 1$ and $p_t \rightarrow 0$ for every mark $t \neq q$; one ordinary jump is coercive by (2.4).

For $t = q$, choose a fixed nonself edge $e : q \rightarrow u$ supplied by strong connectivity. Take one ordinary jump and, only if e fires, one final jump. The edge probability stays bounded below, q remains enabled at the bounded-displacement endpoint, and $p_u(x - q + u) \rightarrow 0$. The exact identity

$$J = D(x, q) + a_e(x)D(x - q + u, u) \quad (5.2)$$

therefore tends to minus infinity. No cap or active-set exit is tested.

38.5.2 5.2 Flat and dormant invariant branches

If both linkages have constant X -degree on their supports, every reaction preserves X , so an escaping one-active sequence cannot lie in one fixed class.

The only dormant residue consists, up to exchanging U, V , of

$$L_D = \{0, X + U, X + V\} \quad (5.3)$$

paired with one of

$$\{U, V\}, \{2U, 2V\}, \{2U, U + V\}, \{2V, U + V\}, \{2U, 2V, U + V\}. \quad (5.4)$$

Every reaction preserves $H = X - U - V$. Equation (5.1) and constancy of H exclude $X \rightarrow \infty$ in the fixed class.

38.5.3 5.3 B/B and B/F0 branches

For a Bellman linkage choose q, c with

$$q_X = 1, \quad c_X = 0, \quad q_U \leq c_U, \quad q_V \leq c_V. \quad (5.5)$$

From an actual mark in that linkage, follow a same-linkage simple path to c , retaining every competing clock, and then take one final ordinary jump. The successful endpoint $x - t + c$ enables q , and

$$p_c(x - t + c) \leq \frac{C(1 + U + V)^2}{X} \rightarrow 0. \quad (5.6)$$

The finite Bellman recursion gives coercive common- W reward. This is unconditional for B/B and for a B/F0 state whose actual mark lies in the B linkage.

If the actual mark lies in the Flat0 linkage, use the cap-free killed resolvent theorem

research_notes/proof_first_remaining_18496_cap_free_bf0_killed_resolvent_theorem.md
SHA-256 5e8ce1d09c794014093bc9b84b9563f9348530acc741bb12b2c8446e2a560783

with independent exact-byte audit SHA-256

7ad5cc50c4e538c0e6e2fd119fc9f01bf29eb9f86c18faf7e1c9592c28145194.

If the B witness is X , it is immediate. Otherwise write it as $X + U$. Before access $U = 0$, and every nonabsorbed source and target is in $\{0, V, 2V\}$, with the actual target degree retained as part of the countable state. The killed marked weight

$$w_\theta(v, s) = ((v - s)!)^\theta \quad (5.7)$$

has strict multiplicative outer drift. Removing closed no-kill components makes the finite core a non-singular M-matrix. The resulting killed Green bound controls the exact relative marked-factorial overshoot and the pre-absorption maximum; physical duration has moments $O((1 + \log(2 + v))^m)$. A closed no-kill

component is a fixed- X one-species binary class and is positive recurrent directly. Otherwise the first access or B-linkage firing is included and its actual-target Bellman payoff appended. The completed estimate is

$$\mathbb{E}[\Delta W + \eta\tau] \longrightarrow -\infty \quad (5.8)$$

along every one-active B/F0 sequence, with uniform positive-reward moments and integrable actual endpoints.

38.6 6. State selection and random-time Foster recurrence

The preceding rules form a finite support-dependent menu: marks, simple paths, certified edges, active-coordinate choices, witnesses, and tie orders all range over finite sets. The Flat0 rule is one countable-state stopping algorithm, not an enumeration of its populations.

Include in the menu the ordinary one-jump rule, which is defined at every marked state. A longer template is declared applicable exactly when its literal first required source is enabled. Every later continuation begins only after the preceding labelled reaction has made its actual target the new mark, so the next prescribed source is then literally enabled. Thus the menu defines at least one positive-duration stopping rule at every state; no zero-time reclassification or asymptotic descriptor is used to define the finite-state dynamics.

Choose one common $\eta > 0$, smaller than all finitely many local values. At a marked state select an applicable rule minimizing its exact score

$$\mathbb{E}_{x,t}[W(X_\tau, T_\tau) - W(x, t) + \eta\tau], \quad (6.1)$$

with deterministic tie breaking. If no finite set K made the selected score at most -1 outside K , a violating sequence would escape by properness of W . Pass to a subsequence with fixed mark, active mask, tier type, source ratios, and support rule. It has one, two, or three active coordinates. Sections 3–5 give a score tending to minus infinity, unless the one-active sequence is excluded by a fixed-class invariant or the class is already positive recurrent by the closed Flat0 alternative. This is a contradiction. Hence

$$\mathbb{E}_{x,t}[W(X_\tau, T_\tau) - W(x, t) + \eta\tau] \leq -1 \quad ((x, t) \notin K). \quad (6.2)$$

Every episode contains at least one actual physical jump, every endpoint is the actual endpoint and target of its last included reaction, and every episode has integrable duration and integrable positive part of its endpoint increment. Complete an episode even if its interior visits K , solely for accounting. At successive episode endpoints S_j , conditional summation of (6.2), localized first to a finite W -sublevel, gives

$$\mathbb{E}W(X_{S_{n \wedge N}}, T_{S_{n \wedge N}}) + \eta \mathbb{E}S_{n \wedge N} + \mathbb{E}(n \wedge N) \leq W(x, t) + 1. \quad (6.3)$$

Here N is the first completed episode whose path visits K . Fatou and monotone convergence give finite expected accounting time and episode count. On nonhitting, the episode endpoints contain an infinite subsequence of genuine physical jump times; nonexplosion forces their times to diverge, contradicting (6.3). Thus the finite marked set K has finite mean hitting time.

From the finitely many physical states below K , irreducibility supplies finite labelled paths to one fixed state. Each has positive success probability and finite mean duration; geometric repetition, using the finite-mean return to K , gives a finite mean return to that state. The closed physical class is positive recurrent. Forgetting the target mark does not change the physical generator.

Binaryity separately gives nonexplosion: population-increasing channels have source degree at most one and aggregate rate $O(1 + |x|_1)$, while population-preserving quadratic clocks cannot accumulate inside a finite population sublevel.

Theorem 6.1 (the exact 6,654 failure-pair theorem). Every ordered support pair in the exact 6,654 failed subset of the outside-mixed 18,496-pair remainder, under every strongly connected orientation and every positive labelled rate vector, is nonexplosive and positive recurrent on each closed irreducible population class.

No orientation, rate vector, population box, or reaction history is enumerated. The finite certificates establish only the exact support and tier scopes used by the analytic alternatives above.

Part VII

Final two-linkage composition

39 Exact 18,496-pair outside-mixed recurrence

Proof-first pair composition, 2026-08-12 PDT. This note composes two standalone fixed-pair recurrence theorems over the exact outside-mixed 18,496-pair remainder. The support split is finite; no stochastic path ever switches between the two branches because its support pair is fixed.

39.1 1. Exact disjoint split

The support and affine-descriptor certificate is frozen at

```
certificate 314f378664052cabe23910e118c9a43acf99884ccb5c63b61daf014a206e4c63
test       28d3cf0087bcd77e24d6dbfa280b226b34d3d026c35e743bc10487c829667769
```

with independent exact-byte audit SHA-256

```
2539d8eee4d1d584ed5566f7494c7262843676dc368078647db12481d6a5822f.
```

It proves the literal disjoint identity

$$\mathcal{R}_{18496} = \mathcal{N}_{11842} \dot{\cup} \mathcal{F}_{6654}, \quad (1.1)$$

where $\mathcal{N}_{11842}$ has no affine-feasible corrected-cut failure and every member of $\mathcal{F}_{6654}$ has at least one. The three pair fingerprints are

```
R18496 eb7db151e42eb9562b1a1d519ea7dad212df52c6df368ffa08edbf79410db4ad
N11842 b425db9040d0836462f4240a4a3acf51d067d356eb4f2bfe4ce2cf648e42db26
F6654 036f9cb8f00f99f78be9cb6c2303208a8ca8b25be8c1bd350b8fac6b35582eed
```

Finite computation certifies only support, tier, affine-feasibility, and set identities. It enumerates no orientation, rate vector, population state, reaction history, or communicating class.

39.2 2. The 11,842 no-failure pairs

The statewide population-Foster theorem target has SHA-256

```
b26742cfe24d40df31da01217050c4141b1955dfaf267e0e2c804000df8ae06b
```

with independent exact-byte audits

```
685206db5ea0e752d1ea812d9bd9dcb8ac250838bc61ee351a84c4111ab0925a
11457192af170d639df1872f44ebb7a5e0914ab41f0aba5984add416118d9b3f.
```

Its load-bearing top-S/global-top-D support identity is frozen at

```
certificate e72da319fd49df6df0c34a9a3090d95f2baf30199c0464966f9081990e347b4a
test       ea8d042875c0c4f5c47bdabdaddc913e70138db7becedddafbe72c171ea1b572
```

and checks 3,010,738 affine-feasible descriptor incidences with zero cut failure, zero empty top S-tier, and zero case in which the top S-tier lies below the global top D-tier.

For each fixed pair, arbitrary strong orientations, positive rates, and a closed irreducible class, the population factorial entropy has statewide generator drift tending to minus infinity along every escaping sequence. The fourth-power lift absorbs all Taylor remainders. This is a direct population Foster theorem and contains no marked-chart or boundary-seam argument.

39.3 3. The 6,654 failure pairs

The complementary common-marked theorem target has SHA-256

69521a82eb05f1287fa43a9224e11e94c6bf0245720225bc777e1f7572144e58

with independent exact-byte audit SHA-256

25c4c8878a1246c04b4979bba3f51742272076774d4788375e0889b7b579b98f.

That theorem uses one actual-target marked factorial on every noninvariant scale. All-active corrected-cut episodes are unconditional because every coordinate diverges; every two-active descriptor uses the unconditional AA actual-target theorem; and the exact globally-nonmixed bridge makes the one-active Q/B/B/F0/invariant exhaustion complete. The Flat0 actual-mark case is closed by its cap-free killed resolvent. The finite state-selected menu therefore yields one random-time Foster inequality under the common marked potential, with no chart exit, cap graph, or weighted seam.

39.4 4. Pairwise composition and nonexplosion

Fix any pair $P \in \mathcal{R}_{18496}$. Identity (1.1) assigns P to exactly one of Sections 2 and 3. Apply that completed pair theorem. No potential comparison between the two branches is needed: a physical network has one fixed support pair and never changes its membership in (1.1).

A population-increasing binary reaction has source degree at most one, so all increasing jumps have aggregate rate $O(1 + |x|_1)$ and bounded size. A linear pure-birth comparison prevents finite-time population escape. Inside a bounded-population sublevel there are finitely many states and a bounded total hazard, so population-neutral quadratic clocks cannot accumulate. Thus every network in the theorem is nonexplosive.

Theorem 4.1. Every ordered support pair in the exact outside-mixed 18,496-pair remainder, under every strongly connected orientation and every positive labelled rate vector, is nonexplosive and positive recurrent on each closed irreducible population class.

This conclusion is the disjoint union of two standalone pair theorems, not an inference from finite counts or a composition of local chart exits.

40 Positive recurrence for every three-species binary two-linkage network

Proof-first final two-linkage composition, 2026-08-12 PDT. This note proves the complete two-linkage theorem by a disjoint union of completed fixed-pair theorems. The finite classification identifies support pairs only. It does not enumerate orientations, rate vectors, populations, reaction histories, or communicating classes.

40.1 1. Theorem

Theorem 1.1. Let a finite stochastic mass-action network have at most three species, exactly two weakly reversible linkage classes, and complexes of total molecularity at most two. Give each linkage support an arbitrary strongly connected labelled orientation and every present label an arbitrary positive rate constant. Then the minimal CTMC is nonexplosive, and every closed irreducible population class is positive recurrent.

The conclusion applies in particular to any reduced network supplied by the exact fixed-class conjugacy of the global composition theorem, provided that the reduction has at most three dynamic species and exactly two active projected linkage classes.

40.2 2. Exact support universe and final disjoint union

After zero-displacement labels are deleted and projected linkages sharing a complex are merged, the two active linkage supports are disjoint subsets of

$$\mathcal{C}_2 = \{0, A, B, C, 2A, 2B, 2C, A + B, A + C, B + C\}, \quad (2.1)$$

each having at least two vertices. The exact ordered support universe has

$$3^{10} - 2\{2^{10} + 10 \cdot 2^9\} + \{1 + 10 + 10 + 90\} = 46,872 \quad (2.2)$$

pairs.

The final finite certificate is

```
certificate 5b249ded4b54801f7eb5ab9ced943ed566216e1228c0e07f3e205b1eef319288
test       dd51ce074aa43bb4722d176ef4c85face956c924150681d5cae32f3b615c5e76
```

It proves the pairwise-disjoint identity

$$46872 = 27462 + 432 + 146 + 336 + 18496. \quad (2.3)$$

The five branches and fingerprints are

branch	pairs	SHA-256
completed mixed orbit	27462	1bf337cf143c6eb4...
active-invariant orbit gap	432	5516d6071b2b9d07...
strict positive invariant	146	d1fc7112f8a08605...
level-set residual	336	ea3d7b08d39c6f9c...
outside-mixed remainder	18496	eb7db151e42eb956...

(2.4)

The full-universe fingerprint is

```
00446e17dca5ce6b75e86cdc755b5660d7c94b68fa4f3e6f028efa40d02c6c60
```

and the branch-manifest fingerprint is

```
bd6ae54bff3aed8fc4fedb9255fe0b7377a28dc67404d6a5bea41c6aa4ac1bba.
```

The code asserts `recurrence_claim=False`; every stochastic conclusion below comes from an analytic theorem.

40.3 3. Analytic branch theorems

Fix one ordered support pair P , one arbitrary pair of strongly connected labelled orientations, one positive rate vector, and one closed irreducible class Γ . Identity (2.3) assigns P to exactly one branch.

40.3.1 3.1 Completed mixed orbit: 27,462 pairs

The exact theorem target and its exact-byte audit have SHA-256 values

```
target a91e8c31f35312ef4b9063e8f5a48af534861145db2236e662ea6cc1eff8e30e
audit  32eec768b2d8d701664f3ace2b1a7c04fd3790a4811eba5e05d56a8fa903e73b
```

with finite seed-partition source/test hashes

```
57d8904dd86cd0bf626e344dbfd7b7f248b239cdeaace48489796058c6875f08
e708c52f6cbc1bbc4dabf33f246d72379dff58c74cc97a38fdd8076ac3ae7d13.
```

It removes the 110 active-only invariant seeds before orbit closure. The remaining 5,059 seeds split into finite invariant, deficiency-zero, literal physical-seam, corrected-tier, and fourteen-row residual theorems. Species permutation and linkage reversal transfer these completed seed theorems to exactly 27,462 pairs. Thus this branch does not inherit the stale atlas claim which treated active-only invariance as global.

40.3.2 3.2 Active-invariant orbit gap: 432 pairs

The seam-free population-Foster theorem target and audit have SHA-256 values

```
target 7edab78daab7e492851efe5326ccc228adfc57f02cd5ff55eaa7056e034c8
audit  1110efc0760ed8714fc4bf203739152820f6f9a18cbdc0e92716638a707140fd
```

It uses one global fourth-power population factorial and the exact 174 deficiency-zero, 234 direct-cut, and 24 exceptional-service split. No chart-exit SCC or potential handoff appears.

40.3.3 3.3 Strictly positive invariant: 146 pairs

For this branch a vector $h \in \mathbb{R}_{>0}^3$ annihilates every reaction vector. Hence $h \cdot x$ is constant on Γ , whose nonnegative integer level set is finite. Therefore Γ is finite and positive recurrent. This is a standalone classwise conclusion, unlike an invariant positive on only two coordinates.

40.3.4 3.4 Level-set residual: 336 pairs

The exact pair theorem and audit have SHA-256 values

```
target 6e9ddcacc03fe64b1c6a57cbaef052e984eaf7b7e2e87c4df52ca1240787a6c
audit  35b18c365ce954594397b4c48ed55f7d11c847af37594f0fb354517434f76d72
```

The 312 homogeneous and 24 anisotropic pairs are proved by classwise physical-time Foster theorems, including the boundary activation and direct death occupation macros. This branch has no chart-local linear-potential seam.

40.3.5 3.5 Outside-mixed remainder: 18,496 pairs

The exact theorem target and audit have SHA-256 values

```
target e7b08be8b6ca3ff604f3975bdae18b526db532ea1168f25bf21170d8248b5106
audit  192dfc3d79401c57416b582b45aeb0140f0c1ad3e0f90ab80acaae48e3b9a090
```

Its disjoint support split is 11,842 no-failure pairs plus 6,654 failure pairs. The first set has a statewise population-factorial Foster theorem. The second has one common actual-target marked factorial covering all-active, unconditional two-active AA, and the exact globally-nonmixed one-active exhaustion, including the cap-free Flat0 killed resolvent. Each is already a standalone fixed-pair recurrence theorem, so their union requires no cross-potential trajectory gluing.

40.4 4. Pairwise composition

The support pair P is fixed for the entire physical network. Therefore the five branches in (2.3) are a classification of *theorems*, not a menu of regions visited by one trajectory. Apply the one completed theorem assigned to P . Potential functions belonging to different support pairs or different branches are never compared.

Every branch theorem gives finite mean return to a finite subset of Γ . From that finite subset, irreducibility supplies finite labelled paths to one fixed state with positive success probabilities and finite mean durations. Geometric retry gives a finite mean positive return to that state. Hence Γ is positive recurrent.

40.5 5. Nonexplosion

Let $N(x) = 1 + |x|_1$. A reaction with a degree-two source cannot increase total population because its target is also binary. All population-increasing channels therefore have source degree at most one and bounded jump size. For a fixed network there is $C < \infty$ such that

$$\sum_r \lambda_r(x) (|z_r| - |y_r|)^+ \leq CN(x). \quad (5.1)$$

Stopping when N reaches R , Dynkin's formula and Gronwall give

$$\mathbb{E}_x N(X_{t \wedge \tau_R}) \leq N(x)e^{Ct}. \quad (5.2)$$

Thus population cannot escape in finite time. Below a fixed population level the state space is finite and total rates are bounded, so neutral quadratic clocks cannot accumulate. The minimal CTMC is nonexplosive.

Sections 2–5 prove Theorem 1.1. $\square$

40.6 6. Claim boundary

This theorem supersedes the stale 2,511-based global composition target; it does not alter that target or its hostile FAIL audit, which remain as provenance. The new proof uses the exact full 46,872-pair universe and cites only standalone pair theorems at their literal support scopes. No orientation, population, rate, or stochastic-history enumeration enters any analytic implication.